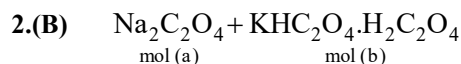
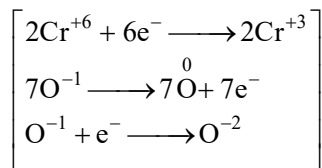
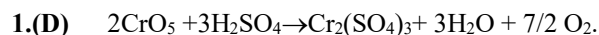




## Solutions - Advanced Problem Package | Chemistry

### Stoichiometry-I & II



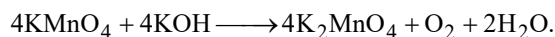
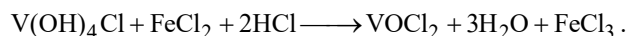
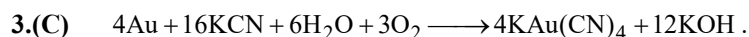
With NaOH : eq of  $\text{Na}_2\text{C}_2\text{O}_4$  + eq of  $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$  = eq of  $\text{KMnO}_4$

$$2a + 4b = 5 \times 0.2 \times V \quad \dots\dots(i)$$

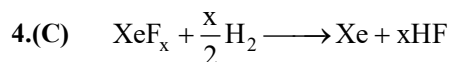
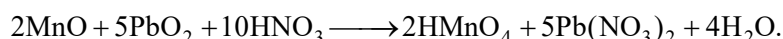
With  $\text{KMnO}_4$  : gmeq of  $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$  (n = 3 as acid) = gmeq of NaOH

$$3b = 1 \times 0.2 \times V \quad \dots\dots(ii)$$

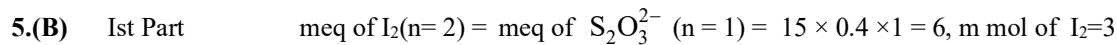
$$\text{Divide (i)/(ii): } \frac{2a + 4b}{3b} = 5 \quad \Rightarrow \quad \frac{a + 2b}{3b} = \frac{5}{2} \quad \Rightarrow \quad \frac{a}{b} = \frac{11}{2}$$



In above reaction out of 4 mol of  $\text{O}^{-2}$  present in  $\text{KOH}$  (R.A) , two are oxidised to  $\text{O}_2$  and other two moles remained as such and went to  $\text{H}_2\text{O}$ .

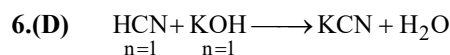


$$\text{m mol of Xe} = \frac{1000}{22400} \times 22.4 = 1 \quad ; \quad \text{m mol of HF (n = 1)} = 6 \Rightarrow x = 6$$

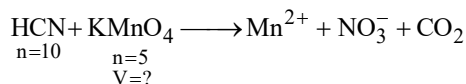


$$\text{m mol of } \text{I}_2 \text{ (n = 2) reacted with NaOH} = \frac{24}{2} = 12$$

$$\text{Total m mol of } \text{I}_2 = 3 + 12 = 15 \quad ; \quad [\text{I}_2] = \frac{15}{150} = 0.1$$



m mol of HCN = meq of HCN = meq of KOH = 100

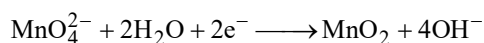
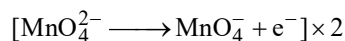


As HCN used in same for  $\text{KMnO}_4$  as is used for KOH.

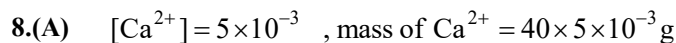
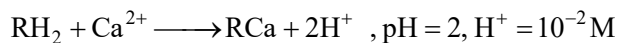
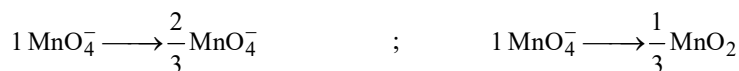
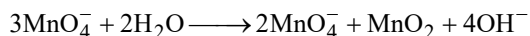
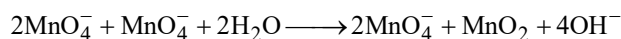
m mol of HCN ( $\text{n} = 1$ ) = 100.

meq of HCN ( $\text{n} = 10$ ) =  $100 \times 10 = 1000$

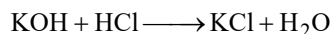
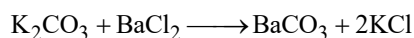
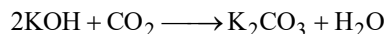
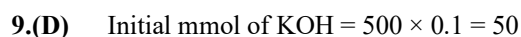
meq of HCN = meq of  $\text{KMnO}_4 \Rightarrow 1000 = 5 \times 5 \times V : \Rightarrow V = 40 \text{ mL}$



On addition, we get :



Hardness in ppm of  $\text{Ca}^{2+} = \frac{40 \times 5 \times 10^{-3}}{10^3} \times 10^6 = 200$



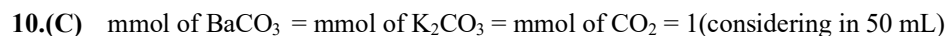
mmol of HCl =  $30 \times 0.1 = 3$  ; mmol of KOH left unreacted in 50 mL solution =  $(50 \times 0.1) - 3 = 2$

millimoles of KOH in 50 mL reacted with  $\text{CO}_2 = 2$  ; millimoles of KOH in 500 mL reacted with  $\text{CO}_2 = 20$

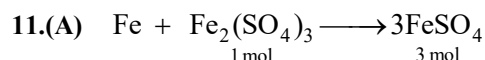
2 mol KOH = 1 mol  $\text{CO}_2$  ; 20 mmol KOH = 10 mmol  $\text{CO}_2$

Volume of  $\text{CO}_2 = 10 \times 10^{-3} \times 22.4 \text{ L}$

ppm of  $\text{CO}_2 = \frac{10 \times 10^{-3} \times 22.4}{224} \times 10^6 = 1000 \text{ ppm}$



$\frac{m_{\text{BaCO}_3}}{197} \times 1000 = 1, \quad m_{\text{BaCO}_3} = \frac{197}{1000} = 0.197 \text{ g}$



1 mol  $\text{Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O} \equiv 1 \text{ mole Fe}_2(\text{SO}_4)_3$

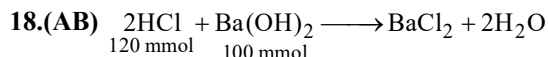
$\frac{2.41}{964} \text{ mol Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 24\text{H}_2\text{O} \equiv \frac{2.41}{964} \text{ mole Fe}_2(\text{SO}_4)_3 \equiv \frac{3 \times 2.41}{964} \text{ mol FeSO}_4$

moles of  $\text{FeSO}_4 = 0.0075$



In basic medium :  $\text{MnO}_4^- \longrightarrow \text{MnO}_4^{2-}$  ( $n = 1$ )

$\text{H}_2\text{O}_2 \longrightarrow \text{O}_2$  ( $n = 2$ )



L.R is HCl (Also 20 mL of water is also added)

mmol  $\text{Ba}(\text{OH})_2$  left =  $(100 - 60) = 40$

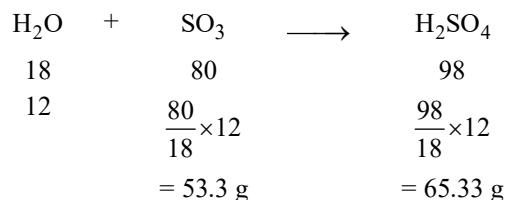
mmol of  $\text{OH}^-$  in excess =  $40 \times 2 = 80$

$$[\text{OH}^-] = \frac{80}{100} = 0.8 \quad ; \quad \text{mmol of BaCl}_2 = 60 \quad ; \quad \text{mmol of Cl}^- = 120$$

$$[\text{Cl}^-] = \frac{120}{100} = 1.2\text{M} \quad ; \quad \text{mmol of Ba}(\text{OH})_2 \text{ left} + \text{mmol of BaCl}_2 = 40 + 60 = 100$$

$$[\text{Ba}^{2+}] = \frac{100}{100} = 1$$

19.(ABD)



$(100 - 53.3) = 46.7$  g of  $\text{H}_2\text{SO}_4$  is initially present and 65.33 g of  $\text{H}_2\text{SO}_4$  is newly formed on add of 12 g of water.

$\text{H}_2\text{SO}_4$  total =  $65.33 + 46.7 = 112$  g

But on addition of 9 g of water, moles of free  $\text{SO}_3 = \frac{12}{18} - \frac{9}{18} = \frac{1}{6}$

Addition of 9 g of water will produce  $\frac{98}{18} \times 9 = 49$  g of newly formed  $\text{H}_2\text{SO}_4$ .

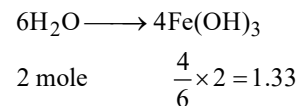
Total  $\text{H}_2\text{SO}_4 = 49 + 46.7 = 95.7$

20.(BD) 2 mol of  $\text{HCO}_3^-$  will make one mole each of  $\text{CaCO}_3$ ,  $\text{CaCl}_2$  and  $\text{MgCl}_2$  to have equal hardness

ppm of  $\text{HCO}_3^- = 61 \times 2 = 122$ g in  $10^6$  mL of  $\text{H}_2\text{O}$

1 mol of  $\text{CaCO}_3 = 100$  ppm ;      1 mol of  $\text{CaCl}_2 = 111$  ppm ;      1 mol of  $\text{MgCl}_2 = 95$  ppm

21.(AB)  $\text{H}_2\text{O}$  is L.R.



22.(AB) Volume strength =  $5.6 \times N$

$$N_f V_f = N_1 V_1 + N_2 V_2 + N_3 V_3$$

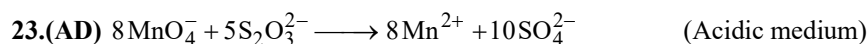
0.5 L of each solution and 1.5 L of water makes final solution 3L

$$N_{\text{final}} \times 3 = \frac{10}{5.6} \times 0.5 + \frac{15}{5.6} \times 0.5 + \frac{20}{5.6} \times 0.5$$

$$N_f = 1.34$$

Volume strength of final solution =  $1.34 \times 5.6 = 7.5$

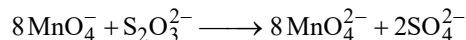




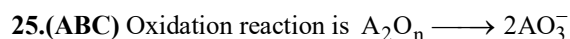
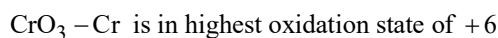
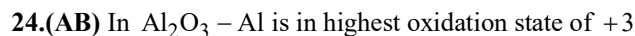
$$\therefore 1 \text{ mole } \text{MnO}_4^- \text{ oxidises } \frac{5}{8} \text{ mole } \text{S}_2\text{O}_3^{2-}$$

Also, oxidation number of S changes from +2 to +6

In basic medium,

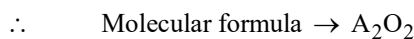


Change in medium from acidic to basic will change the nature of product & stoichiometry of reaction.



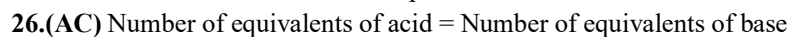
$$1.34(10 - 2n) = 32.2 \times 0.05 \times 5$$

$$n = 2$$



Since oxidation state of A changes from +2 to +5,

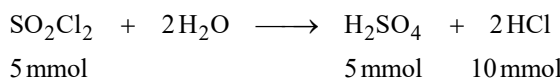
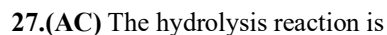
$\therefore$  It cannot be a Group-II metal



$$\therefore \frac{1.25}{E_{\text{acid}}} \times 1000 = 25 \times 0.25 \times 2 \Rightarrow E_{\text{acid}} = 100 \text{ g}$$

If n-factor = 2,  $M_0 = 200 \text{ g}$

n-factor = 1,  $M_0 = 100 \text{ g}$



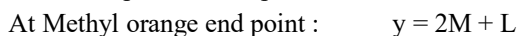
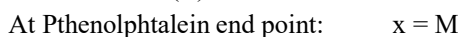
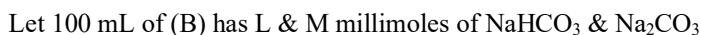
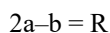
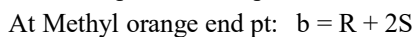
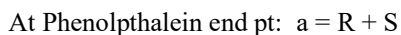
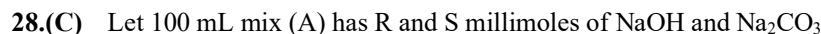
$$M_{\text{H}_2\text{SO}_4} = \frac{5 \text{ mmol}}{100 \text{ mL}} = 0.05 \text{ M}$$

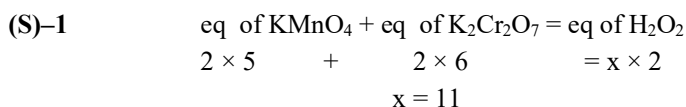
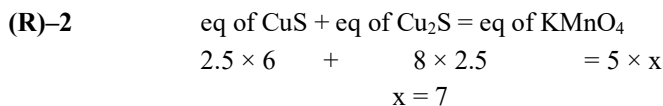
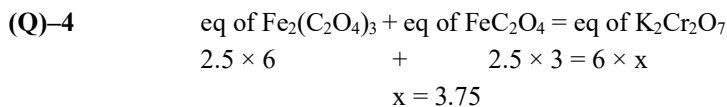
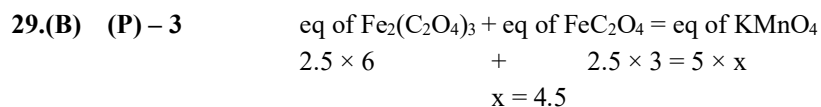
$$M_{\text{HCl}} = \frac{10 \text{ mmol}}{100 \text{ mL}} = 0.10 \text{ M}$$

10 mL stock solution has

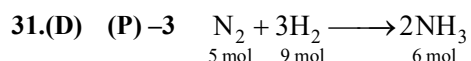
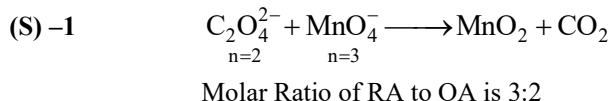
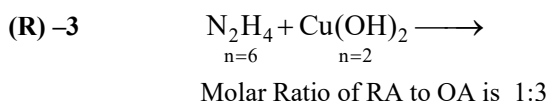
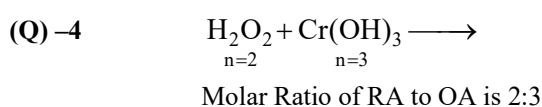
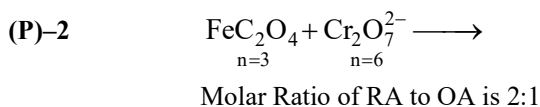
$$= \frac{10 \times 0.05 \times 2}{\text{H}_2\text{SO}_4} + \frac{10 \times 0.10}{\text{HCl}} = 2 \text{ meq of acid}$$

$$\therefore \text{Volume of NaOH req.} = \frac{2}{0.2} = 10 \text{ mL}$$

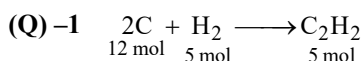




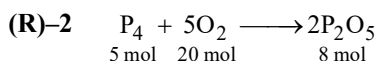
30.(C) Reciprocal of n-factor is the molar ratio.



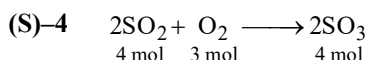
L.R is  $\text{H}_2$ , % yield is 80. Therefore, moles of  $\text{NH}_3$  produced,  $\frac{x}{6} \times 100 = 80$ ,  $x = 4.8$



L.R is  $\text{H}_2$ , % yield is 70. Therefore, moles of  $\text{NH}_3$  produced,  $\frac{x}{5} \times 100 = 70$ ,  $x = 3.5$

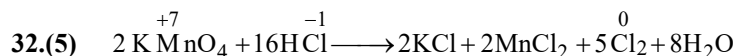


L.R is  $\text{O}_2$ , % yield is 50. Therefore, moles of  $\text{P}_2\text{O}_5$  produced,  $\frac{x}{8} \times 100 = 50$ ,  $x = 4$



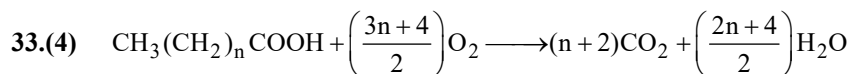
L.R is  $\text{SO}_2$ , % yield is 75.

Therefore, moles of  $\text{SO}_3$  produced,  $\frac{x}{4} \times 100 = 75$ ,  $x = 3$



2 mole of  $\text{KMnO}_4$  gained  $10e^-$ .

Therefore 10 mol of  $\text{HCl}$  has lost  $10e^-$ . For this to happen 10 mole of  $\text{HCl}$  is acting as reducing agent. So for per mole of  $\text{KMnO}_4$  5 mole of  $\text{HCl}$  as RA is required.

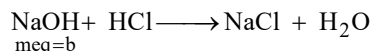
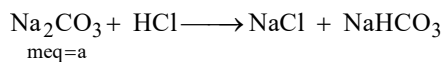


1 mole of acid =  $(n+2)$  mol of  $\text{CO}_2$ .

$$\frac{0.58}{60+14n} \text{ mol of acid} = \frac{n+2}{(60+14n)} \times 0.58 \text{ mol of CO}_2$$

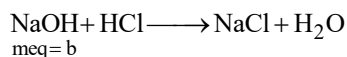
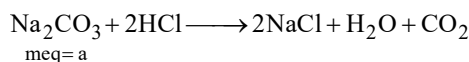
This CO<sub>2</sub> is passed through excess of NaOH where CO<sub>2</sub> is converted to Na<sub>2</sub>CO<sub>3</sub> and some NaOH is left.

With Phenolphthalin



$$\frac{a}{2} + b = 50 \times 1 \quad \dots\dots\dots (i)$$

With methyl orange



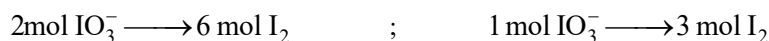
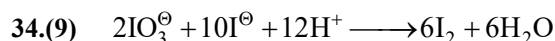
$$a + b = 80 \quad \dots\dots\dots (ii)$$

From (i) to (ii)

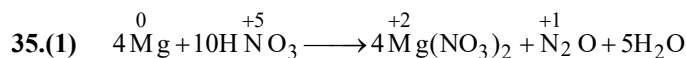
$$a(\text{Mg of Na}_2\text{CO}_3) = 60, b(\text{meq of NaOH left}) = 20$$

$$\text{meq of Na}_2\text{CO}_3 = \text{meq of CO}_2$$

$$60 = \frac{(n+2)0.58}{60+14n} \times 2 \times 10^3 = n = 4$$



$$\% \text{ yield} = \frac{0.27}{3} \times 100 = 9$$

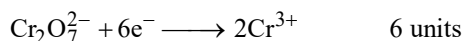
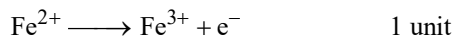
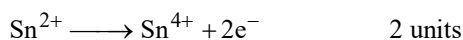


4 mole of Mg required = 2 mole of HNO<sub>3</sub> as RA

2 mole of Mg required = 1 mole of HNO<sub>3</sub>

36.(0.82) Reaction

Change in ON



$$\text{Thus, } 0.04 \text{ M Sn}^{2+} = 0.08 \text{ N Sn}^{2+}$$

$$x \text{ M Fe}^{2+} = x \text{ N Fe}^{2+}$$

$$0.125 \text{ M Cr}_2\text{O}_7^{2-} = 0.125 \times 6 = 0.750 \text{ N Cr}_2\text{O}_7^{2-}$$

$$\text{Equivalent of Fe}^{2+} + \text{Equivalent of Sn}^{2+} = \text{Equivalent of Cr}_2\text{O}_7^{2-}$$

$$\frac{15 \times 0.08}{1000} + \frac{15x}{1000} = \frac{18 \times 0.750}{1000}$$

$$1.2 + 15x = 13.5 \Rightarrow 15x = 12.3 \Rightarrow x = 0.82 \text{ M}$$

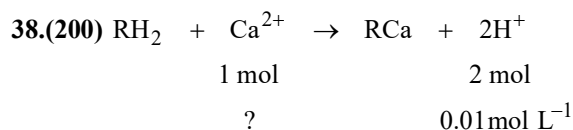
37.(1.75) Fe<sup>2+</sup> is 1.0 ppm

Thus, 10<sup>6</sup> g solution has Fe<sup>2+</sup> = 1 g

$$250 \text{ mL} (= 250 \text{ g}) \text{ solution has Fe}^{2+} = \frac{250}{10^6} \text{ g} \quad (\text{Note } d = 1 \text{ g/mL})$$

$$56 \text{ g of Fe}^{2+} \text{ is in } = 392 \text{ g salt}$$

$$\therefore \frac{250}{10^6} \text{ g of Fe}^{2+} \text{ is in } = \frac{392}{56} \times \frac{250}{10^6} \text{ g} = 1.75 \times 10^{-3} \text{ g}$$



When  $[\text{H}^+]$  is 2 mol then,  $[\text{Ca}^{2+}]$  is = 1 mol

When  $[\text{H}^+]$  is 0.01 mol then  $[\text{Ca}^{2+}]$  is =  $\frac{0.01}{2} \text{ mol L}^{-1}$

$$\therefore [\text{Ca}^{2+}] = 0.005 \text{ M} = 0.005 \times 40 \text{ g L}^{-1} = 0.2 \text{ g L}^{-1}$$

1000 mL of water has  $[\text{Ca}^{2+}] = 0.2 \text{ g}$

$$10^6 \text{ mL (parts per million) has } [\text{Ca}^{2+}] = \frac{0.2 \times 10^6}{10^3} = 200$$

**39.(3.12)** Given reaction is not balanced

Based on stoichiometry

1 mol  $\text{Na}_2\text{CO}_3$  is from 2 mol  $\text{NaHCO}_3$

2 mol  $\text{NaHCO}_3$  is from 2 mol  $\text{NH}_4\text{HCO}_3$

2 mol  $\text{NH}_4\text{HCO}_3$  is from 2 mol  $\text{NH}_3$

Thus, 1 mol  $\text{Na}_2\text{CO}_3$  is form = 2 mol  $\text{NH}_3$

106 g  $\text{Na}_2\text{CO}_3$  is from =  $2 \times 17 \text{ g NH}_3$

34 g  $\text{NH}_3$  gives = 106 g  $\text{Na}_2\text{CO}_3$

$$\text{Thus, } 1000 \text{ g NH}_3 \text{ gives } = \frac{106 \times 1000}{34} \text{ g Na}_2\text{CO}_3 = 3118 \text{ g Na}_2\text{CO}_3 = 3.118 \text{ kg Na}_2\text{CO}_3$$

**40.(44.25)** 2 moles  $\text{Na}_2\text{CO}_3$  give = 4 moles  $\text{NaNO}_2$

0.3375 mole  $\text{Na}_2\text{CO}_3$  gives

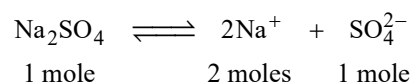
$$= 2 \times 0.3375 = 0.6750 \text{ mol NaNO}_2$$

$$= 0.6750 \times 69 \text{ g NaNO}_2$$

$$= 46.575 \text{ g NaNO}_2 \text{ with } 100\% \text{ yield}$$

$$= 46.575 \times 0.95 \text{ g NaNO}_2 \text{ with } 95\% \text{ yield} = 44.2462 \text{ g}$$

**41.(1.104)**



$$\text{Na}_2\text{SO}_4 = 6.0 \text{ micromoles} = 6.0 \times 10^{-6} \text{ mole}$$

$$\therefore \text{Na}^+ \text{ ions} = 2 \times 6.0 \times 10^{-6} \text{ mole} = 2 \times 6.0 \times 10^{-6} \times 23 \text{ g}$$

In 250 mL solution (250 g  $\text{H}_2\text{O}$  being dilute solution)

$$250 \text{ g H}_2\text{O} \text{ has Na}^+ \text{ ions} = 2 \times 6.0 \times 10^{-6} \times 23 \text{ g}$$

$$10^6 \text{ g (per million) H}_2\text{O} \text{ has Na}^+ \text{ ions} = \frac{2 \times 6.0 \times 10^{-6} \times 23 \times 10^6}{250} = 1.104 \text{ g} = 1.104 \text{ ppm}$$

## Solutions - Advanced Problem Package | Chemistry

### Atomic Structure

1.(B) As we know at radial nodes

$$\psi_{2s}^2 = 0 \Rightarrow 2 - \frac{r}{a_0} = 0 \Rightarrow r = 2a_0$$

2.(C)  $n_1 \rightarrow n_2;$                    $n_2 \rightarrow 2$

$$10 \cdot 2 \text{ eV} \quad 17 \cdot 0 \text{ eV}$$

$n_1 \rightarrow n_3;$                    $n_3 \rightarrow 3$

$$4 \cdot 25 \text{ eV} \quad 5.95 \text{ eV}$$

$$\text{For } n_1 \rightarrow 2 \quad \Delta E_1 = 13 \cdot 6 \times Z^2 \times \left( \frac{1}{2^2} - \frac{2}{n_1^2} \right) = 10 \cdot 2 + 17$$

$$\text{For } n_1 \rightarrow 3 \quad \Delta E_2 = 13 \cdot 6 \times Z^2 \times \left( \frac{1}{3^2} - \frac{2}{n_1^2} \right) = 4 \cdot 25 + 5 \cdot 95$$

Use subtraction to eliminate  $\frac{1}{n_1^2}$  & get  $Z = 3$

3.(B) First line is  $3 \rightarrow 2$ .

$$\Rightarrow h\nu_0 = 13.6 \cdot (1)^2 \left( \frac{1}{2^2} - \frac{1}{3^2} \right)$$

For  $\text{He}^+$

$$h\nu = 13.6 \times 2^2 \left( \frac{1}{2^2} - \frac{1}{3^2} \right) \quad \text{Hence } \nu = 4\nu_0$$

4.(A) Angular Momentum:  $mvr = \frac{nh}{2\pi}$

$$3.1652 \times 10^{-34} = \frac{n \times 6.6 \times 10^{-34}}{2\pi}$$

$$n = 3$$

$$\therefore \bar{\nu} = R \cdot Z^2 \cdot \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\bar{\nu} = R \cdot 1^2 \left( \frac{1}{1^2} - \frac{1}{3^2} \right) \Rightarrow R \left( \frac{8}{9} \right)$$

5.(A) For H :  $\Delta E_{3 \rightarrow 1} = E_3 - E_1 = 13.6 \left[ \frac{1}{1} - \frac{1}{9} \right] \text{ eV} \quad \dots(1)$

For  $\text{He}^+$  :  $\Delta E = 13.6 \times 4 \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ eV} = 13.6 \times 4 \left[ \frac{1}{4} - \frac{1}{36} \right] \text{ eV} \quad \dots(2)$

On comparison :  $n_1 = 2, n_2 = 6$

6.(D) Check by option. Only  $4 \rightarrow 1$  overall transition satisfies all the three condition  $4 \rightarrow 1$  can be broken into

(i)  $\frac{4 \rightarrow 2}{\text{Balmer}}; \frac{2 \rightarrow 1}{\text{Lyman}}$

(ii)  $\frac{4 \rightarrow 3}{\text{Paschen}}; \frac{3 \rightarrow 2}{\text{Balmer}}; \frac{2 \rightarrow 1}{\text{Lyman}}$

(iii)  $\frac{4 \rightarrow 3}{\text{Paschen}}; \frac{3 \rightarrow 1}{\text{Lyman}}$

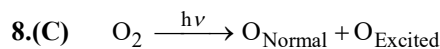
$$7.(A) \quad E_n = -\frac{13.6}{n^2} \text{ eV}; E_2 = -\frac{13.6}{2^2}; E_4 = -\frac{13.6}{4^2} \text{ eV/atom}$$

$$\Delta E = E_4 - E_2 = 2.55 \text{ eV}$$

Absorbed energy = work function of metal +  $KE_{\text{max}}$

$$2.55 = 2 + KE_{\text{max}}$$

$$KE_{\text{max}} = 0.55 \text{ eV}$$



Energy required for simple dissociation of  $O_2$  into the normal atoms  $498 \times 10^3 \text{ J/mol}$

$$= \frac{498 \times 10^3}{6.023 \times 10^{23}} \text{ J/mol} = 82.68 \times 10^{-20} \text{ J/mol}$$

If one atom in excited state has more energy i.e.  $2.5 \text{ eV}$

$$= 2.5 \times 1.602 \times 10^{-19} = 4 \times 10^{-19} = 40 \times 10^{-20} \text{ J}$$

$$\text{Total energy} = (82.68 \times 10^{-20}) + (40 \times 10^{-20}) \text{ J} = 122.68 \times 10^{-20} \text{ J}$$

$$\therefore E = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{122.68 \times 10^{-20}} = 163 \text{ nm}$$

$$9.(B) \quad \frac{1}{\lambda} = R(z)^2 \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \Rightarrow \frac{1}{x} = R(2)^2 (1-0)$$

$$\frac{1}{y} = R(3)^2 \left( \frac{1}{4} - 0 \right) \Rightarrow y = \frac{16}{9} x \text{ \AA}$$

$$10.(B) \quad \text{Using formula } \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2} = \text{number of spectral lines}$$

$$\lambda_{\text{min}} = \Delta E_{\text{max}} \Rightarrow 4 \rightarrow 1 \text{ transition}$$

$$\frac{1}{\lambda} = RZ^2 \left( \frac{1}{1^2} - \frac{1}{4^2} \right)$$

$$\lambda_{\text{min}} = \frac{4}{15R}$$

$$11.(D) \quad \Delta E_{4 \rightarrow 2}(\text{He}^+) = 13.6 \times 2^2 \times \left( \frac{1}{2^2} - \frac{1}{4^2} \right) = 13.6 \times 1^2 \times \left( \frac{1}{1^2} - \frac{1}{2^2} \right) = \Delta E_{2 \rightarrow 1}(\text{H})$$

$$12.(C) \quad M \text{ shell} \Rightarrow n = 3$$

Transitions possible are  $3 \rightarrow 2; 2 \rightarrow 1; 3 \rightarrow 1$

13.(D) Lyman lines can still be observed if  $e^-$  falls to  $2^{\text{nd}}$  and  $3^{\text{rd}}$  orbit

14.(C) Range for Brackett series for H-atom is  $1500 - 4000 \text{ nm}$

Similar range for  $\text{Li}^{2+}$  would be  $166$  to  $450 \text{ nm}$

15.(D) Range for Brackett series for H-atom is  $1500 - 4000 \text{ nm}$

Similar range for  $\text{He}^+$  would be  $375$  to  $1000 \text{ nm}$

16.(B) For H atom,

$$\Delta E = 13.6 \times 1^2 \times \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = 13.6 \times 3^2 \times \left( \frac{1}{(3n_1)^2} - \frac{1}{(3n_2)^2} \right)$$

For corresponding transition in H is atom transition state should be divisible by 3.

17.(D) Energy of photon incident =  $\frac{1242}{12.42} = 100 \text{ eV}$

18.(B)  $\lambda = \frac{h}{\sqrt{2mKE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 75 \times 1.6 \times 10^{-19}}} = 1.4 \text{ \AA}$

19.(A)  $\lambda = \frac{h}{p} \Rightarrow d\lambda = -\frac{h dp}{p^2} \Rightarrow |\Delta\lambda| = \frac{h\lambda^2}{h^2} \Delta p = \frac{(1.4 \times 10^{-10})^2 \times 6.62 \times 10^{-28}}{6.62 \times 10^{-34}} \Rightarrow |\Delta\lambda| = 1.96 \times 10^{-14} \text{ m}$

20.(AB) Radius =  $0.529 \frac{n^2}{Z}$

$$r \propto n^2$$

$$r \propto \frac{1}{Z}$$

So,  $r \propto n^2$  is

$$\Rightarrow \text{velocity} = 2.18 \times 10^6 \frac{Z}{n} \text{ m/s}$$

$$V \propto Z$$

$$V \propto \frac{1}{n}$$

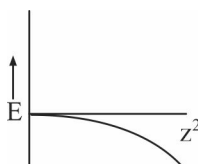
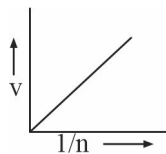
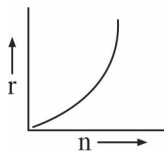
So,  $v \propto 1/n$  is

$$\Rightarrow E = -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$$

$$E \propto Z^2$$

$$E \propto \frac{1}{n^2}$$

So,  $E \propto \frac{1}{n^2}$  is



21.(ABD) Since six different wavelengths are emitted highest state is 4.

Since three wavelengths are shorter

$\Rightarrow$  Three transitions are releasing more energy than what was absorbed

Hence initial orbit may be 2.

Initially  $e^-$  were in  $n = 2$  and were excited to  $n = 4$ . Six possible lines

$$\begin{array}{l} 4 \rightarrow 3 \\ 4 \rightarrow 2 \\ 3 \rightarrow 2 \end{array} \left\{ \begin{array}{l} 3 \rightarrow 1 \\ 4 \rightarrow 1 \\ 2 \rightarrow 1 \end{array} \right\} \rightarrow \lambda < \lambda_0$$

22.(ABD) BE for  $(n = 3) = 1.51 Z^2 = 12 \text{ eV}$  (given)

$$\therefore Z^2 = 12/1.51$$

$$\text{I Excitation potential} = 10.2 Z^2 = 10.2 \times (12/1.51) = 81 \text{ V}$$

$$\text{II Excitation potential} = 12.09 Z^2 = 12.09 \times (12/1.51) = 96 \text{ eV}$$

$$\text{Ionisation potential} = 13.6 Z^2 = 13.6 \times (12/1.51) = 108 \text{ V}$$

$$BE \text{ of } (n = 2) = 3.4Z^2 = 3.4 \times (12 \times 1.51) = 27 \text{ eV}$$

23.(ABC)  $4.25 = (W_0)_A + (K.E.)_A$

$$4.70 = (W_0)_B + (K.E.)_A - 1.5$$

$$\text{So } (W_0)_B - (W_0)_A = 0.45 + 1.5 = 1.95$$

$$\text{Now, } \lambda_B = 2\lambda_A$$

$$\frac{h}{\sqrt{2m(K.E)_B}} = \frac{2h}{\sqrt{2m(K.E)_A}}$$

$$\text{So } (K.E.)_A = 4(K.E.)_B$$

$$4.25 - (W_0)_A = 4[4.7 - (W_0)_B]$$

$$4(W_0)_B - (W_0)_A = 14.55$$

$$\text{So } (W_0)_B = 4.2 \text{ eV}$$

$$\text{So } (W_0)_A = 2.25 \text{ eV}$$

$$(K.E.)_A = 2 \text{ eV}$$

$$(K.E.)_B = 0.5 \text{ eV}$$

24.(AB) (A) Only first four spectral lines belonging to Balmer series in hydrogen spectrum lie in visible region.

(B) If a light of frequency  $\nu$  falls on a metal surface having work functional  $h\nu$ , photoelectric effect will take place only if  $\nu \geq \nu_0$ , since  $\nu_0$  is the minimum frequency required for photoelectric effect.

25.(ABCD) Refer to Theory.

26.(AB) Refer to Theory

27.(6)  $\frac{\lambda_1}{\lambda_2} = \frac{n_1}{n_2} \Rightarrow \frac{3}{1} = \frac{n}{2} \Rightarrow n = 6$

28.(4) Radial node  $= n - \ell - 1$

$$\text{Angular node} = \ell$$

$$\text{Total node} = n - 1$$

$$5f \text{ orbital } = n = 5, \ell = 3$$

$$\text{Total node } 5 - 1 = 4$$

29.(4) Radius  $= 0.529 \frac{n^2}{z}$

$$r_{(n+1)} - r_n = r_{(n-1)}$$

$$0.529 \frac{(n+1)^2}{z} - 0.529 \frac{n^2}{z} = 0.529 \frac{(n-1)^2}{z}$$

$$\text{For hydrogen atom } z = 1$$

$$(n+1)^2 - n^2 = (n-1)^2$$

$$n^2 + 1 + 2n - n^2 = n^2 + 1 - 2n$$

$$4n = n^2$$

$$n = 4$$

30.(4)  $2\pi r_n = n\lambda$

$$2n\pi r_1 = \lambda$$

$$n = \frac{\lambda}{2\pi r_1} = \frac{1340}{(2)(3.14)(53)} = 4$$



31.(199.26)

Photoelectric effect is represented by

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

$$h\nu = W_0 + \frac{1}{2}mv^2 = (\text{work function}) + \text{KE}$$

$$\therefore \nu = \frac{\text{work function}}{h} + \frac{\text{KE}}{h}$$

Graph between  $\nu$  (y-axis) and  $\frac{\text{KE}}{h}$  (x-axis) is a straight line with

$$\text{OA} = 5 \times 10^{14} \text{ s}^{-1}$$

$$\text{Thus, } \frac{W_0}{h} = 5 \times 10^{14} \text{ s}^{-1} \text{ atom}^{-1}$$

$$W_0 = h \times 5 \times 10^{14} \text{ s}^{-1} \text{ atom}^{-1}$$

$$W_0 = 6.62 \times 10^{-34} \text{ Js} \times 5 \times 10^{14} \text{ s}^{-1} \times 6.02 \times 10^{23} \text{ J mol}^{-1}$$

$$= 199262 \text{ J mol}^{-1} = 199.262 \text{ kJ mol}^{-1}$$

32.(–1.22)

Binding energy is the minimum energy to which electron is bound to nucleus or which is required to eject the electron.

Thus, binding energy = – threshold energy

$$E = h\nu = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \text{ Js} \times 3.0 \times 10^8 \text{ ms}^{-1}}{150 \times 10^{-12} \text{ m}} = 1.324 \times 10^{-16} \text{ J}$$

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2} \times 9.11 \times 10^{-31} \text{ kg} \times (1.5 \times 10^7)^2 \text{ m}^2 \text{ s}^{-2} = 1.025 \times 10^{-16} \text{ J}$$

$$\text{Thus, } E = h\nu_0 + \frac{1}{2}mv^2 = W_0 + \frac{1}{2}mv^2 = W_0 + \text{KE}$$

$$\therefore W_0 = (E - \text{KE}) = 1.324 \times 10^{-16} \text{ J} - 1.025 \times 10^{-16} \text{ J} = 2.99 \times 10^{-17} \text{ J}$$

$$\therefore \text{Binding energy} = -2.99 \times 10^{-17} \text{ J}$$

33.(5) For He atom  $(\Delta x \cdot \Delta p)_{\text{He}} \geq \frac{h}{4\pi}$  and for electron  $(\Delta x \cdot \Delta p)_{\text{electron}} \geq \frac{h}{4\pi}$ 

$$(\Delta x)_{\text{helium}} = (\Delta x)_{\text{electron}} = \pm 1.00 \text{ mm} = \pm 100 \times 1.00 \times 10^{-3} \text{ m}$$

$$\text{Hence, } (\Delta p)_{\text{helium}} = (\Delta p)_{\text{electron}} = 5.0 \times 10^{-26} \text{ kg ms}^{-1}$$

34.(20.4)

 $E_1$  = Energy of H-atom in the ground state = 13.6 eV

$$E_2 = 13.6 + (13.6 \times 1.5) \text{ eV}$$

$$E_1 = 13.6 \text{ eV}$$

$$\text{Energy absorbed} = (13.6 \times 1.5) = 20.4 \text{ eV}$$

$$E_2 = \text{Energy of the excited state} = 13.6 + 20.4 = 34.0 \text{ eV}$$

$$\Delta E = \text{KE} = (E_2 - E_1) = 34.0 - 13.6 = 20.4 \text{ eV}$$



$$\Delta H^\circ = 498 \text{ kJ mol}^{-1} = \frac{498 \times 10^3 \text{ J}}{6.02 \times 10^{23}} \text{ J molecule}^{-1} = 8.27 \times 10^{-19} \text{ J molecule}^{-1}$$

Energy required to convert normal (O) into (O\*) = 1.967 eV

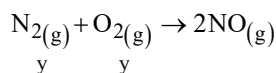
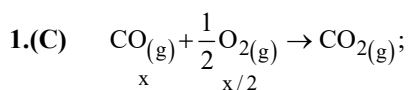
$$= 1.967 \times 1.6 \times 10^{-19} \text{ J} = 3.147 \times 10^{-19} \text{ J}$$

$$\text{Total Energy} = (8.27 \times 10^{-19} + 3.15 \times 10^{-19}) = 1.142 \times 10^{-18} \text{ J}$$

$$\begin{aligned} \text{Thus, } \lambda &= \frac{hc}{E} = \frac{6.62 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{1.142 \times 10^{-18} \text{ J}} \\ &= 1.739 \times 10^7 = 173.9 \times 10^{-9} \approx 174 \text{ nm} \end{aligned}$$

## Solutions - Advanced Problem Package | Chemistry

## Gaseous State



$$x + y = 10 \quad \dots(i)$$

$$\frac{x}{2} + y = 7 \quad \dots(ii)$$

Solving equation (i) and (ii)  $x = 6$  &  $y = 4$

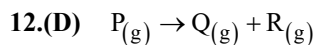
$$4.(D) \quad \text{K.E.} \propto T \text{ in K.}$$

5.(B) At critical point pressure applied is minimum and below this point gas can be liquified easily.

6.(C) Vapour pressure is pressure corresponding to the liquifaction curve.

7.(B) At temperature above critical temperature gas can't be liquefied whatever pressure is applied.

8.(D) For liquifaction 'a' should be greater and 'b' should be smaller.



$$n_{\text{He}} = \frac{V \times 1}{R \times 300}$$

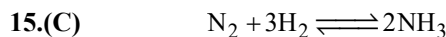
$$n_{\text{He}} + n_p = \frac{2 \times V}{R \times 400}$$

$$n_p = \frac{2V}{400R} - \frac{V}{300R} = \frac{600V - 400V}{120000R} = \frac{200V}{120000R} = \frac{1V}{600R}$$

$$\text{After heating moles} = \frac{2V}{600R} \quad \therefore \quad n_T = \frac{2V}{600R} + \frac{V}{300R} = \frac{600V + 600V}{180000R} = \frac{1200V}{180000R}$$

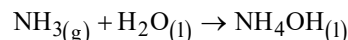
$$P = \left( \frac{12V}{1800R} \right) \times \frac{R \times 600}{V} = 4$$

$$13.(D) \quad \frac{P_1 V_1}{P_2 V_2} = \frac{Z_1 n R T_1}{Z_2 n R T_2}; V_2 = 4$$



$$t = 0 \quad 11 \quad 12 \quad -$$

$$t = \text{eq} \quad 9 \quad 6 \quad 4$$



3.58 L

Free volume for gases  $20 - 3.58 = 16.42$

Gaseous moles =  $n_{\text{N}_2} + n_{\text{H}_2}$  at  $t_{eq}$

$$PV = nRT \rightarrow 22.5 \text{ atm}$$

$$2.(B) \quad P.L = W_{\text{air displaced}} - (\text{weight of balloon} + \text{weight of gas})$$

$$W_{\text{air displaced}} = v \times d = \frac{4}{3} \times \frac{22}{7} \times (10)^3 \times 1.2 = 5028.6 \text{ kg}$$

$$PL = 5028.6 - \left( 100 + \frac{PVM}{RT} \right) \left[ PV = \frac{wRT}{M} \text{ for He} \right]$$

$$= 5028.6 - (100 + 1117.5) = 3811.1 \text{ kg}$$

$$3.(C) \quad \text{KE per mole} = \frac{3}{2} RT$$

$$\text{KE per molecule} = \frac{3}{2} \frac{RT}{N} = \frac{3}{2} KT \quad \left[ K = \frac{R}{N} \right]$$

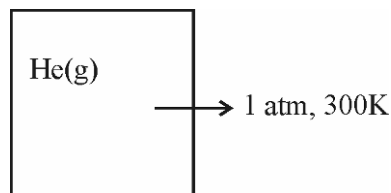
$$9.(C) \quad PV = nRT \quad \text{If } V \text{ constant } P \propto T \therefore T_1 > T_2 > T_3$$

$$10.(C) \quad PM = dRT,$$

$$\frac{P_A}{P_B} = \frac{d_A}{M_A} \times \frac{M_B}{d_B}; \frac{P_A}{P_B} = \frac{3}{1.5} \times \frac{M_B}{\frac{M_B}{2}} = 4$$

$$11.(C) \quad E_K = \frac{3}{2} KTN$$

$$\frac{E_1}{E_2} = \frac{T_1 N_1}{T_2 N_2}; \frac{E_k}{2E_k} = \frac{150N}{300N_2}; N_2 = N$$



$$14.(B) \quad PM = dRT$$

16.(B)  $T_1 > T_2$

$\therefore$  x moles will move to B

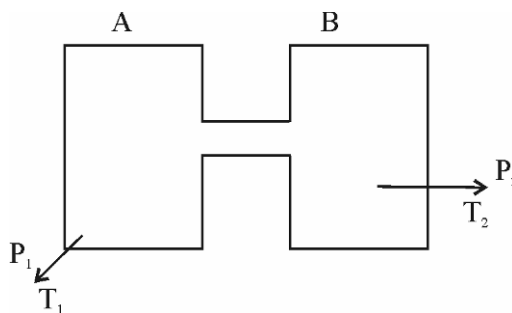
$$P_A V = (n - x)RT_1$$

$$P_A V = (n + x)RT_2$$

$$x = \frac{n(T_1 - T_2)}{T_1 + T_2}$$

Finally  $P_1 \times 2V = 2nRT_1$

$$V = \frac{nRT_1}{P_1} \text{ substitute in above and calculate } P_A$$



17.(ABC)  $V \propto T$  (constant n and P)

18.(ABD) As temperature increased number of collision increases, hence presence of gas molecules increases. The energy of gas molecules also increases due to increase of T.

19.(BD)  $U = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3RT}{M}}$ , U will be doubled when P or T is made 4 times at constant volume

20.(ABC) Refer to theory

21.(BCD) With increase in T molecules with less speed will become less and with more speed will increase and with near about mps will also increase.

22.(BC)  $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ ,  $T_b = \frac{a}{Rb}$

$$T_b = \frac{a}{Rb} \text{ and } T_c = \frac{8a}{27Rb}$$

$$\therefore T_b > T_c$$

23.(AC) For negligible size,  $U_m \gg b$

For negligible intermolecular forces,  $P \gg \frac{a}{V_m^2}$

$$\therefore \text{when } V_m \gg b, \left(P + \frac{a}{V_m^2}\right)(V_m) = RT$$

When  $V_m \gg b$  &  $P \gg \frac{a}{V_m^2}$ , then  $(P)(V_m) = RT$

$$\Rightarrow \frac{PV_m}{RT} = 1 \Rightarrow Z = 1$$

When  $P \gg \frac{a}{V_m^2}$ , then  $(P)(V_m - b) = RT$

$$\Rightarrow PV_m - Pb = RT \Rightarrow \frac{PV_m}{RT} - \frac{Pb}{RT} = 1$$

$$\Rightarrow Z = 1 + \frac{Pb}{RT}$$

27.(ABD)  $15 \times 1.24 = Z \times 0.0821 \times 250$

$$PV = nRT$$

$$15V = 1 \times 0.0821 \times 250$$

$$V = 1.366$$

$$V = 1.366 - 0.12 = 1.24$$

28.(B) Refer to theory

24.(BC)  $\lambda = \frac{U_{av}}{Z_1}$ ,  $Z_1 = \sqrt{2}\pi\sigma^2 V_{av} N^*$ ;  $\left[N^* = \frac{N}{V}\right]$

25.(CD)  $n_f + \frac{2}{3}n_f = \frac{3n_f + 2n_f}{3} = \frac{5}{3}n_f$

$$\frac{5}{3}n_f \times 300 = n_f T$$

$$T = 500 \text{ K}$$

26.(AD) At very low pressure,  $V_m \gg b$

$$\therefore z = 1 - \frac{a}{RTV_m}$$

At high pressures,  $P \gg \frac{a}{V_m^2}$

$$\therefore z = 1 + \frac{Pb}{RT}$$

29.(AC) Equal no. of moles of both He & Ne shall exert equal pressure.

$$\text{Average molecular speed, } C_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$

$\therefore$  Helium has higher  $C_{\text{avg}}$  than Neon

30.(CD) At constant V & n,

$$P \propto T$$

$$\frac{P_2}{P_1} = \frac{T_2}{T_1} \Rightarrow \frac{P_2}{1 \text{ atm}} = \frac{273}{546} \Rightarrow \boxed{P_2 = \frac{1}{2} \text{ atm}}$$

31.(AD)  $Z = \frac{(V_m)_{\text{real}}}{(V_m)_{\text{ideal}}} > 1$

At STP,  $(V_m)_{\text{ideal}} = 22.4 \text{ L}$

$\therefore (V_m)_{\text{real}} > 22.4 \text{ L}$

As  $Z > 1$ , the gas will become less liquefiable.

32.(ABC) At critical temperature,  $T_C$ , densities of liquid and gas become same.

33.(AC) For H & He, Z is always greater than 1

For all other real gases,

$Z < 1$  at low pressure

34. [A-s]  $\rightarrow$  [B-p]  $\rightarrow$  [C-r]  $\rightarrow$  [D-q]

$$A \rightarrow T = \frac{PV}{nR}$$

On moving along the line from i to f,  $P \times V$  will increase or decrease depending upon the equation of straight line.

B  $\rightarrow$  At constant temperature,  $P \propto 1/V$

C  $\rightarrow$  At constant volume,  $P \propto T$

D  $\rightarrow$  At constant pressure,  $V \propto T$

35. [A-p, q, r]  $\rightarrow$  [B-s]  $\rightarrow$  [C-s]  $\rightarrow$  [D-s]

Molecular speed of gas molecules is dependent on temperature and molecular mass of molecules.

$\therefore$  At constant temperature, speed of gas molecules will not change Also,  $PM = dRT$

If density S is lowered, pressure also decreases in same proportion hence speed remains constant  $\left[ C \propto \sqrt{\frac{P}{\text{density}}} \right]$

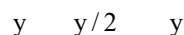
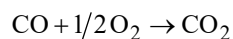
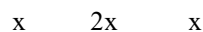
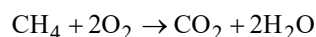
36.(6) Volume of  $\text{CH}_4$ ; x mL

Volume of  $\text{CO}$ ; y mL

Volume of Ne: z mL

Final reduction of volume when passed through alkaline pyrogallol indicates there was residual oxygen.

Pyrogallol in alkaline medium absorbs oxygen. Thus,  $\text{O}_2$  was not the limiting reagent.



Reduction of volume when passed through KOH means:

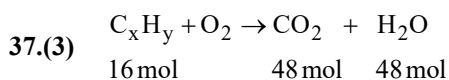
$$x + y = 9 \text{ mL (must be CO}_2\text{)} \dots\dots(i)$$

Total volume initially used  $40 + 10 = 50 \text{ mL}$

On cooling 36.5 mL means ( $\text{H}_2\text{O}$  gets condensed)

$$\underbrace{2x + \frac{y}{2}}_{\text{Oxygen consumed}} = 50 - 36.5 = 13.5 \text{ mL}$$

Thus solving (i) and (ii) equations  $x = 6 \text{ mL}$ ,  $y = 3 \text{ mL}$



Applying POAC for 'C'

$$x \times 16 = 1 \times 48$$

$$x = 3$$

Applying POAC for 'H'

$$y \times 16 = 2 \times 48$$

$$y = 6$$

So, the formula of hydrocarbon is  $\text{C}_3\text{H}_6$ .

38.(2)  $\frac{P_1 V_1}{P_2 V_2} = \frac{n_1 T_1}{n_2 T_2}$

$$\frac{n_1}{n_2} = \frac{P_1 V_1 T_2}{P_2 V_2 T_1}$$

$$\frac{n_1}{n_2} = \frac{1.5 \times 16.4 \times 500}{4.1 \times 5 \times 300} = \frac{2}{1}$$

39.(4) Volume of molecule in motion =  $4 \times$  volume of molecule in rest

40.(7) Let  $m$  be the mass of  $\text{SO}_3$  and  $\text{O}_2$  enclosed in the vessel.

$$\text{Number of moles of } \text{SO}_3 = \frac{m}{80}$$

$$\text{Number of moles of } \text{O}_2 = \frac{m}{32}$$

$$\text{Partial pressure of } \text{SO}_3, P_A = \frac{m}{80} \times \frac{R \times T}{V}$$

$$\text{Partial pressure of } \text{O}_2, P_A = \frac{m}{32} \times \frac{R \times T}{V}$$

$$\text{Now } \frac{P_A}{P_B} = \frac{m}{80} \times \frac{32}{m} = \frac{2}{5}$$

Hence, the sum of the ratio of partial pressures of  $\text{SO}_3$  and  $\text{O}_2$  is  $2 + 5 = 7$

41.(4) Under identical conditions of  $T$  and  $P$ , volumes are in the ratio of their number of moles. Moles of  $\text{CH}_4 = 1.2/16$ .

$$\text{Moles of hydrocarbon, } \text{C}_n\text{H}_{2n-2} = 8.1/M$$

$$\therefore \frac{1.2/16}{8.1/M} = \frac{1\text{L}}{2\text{L}} \text{ or } M = 54$$

$$\text{Molecular mass of } \text{C}_n\text{H}_{2n-2} = 12n + 2n - 2 = 54$$

$$14n = 56$$

$$n = 4.$$

42.(7)  $P_1 V_1 = n_1 RT$ , for the first bulb

$$P_2 V_2 = n_2 RT, \text{ for the second bulb}$$

$$\text{or } P_1 V_1 + P_2 V_2 = (n_1 + n_2) RT, \quad \dots(i)$$

When stopcock is opened; the total volume is  $(V_1 + V_2)$  and total no. of moles is  $(n_1 + n_2)$ . Suppose the equilibrium pressure at each bulb is  $P$  atm. Then,

$$P(V_1 + V_2) = (n_1 + n_2)RT \quad \dots(ii)$$

From equations (i) and (ii), we have,

$$P_1 V_1 + P_2 V_2 = P(V_1 + V_2)$$

$$\text{or } 9 \times 5 + 6 \times 10 = P \times 15$$

$$P = 7 \text{ atm}$$

**43.(4)** Given,  $T_1 = 400\text{K}$ ,  $T_2 = 60\text{K}$

Molecular weight of X,  $M_1 = 40$

Molecular weight of Y,  $M_2 = ?$

$$U_{\text{rms}(x)} = \sqrt{\frac{3RT_1}{M_1}}, U_{\text{mp}(y)} = \sqrt{\frac{2RT_2}{M_2}}$$

Given,  $v_{\text{rms}(x)} = v_{\text{mp}(y)}$

$$\therefore \sqrt{\frac{3R \times 400}{40}} = \sqrt{\frac{2R \times 60}{M_2}}$$

$$30 = \frac{120}{M_2} \Rightarrow M_2 = 4$$

**44(5.26)**  $\text{CO} + \text{H}_2\text{O} \longrightarrow \text{CO}_2 + \text{H}_2$

1 mol of CO will produce 1 mol  $\text{H}_2$

$\Rightarrow$  22.4 L of CO at STP will produce 22.4 L of  $\text{H}_2$

$\Rightarrow$  Water gas sample contains 50%  $\text{H}_2$ , 45% CO and 5%  $\text{CO}_2$

1 L of sample contains 0.45 L CO that will produce 0.45 L  $\text{H}_2$  on reaction with steam. +0.5 L of  $\text{H}_2$  already present.

$\Rightarrow$  1 L of sample will give 0.95 L  $\text{H}_2$

$\Rightarrow$  5 L of  $\text{H}_2$  will be produced by  $= \frac{1}{0.95} \times 5 = 5.263 \text{ L}$

**45(8.25)**  $r_{\text{sample}} = 0.98 r_{\text{O}_2}$

$$\Rightarrow \frac{r_{\text{sample}}}{r_{\text{O}_2}} = 0.98 = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{sample}}}} \Rightarrow M_{\text{sample}} = 33.33 \text{ g mol}^{-1}$$

Sample contains  $\text{O}_2$  and  $\text{O}_3$

$$x_{\text{O}_2} \cdot M_{\text{O}_2} + x_{\text{O}_3} \cdot M_{\text{O}_3} = 33.33$$

$$\Rightarrow (1 - x_{\text{O}_3})32 + x_{\text{O}_3}48 = 33.33 \quad \text{Total 1 mole} \Rightarrow 0.33 \text{ mol}$$

$$\Rightarrow x_{\text{O}_3} = \frac{33.33 - 32}{16} = 0.0825$$

$$\Rightarrow \% \text{O}_3 = \frac{0.0825}{1} \times 100 = 8.25\%$$

**46(1.64)**  $2\text{X(g)} + 3\text{Y(g)} \longrightarrow \text{Z(g)}$

|         |   |   |   |       |      |
|---------|---|---|---|-------|------|
| Initial | 3 | 3 | – | 500 K | 15 L |
| Final   | 1 | 0 | 1 | 300 K | 15 L |

After adding water, Z(g) will dissolve to form a solution and 1 mol of X is left in 15L flask at 300K.

$$PV = nRT$$

$$P \times 15 \text{ L} = 1 \text{ mol} \times 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}$$

$$P = 1.642 \text{ atm}$$

**47(0.29)** Density of mixture (CO + CO<sub>2</sub>) = 1.7 g L<sup>-1</sup> at STP.

$$PM = \rho RT$$

$$1 \text{ atm} \times M = 1.76 \text{ g L}^{-1} \times 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 273 \text{ K}$$

$$M = 39.44 \text{ g mol}^{-1}$$

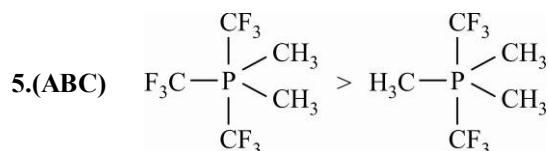
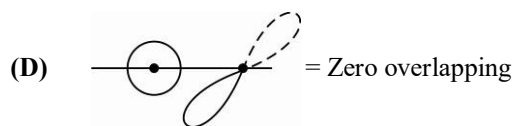
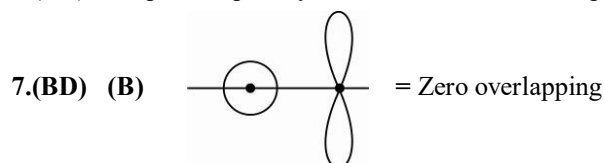
$$x_{\text{CO}} \cdot 28 + (1 - x_{\text{CO}}) 44 = 39.45$$

$$16x_{\text{CO}} = 4.55 \quad \Rightarrow \quad x_{\text{CO}} = 0.285$$

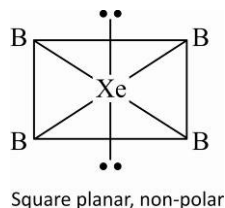
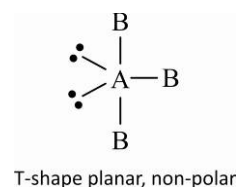
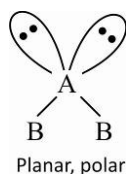


## Solutions - Advanced Problem Package | Chemistry

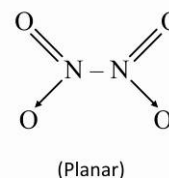
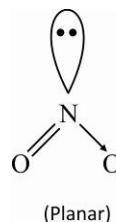
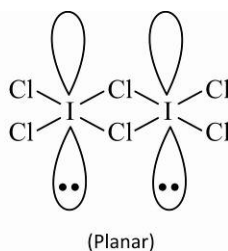
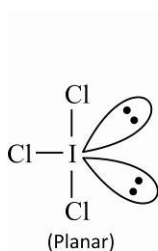
## Chemical Bonding

1.(B) Bond angle :  $\text{ClO}_3^- > \text{BrO}_3^- > \text{IO}_3^-$ 2.(A) % s-character  $\text{BeCl}_2 > \text{BeCl}_3^- > \text{BeCl}_4^{2-}$ 3.(C)  $\text{NH}_3 = 106.6^\circ$  (VSEPR) $(\text{CH}_3)_3\text{N} = 110.9^\circ$  (size of  $-\text{CH}_3$  group) $(\text{SiH}_3)_3\text{N} = 120^\circ$  (due to back bonding)4.(B)  $\text{O}_2^+$  = paramagnetic $\text{O}_2^{2-}$  = diamagnetic $\text{O}_2^-$  = paramagnetic $\text{O}^{2-}$  = diamagnetic6.(AB) In  $\text{sp}^3$ - and  $\text{sp}^3$  d-hybridisation, there is no scope for opp. position of lone pair.8.(BD)  $\pi$  - bonding and  $\pi$  - antibonding molecular orbitals are not affected by s-p mixing.

9.(AC)

(A) When  $n = 4$ ;(B) When  $n = 3$ ;(C) When  $n = 2$ ;

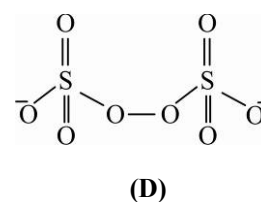
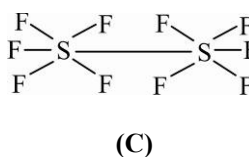
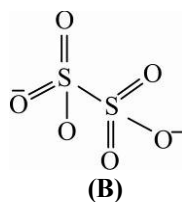
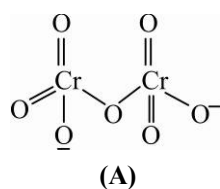
10.(AC)



11.(AC)

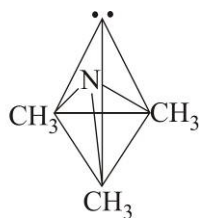
B.E. =  $\text{Cl}_2 > \text{Br}_2 > \text{F}_2 > \text{I}_2$  ; B.E. =  $\text{N}_2 > \text{O}_2 > \text{C}_2 > \text{F}_2$ 

12.(ABCD)

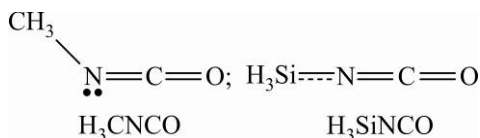


13.(D)

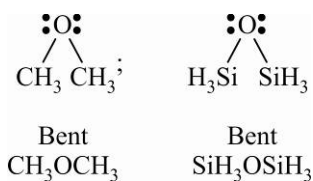
(A)

Structure of  $(\text{CH}_3)_3\text{N}^+$ , Hybridisation of N  $\rightarrow sp^3$ Structure of  $(\text{SiH}_3)_3\text{N}$ , Hybridisation of N  $\rightarrow sp^2$ 

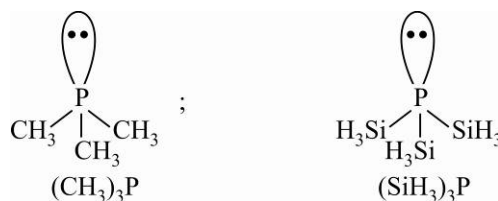
(B)



(C)

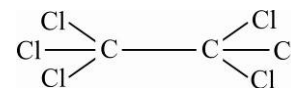
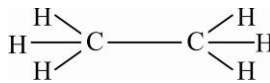
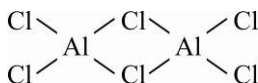
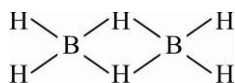


(D)



14.(ABC) All are isoelectronic.

15.(CD)

16.(C)  $\text{Be}_2$  does not exist because B.O. = 017.(A) B.O. of  $\text{N}_2^+$  = 2.5 ; B.O. of  $\text{O}_2^+$  = 2.518.(C)  $\text{C}_2$ ,  $\text{N}_2$  and  $\text{F}_2$  are diamagnetic.

19.(A)

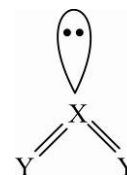
|                   | B.O. |
|-------------------|------|
| $\text{O}_2$      | 2    |
| $\text{O}_2^{2+}$ | 3    |
| $\text{O}_2^+$    | 2.5  |
| $\text{O}_2^{2-}$ | 1    |

20.(C)  $\text{O}_2^{2-}$  peroxide ion is diamagnetic ion21.(CD) (A)  $\text{C} \equiv \text{C}$  (One  $\sigma$  and Two  $\pi$  - bonds)(B)  $\text{O}_2^-$  (One  $\sigma$  and 0.5  $\pi$  - bonds)(C)  $\text{O}_2^{2-}$  (One  $\sigma$  - bond)(D)  $\text{F}-\text{F}$  (One  $\sigma$  - bond)22.(A)  $\text{N}_2 \rightarrow \text{N}_2^+$ 

(B.O. = 3) (B.O. = 2.5)

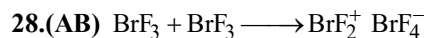
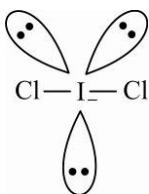
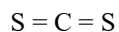
 $\text{O}_2 \rightarrow \text{O}_2^+$ 

(B.O. = 2) (B.O. = 2.5)

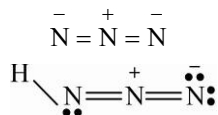
23.(BC)  $\text{SF}_4$ ,  $\text{PF}_5$  have  $sp^3 d$ -hybridised central atom.24.(C)  $\text{Y} = \ddot{\text{X}} = \text{Y}$ 25.(C)  $\text{XeF}_4$  and  $\text{ICl}_4^-$  both are square planar.

26.(ABD)

27.(ABC)



29.(ABCD)

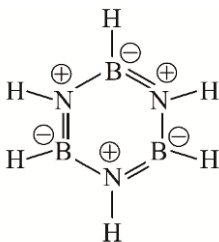


30.(D)  $\sigma$  bond overlapping is greater for p-orbital because p-orbital having good directional character as compared to the s-orbital that is oriented toward internuclear axis.

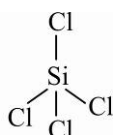
31.(C)

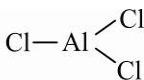
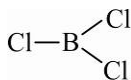
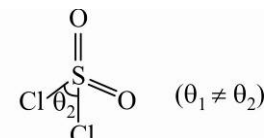
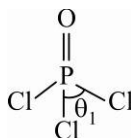
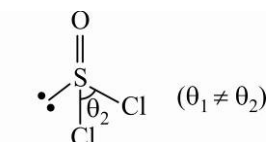
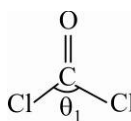
**Molecule****Bond angle**less than  $109^\circ$  $109^\circ$  $90^\circ, 120^\circ$  $90^\circ, 180^\circ$ 

32.(C)

Inorganic Benzene ( $\text{B}_3\text{N}_3\text{H}_6$ )Hybridization of central atom B =  $\text{sp}^2$ , N =  $\text{sp}^2$  planar.

33.(C)

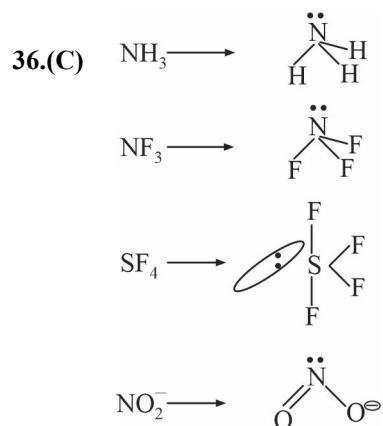


$$\left[ \begin{array}{l} \text{Cl}-\text{C}-\text{Cl} \\ \text{Cl}-\text{Si}-\text{Cl} \end{array} \right] \text{Bond angle} = 109^\circ 28'$$
[Bond angle =  $120^\circ$ ] $(\theta_1 \neq \theta_2)$  $(\theta_1 \neq \theta_2)$ 

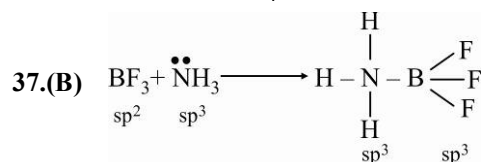
34.(B)  $\text{ICl}_4^- - \text{sp}^3\text{d}^2$  - hybridised no effect of lone pair on bond pair. Lone pairs are on axial position.

$\text{XeF}_2 - \text{sp}^3\text{d}$  - hybridised no effect of lone pair on bond pair. Lone pairs are on equatorial.

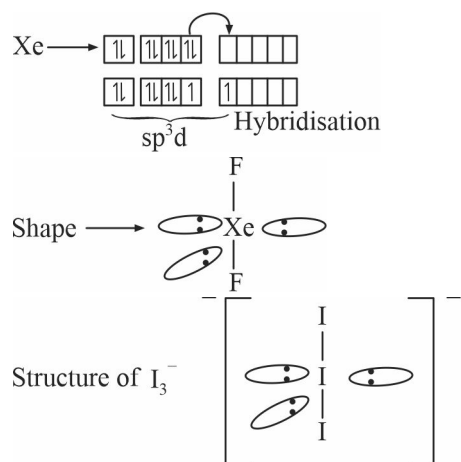
35.(B)



Among all,  $\text{SF}_4$  having maximum bond pair – lone pair repulsion because of least bond angle between them.

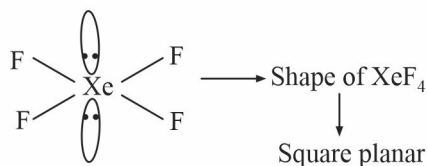
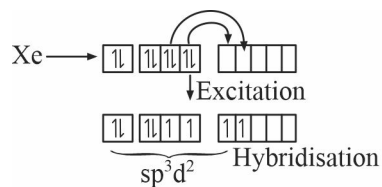


38. (C)  $\text{XeF}_2$



$\Rightarrow \text{XeF}_2$  is linear and also isostructural with  $\text{I}_3^-$

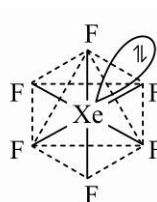
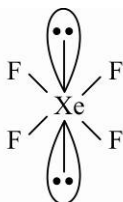
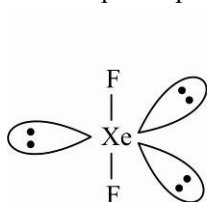
39.(D)  $\text{XeF}_4$



40.(B)  $\text{XeF}_2 = \text{sp}^3\text{d}$  linear

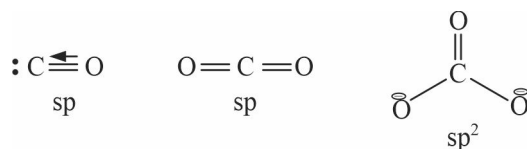
$\text{XeF}_4 = \text{sp}^3\text{d}^2$  square planar

$\text{XeF}_6 = \text{sp}^3\text{d}^3$  capped octahedral



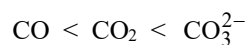
41.(A) Extent of back bonding in B–F bond is decreased if –NH<sub>2</sub> or –OH groups are bonded with boron

42.(B) (I)



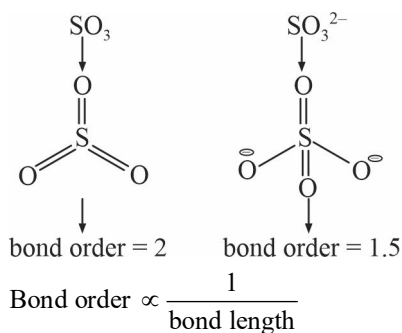
| Molecule                      | Bond order | No. of sigma bond | % $\sigma$ bonding |
|-------------------------------|------------|-------------------|--------------------|
| CO                            | 3          | 1                 | 33.3%              |
| CO <sub>2</sub>               | 4          | 2                 | 50%                |
| CO <sub>3</sub> <sup>2-</sup> | 1.33       | 1                 | 75%                |

Order of %  $\sigma$  bonding



(II) Back bonding decreases down the group BF<sub>3</sub> > BCl<sub>3</sub> > BBr<sub>3</sub> > BI<sub>3</sub>

(III)



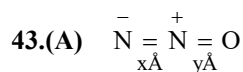
⇒ Order of bond length is SO<sub>4</sub><sup>2-</sup> > SO<sub>3</sub>

(IV)

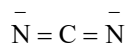
| Molecule                      | Structure | Bond order |
|-------------------------------|-----------|------------|
| ClO <sub>2</sub> <sup>-</sup> |           | 1.5        |
| ClO <sub>3</sub> <sup>-</sup> |           | 1.6        |
| ClO <sub>4</sub> <sup>-</sup> |           | 1.75       |

Order of  $\pi$ -bond order

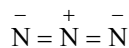




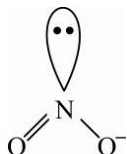
(Two type of bond)



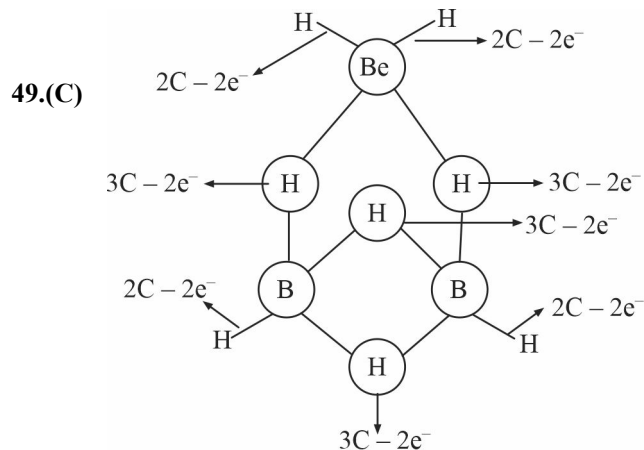
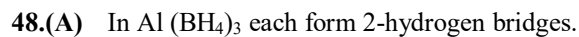
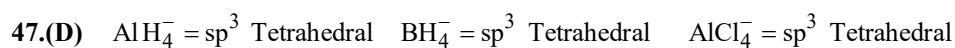
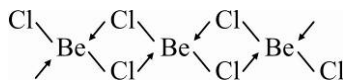
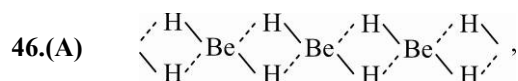
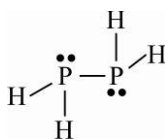
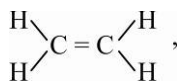
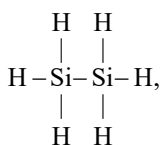
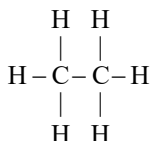
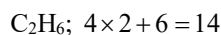
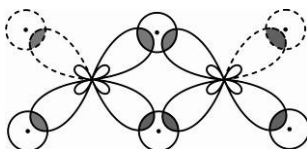
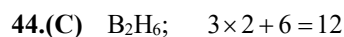
(Both bonds are identical)



(Both bonds are identical)

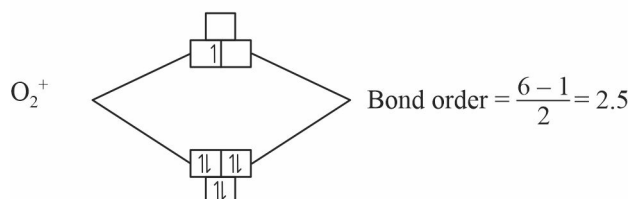
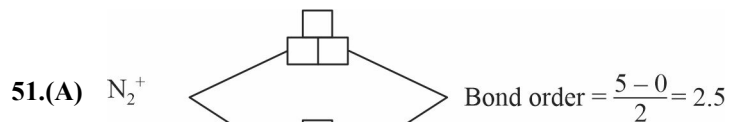


(Both bonds are identical)

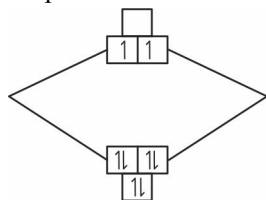


- 50.(C) (A)  $N_2^+ \Rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi p_x^2 = \pi 2p_z^2, \sigma 2p_y^1$
- (B)  $O_2^+ \Rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \pi 2p_y^2 = \pi 2p_z^2, \pi 2p_y^1 = \pi 2p_z^1$
- (C)  $O_2^- \Rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_x^2, \pi 2p_y^2 = \pi 2p_z^2, \pi^* 2p_y^2 = \pi^* 2p_z^1$
- (D)  $CN^- \Rightarrow \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_y^2 = \pi 2p_z^2, \sigma 2p_x^2$

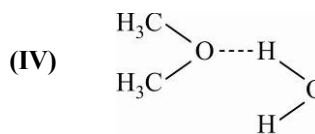
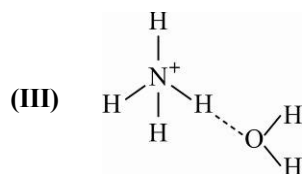
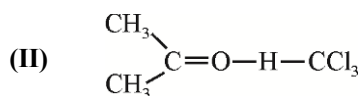
n n : No. of unpaired electron.



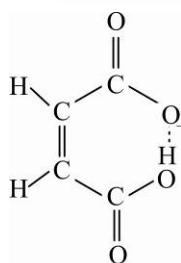
- 52.(D) Mot diagram of  $O_2$   
Having 2 unpaired electron



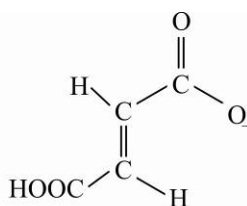
So, magnetic moment =  $\sqrt{n(n+2)} = \sqrt{2(4)} = \sqrt{8}$



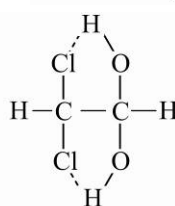
- 54.(B)



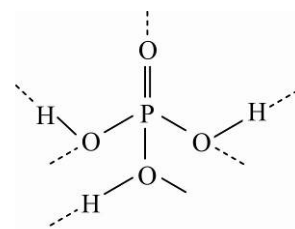
(Maleate ion)



(Fumarate ion)

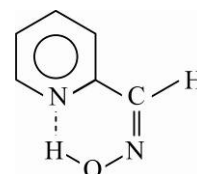


(Chloral hydrate)



(Intermolecular H-bonding responsible for more viscosity of  $\text{H}_3\text{PO}_4$ )

55.(D) Correct order of basic strength in (D) is reversed because L.P. of N in given molecule is involved in hydrogen bonding.

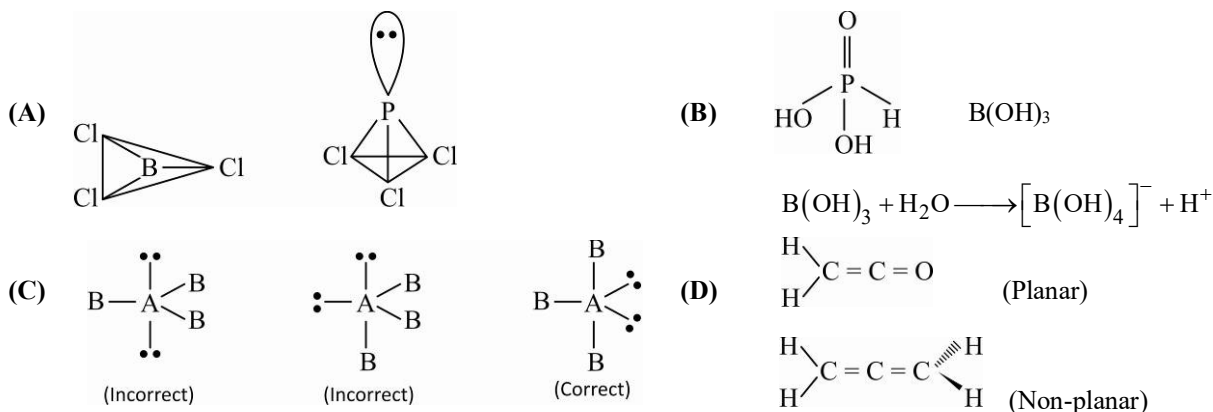


56.(B) Strength of hydrogen bond depends on EN of atom.

57.(D) Liquidation of Xe is explained by London forces which arise as a result of momentary imbalance in electronic distribution.

58.(D)  $\text{AlF}_3$  has highest B.P. in option (D) because it is ionic but other are covalent.

59. [A-p, q, r]  $\rightarrow$  [B-p, s]  $\rightarrow$  [C-s, t]  $\rightarrow$  [D-q, t]



60. [A-p, q, r]  $\rightarrow$  [B-r, s, t]  $\rightarrow$  [C-s, t]  $\rightarrow$  [D-s, t]

(A)  $\text{CsCl}$ ,  $\text{CsBr}$ ,  $\text{CsI}$

$\Rightarrow$  According to Fajan's rule larger size of anion  $\rightarrow$  larger covalent character

$\Rightarrow$  Order of covalent character  $\text{CsCl} < \text{CsBr} < \text{CsI}$

$\Rightarrow$  Covalent character  $\propto \frac{1}{\text{Thermal stability}}$

Order of thermal stability  $\text{CsCl} > \text{CsBr} > \text{CsI}$

$\Rightarrow$  Order of lattice energy  $\rightarrow$  bond between more electron positive and more electron negative  
 $\text{CsCl} > \text{CsBr} > \text{CsI}$

(B)  $\text{LiOH}$ ,  $\text{NaOH}$ ,  $\text{KOH}$

According to Fajan's rule

Smaller size of cation  $\rightarrow$  larger covalent character.

Order of covalent character  $\text{LiOH} > \text{NaOH} > \text{KOH}$

$\Rightarrow$  Covalent character  $\propto \frac{1}{\text{Ionic character}} \propto \frac{1}{\text{Thermal stability}}$

$\Rightarrow$  Order of thermal stability and ionic character  $\text{LiOH} < \text{NaOH} < \text{KOH}$

$\Rightarrow$  More lattice energy  $\rightarrow$  Bond between electropositive and electronegative atom  
 $\text{LiOH} > \text{NaOH} > \text{KOH}$

(C)  $\text{LiH}$ ,  $\text{NaH}$ ,  $\text{KH}$

(D)  $\text{Mg}_3\text{N}_2$ ,  $\text{Ca}_3\text{N}_2$ ,  $\text{Sr}_3\text{N}_2$

Lesser size of cation  $\rightarrow$  More covalent character.

$\Rightarrow$  Order of covalent character  $\text{LiH} > \text{NaH} > \text{KH}$  and  $\text{Mg}_3\text{N}_2 > \text{Ca}_3\text{N}_2 > \text{Sr}_3\text{N}_2$



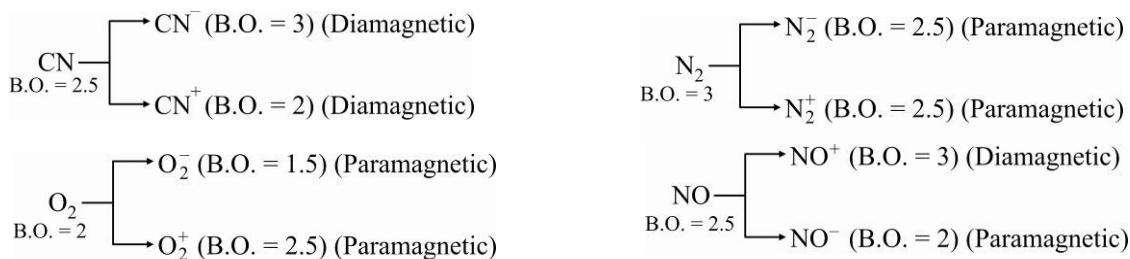
$$\Rightarrow \text{Covalent character} \propto \frac{1}{\text{Ionic character}} \propto \frac{1}{\text{Thermal stability}} \text{ order of Thermal stability and Ionic character}$$



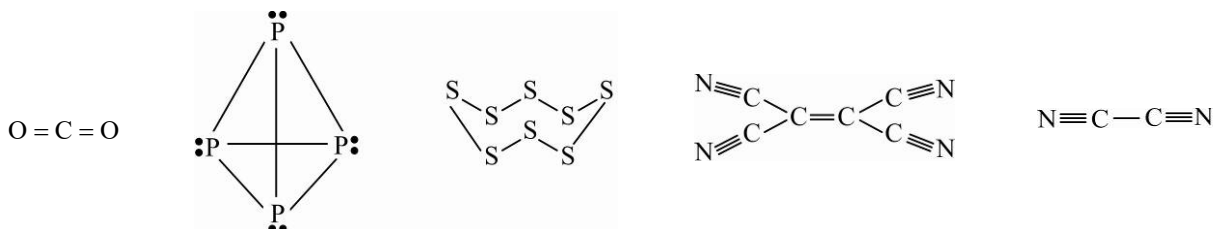
61. [A-s] → [B-p, q, t] → [C-r] → [D-p, q, t]



62. [A-p] → [B-p, r, s] → [C-p, q, r, s] → [D-p, q, r]



63.(5) Five molecules have zero dipole moment



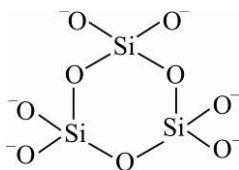
64.(5) B.P. depends molecular weight or hydrogen bonding.

65.(6) Non-existing molecule/ion  $\text{BF}_6^{3-}$ ,  $\text{BiCl}$ ,  $\text{Si}_6$ ,  $\text{ClF}_7$ ,  $\text{NF}_6^-$ ,  $\text{KH}_3$

66.(6) Six statements are correct which are II, III, IV, VI, VII, IX.

67.(6)  $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_4$

68.(3)  $\text{Si}_3\text{O}_x^{6-}$   $x = 9$



69.(5)  $\text{PCl}_4^+$  ( $\text{sp}^3$ ),  $\text{ICl}_2^+$  ( $\text{sp}^3$ )  
 $\text{SF}_2$  ( $\text{sp}^3$ ),  $\text{SiF}_4$  ( $\text{sp}^3$ ),

70.(4)  $\text{NH}_2^-$  (Bent)  $\text{OH}_2$  (Bent)  $\text{SO}_2$  (Bent)  $\text{SF}_2$  (Bent)

71.(3)  $\text{PF}_3$ ,  $\text{CF}_4$  and  $\text{BeF}_2$  have all the possible B.A. identical.

- 72.(4) Correct: Li > Be (size)  
 C > O (size)  
 F > Cl (IE)  
 N > O (IE)

73.(84.4) Percentage ionic character =  $\frac{\text{Observed value of dipole moment} \times 100}{\text{Calculated value of dipole moment}}$

Observed dipole moment = 6.32 D

$$0.156 \text{ nm} = 0.156 \times 10^{-9} \text{ m} = 0.156 \times 10^{-7} \text{ cm}$$

Calculated value of dipole moment = charge  $\times$  distance

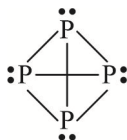
$$= 4.8 \times 10^{-10} \text{ esu} \times 0.156 \times 10^{-7} \text{ cm}$$

$$= 7.488 \times 10^{-18} \text{ esu cm} = 7.488 \text{ D}$$

Thus, percentage ionic character =  $\frac{6.32}{7.48} \times 100 = 84.4\%$

74.(75)  $\text{P}(15) = 1s^2 2s^2 2p^6 3s^2 3p^3$

Valence electron = 5



Each P-atom has three (P-P) bonds and one lone pair and is  $sp^3$ -hybridised.

Thus, p-character is  $\frac{3}{4} \times 100 = 75\%$

75.(4)  $p = \frac{n}{V} RT = \frac{w}{mV} RT$

Since  $\frac{w}{V} = gL^{-1} = \text{density}$

Hence  $p = \frac{d}{m} RT$

$$m (\text{HF-associated}) = \frac{dRT}{p} = \frac{3.17 \times 0.0821 \times 300}{100} = 78.08 \text{ gL}^{-1}$$

$m (\text{HF})$  of single unit =  $20 \text{ gL}^{-1}$

HF molecules associated by H-bonding =  $\frac{78}{20} \approx 4$

76.(7)  $\text{Na}(11) : 1s^2 2s^2 2p^6 3s^1$

Electron in (3s) orbital is a conduction band. Each orbital has one electron, thus one energy level.

$$26.8 \text{ mg Na} = 26.8 \times 10^{-3} \text{ g Na - atoms} = \frac{26.8 \times 10^{-3}}{23} \text{ g mol Na}$$

$$= \frac{26.8 \times 10^{-3} \times 6.02 \times 10^{23}}{23} \text{ Na - atoms}$$

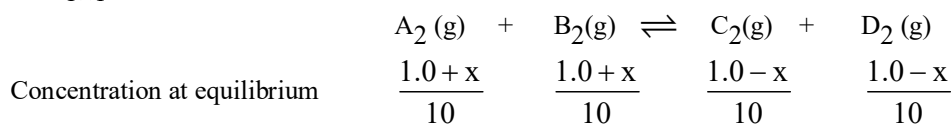
$$= 7.0 \times 10^{20} \text{ atoms} = 7.0 \times 10^{20} \text{ conduction bands}$$

Thus,  $n = 7$

## Solutions - Advanced Problem Package | Chemistry

### Chemical Equilibrium

1.(B)  $Q = \frac{1 \times 1}{1 \times 1} \Rightarrow 1$  ( $Q > K_c$  so reaction will proceed in backward direction)

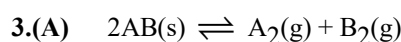


$$0.25 = \frac{\left(\frac{1-x}{10}\right)^2}{\left(\frac{1+x}{10}\right)^2}$$

$$0.5 = \frac{1-x}{1+x} \Rightarrow 0.5 + 0.5x = 1-x, \quad 1.5x = 0.5, \quad x = 0.333$$

$$[A_2(g)] = \frac{1+x}{10} \Rightarrow \frac{1.333}{10} \Rightarrow 0.13$$

2.(B) After  $t = 3$  min reaction is going forward which should be due to increase in temperature



$$0.5 + x \quad x$$

$$K_p = P_{A_2} \cdot P_{B_2} \Rightarrow 0.06 = (0.5+x)x$$

$$x^2 + 0.5x - 0.06 = 0$$

$$x = \frac{-0.5 \pm \sqrt{(0.5)^2 + 4(0.06)}}{2} \Rightarrow \frac{-0.5 + 0.7}{2}$$

$$\Rightarrow 0.1$$

$$P_{\text{total}} = P_{A_2} + P_{B_2} \Rightarrow 0.6 + 0.1 \Rightarrow 0.70 \text{ atm}$$



$$t = t_{\text{eq}} \quad 1-\alpha \quad \alpha \quad \alpha/2$$

$$K_p = \frac{\left(\frac{\alpha}{1+\alpha/2} \cdot P_T\right)^2 \left(\frac{\alpha/2}{1+\alpha/2} \cdot P_T\right)}{\left(\frac{1-\alpha}{1+\alpha/2} \cdot P_T\right)^2} = P_T$$

$$\Rightarrow \alpha = \frac{2}{3}$$

Initial moles = 4 has no use in calculation of  $\alpha$ .



$$\text{At equilibrium} \quad a(1-\alpha) \quad \frac{a\alpha}{2} \quad \frac{a\alpha}{2}$$

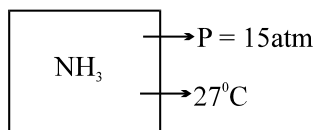
$$a = \frac{0.96}{128} = 0.0075 \Rightarrow$$

$$n_{I_2} = \frac{a\alpha}{2} = \frac{0.0075 \times 0.21}{2}$$

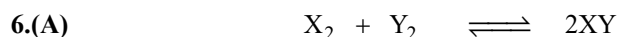
Meq of hypo required = Meq of  $I_2$

$$N_{I_2} \times n_f = 2 \times 1000 = \left(\frac{0.0075 \times 0.21}{2}\right) \times 2 \times 1000 = 1.575 \text{ Meq}$$

7.(A)



$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$



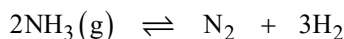
$$\text{At initial} \quad 1 \quad 2 \quad 0$$

$$\text{At equilibrium} \quad 1-x \quad 2-x \quad 2x$$

$$\text{Equilibrium conc.} \quad \frac{1-x}{3} \quad \frac{2-x}{3} \quad \frac{2x}{3}$$

$$\frac{2x}{3} = 0.6, \quad \frac{x}{3} = 0.3$$

$$\frac{15}{P_2} = \frac{300}{620}, \quad P_2 = 15 \times \frac{620}{300} = 31 \text{ atm}$$



$$\text{Initial Pressure} \quad 31 \quad 0 \quad 0$$

$$\text{Equil. Pressure} \quad 31-2x \quad x \quad 3x$$

$$31 - 2x + x + 3x = 50$$

$$31 + 2x = 50$$

$$2x = 19$$

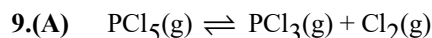
$$P_{\text{NH}_3} \text{ decomposed} = 2x = 19$$

$$P_{\text{NH}_3} \text{ taken} = 31$$

$$\% \text{ of } \text{NH}_3 \text{ decomposed} = \frac{19}{31} \times 100 = 61.29\%$$

$$8.(A) \quad \Delta G = \Delta_r G^\circ + RT \ln Q$$

$$Q = \frac{2 \cdot 4}{(1 \cdot 5)^2}$$



$$PV = \frac{\omega}{M} RT$$

$$P = \left( \frac{\omega}{V} \right) \frac{RT}{M}$$

$$P = \rho \frac{RT}{M}$$

$$\frac{P}{\rho} = \frac{RT}{M_0} = \frac{0.082 \times 503}{(2 \times 62)}$$

$$\frac{P}{\rho} = 0.3326$$

$$10.(C) \quad Q = \frac{P_{\text{COCl}_2}}{P_{\text{CO}} \times P_{\text{Cl}_2}} = \frac{0 \cdot 25}{0 \cdot 1 \times 0 \cdot 2} = 12 \cdot 5 > K$$

$\Rightarrow$  reaction goes forward.

$$11.(A) \quad \Delta G^\circ = \Delta H^\circ - (298) \Delta S^\circ = -RT \ln K_{\text{eq}}$$

$$\text{Take } K_{\text{eq}} = K_1 \text{ and } T_1 = 298\text{K}$$

Find  $K_2$  at  $T_2 = 400\text{K}$  using

$$\ln \frac{K_2}{K_1} = \frac{\Delta H}{R} \left( \frac{T_2 - T_1}{T_1 T_2} \right)$$

$$12.(C) \quad k = A \cdot e^{-\frac{\Delta H}{RT}}$$

$$13.(D) \quad K_c = \frac{(0 \cdot 4)^1 (0 \cdot 4)^1}{(0 \cdot 2)^1 (0 \cdot 2)^1} = 4$$

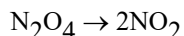
Use  $K_c = 4$  to define equilibrium conditions

14.(A) As reaction goes forward molecular mass of mixture decreases. Hence vapour density also decreases

15.(A) Refer module OWS in chemical equilibrium

$$16.(C) \quad K_p = \frac{4\alpha^2}{1-\alpha^2} \cdot P_T = 640$$

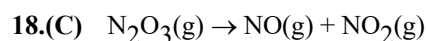
17.(A) Let initial moles of  $\text{N}_2\text{O}_4$  are 1



$$1 - 0.4 \quad 0.8$$

$$= 0.6$$

$$K_p = \frac{0.8 \times 0.8}{0.6} \times \frac{1}{1.4} \Rightarrow \frac{1}{1.4} \times \frac{0.8 \times 0.8}{0.6} = \frac{8 \times 8}{1} \times \frac{P}{9} \Rightarrow P = 0.107 \text{ atm}$$



$$1 - \alpha \quad \alpha \quad \alpha$$

$$\therefore \alpha = \frac{D-d}{d(n-1)} = \frac{38-23.75}{23.75(2-1)} = 0.6$$

$$\text{Mass \% of } \text{N}_2\text{O}_3 \text{ in the equilibrium mixture} = \frac{\text{wt. of } \text{N}_2\text{O}_3}{\text{Total wt.}} \times 100 = \frac{(0.4 \times 76)}{(0.6 \times 30) + (0.6 \times 46) + (0.4 \times 76)} \times 100 = 40 \%$$

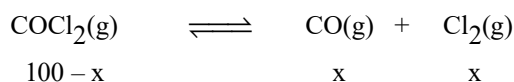
19.(D)  $\text{H}_2\text{O} + \text{CO}_2 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$ . If  $\text{CO}_2$  is increased,  $[\text{H}^+]$  will increase  $\Rightarrow$  pH will decrease. Body very minutely maintains pH of the blood. The lungs will work faster so as to exhale the excess of  $\text{CO}_2$  & maintain its at pH 7.4 again.

20.(B) Slope  $= -\frac{\Delta H}{2 \cdot 3R} = 230$  (use  $R = 2 \text{ cal/mol/K}$ )      21.(B) y-intercept  $= \frac{\Delta S^\circ}{2 \cdot 3R} < 0$

22.(B) T increases  $\Rightarrow K_c$  increases  $\Rightarrow$  On increasing temperature the reaction is going forwards  $\Rightarrow \Delta H > 0$

23 - 25 Pressure of  $\text{H}_2\text{O} = 20$  torr

pressure due to  $\text{CO}$ ,  $\text{Cl}_2$  and  $\text{COCl}_2$  in final state = 80 torr



$$\text{Total pressure} = (100 + x) \times \frac{2}{3} = 80$$

$$x = 20 \text{ torr.}$$

23.(C)  $P_{\text{COCl}_2}(\text{final}) = 80 \times \frac{2}{3} = \frac{160}{3} \text{ torr}$

24.(D)  $P_{\text{CO}}(\text{final}) = P_{\text{Cl}_2}(\text{final}) = \frac{20 \times 2}{3} = \frac{40}{3} \text{ torr}$

$$P_{\text{H}_2\text{O}(\text{g})} = 20 \text{ torr} \quad ; \quad x_{\text{CO}} = \frac{\frac{40}{3}}{100} = \frac{4}{30} = \frac{2}{15}$$

25.(C)  $\frac{P_{\text{CO}}}{P_{\text{COCl}_2}} = \frac{40/3}{160/3} = 1:4$

26.(ABC)      27.(ABC) As reaction is endothermic on increasing temperature reaction goes forward.

28.(ABC) Use  $\Delta_r G^\circ = -RT \ln K$

$$K_p = 1.8 \times 10^5$$

Compare  $Q_p$  and  $K_p$

29.(AB)  $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$

|                                                                    |                                                                   |                  |
|--------------------------------------------------------------------|-------------------------------------------------------------------|------------------|
| $W_1 \text{ gm}$                                                   | $W_2 \text{ gm}$                                                  | $0$              |
| $\left( W_1 - \frac{3.4}{17} \times \frac{1}{2} \times 28 \right)$ | $\left( W_2 - \frac{3.4}{17} \times \frac{3}{2} \times 2 \right)$ | $3.4 \text{ gm}$ |
| $= 3.4 \text{ gm}$                                                 | $= 3.4 \text{ gm}$                                                |                  |
| $W_1 = 6.2 \text{ gm}$                                             | $W_2 = 4 \text{ gm}$                                              |                  |

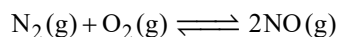
$$\text{Limiting reagent is } \text{N}_2, \therefore \text{Maximum possible mass of } \text{NH}_3 = \frac{6.2}{28} \times \frac{2}{1} \times 17 = 7.528 \text{ gm}$$

30.(AC)  $K_p = K_c$  when  $\Delta n_g$  is zero

31.(BD) Indicator should have a small transition range. pH at equivalence point  $\approx \text{pK}_{\text{In}}$  of the indicator  
pH at equivalence point & at end point may be different.

32.(CD) Addition of solids have no effect on equilibrium and temperature favours endothermic direction while increasing pressure will shift equilibrium in backward direction as  $\Delta n_g$  is +ve.

33.(ABCD)



- (A) For changing pressure volume has to be changed, though number of moles of NO(g) do not get changed but its concentration will get changed
- (B) Temperature change will change  $K_p$  and hence concentration
- (C) Volume change will change concentration, not the number of moles
- (D) Catalyst does not change equilibrium concentrations

34.(AB) Number of moles will remain unchanged but due to decreased volume pressure will get increased and also the concentrations.

35.(ABC) Positive catalyst only alters the equilibrium time. The amount of product, heat of reaction and equilibrium constant remain unaffected by the catalyst.

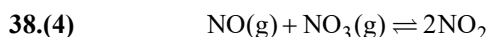
36.(A) Rate constant of a reaction and ionic product of water both increase with increase in temperature. Equilibrium constant of exothermic reaction will decrease with rise in temperature

$$k = Ae^{-E_a/RT} \quad \dots(i)$$

$$\log_{10} \left( \frac{K_2}{K_1} \right) = \frac{\Delta H}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right] \quad \dots(ii)$$

Equation (i) shows the relation between temperature and rate constant, while equation (ii) shows relation between temperature and equilibrium constant.

37. [A-r]  $\rightarrow$  [B-r]  $\rightarrow$  [C-q s]  $\rightarrow$  [D-p]



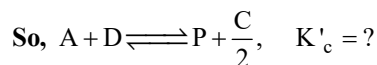
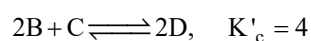
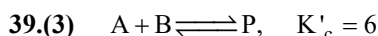
$$t = t_{eq} \quad \begin{matrix} 1-x/2 & 3-x/2 & x \end{matrix}$$

$$t = t_{eq} \quad \begin{matrix} 3-x & 3-x & 2x \end{matrix}$$

$$K_p = \frac{x^2}{(1-x/2)(3-x/2)} = \frac{(2x)^2}{(3-x)^2}$$

$$x = 3/2$$

$$K_p = 4$$

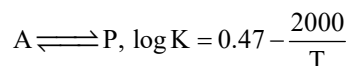


$$K'_c \times \frac{1}{\sqrt{K'_c}} = K'_c$$

$$6 \times \frac{1}{\sqrt{4}} = K'_c$$

$$K'_c = 3$$

40.(9) For the reversible reaction,



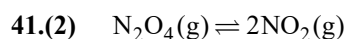
$$\Delta G^\circ = -2.303 RT \log K$$

$$T\Delta S^\circ - \Delta H^\circ = 2.303 RT \log K$$

$$\log K = \frac{T\Delta S^\circ - \Delta H^\circ}{RT \times 2.303}$$

$$\frac{\Delta S^\circ}{R \times 2.303} = 0.47$$

$$\Delta S^\circ = 0.47 \times 8.314 \times 2.303 = 9$$



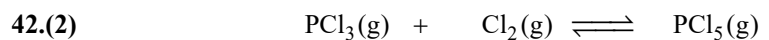
$$K_p = \frac{2^2}{2} = 2$$

At new equilibrium if  $p_{\text{N}_2\text{O}_4} = 0.85 \text{ atm}$

$$\frac{(p_{\text{NO}_2})^2}{0.85} = 2 \Rightarrow p_{\text{NO}_2} = 1.3 \text{ atm}$$

$$P_T = 1.3 + 0.85 = 2.15 \text{ atm}$$

Ratio of  $P_T$  before and after expansion is  $\frac{4}{2.15} \approx 2$



Given  $n \text{ moles} \quad n \text{ moles} \quad n \text{ moles}$

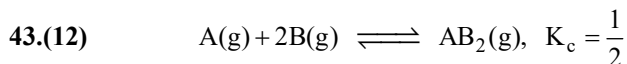
At equilibrium  $n - \alpha \quad n + \frac{20}{3} - \alpha \quad n + \alpha$

According to Gay Lussac's law, at constant temperature & pressure  $v \propto n$

$$\text{So, } \frac{n - \alpha + n + \frac{20}{3} - \alpha + n + \alpha}{3n} = 2 \quad \dots\dots\dots (1)$$

$$\text{Also, as the temperature is constant, } K_c \text{ does not change } \frac{1}{n} = \frac{n + \alpha}{(n - \alpha) \left( n + \frac{20}{3} - \alpha \right)} \quad \dots\dots\dots (2)$$

$$\text{Solve equation (1) and (2) } n = \frac{50}{21} \approx 2$$



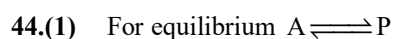
Given 2 M 2M ?

$$K_c = \frac{(AB_2)}{[A][B]^2}$$

$$\frac{1}{2} = \frac{[AB_2]}{2 \times (2)^2}$$

$$\text{So, } [AB_2] = 4$$

On further addition of two moles of B,  $AB_2$  should be 16M

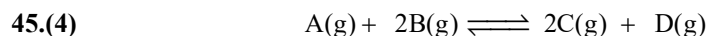


$$\Delta G^\circ = -2.303RT \log K_c$$

$$0 = -2.303RT \log K_c$$

$$\log K_c = 0$$

$$K_c = 1$$



Initial conc. x 1.5x 0 0

At equilibrium  $x - \alpha$   $1.5x - 2\alpha$   $2\alpha$   $\alpha$

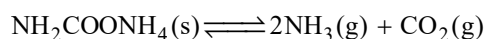
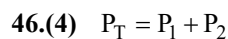
$$x - \alpha = 1.5x - 2\alpha$$

$$-0.5x = -\alpha$$

$$\alpha = 0.5x$$

$$K_p = K_c \quad \text{as } \Delta n_g = 0$$

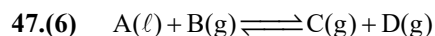
$$K_p = \frac{(2\alpha)^2 \times \alpha}{(x - \alpha)(0.5x)^2} = \frac{1}{0.25} = 4$$



2x + x = 3 2x x

$$x = 1$$

$$K_p = (p_1)^2 (p_2) = (2)^2 (1) = 4$$



$$K_p = \frac{p_c \cdot p_d}{p_B} = \frac{2 \times 3}{1} = 6$$



t = 0 0.1 M, V 0.2 M, V

moles 0.1 V 0.2 V 0.1 0.1

$$0 \quad \frac{(0.2 - 0.1)V}{2V}$$

$$K_C = \frac{[\text{CH}_3\text{OH}][\text{Cl}^-]}{[\text{CH}_3\text{Cl}][\text{OH}^-]} = 1 \times 10^{16}$$

Due to very large value of  $K_C$ , the reaction will go to completion

$$[\text{OH}^-]_{\text{left}} = \frac{(0.2\text{ V} - 0.1\text{ V})}{2\text{ V}}$$

49(22.62)

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

$$-534.52 \times 10^3 \text{ J} = -8.314 \text{ JK}^{-1} \text{ mol}^{-1} \times 2000 \text{ K} \ln K_{\text{eq}}$$

$$K_{\text{eq}} = 1.03 \times 10^{14}$$

Since the value of  $K_{\text{eq}}$  is very high, reaction will go to completion



$$K_P = \frac{P_{\text{O}_2}^3}{P_{\text{O}_3}^2}$$

$$V_T = 10 \text{ L}; \quad P_T = 8 \text{ atm}; \quad T = 2000 \text{ K}$$

$$1.03 \times 10^{14} = \frac{(8)^3}{(P_{\text{O}_3})^2} \Rightarrow P_{\text{O}_3} = 22.62 \times 10^{-7} \text{ atm}$$

50(0.66)



$$\begin{array}{ccc} \text{C} & - & - \\ \text{C-P} & \text{P} & \text{P} \end{array}$$

$$K_P = 0.109 \text{ at } 300\text{K}.$$

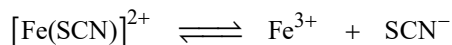
$$K_P = P_{\text{NH}_3} \cdot P_{\text{H}_2\text{S}} = P^2 \Rightarrow P = 0.33 \text{ atm}$$

$$\text{Total pressure} = 2P = 0.66 \text{ atm}$$

51.(0.0036)

$$\text{Initial concentration of } \text{Fe}^{3+} = \frac{10^{-3} \text{ g}}{10^3 \text{ L}} = \frac{10^{-3}}{56} \text{ mol L}^{-1}$$

Let initial concentration of  $\text{SCN}^-$  be 'x'



$$t = 0 \quad 0 \quad 1.78 \times 10^{-5} \quad x$$

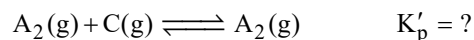
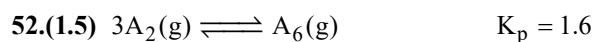
$$t = t_{\text{eq}} \quad y \quad (1.78 \times 10^{-5} - y) \quad (x - y)$$

$$y = 6 \times 10^{-6} \text{ M}$$

$$\Rightarrow K = \frac{[\text{Fe}^{3+}][\text{SCN}^-]}{[\text{Fe(SCN)}]^{2+}} = \frac{(1.78 \times 10^{-5} - y)(x - y)}{y}$$

$$7.142 \times 10^{-3} = \frac{(1.78 \times 10^{-5} - 6 \times 10^{-6})(x - 6 \times 10^{-6})}{6 \times 10^{-6}} \Rightarrow x = 0.0036$$



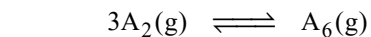


At equilibrium,

Total  $p = 1.4 \text{ atm}$

$p_{A_6} = 0.2 \text{ atm}$

Initially, 2 : 1 molar ratio of  $A_2$  and C are mixed



$t = 0 \quad 2p$

$t = t_{eq} \quad (2p - 3p' - p'') \quad p'$



$t = 0 \quad 2p \quad 1p$

$t = t_{eq} \quad (2p - p'' - 3p') \quad (p - p'') \quad p''$

$$K_p = \frac{p_{A_6}}{p_{A_2}^3} \Rightarrow 1.6 = \frac{0.2}{(p_{A_2})^3}$$

$$\Rightarrow p_{A_2} = 0.5 \text{ atm}$$

$$\Rightarrow 2p - 3p' - p'' = 0.5 \text{ atm} \quad (p'' = 0.2)$$

$$\Rightarrow 2p - 0.6 - p'' = 0.5 \quad \dots\dots (i)$$

Total pressure =  $p_{A_2} + p_{A_6} + p_{A_2C} + p_C$

$$\Rightarrow 1.4 = 0.5 + 0.2 + p'' + p - p'' \Rightarrow p = 0.7 \text{ atm}$$

Subs in (i)

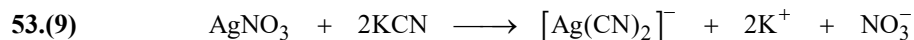
$$\Rightarrow 2(0.7) - 0.6 - p'' = 0.5 \Rightarrow 1.4 - 0.6 - p'' = 0.5 \Rightarrow p'' = 0.3$$

$$p_{A_2C} = 0.3 \text{ atm}$$

$$p_C = p - p'' = 0.7 - 0.3 = 0.4 \text{ atm}$$

$$K'_p = \frac{p_{A_2C}}{p_{A_2} \cdot p_C} = \frac{0.3}{0.5 \times 0.4}$$

$$K'_p = 1.5$$

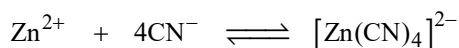


$t = 0 \quad 0.1 \quad 0.5 \quad -$

$t = t_{eq} \quad 10^{-6} \quad 0.5 - 2(0.1 - 10^{-6}) \quad (0.1 - 10^{-6})$

$$K_{eq} = \frac{[Ag(CN)_2]^-}{[Ag^+][CN^-]^2} = \frac{0.1}{10^{-6} \times 0.3}$$

$$K_1 = 0.33 \times 10^6$$



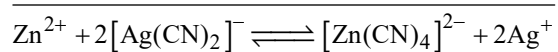
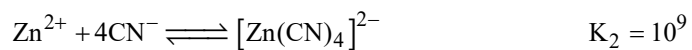
$t = 0 \quad 0.1 \quad 0.5 \quad -$

$t = t_{eq} \quad 10^{-12} \quad 0.5 - 4(0.1 - 10^{-12}) \quad (0.1 \times 10^{-12})$

$10^{-12} \quad 0.1 \quad 0.1$

$$K_2 = \frac{[\text{Zn}(\text{CN})_4]^{2-}}{[\text{Zn}^{2+}][\text{CN}^-]^4} = \frac{0.1}{10^{-12} \times (0.1)^4} = 1 \times 10^9$$

Multiply equation (1) by 2 and subtract from equation (2)



$$K_{\text{eq}} = K_1 K_2 = (9 \times 10^{12})(10^9) \Rightarrow K_{\text{eq}} = 9 \times 10^{21}$$

## Solutions - Advanced Problem Package | Chemistry

### Ionic Equilibrium

- 1.(C) At pH = 3,  $[H^+] = 10^{-3}$  M  
At pH = 4,  $[H^+] = 10^{-4}$  M

When equal volume of the two solutions are mixed, the  $[H^+] = \frac{10^{-3} + 10^{-4}}{2} = \frac{10^{-3}[1 + 0.1]}{2} = \frac{1.1}{2} \times 10^{-3}$

$$[H^+] = 5.5 \times 10^{-4}$$

$$-\log [H^+] = -\log (5.5) - \log 10^{-4}$$

$$\text{pH} = -0.7404 + 4 = \mathbf{3.26}$$

- 2.(C)  $\text{CH}_3\text{COOH} + \text{NaOH} \longrightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$

|            |       |   |     |   |   |
|------------|-------|---|-----|---|---|
| Millimoles | t = 0 | 5 | 2.5 | 0 | 0 |
|------------|-------|---|-----|---|---|

|            |        |     |   |     |  |
|------------|--------|-----|---|-----|--|
| Millimoles | t = t' | 2.5 | 0 | 2.5 |  |
|------------|--------|-----|---|-----|--|

Here, millimoles of  $\text{CH}_3\text{COOH}$  and  $\text{CH}_3\text{COONa}$  are same. Together they constitute acidic buffer and for acidic buffer:

$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

Because  $[\text{salt}] = [\text{acid}]$ , so  $\text{pH} = \text{pK}_a$

$$\text{pH} = -\log 2 \times 10^{-5}$$

$$\text{pH} = \mathbf{4.7}$$

- 3.(B)  $\text{CH}_3\text{COO}^-$  when hydrolysed generates  $\text{OH}^-$  ions making the solution basic.  $\text{CH}_3\text{COONH}_4$  results in a neutral solution since  $K_a(\text{CH}_3\text{COOH}) = K_b(\text{NH}_4\text{OH})$

- 4.(C)  $\text{pH} = \frac{1}{2} [\text{pK}_{a_1} + \text{pK}_{a_2}] = \frac{1}{2} [7 + 11] = 9$ .

- 5.(B)  $\alpha_1^2 = \frac{K_{a_1}}{C_1}$   $[\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-]$  where  $\alpha_1$  is degree of dissociation of HA.

$$\alpha_2^2 = \frac{K_{a_2}}{C_2} \quad [\text{HB} \rightleftharpoons \text{H}^+ + \text{B}^-] \text{ where } \alpha_2 \text{ is degree of dissociation of HB.}$$

$$\text{or, } \frac{C_2}{C_1} = \left( \frac{\alpha_1}{\alpha_2} \right)^2 \times \frac{K_{a_2}}{K_{a_1}} = \left( \frac{1}{10} \right)^2 \times 50 = \frac{1}{2} \quad \therefore \quad C_1 : C_2 = \mathbf{2 : 1}$$

- 6.(D)  $K_{sp}$  of  $\text{PbCl}_2 = 4s^3 = 4 \times (0.01)^3 = 4 \times 10^{-6}$

$$\therefore \quad \text{Concentration of } \text{Pb}^{2+} \text{ in NaCl solution} = \frac{K_{sp}}{[\text{Cl}^-]^2} = \frac{4 \times 10^{-6}}{10^{-2}} = \mathbf{4 \times 10^{-4} \text{ M}}$$

- 7.(D)  $\text{CaF}_2(\text{s}) \rightleftharpoons \text{Ca}^{2+}(\text{aq}) + 2\text{F}^-(\text{aq})$

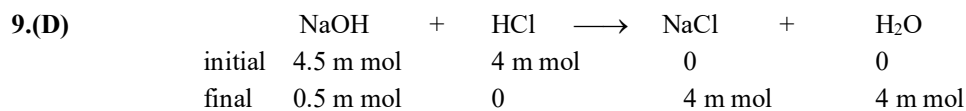
$$K_{sp} = (s)(2s)^2 = 4s^3 = 4 \times (2 \times 10^{-4})^3$$

$$K_{sp} = 4 \times 8 \times 10^{-12} = 32 \times 10^{-12}$$

- 8.(D)  $[H^+] = \sqrt{K_a C}$

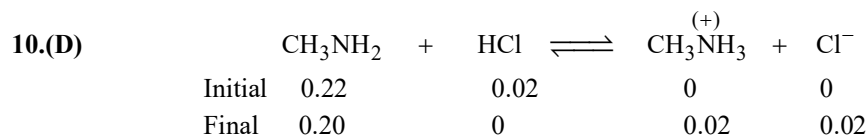
$$\sqrt{K_1 \times C_1} = \sqrt{K_2 \times C_2}$$

$$\sqrt{\frac{C_1}{C_2}} = \sqrt{\frac{K_2}{K_1}} = \sqrt{\frac{1}{4}} \quad \therefore \quad \frac{C_1}{C_2} = \frac{1}{4} = \mathbf{0.25}$$



So solution is basic hence  $[\text{OH}^-] = \frac{0.5}{50} = 10^{-2} \text{ M}$

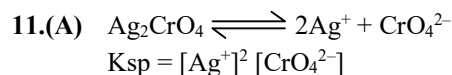
$\text{pOH} = 2$  ; so  $\text{pH} = 12$ .



It's a basic buffer

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} = 6 + \log \frac{[.02]}{[.20]}$$

$\text{pOH} = 6 - 1 = 5$ ; So  $\text{pH} = 9$  or  $[\text{H}^+] = 10^{-9} \text{ M}$ .



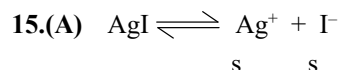
12.(C)  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$  is an acidic Buffer.

13.(B)  $\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} = 4.76 + \log \left( \frac{20 \times 0.5}{10} \right) = 4.76$

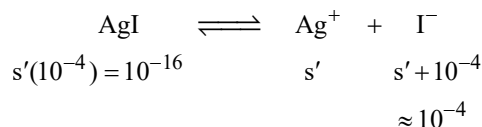
14.(A) When half of the ammonium hydroxide is neutralised a basic buffer is formed.

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} = 4.75 + \log(1)$$

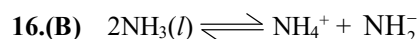
$\text{pH} = 9.25$



In presence of  $10^{-4} \text{ N KI}$ , solubility of  $\text{AgI}$  decreases due to common ion effect. Let it be  $S$ .



$s' = 1 \times 10^{-12} \text{ M}$ .



$$K_{\text{NH}_3} = [\text{NH}_4^+] [\text{NH}_2^-] = [\text{NH}_2^-]^2 = 10^{-30} \text{ M}^2$$

$$[\text{NH}_2^-] = 10^{-15} \text{ M}$$

$$\text{NH}_2^- \text{ ions per cm}^3 \text{ of liquid NH}_3 = \frac{10^{-15}}{1000} \times 6 \times 10^{23} = 6 \times 10^5 \text{ ions/cm}^3.$$

17.(D) For preparing a buffer solution of given pH, we prefer to choose an acid/base having  $\text{pK}_a/\text{pK}_b$  value as close as the pH value.

These are all acidic buffers, so pH is given by  $\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$

In (d),  $\text{pK}_a = -\log K_a = -\log(8 \times 10^{-8}) = 8 - 3 \log 2 = 8 - 0.9 = 7.1$ .

18.(D)  $\text{pH}_{\text{CH}_3\text{COOH}} = 3$

$$\therefore [\text{H}_3\text{O}^+] = \sqrt{c \times K_a} = 10^{-3}$$

$$\sqrt{c \times 10^{-5}} = 10^{-3}$$

$$c = 10^{-1} \text{ M}$$

pH of  $10^{-1} \text{ M}$  solution of  $\text{NH}_4\text{OH}$

$$[\text{OH}^-] = \sqrt{10^{-1} \times 10^{-5}} = 10^{-3}$$

$$\text{pOH} = 3$$

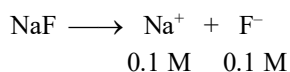
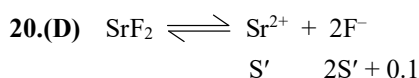
$$\therefore \text{pH} = 11.$$

19.(C)  $\text{HIn} \rightleftharpoons \text{H}^+ + \text{In}^-; \text{pH} = \text{pK}_{\text{in}} + \log \frac{[\text{In}^-]}{[\text{HIn}]}$

$$(\text{pH})_1 = \text{pK}_{\text{in}} + \log \frac{20}{80} = \text{pK}_{\text{in}} - 2 \log 2$$

$$(\text{pH})_2 = \text{pK}_{\text{in}} + \log \frac{80}{20} = \text{pK}_{\text{in}} + 2 \log 2$$

$$\text{Hence, } (\text{pH})_2 - (\text{pH})_1 = \text{pK}_{\text{in}} + 2 \log 2 - (\text{pK}_{\text{in}} - 2 \log 2) = 4 \log 2 = 1.20$$



Let the solubility of  $\text{SrF}_2$  in  $0.1 \text{ M}$   $\text{NaF}$  solution be  $\text{S}'$

$$\begin{aligned} K_{\text{sp}} &= [\text{Sr}^{2+}] [\text{F}^-]^2 \\ 8 \times 10^{-10} &= \text{S}'(2\text{S}' + 0.1)^2 \\ (2\text{S}' + 0.1) &\simeq 0.1 \end{aligned}$$

$$\text{S}' = \frac{8 \times 10^{-10}}{10^{-2}}$$

$$\text{S}' = 8 \times 10^{-8}.$$

21.(D) For precipitation, Ionic product > Solubility product.

24.(A)  $\text{pH} = -\log [\text{H}^+] = -\log [C\alpha] = -\log (0.5 \times 6 \times 10^{-3})$   
 $= 2.52$

25.(C) We want to double the pH i.e. = 5.04

$$\text{pH} = 5.04 = -\log [\text{H}^+] = -\log [c_1 \alpha_1]$$

$$10^{-6} = c_1 \alpha_1$$

$$K_a = \frac{[\text{H}^+] [\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

$$K_a = \frac{c_1 \alpha_1 \times c_1 \alpha_1}{c_1 (1 - \alpha_1)} = \frac{c_1 \alpha_1 \times \alpha_1}{1 - \alpha_1}$$

$$1.8 \times 10^{-5} = \frac{9.12 \times 10^{-6} \times \alpha_1}{1 - \alpha_1}$$

22.(C)  $K_{\text{sp}} = [\text{Mg}^{2+}] [\text{OH}^-]^2$   
 $1 \times 10^{-11} = \text{S} \times (2\text{S})^2$

$$\text{S} = \sqrt[3]{\frac{1 \times 10^{-11}}{4}}$$

$$\text{S} = 1.35 \times 10^{-4}$$

$$[\text{OH}^-] = 2\text{S} = 2.7 \times 10^{-4}$$

$$\text{pOH} = -\log [\text{OH}^-] = -\log 2.7 \times 10^{-4} = 4 - 0.43 = 3.7$$

$$\text{pH} = 14 - 3.7 = 10.43$$

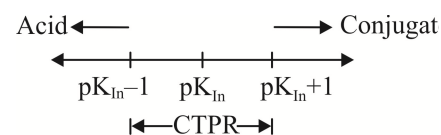
23.(B)  $K_a = C\alpha^2$

$$\begin{aligned} \alpha &= \sqrt{\frac{K_a}{C}} = \sqrt{\frac{1.8 \times 10^{-5}}{0.5}} = \sqrt{3.6 \times 10^{-5}} \\ &= 6 \times 10^{-3} \end{aligned}$$

26.(B)  $\text{pK}_{\text{In}} = \frac{4+6}{2} = 5 \Rightarrow$

$$K_{\text{In}} = 10^{-5}$$

27.(A) As per pH scale:



$\therefore$  In mid way  $\text{pH} = \text{pK}_{\text{In}}$

$$1.8 \times 10^{-5} - 1.8 \times 10^{-5} \alpha_1 = 9.12 \times 10^{-6} \alpha_1$$

$$1.8 \times 10^{-5} = 9.12 \times 10^{-6} \alpha_1 + 1.8 \times 10^{-5} \alpha_1 = \alpha_1 \times 2.712 \times 10^{-5}$$

$$\alpha_1 = \frac{1.8 \times 10^{-5}}{2.712 \times 10^{-5}} = 0.6637$$

$$c_1 \alpha_1 = 9.12 \times 10^{-6}$$

$$c_1 = \frac{9.12 \times 10^{-6}}{0.6637} = 1.37 \times 10^{-5} \text{ M}$$

30.(B) When  $V_{\text{NaOH}} = 0 \text{ mL}$ ;  $\text{pH} = 3$  or  $[\text{H}^+] = 0.001$

$$K_{a(\text{HB})} = \frac{C\alpha^2}{(1-\alpha)} = \frac{(0.001)\alpha}{1-\alpha} = 10^{-5}$$

$$\alpha = 0.01$$

$$C = \frac{0.001}{0.01} = 0.1$$

$$C = \frac{n_{\text{HB}}}{V}$$

$$0.1 = \frac{5}{V}; V = 50 \text{ mL}$$

$$\therefore [\text{NaB}] = \frac{\text{milli-moles of acid}}{\text{total volume}} = \frac{5}{50+50} = 0.05$$

$$\text{pH} = \frac{1}{2} [\text{pK}_w + \text{pK}_a + \log C] = \frac{1}{2} [14 + 5 + \log(0.05)] = 8.85$$

31.(A) For best indicator,  $\text{pH} = \text{pK}_{\text{In}}$  of an indicator as well as pH transition range of the indicator must coincide with the steep portion of the titration curve.

32.(A)  $[\text{S}^{2-}]_{\text{min}}$  for  $\text{SnS} = \frac{K_{\text{SP}}(\text{SnS})}{[\text{Sn}^{2+}]}$ ;  $[\text{S}^{2-}]_{\text{min}}$  for  $\text{CdS} = \frac{K_{\text{SP}}(\text{CdS})}{[\text{Cd}^{2+}]}$ ;  $[\text{S}^{2-}]_{\text{min}}$  for  $\text{NiS} = \frac{K_{\text{SP}}(\text{NiS})}{[\text{Ni}^{2+}]}$

and  $K_{\text{sp}}(\text{NiS}) > K_{\text{sp}}(\text{CdS}) > K_{\text{sp}}(\text{SnS}) \therefore \text{SnS}$  will precipitate first

33.(C)  $\text{H}_2\text{S} \rightleftharpoons 2\text{H}^+ + \text{S}^{2-} \Rightarrow K_1 K_2 = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]} = 10^{-21}$

$$[\text{S}^{2-}]_{\text{min}}(\text{NiS}) = 3 \times 10^{-20} \text{ M}, \quad [\text{H}^+] = \sqrt{\frac{10^{-22}}{3 \times 10^{-20}}} = 5.77 \times 10^{-2}, \quad \text{pH} = 2 - \log 5.77 = 2 - 0.76 = 1.24$$

34.(B) Solubility of either  $\text{CdS}$ ,  $\text{NiS}$ ,  $\text{SnS} = \sqrt{K_{\text{SP}}} \Rightarrow$  look for highest  $K_{\text{SP}}$ .

35.(A)  $K_1 K_2 = \frac{[\text{H}^+]^2 [\text{S}^{2-}]}{[\text{H}_2\text{S}]} \Rightarrow [\text{S}^{2-}]_{\text{sat.}} = 10^{-20} \text{ M} \Rightarrow [\text{Cd}^{2+}]_{\text{allowed}} = \frac{K_{\text{SP}}(\text{CdS})}{[\text{S}^{2-}]_{\text{sat.}}} = 10^{-8} \text{ M}$

36.(ACD)  $\text{NH}_3 / \text{NH}_4^+$ ,  $\text{HCO}_3^- / \text{CO}_3^{2-}$ ,  $\text{H}_2\text{S} / \text{HS}^-$  are conjugate acid-base pairs.

37.(ABC) (A)  $C = 10^{-3}$ ,  $\alpha = 10^{-1}$

$$[\text{H}^+] = C\alpha = 10^{-4} \Rightarrow \text{pH} = 4$$

(B)  $[\text{H}^+] = \sqrt{K_a \cdot c} = \sqrt{10^{-2} \times 10^{-6}} = 10^{-4} \Rightarrow \text{pH} = 4$

(C)  $[\text{H}^+] = [\text{A}^-] = 10^{-4} \Rightarrow \text{pH} = 4$

28.(C) When acid is half neutralized  $\text{pH} = \text{pK}_a$   
 $\therefore \text{pK}_a = 5$

29.(A)  $K_{\text{eq}} = \frac{K_a(\text{HB})}{K_a(\text{HA})} = \frac{10^{-5}}{10^{-6}} = 10$

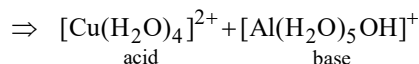
38.(BC)  $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$  is conjugate acid base pair.

pH = 7, salt of strong acid and strong base

39.(BC)  $\text{Cu}[(\text{H}_2\text{O})_3(\text{OH})]^+ + [\text{Al}(\text{H}_2\text{O})_6]^+$   
(A)

base

acid



40.(AB) Option (A) is a basic buffer while option (B) is an acidic buffer. The pH of a buffer solution does not change on dilution.

41.(ABC) (A) 0.1mol NaOH + 0.15mol of  $\text{CH}_3\text{COOH}$

↓

0.1mol  $\text{CH}_3\text{COONa}$  + 0.05mol of  $\text{CH}_3\text{COOH}$

↓

Buffer solution

(B) Salt of WA & WB acts as a mixed buffer solution.

(C) Conjugate acid – base pair

42.(A) Concentration of  $\text{Cl}^-$  in solution when pure  $\text{AgCl(s)}$  is added to 0.01 M  $\text{AgNO}_3$  is given by  $[\text{Cl}^-] = \frac{K_{\text{sp}}}{[\text{Ag}^+]} = \frac{K_{\text{sp}}}{0.01}$

Similarly, concentration of  $\text{Ag}^+$  in the second solution when pure  $\text{AgCl(s)}$  is added to 0.025 M  $\text{KCl}$  is given by

$$[\text{Ag}^+] = \frac{K_{\text{sp}}}{[\text{Cl}^-]} = \frac{K_{\text{sp}}}{0.025}$$

The ratio of  $[\text{Cl}^-]$  in the first solution to  $[\text{Ag}^+]$  in the second solution is given by  $\frac{[\text{Cl}^-]_1}{[\text{Ag}^+]_2} = \frac{0.025}{0.01} = 2.5$ .

43.(ACD)  $\frac{[\text{H}^+]_1}{[\text{H}^+]_2} = \frac{C\alpha_1}{C\alpha_2} = \frac{\alpha_1}{\alpha_2} = \sqrt{\frac{K_1}{K_2}}$

44.(ABCD) (A)  $[\text{CH}_3\text{COOH}] > [\text{NaOH}]$  : Final solution  $[\text{CH}_3\text{COOH}] + [\text{CH}_3\text{COONa}]$

(B)  $[\text{CH}_3\text{COONa}] > [\text{HCl}]$  : Final solution  $[\text{CH}_3\text{COOH}] + [\text{CH}_3\text{COONa}]$

(C) (weak acid + weak base) salt

(D) Acid + conjugate base pair

45.(CD) Bronsted acid is  $\text{H}^+$  donor. Lewis acid is  $e^-$  pair accept D stronger the acid, weaker is conjugate base

46.(ABCD) Refer theory

47.(AB)

Colour P  $\leftarrow$   $\xrightarrow$  Colour Q ( $\text{In}^-$ )  
(HIn)  $\leftarrow$   $\xrightarrow$   
 $\text{pK}_{\text{In}}-1$   $\text{pK}_{\text{In}}$   $\text{pK}_{\text{In}}+1$

49.(ABCD) All statements are correct.

48.(BC)  $\text{pH} = \text{pK}_{\text{In}} + \log \frac{[\text{In}^-]}{[\text{HIn}]}$

75% red  $\Rightarrow \frac{[\text{HIn}]}{[\text{In}^-]} = \frac{3}{1}$

$\text{pH} = (5 - \log 3) + \log \left( \frac{1}{3} \right)$

$\text{pH} = 4.05$

75% blue  $\Rightarrow \frac{[\text{HIn}]}{[\text{In}^-]} = \frac{1}{3}$

$\text{pH} = (5 - \log 3) + \log 3$

$\text{pH} = 5$

50.(ACD) Two weak acids  $HA_1$  &  $HA_2$ ,



$$[H^+]_1 = C_0 \alpha_1 = \sqrt{K_1 C_0} \quad [H^+]_2 = C_0 \alpha_2 = \sqrt{K_2 C_0}$$

$$\text{Relative acidic strengths} = \frac{[H^+]_1}{[H^+]_2} = \frac{\alpha_1}{\alpha_2} = \frac{K_1}{K_2}$$

51.(ABCD)

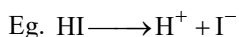
- (A)  $[CH_3COOH] > [NaOH]$   
 $\rightarrow$  gives a mixture of  $CH_3COOH + CH_3COONa$
- (B)  $[CH_3COONa] > [HCl]$   
 $\rightarrow$  gives a mixture of  $CH_3COONa + CH_3COOH$
- (C)  $NH_4CN$  – Salt of weak acid & weak base  
 $\rightarrow$  Acts as simple buffer
- (D)  $HCN + NaCN$   
 $\rightarrow$  weak acid & its salt

52.(CD) All Arrhenius acids & bases are Bronsted acids and bases

All Bronsted acids & bases are Lewis acids & bases but not vice-versa

Eg.  $BF_3$  is a Lewis acid (electron acceptor) but not Bronsted acid

Conjugate base of a strong acid is a weak base



$HI$  is a strong acid &  $I^-$  is a weak base

53.(AB)  $\underset{\text{Colour P}}{HIn} \rightleftharpoons H^+ + \underset{\text{Colour Q}}{In^-}$

Solution assumes colour P, when  $[HIn] \geq 10[In^-]$

$$\therefore pH \leq pK_{In} - 1$$

Solution assumes colour Q, when  $[In^-] \geq 10[HIn]$

$$pH \geq pK_{In} + 1$$

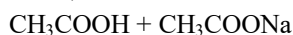
54.(ABC)  $KCN$  is a basic salt so its addition in water increases the pH of water.

55.  $[A-p, s] \rightarrow [B-q, r] \rightarrow [C-p, r] \rightarrow [D-p]$

- (A) As  $CH_3COONa$  is formed
- (B) As  $CH_3COOH + CH_3COONa$  is left
- (C) As  $NH_4Cl + NH_4OH$  is left
- (D) As  $NaOH$  is left

56.  $[A-p, s] \rightarrow [B-q] \rightarrow [C-p, r] \rightarrow [D-p, q]$

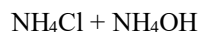
(A) 50 ml of 0.1 M  $CH_3COOH$  + 25 ml of 0.1 M  $NaOH$



(B) 50 ml of 0.1 M  $NaOH$  + 50 ml of 0.1 M  $HCl$



(C) 50 ml of 0.1 M  $NH_4OH$  + 25 ml of 0.1 M  $HCl$

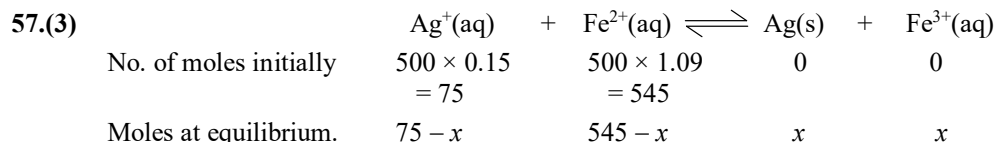




(D) 50 ml 0.1 M CH<sub>3</sub>COOH + 50 ml of 0.1 M NH<sub>4</sub>OH



CH<sub>3</sub>COONH<sub>4</sub> (salt of weak acid & weak base)



On titration of reaction mixture with KMnO<sub>4</sub>, only Fe<sup>2+</sup> reacts with it.

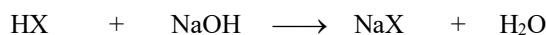
∴ Equivalents of Fe<sup>2+</sup> in 25 ml = Equivalents of KMnO<sub>4</sub>

$$\frac{(545 - x)}{1000} \times 25 \times 10^{-3} \times 1 = 30 \times 10^{-3} \times 0.0833 \times 5$$

$$\frac{(545 - x) \times 25}{1000} \approx 12.5 ; 545 - x = 500 ; x = 45$$

$$\therefore K_c = \frac{[\text{Fe}^{3+}]}{[\text{Ag}^+][\text{Fe}^{2+}]} = \frac{\frac{45}{1000}}{\frac{(75 - 45)}{1000} \times \frac{(545 - 45)}{1000}} = \frac{45 \times 1000}{30 \times 500} = 3.0$$

58.(6) Let the molarity and volume of HX be M<sub>1</sub> and V<sub>1</sub> ml respectively while the molarity of NaOH be M<sub>2</sub>.



Initial millimoles M<sub>1</sub>V<sub>1</sub> 10M<sub>2</sub> – –

After the addition M<sub>1</sub>V<sub>1</sub> – 10M<sub>2</sub> 0 10 M<sub>2</sub> –

of 10 c.c of NaOH

Since weak acid HX and NaX are left after the reaction, they will constitute an acidic buffer.

$$\therefore \text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$5.7 = \text{pK}_a + \log \frac{10 M_2}{(M_1 V_1 - 10 M_2)} \quad \dots(1)$$

Let V<sub>2</sub> ml be the volume of NaOH required to neutralize given HX completely.

∴ At equivalence point, M<sub>1</sub>V<sub>1</sub> = M<sub>2</sub>V<sub>2</sub>

$$V_2 = \frac{M_1 V_1}{M_2}$$

Dividing the numerator and denominator of log term by M<sub>2</sub>, we get

$$\therefore 5.7 = \text{pK}_a + \frac{\frac{10 M_2}{M_2}}{\frac{M_1 V_1 - 10 M_2}{M_2}} = \text{pK}_a + \log \frac{10}{V_2 - 10} \quad \dots(2)$$

Similarly, after the addition of 20 c.c of NaOH, we have

$$6.3 = \text{pK}_a + \log \frac{20}{V_2 - 20} \quad \dots(3)$$

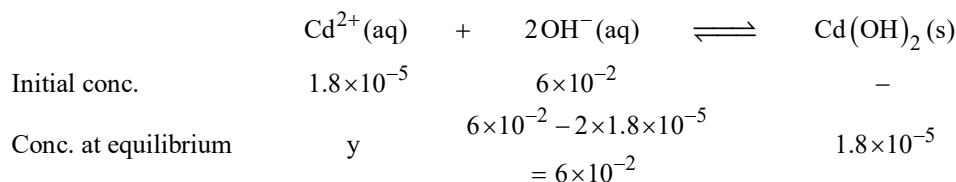
Subtracting equation (2) from equation (3)  $6.3 - 5.7 = \log \frac{20}{V_2 - 20} - \log \frac{10}{V_2 - 10}$

$$0.6 = \log \left( \frac{20}{V_2 - 20} \times \frac{V_2 - 10}{10} \right)$$

Taking antilog,  $4 = \frac{2(V_2 - 10)}{(V_2 - 20)} \quad \therefore V_2 = 30 \text{ ml.}$

Putting  $V_2$  in equation (2),  $5.7 = pK_a + \log \frac{10}{30-10} = pK_a + \log \frac{1}{2}$   
 $pK_a = 5.7 + \log 2 = 5.7 + 0.3 = 6.0$ .

59. The equilibrium constant for the precipitation reaction is the inverse of  $K_{sp}$  for  $Cd(OH)_2$ .



$$K_{eq} = \frac{1}{[Cd^{2+}]_{eq} [OH^-]_{eq}^2} = \frac{1}{K_{sp}} = \frac{1}{2.5 \times 10^{-14}} = 4.0 \times 10^{13}$$

$$M_2 = \frac{M_1 V_1}{V_2} = \frac{(6.0 M)(1.0 L)}{1.00 \times 10^2 L} = 6.0 \times 10^{-2} M = [OH^-]_{initial}$$

$$K_{sp} = (y)(6.0 \times 10^{-2} + 2y^2) \approx (y)(6.0 \times 10^{-2})^2 = 2.5 \times 10^{-14}$$

$$y = \frac{2.5 \times 10^{-14}}{(6.0 \times 10^{-2})^2} = 6.9 \times 10^{-12}$$

$$[Cd^{2+}]_{eq} = 6.9 \times 10^{-12} M$$

$$\text{mol of } Cd(OH)_2 = \text{mol of } Cd^{2+} = MV = (1.8 \times 10^{-5} M)(1.00 \times 10^2 L) = 1.8 \times 10^{-3} \text{ mol}$$

$$\text{Mass of } Cd(OH)_2 = (\text{mol})(MM) = (1.8 \times 10^{-3} \text{ mol})(146 \text{ g/mol}) = 0.26 \text{ g}$$

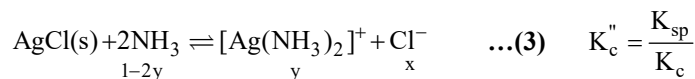
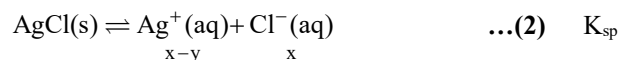
60.  $[Cl^-] = [Ag^+] = \sqrt{K_{spAgCl}} = \sqrt{1 \times 10^{-10}} = 10^{-5} M$

Mixing with NaBr has  $[Ag^+] = \frac{10^{-5} \times 100}{200}$  and  $[Br^-] = \frac{100 \times 0.03}{200}$

$$[Ag^+][Br^-] = \left[ \frac{10^{-5} \times 100}{200} \right] \times \left[ \frac{100 \times 0.03}{200} \right] = 7.5 \times 10^{-8}$$

The product of ionic concentration is greater than  $K_{sp}$  and thus AgBr will be precipitated.

61.  $Ag^+ + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^+ \quad \dots(1) \quad K'_c = \frac{1}{K_c}$



As the value of  $K'_c = \frac{1}{K_c}$  is very large, so  $x \approx y$

$$K''_c = \frac{[Ag(NH_3)_2]^+ [Cl^-]}{[NH_3]^2}$$

$$\frac{K_{sp}}{K_c} = \frac{y^2}{(1-2y)^2}$$

$$\frac{y}{1-2y} = \sqrt{\frac{1.8 \times 10^{-10}}{6.2 \times 10^{-8}}}$$

$$\frac{y}{1-2y} = 0.05$$

$$y = 0.05 - 0.1y$$

$$1.1y = 0.05$$

$$y = \frac{0.05}{1.1} = 0.0489$$

62.(9)  $[\text{Cl}^-] = 0.1\text{M}$   $[\text{Br}^-] = 0.01\text{M}$   $[\text{I}^-] = 0.001\text{M}$



$$[\text{Ag}^+][\text{Cl}^-] = K_{\text{sp}} \Rightarrow [\text{Ag}^+] = \frac{10^{-10}}{0.1} = 10^{-9}\text{M}$$

AgCl will start precipitating after  $[\text{Ag}^+]$  is greater than  $10^{-9}$ .



$$[\text{Ag}^+][\text{Br}^-] = K_{\text{sp}} \Rightarrow [\text{Ag}^+] = \frac{10^{-13}}{0.01} = 10^{-11}$$

AgBr starts precipitating after the concentration of  $[\text{Ag}^+]$  is greater than  $10^{-11}$ .



$$[\text{Ag}^+][\text{I}^-] = K_{\text{sp}} \Rightarrow [\text{Ag}^+] = \frac{10^{-17}}{0.001} = 10^{-14}$$

AgI precipitate when  $[\text{Ag}^+]$  is greater than  $10^{-14}$ .

Order of precipitation :  $\text{AgI} > \text{AgBr} > \text{AgCl}$

Concentration of  $\text{Ag}^+$  required to ppt. all three ions =  $10^{-9}$ .

63.(5.97)  $\text{pK}_{\text{a}1} = 2.34$

$$\text{pK}_{\text{a}2} = 9.60$$

$$\text{The first inflection point } \text{pH} = \frac{\text{pK}_{\text{a}1} + \text{pK}_{\text{a}2}}{2} = \frac{2.34 + 9.60}{2} = 5.97$$

64.(3.88)

At equivalence point,

$$\text{pH} = 8.15$$

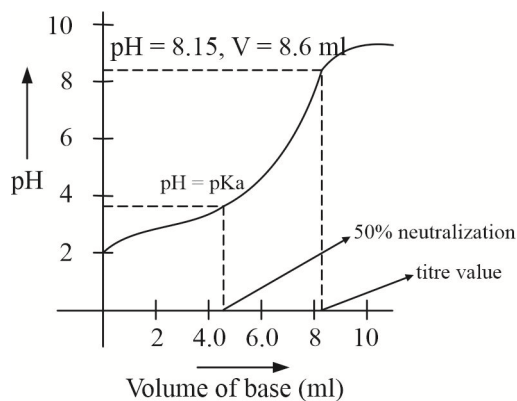
$$V_{\text{base}} = 8.6\text{mL}$$

At half equivalence point,

$$[\text{salt}] = [\text{acid}]$$

$$\Rightarrow \text{pH} = \text{pK}_{\text{a}}$$

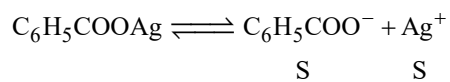
$$\Rightarrow \text{pH} = 3.88$$



65.(3.31)  $K_a = 6.46 \times 10^{-5} (\text{C}_6\text{H}_5\text{COOH})$

$$K_{\text{SP}}(\text{C}_6\text{H}_5\text{COOAg}) = 2.5 \times 10^{-13}$$

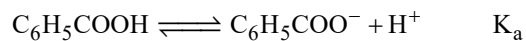
Solubility in water



$$K_{\text{sp}} = [\text{C}_6\text{H}_5\text{COO}^-][\text{Ag}^+]$$

$$S^2 = 2.5 \times 10^{-13} \Rightarrow S = 5 \times 10^{-7}$$

Solubility in a buffer of pH = 3.19



$$S = \left\{ \frac{K_{\text{SP}}}{K_a} [\text{H}^+] + K_a \right\}^{1/2}$$

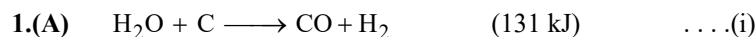
$$S = \left\{ \frac{2.5 \times 10^{-13}}{6.46 \times 10^{-6}} (64.6 \times 10^{-5} + 6.46 \times 10^{-5}) \right\}^{1/2}$$

$$S = 16.5 \times 10^{-7}$$

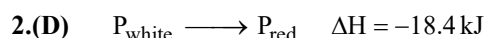
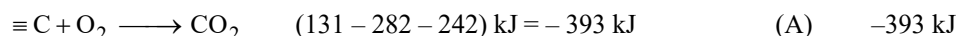
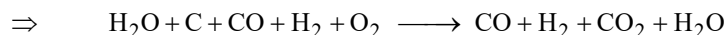
$$\frac{S_{\text{buffer}}}{S_{\text{water}}} = \frac{16.55 \times 10^{-7}}{5 \times 10^{-7}} = 3.31$$

## Solutions - Advanced Problem Package | Chemistry

### Thermochemistry & Thermodynamics



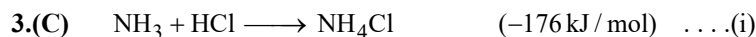
(i) + (ii) + (iii)



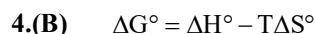
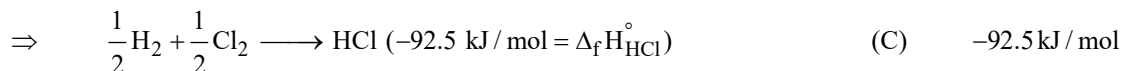
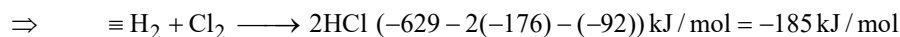
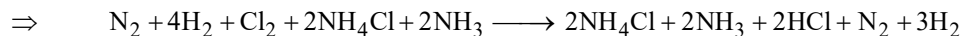
The energy considerations, i.e., the thermodynamics of reaction does not tell us of the kinetics of the reaction. We can not say that  $\text{P}_{\text{red}}$  is readily formed from  $\text{P}_{\text{white}}$ .

But, it can be said that  $\text{P}_{\text{red}}$  is more stable as energy is released in its production

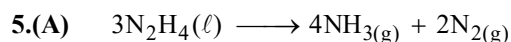
(D) White P can be converted to red P and red P is more stable.



(iii) - 2(i) - (ii)



At equilibrium,  $\Delta G^\circ \Rightarrow \Delta H^\circ - T\Delta S^\circ \Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{30000 \text{ cal}}{40 \text{ cal/K}} = 750 \text{ K} \quad (\text{B}) \quad 750 \text{ K}$



1 mol  $\text{N}_2\text{H}_4(\ell)$  would produce 2 mol gases

(3 mol  $\text{N}_2\text{H}_4$  produces 6 mol of gaseous products)

$P = 1.0 \text{ atm}, T = 27^\circ\text{C} = 300\text{K}$

Work done by system =  $P\Delta V = RT\Delta n = (8.314 \text{ J/mol/K}) (300\text{K}) (2\text{mol}) = 4988.4 \text{ J} \quad (\text{A}) \quad -4988.4 \text{ J}$



$\Delta H_f^\circ = \Delta H_f^\circ(\text{CaCO}_3) - \Delta H_f^\circ(\text{CaO}) - \Delta H_f^\circ(\text{CO}_2)$

$= -1207 - (-635) - (-394)$

$\Delta H_f^\circ = -178 \text{ kJ/mol}$

$\therefore \Delta E = \Delta H - \Delta n_{(\text{g})}RT$

$$\Delta E = -178 - \frac{(-1) \times 8.3 \times 300}{1000} = -175.51$$

$$n_{\text{CaO}} = \frac{280}{56} \Rightarrow 5.0 \quad \therefore \quad q_v = \Delta E$$

For  $n$  moles  $q_v = \Delta E = 5 \times (-175.51) = -877.55 \text{ kJ}$

$$|q| \Rightarrow 877.55 \text{ kJ}$$

$$7.(A) \quad P_1 V_1 = nRT \Rightarrow V_1 = \frac{nRT}{P_1} = \frac{(0.5)(0.08)(300)}{(2)} = 6 \text{ L initial}$$

As  $T$  is constant,  $PV$  is also constant at the end of each step.

$$P_1 V_1 = P_1 V_1 = P_2 V_2 = P_3 V_3 \quad \Rightarrow \quad V_1 = \frac{P_1 V_1}{P_1} - V_2 = \frac{P_1 V_1}{P_2}, \quad V_3 = \frac{P_1 V_1}{P_3}$$

$$\therefore \quad W_1 = -P_1(V_1 - V_1) = -(P_1 - P_1)V_1$$

Similarly,

$$W_2 = -P_2(V_2 - V_1) = -(P_1 - P_2)V_1$$

$$W_3 = -P_3(V_3 - V_2) = -(P_2 - P_3)V_2$$

$$\therefore \quad |W| = (2 - 1.6)(6) + (1.6 - 1.2)\left(\frac{12}{1.6}\right) + (1.2 - 1)\left(\frac{12}{1.2}\right) = (0.4)(6) + (0.4)(7.5) + (0.2)(10) \\ = 2.4 + 3 + 2 = 7.4 \text{ atm-L}$$

$$(A) \quad 7.4 \text{ atm-L}$$

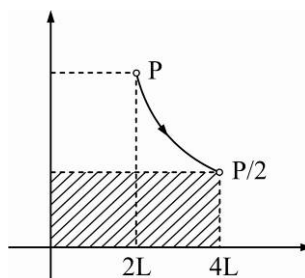
$$\text{Alter : } |W| = nRT \left( \frac{P_1 - P_1}{P_1} + \frac{P_1 - P_2}{P_1} + \frac{P_2 - P_3}{P_2} \right) = 12 \left( \frac{0.4}{2} + \frac{0.4}{1.6} + \frac{0.2}{1.2} \right) = 2.4 + 3.2 = 7.4 \text{ atm-L}$$

8.(B) Work calculated by student

$$= \frac{P}{2} \cdot 4L = P \cdot 2L = W_0$$

$$\text{Work required} = P \cdot 2L \cdot \ln \left( \frac{4L}{2L} \right) = W_0 \ln 2$$

$$= 49.26 \text{ L atm} \times 0.69 = 34.14 \text{ L-atm}$$



Since, this work is done by the system, work done on system =  $-34.14$ .

$$(B) \quad -34.14$$

9.(C) Pressure is constant,  $P$

Volume doubles from  $V$  to  $2V$

$\Rightarrow$

Temperature double from  $T$  to  $2T$

$$\Delta u = nC_V \Delta T = \frac{C_V P \Delta V}{R} = \frac{C_V P V}{R}$$

$$\gamma = \frac{C_P}{C_V} = \frac{C_V + R}{C_V} = 1 + \frac{R}{C_V} \quad \Rightarrow \quad \frac{R}{C_V} = \gamma - 1 \quad \Rightarrow \quad \frac{C_V}{R} = \frac{1}{\gamma - 1}$$

$$\therefore \quad \Delta u = \frac{P V}{\gamma - 1} \quad (C) \quad \frac{P V}{\gamma - 1}$$

10.(B) Work done by gas in A :  $2W$  ; Work done by gas in B :  $W$

Temperature rise is same in both processes.

$\therefore$  More heat is absorbed in A than B for same temperature rise  $\Rightarrow C_A > C_B$

$$(B) \quad C_A > C_B$$

11.(A) In AB : 600 J heat is added

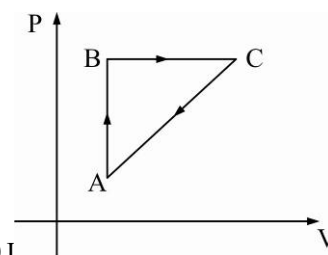
$$q_{AB} = 600 \text{ J}, W_{AB} = 0 \Rightarrow 600 \text{ J}$$

For BC, 200 J heat is added

$$W_{BC} = P_B(V_C - V_B) = P_B(V_C - V_A) = 8 \times 10^4 (5 - 2) \times 10^3 = 240 \text{ J}$$

$$q_{BC} + W_{BC} = \Delta U_{BC} \Rightarrow \Delta u_{BC} = 200 \text{ J} - 240 \text{ J} = -40 \text{ J}$$

$$\Delta U_{AB} + \Delta U_{BC} = \Delta U_{AC} \text{ internal energy is state function} \Rightarrow \Delta U_{AC} = 560 \text{ J}$$



12.(D) Proceeding without external influence once it has begun. (Defn)



$$2(i) + (ii) - \frac{1}{2}(iii)$$

$$= 2C + H_2 \longrightarrow C_2H_2 \left( 2(-393.5) + (-285.8) - \frac{1}{2}(-2598.8) \right) \text{ kJ/mol}$$

$$= 226.6 \text{ kJ/mol}$$

14.(D) All spontaneous processes are those where total entropy i.e. system + surroundings increases.

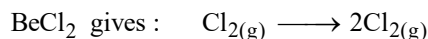
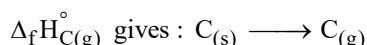
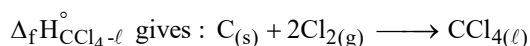
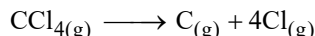
Thus, (a), (c), (d) are true.

Basically, time moves in the direction where entropy of universe increases.

Now, in (b), we have a closed system, i.e. even though matter can't move into or out of system, energy can.

Clearly, (b) can also be true of entropy of surroundings increase more than decrease of entropy of system.

15.(D) C – Cl energy is calculated from



Clearly, we need all these values to calculate C – Cl bond energy.

16-18. Heat of formation : Heat absorbed/evolved when one mol of a compound is formed from constituent element in standard state.

Heat of combustion : Heat absorbed/evolved when one mol of compound is burnt completely in excess of oxygen.

16.(A) Energy released = x

$$\therefore \Delta H = -x$$

$$(A) - x$$

17.(C)  $\Delta H_{\text{combustion, } C_6H_{12}O_6}^\circ$

18.(B)  $x + z$  (Hess' law)

19-21.  $\gamma = \frac{C_P}{C_V} = \frac{5}{3}$  for monoatomic gas

19.(A) 1.66

20.(B) Argon : molar mass 40 g

$$\text{Energy required} = \frac{20\text{g}}{40\text{g/mol}} \times 5\text{ cal/mol/}^\circ\text{C} \times 1^\circ\text{C} = 2.5\text{ cal}$$

21.(C) Constant volume, specific heat

$$\Rightarrow \left[ \frac{\delta E}{\delta T} \right]_V$$

22-24.

22.(A) -ve  $\Delta S \Rightarrow$  randomness of system decreases

Clearly in adsorption, the movement, i.e. randomness of absorbed particles decreases.

(A) adsorption

23.(A) Clearly,  $\Delta S_{\text{fusion}}$  is at melting point.

$\Delta S_{\text{vap.}}$  is at boiling point.

(A)  $T_1$  is MP,  $T_2$  is BP

24.(D) Absolute entropy is determined using 3<sup>rd</sup> law which states that entropy of system at 0K is zero.

(D) 3<sup>rd</sup> law.

25-27.

25.(D) Isochoric  $\Rightarrow \Delta V = 0$

$\Delta u = q + w$  is always true (1<sup>st</sup> law)

As  $\Delta V = 0$ ,  $\Delta u = q_V$

(D) all of these

26.(D) In isothermal process,  $\Delta T = 0$

$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$  is not true for isothermal processes. It is true for adiabatic processes.

(D)  $T_1 V_1^{-1} = T_2 V_2^{\gamma-1}$

27.(C)

28.(ABC) Isolated system

$\Rightarrow$  matter and energy cannot be exchanged through it.

(BC) impermeable, adiabatic

29.(AB)  $\Delta E = 0 \Rightarrow \Delta T = 0$

(A) cyclic process, isothermal expansion.

30.(AB) If both temperature are less than critical temperature on increasing temperature, Z would increase. Gas would become less compressible and volume would become more than double.

However, above Boyle temperature, on increasing temperature, Z decreases. Gas becomes more compressible and volume would be less than doubled.

31.(ABC) Molar internal energy is intensive property. Heat capacity is extensive property

Reversible processes take place quasistatically.

32.(AD) Reversible adiabatic process has  $\Delta S = 0$  as  $\Delta G = 0$

$\Delta S_{\text{surr}} = 0$ ,  $\Delta S_{\text{sys}} > 0$  : Irreversible adiabatic expansion

$\Delta S_{\text{surr}} = 0$ ,  $\Delta S_{\text{sys}} > 0$  : Free expansion

$\Delta S_{\text{surr}} > 0$ ,  $\Delta S_{\text{sys}} < 0$  : Irreversible isothermal compression

33.(ACD) All spontaneous processes are irreversible (ABC)



At 0.1 atm, BP of water < 373 K.

So, at T = 373 K vaporization is spontaneous

- 34.(ACD) 3<sup>rd</sup> law: At 0 K, entropy of perfectly crystalline substance = 0  
At all temperatures above 0 K, entropy > 0

(ACD) are false

- 35.(AD)  $\Delta_f H^\circ \cdot H^+(aq), Br_2(l) = 0$

- 36.(ABC) Isothermal irreversible expansion  $\Delta U = 0$

$$w = -p_{\text{ext.}}(V_2 - V_1)$$

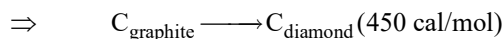
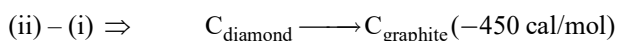
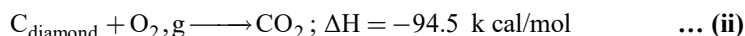
$$p_{\text{ext}} = p_2,$$

$$p_2 V_2 = p_1 V_1 = RT$$

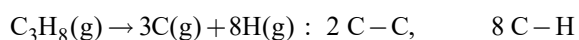
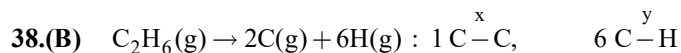
$$\therefore w = -p_2(V_2 - V_1) = (p_2 - p_1)V_1 = -\left(1 - \frac{p_2}{p_1}\right)RT$$

$$q = -w = RT\left(1 - \frac{p_2}{p_1}\right) = \Delta H$$

- 37.(BC)  $C_{\text{graphite}} + O_2, g \longrightarrow CO_2; \Delta H = -94.05 \text{ k cal/mol} \dots(i)$



Clearly, graphite is more stable allotrope



$$x + 6y = 620 \quad \dots(i)$$

$$2x + 8y = 880 \quad \dots(ii)$$

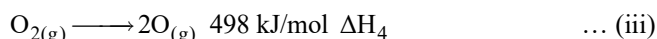
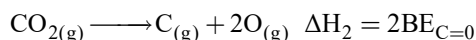
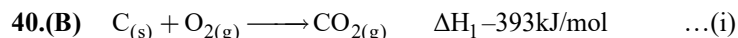
$$2(i) - (ii) \Rightarrow 4y = 360 \Rightarrow y = 90 \text{ kJ/mol}$$

$$\Rightarrow x = 80 \text{ kJ/mol} \quad (\text{B}) 80 \text{ and } 90 \text{ kJ/mol}$$

39.(AB)  $ds = \frac{dq}{T} = \frac{du - dw}{T} = \frac{nC_v dT + pdV}{T} = nC_v \frac{dT}{T} + nR \frac{dV}{V}$

$$\Rightarrow \int ds = nC_v \int \frac{dT}{T} = nR \int \frac{dV}{V} \Rightarrow \Delta S = nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right)$$

$$(A) nC_v \ln\left(\frac{T_2}{T_1}\right) + nR \ln\left(\frac{V_2}{V_1}\right) \quad \text{Modify the equal to get (B) also}$$



Thus expected value  $\Delta H_2 = \Delta H_3 + \Delta H_4 - \Delta H_1 = 1609 \text{ kJ/mol}$

Actual value of  $\Delta H_2 = 2BE_{C=O} = 678 \text{ kJ/mol}$

Difference:  $(678 - 1609) \text{ kJ/mol} = -931 \text{ kJ/mol}$

41.(ABD)  $-\Delta G^\circ = RT \ln k$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At equilibrium,  $\Delta G = 0$

Thus, (ABC) are true

42.(BCD) According to II<sup>nd</sup> law  $dS_m = C_V \frac{dT}{T} + R \frac{dV}{V}$

$$dG = dH - SdT$$

$$dH = dq \quad (D) \text{ are true}$$

43.(ACD) Free expansion :  $\Delta U = 0, \Delta T = 0, q = 0$

$$\Delta S_{\text{sys}} > 0, \Delta S_{\text{surr}} = 0, W_{\text{sys}} = 0,$$

$$\Delta G < 0$$

(ACD)

44.(BC)  $\Delta G_{\text{sys}} < 0 \Rightarrow \Delta H_{\text{sys}} - T\Delta S_{\text{sys}} < 0$

Reaction need not always be exothermic.

$$\Delta_f H^\circ_{\text{s, rhombic}} = 0, \Delta_f H^\circ_{\text{s, monoclinic}} \neq 0$$

$$\text{CH}_4 : 4\text{C-H} : 1656 \text{ kJ/mol}$$

$$\Rightarrow \text{C-H} : 414 \text{ kJ/mol}$$

$$\text{C}_2\text{H}_6 : \text{C-C}, 6 \text{ C-H} : 2812 \text{ kJ/mol}$$

$$6 \text{ C-H} : 2484 \text{ kJ/mol}$$

$$\Rightarrow \text{C-C} : 328 \text{ kJ/mol}$$

$$\text{H}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})} \rightarrow \text{H}_2\text{O}(\text{l}) -56 \text{ kJ/mol}$$

$$\Delta_f H^\circ \text{H}_2\text{O}_g = -242 \text{ kJ/mol}$$

$$\Delta_{\text{vap}} H^\circ \text{H}_2\text{O} = 44 \text{ kJ/mol}$$

$$\therefore \Delta_f H^\circ \text{H}_2\text{O}, \text{l} = (-242 - 44) \text{ kJ/mol} = -286 \text{ kJ/mol}$$

$$\Delta_r H^\circ = \Delta_f H^\circ_{\text{H}_2\text{O}, \text{l}} - \Delta_f H^\circ_{\text{OH}^-, \text{aq}} \Rightarrow \Delta_f H^\circ_{\text{OH}^-, \text{aq}} = (-286 \text{ kJ/mol}) - (-56 \text{ kJ/mol}) = -230 \text{ kJ/mol}$$

45.(ABD) We know,

$$\Delta G = \Delta H - T\Delta S; \Delta G \text{ will be negative for spontaneous process.}$$

$\therefore$  When  $\Delta H = -\text{ve}$  and  $\Delta S = +\text{ve}$ , then  $\Delta G = -\text{ve}$  and process will be spontaneous.

In all other conditions, the process may be non spontaneous.

46.(ABC) During adiabatic expansion or compression of an ideal gas,  $q = 0$

$$\therefore w = \Delta U = nC_V \Delta T$$

$$\text{As, } C_V = \frac{R}{\gamma - 1}$$

$$\Rightarrow W = \frac{nR}{\gamma - 1} (T_2 - T_1)$$

47.(BC) For the cyclic process, ABCA,

$$\Delta H = 0, \Delta U = 0, \Delta S = 0$$

Total work done = Total heat exchange

$$\text{i.e., } W = q$$

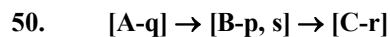
48.(BC)  $\Delta H = \Delta U + (\Delta n_g)RT$

For  $\Delta H < \Delta U$ ,  $\Delta n_g > 0$

In options A & D,  $\Delta n_g = 0$

In options B & C,  $\Delta n_g > 0$

49.(ABCD) Refer to Theory



Isochoric :  $w = 0$ ,  $\Delta V = 0$

Isothermal reversible expansion :  $\Delta T = 0$ ,  $\Delta E = 0$ ,  $\Delta H = 0$

Adiabatic work :  $w = \frac{nR}{\gamma - 1}(T_2 - T_1) = nC_V\Delta T$



10 mL HA + 10 mL BOH produces  $5^\circ\text{C}$  temperature rise

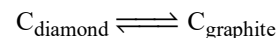
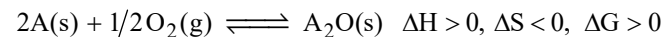
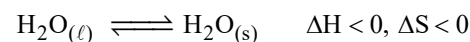
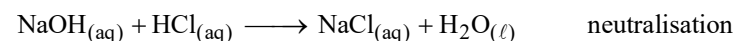
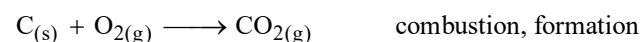
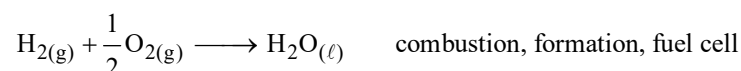
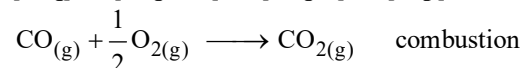
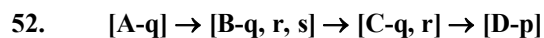
Heat evolved :  $5^\circ\text{C}$  from 20 mL  $q = 20 \times s \times 5 = 20 \times 4.2 \times 5 = 420 \text{ cal}$

100 mL HA + 100 mL BOH :  $\frac{420}{20} \times 200 = 200 \times 4.2 \times \Delta T$ ,  $\Delta T = 5^\circ\text{C}$

10 mL HA + 20 mL BOH :  $\frac{420}{20} \times 20 = 30 \times 4.2 \times \Delta T$ ,  $\Delta T = 33.3^\circ\text{C}$

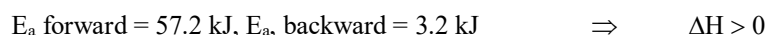
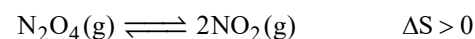
20 mL HA + 30 mL BOH :  $\frac{420}{20} \times 40 = 50 \times 4.2 \times \Delta T$ ,  $\Delta T = 4^\circ\text{C}$

50 mL HA + 150 mL BOH :  $\frac{420}{20} \times 100 = 200 \times 4.2 \times \Delta T$ ,  $\Delta T = 2.5^\circ\text{C}$



Diamond forms at high T, P

$\therefore \Delta S > 0$ ,  $\Delta H > 0$  formation of diamond  $\therefore \Delta H > 0$  formation of diamond,  $\Delta S < 0$  formation of graphite.



54.(8) 2 kcal heat added to system and 6 kcal heat added to system

55.(5) T : 300 K to 400 K

$P_{\text{ext}} = \text{constant}$

Work done by gas = 4.157 kJ

$W = -P_{\text{ext}} \Delta V = -nR\Delta T \Rightarrow n = \frac{W}{R\Delta T} = 5$

56.(6)  $\Delta H = \Delta U + \Delta PV = 14 + ((2)(1) - (5)(2)) = 14 + 2 - 10 = 6$

57.(8) 4.0 L                      20 K  
Adiabatic expansion to 25 L  
 $\gamma = 1.5$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1} \quad \Rightarrow \quad (4)^{1/2} (20) = T_2 (25)^{1/2} \quad \Rightarrow \quad T_2 = \frac{(20)(2)}{(5)} = 8$$

58.(8)  $\Delta H_{\text{vap}} = 3600 \text{ J}$   
BP =  $177^\circ\text{C} = 450\text{K}$   
 $\Delta S = \frac{\Delta H}{T} = 8 \text{ J/K}$

59.(27.20)

$$\begin{array}{ccc} P_1 = 20 \text{ atm} & \longrightarrow & P_2 = 10 \text{ atm} \\ V_1 = 15 \text{ L} & & V_2 = 60 \end{array}$$

$$C_{p,m} = 30.96$$

$$\begin{aligned} \Delta S &= nC_p \ln \frac{T_2}{T_1} - nR \ln \frac{P_2}{P_1} \\ &= 30.96 \ln \left( \frac{7308}{3654} \right) - 8.314 \ln \left( \frac{10}{20} \right) = 21.46 + 5.76 \quad (\text{temperature calculated using } PV = nRT) \end{aligned}$$

$$\Delta S = 27.22 \text{ JK}^{-1} \text{ mol}^{-1}$$

60.(15)  $\begin{array}{ccc} 1 \text{ L} & \longrightarrow & 4 \text{ L} \\ 10 \text{ atm} & & 5 \text{ atm} \end{array} \quad P_{\text{ext}} = 1 \text{ atm}$   
 $T = 300 \text{ K} \quad C = 50 \text{ J/}^\circ\text{C}$

Work done during process,

$$w = -P_{\text{ext}}(V_2 - V_1) = -1 \text{ atm}(4 \text{ L} - 1 \text{ L}) = -3 \text{ L atm} = -300 \text{ J}$$

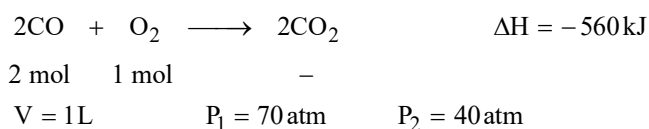
Using ideal gas equation,

$$\begin{aligned} \frac{P_1 V_1}{T_1} &= \frac{P_2 V_2}{T_2} \\ \Rightarrow \quad T_2 &= \frac{4 \times 5 \times 300}{1 \times 10} \quad \Rightarrow \quad T_2 = 600 \text{ K} \end{aligned}$$

$$\Delta T = (600 \text{ K} - 300 \text{ K}) = 300 \text{ K} = 300^\circ\text{C}$$

$$\Delta H = C \Delta T = 50 \text{ J/}^\circ\text{C} \times 300^\circ\text{C} = 15000 \text{ J} = 15 \text{ kJ}$$

61. (-557)

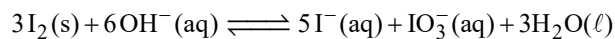


$$\Delta H = \Delta U + \Delta(PV) \quad (\text{Since volume remains constant})$$

$$-560 \text{ kJ} = \Delta U + V \Delta P$$

$$\Delta U = -560 \text{ kJ} - (1 \times -30 \times 0.1) \text{ kJ} = -557 \text{ kJ}$$

62.(10.9) The complete reaction is



$$K_{\text{eq}} = \frac{[\text{IO}_3^-][\text{I}^-]^5}{[\text{OH}^-]^6} \quad \dots\dots (i)$$

$$\begin{aligned}\Delta G_{\text{rxn}}^\circ &= 3\Delta G_{\text{f}(\text{I}^-)}^\circ + \Delta G_{\text{f}(\text{IO}_3^-)}^\circ + 3\Delta G_{\text{f}(\text{H}_2\text{O})}^\circ - \left\{6\Delta G_{\text{f}(\text{OH}^-)}^\circ\right\} \\ &= 3(-50\text{ kJ mol}^{-1}) + (-123.5\text{ kJ mol}^{-1}) + 3(-233\text{ kJ mol}^{-1}) - 6(-150\text{ kJ mol}^{-1}) \\ &= -72.5\text{ kJ mol}^{-1}\end{aligned}$$

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

$$-72.5\text{ kJ mol}^{-1} = -2.3 \times \frac{25}{3} \text{ JK}^{-1}\text{mol}^{-1} \times \log K_{\text{eq}} \times 300\text{ K}$$

$$\Rightarrow \log K_{\text{eq}} = 12.6 \quad \Rightarrow \quad K_{\text{eq}} = 3.98 \times 10^{12}$$

Substitution in (i)

$$3.98 \times 10^{12} = \frac{(0.1)(0.1)^5}{[\text{OH}^-]^6}$$

$$\Rightarrow [\text{OH}^-]^6 = 2.51 \times 10^{-19} \quad \Rightarrow \quad [\text{OH}^-] = 0.00079$$

$$\Rightarrow \text{pOH} = -\log(0.00079)$$

$$\text{pOH} = 3.09$$

$$\Rightarrow \text{pH} = 14 - \text{pOH} = 10.9$$

## Solutions - Advanced Problem Package | Chemistry

### Chemical Kinetics

1.(B)  $M \rightarrow N$

$$\text{Rate} \propto [M]^x$$

$$\text{Rate} = k [M]^x$$

$$\text{Rate} \times 8 = k [2M]^x$$

$$\frac{8R}{R} = \frac{k[2M]^x}{k[M]^x} \Rightarrow [2]^3 = [2]^x$$

$$\Rightarrow x = 3$$

2.(D) We know,  $\log_{10} k = \log_{10} A - \frac{E_a}{2.303R} \times \frac{1}{T}$  .... (i)

The given equation is  $\log_{10} k = 6 - \frac{2000}{T}$  .... (ii)

Comparing eqns. (i) and (ii)

$$\log_{10} A = 6 \therefore A = 10^6 \text{ sec}^{-1} \text{ (for first order reaction)}$$

$$\frac{E_a}{2.303R} = 2000$$

$$E_a = 2000 \times 2.303 \times 8.314 = 38294 \text{ J mol}^{-1} = 38.3 \text{ kJ mol}^{-1}$$

3.(A)  $2A \xrightleftharpoons{k} B$  ;  $B + C \xrightarrow{k_f} P$

$$K = \frac{[B]}{[A]^2} \quad \dots \text{(i)} \quad \frac{d[P]}{dt} = k_f[B][C] \quad \dots \text{(ii)}$$

$$\text{From (i) and (ii), } \frac{d[P]}{dt} = Kk_f[A]^2[C]$$

4.(A)  $t = \frac{0.693}{k}$  for first order reaction

$$k = \frac{2.303}{\text{time}} \log \left( \frac{a}{a-x} \right)$$

$$\frac{0.693}{t} = \frac{2.303}{\text{time}} \log \left( \frac{100}{100 - 99.9} \right)$$

$$\text{time} = \frac{6.909}{0.693} \approx 10t$$

5.(A)  $\text{Rate} \propto \frac{kP}{1 + kP}$

At high pressure,  $kP \gg 1 \therefore kP + 1 \approx kP$

$\therefore$  Rate = Constant and the reaction will be a zero order reaction.

6.(B)  $PV = nRT$

$$P = \frac{n}{V}RT = CRT$$

$$\frac{1}{RT} \frac{dP}{dt} = \frac{1}{V} \frac{dn}{dt} = \frac{dC}{dt}$$

7.(A) For the given reaction,  $-\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$

$$\frac{1}{2}k_1 = k_2 = \frac{1}{3}k_3$$

$$1.5k_1 = 3k_2 = k_3$$

8.(D)  $\log_{10} \left( \frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left[ \frac{1}{T_1} - \frac{1}{T_2} \right]$

$$\log \frac{4.5 \times 10^7}{1.5 \times 10^7} = \frac{E_a}{2.303 \times 8.314} \left[ \frac{1}{323} - \frac{1}{373} \right]$$

$$E_a = 2.2 \times 10^4 \text{ J mol}^{-1}$$

- 9.(D) (i) Molecularity can never be fractional  
 (ii) Zero order reaction completes  
 (iii) first order reaction may be homogeneous or heterogenous  
 (iv)  $K = \frac{A}{e^{E_a/RT}}$  as temperature increases, K increases

10.(B) Photo chemical reactions generally are zero order

11.(A)  $k_2 = Ae^{-E_{a2}/RT}$ ,  $k_1 = Ae^{-E_{a1}/RT}$   
 $\therefore \frac{k_2}{k_1} = e^{(-E_{a2} + E_{a1})/RT}$   
 $\frac{k_2}{k_1} = e^{-E_{a1}/RT}$  (Since,  $E_{a2} = 2E_{a1}$ )  
 $k_2 = k_1 e^{-E_{a1}/RT}$

12.(C) Collision theory is given for Biomolecular reaction.

13.(C) Statement 1 incorrect because, rate does not depend on concentration for zero order reactions.

14.(B) Half - life of the given reaction is independent of concentration. Hence, it will be a first order reaction.

15.(C) Mechanism-I

$$\text{rate} \propto [\text{NO}_2]^1 [\text{O}_3]^1$$

Mechanism-II

$$\text{rate} \propto [\text{NO}_2]^1 [\text{O}] \quad \dots(1)$$

Apply steady state approximation.

$$K_{\text{eqm}} = \frac{[\text{O}_2][\text{O}]}{[\text{O}_3]}$$

$$[\text{O}] = K_{\text{eqm}} \frac{[\text{O}_3]}{[\text{O}_2]} \quad \dots(2)$$

on putting (2) in (1)

$$\text{rate} \propto \frac{[\text{NO}_2]^1 [\text{O}_3]^1}{[\text{O}_2]}$$

16.(B) For maximum increase of k,  $E_a$  should be high &  $\Delta T$  as high as possible

17.(B)  $\frac{t_{1/2}}{t_{3/4}} = \frac{2^{n-1} - 1}{4^{n-1} - 1}$

It depends only on 'n'

18.(B)  $t_{1/2} \propto \frac{1}{a^{n-1}}$

n = 2, for second order reaction

$\therefore t_{1/2} \propto \frac{1}{a}$

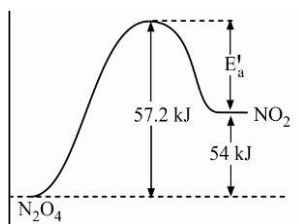
19.(A)  $[C_t] = \frac{[C_0]}{2^n} \Rightarrow 1 = \frac{16}{2^n} \Rightarrow n = 4$  where n is the no. of half lives

$\Rightarrow n = \frac{\text{total time taken}}{\text{Half life}}$

$4 = \frac{\text{total time}}{30}$

Total time = 120 mins.

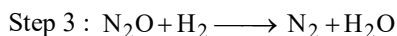
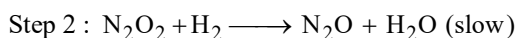
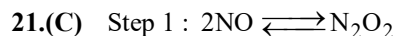
20.(B)



$E'_a = E_a - \Delta H$

$= 57.2 - 54$

$= 3.2 \text{ kJ}$



Rate =  $k[\text{N}_2\text{O}_2][\text{H}_2]$

From step 1,  $k = \frac{[\text{N}_2\text{O}_2]}{[\text{NO}]^2}$  or  $[\text{N}_2\text{O}_2] = k[\text{NO}]^2$

$\therefore \text{Rate} = kK[\text{NO}]^2[\text{H}_2] = k'[\text{NO}]^2[\text{H}_2]$

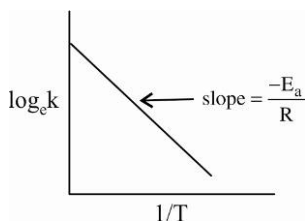
22.(C)  $-\frac{d[A]}{dt} = k_1[A] - k_2[B]$

23.(B)  $k = Ae^{-E_a/RT}$

$\log_e k = \log_e A - \frac{E_a}{RT}$

$y = c + mx$

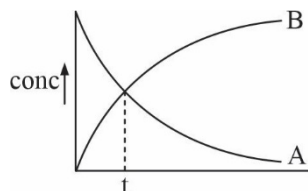
Slope =  $\frac{-E_a}{R}$





**24.(C)** Catalyst forms new intermediate in the reaction, therefore, provides alternative path for climbing the activation energy barrier. In other words. Catalyst alters the reaction mechanism.

**25.(A)**



At the point of intersection

$$[A]_t = \frac{[A]}{2}$$

$\therefore$  Point of intersection of two curves represents  $t_{1/2}$ .

**26.(D)** Specific rate constant depends only on temperature

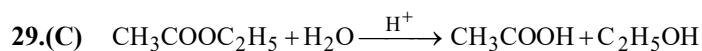
$$K = A e^{-E_a/RT}$$

**27.(C)** Statement-I For complex reaction molecularity is not applicable.

Statement-II Molecularity & order are same only for elementary reaction & not for complex reaction

**28.(B)** Statement-I Order of reaction may be, zero, fractional positive & negative

Statement-II Order of reaction is the mathematical number that describes the effect of change of concentration on rate of reaction.



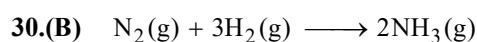
$$\text{Rate} = k[\text{CH}_3\text{COOC}_2\text{H}_5][\text{HOH}]$$

Water is present in large excess, therefore, concentration of water will be almost constant.

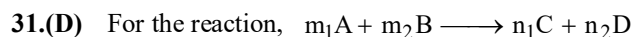
Thus, rate law may be given as

$$\text{Rate} = k'[\text{CH}_3\text{COOC}_2\text{H}_5]$$

$\therefore$  It is pseudo first order reaction.



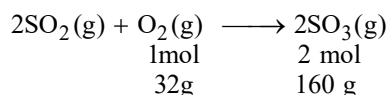
$$-\frac{d[\text{N}_2]}{dt} = -\frac{1}{3} \frac{d[\text{H}_2]}{dt} = +\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \text{Rate of reaction}$$



$$-\frac{1}{m_1} \frac{d[\text{A}]}{dt} = +\frac{1}{n_1} \frac{d[\text{C}]}{dt}$$

$$-\frac{d[\text{A}]}{dt} \bigg/ \frac{d[\text{C}]}{dt} = \frac{m_1}{n_1}$$

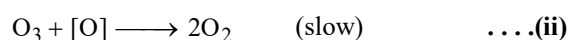
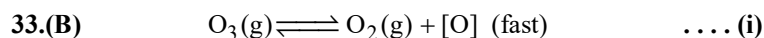
**32.(B)** In the reaction



$\therefore$  160 g of  $\text{SO}_3$  is formed by 32 g of  $\text{O}_2$

$\therefore$  100 g of  $\text{SO}_3$  will be formed by  $\frac{32}{160} \times 100$  or 20 g of  $\text{O}_2$

$\therefore$  Rate of disappearance of  $\text{O}_2 = 20 \text{ g min}^{-1}$



$$\text{Rate} = k'[\text{O}_3][\text{O}] \quad \dots \text{(iii)}$$

$$\text{From eqn. (i)} \quad K = \frac{[\text{O}_2][\text{O}]}{[\text{O}_3]}$$

$$[\text{O}] = \frac{K[\text{O}_3]}{[\text{O}_2]}$$

$$\text{From eqn. (ii) rate} = k'[\text{O}_3] \frac{K[\text{O}_3]}{[\text{O}_2]} = k''[\text{O}_3]^2 [\text{O}_2]^{-1}$$

34.(C) Step 1 :  $\text{NO} + \text{Br}_2 \rightleftharpoons \text{NOBr}_2$  (fast)

Step 2 :  $\text{NOBr}_2 + \text{NO} \longrightarrow 2\text{NOBr}$  (slow)

$$\text{Rate} = k' [\text{NOBr}_2] [\text{NO}] \quad \dots \text{(i)}$$

$$\text{From step 1,} \quad K = \frac{[\text{NOBr}_2]}{[\text{NO}] [\text{Br}_2]}$$

$$[\text{NOBr}_2] = K [\text{NO}] [\text{Br}_2] \quad \dots \text{(ii)}$$

$$\begin{aligned} \text{From eqns. (i) and (ii),} \quad \text{Rate} &= k'K[\text{NO}] [\text{Br}_2][\text{NO}] \\ &= k[\text{NO}]^2 [\text{Br}_2] \end{aligned}$$

$$\text{Order} = 2 + 1 = 3$$

35.(D) Slowest step is rate determining.

Thus Rate =  $k[\text{A}_2]$  and Order = 1

36.(B) Slowest step is rate determining, therefore  $\text{B} \rightarrow \text{C}$  will be rate determining. Its rate constant is lowest

37.(AD) (A)  $\text{A} \longrightarrow \text{B}$

$$t = 0 \quad \text{C} \quad -$$

$$t = t \quad \text{C} - C\alpha \quad C\alpha$$

for first order reaction

$$Kt = \ln \frac{a}{a-x}$$

$$Kt = \ln \frac{C}{C-C\alpha}$$

$$e^{Kt} = \frac{1}{1-\alpha}$$

$$1-\alpha = e^{-Kt}$$

$$-\alpha = e^{-Kt} - 1$$

$$\alpha = 1 - e^{-Kt}$$

(B)  $\frac{1}{C_t}$  vs  $t$  is straight-line for second order reaction

$$(C) \quad t_{3/4} = \frac{3}{2} \times t_{1/2}$$

(D)  $K = Ae^{-E_a/RT}$ , exponential factor is unit less. &  $K$  has unit of  $(\text{time})^{-1}$ . So  $A$  will also has unit of  $(\text{time})^{-1}$ .

38.(BC) Rate =  $k[\text{RCl}]$ . It is of first order, therefore, rate of reaction will be halved when the concentration of halide is reduced to half.

Rate of reaction increases with increase in temperature. We have to consider only forward reaction.

As  $T \uparrow \Rightarrow k \uparrow \Rightarrow r \uparrow$

39.(ABD) Only option (C) is incorrect because half - life order reaction is independent of initial concentration of reactant.

$$t_{1/2} = \frac{0.693}{k}$$

40.(ACD) Catalyst does not alters the equilibrium constant because it affects the rate constant of forward and backward reaction equally.

41.(AB) In the equation :  $k = Ae^{-E_a/RT}$ ;  $k$  will be equal to 'A' when either  $E_a = 0$  or  $T = \infty$  K

42.(ABD) Rate =  $k[A]^{2/3}[B]$

$$\text{Order} = \frac{2}{3} + 1 = \frac{5}{3}$$

$$\text{Unit of } k = \left[ \frac{\text{L}}{\text{mol}} \right]^{n-1} \times \text{sec}^{-1} = \left[ \frac{\text{L}}{\text{mol}} \right]^{\frac{5}{3}-1} \times \text{sec}^{-1} = \text{L}^{2/3} \text{ mol}^{-2/3} \text{ sec}^{-1}$$

43.(CD) Arrhenius equation is :  $k = Ae^{-E_a/RT}$

$$\ln k = \ln A - \frac{E_a}{RT} \quad \dots \text{(i)}$$

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303 RT} \quad \dots \text{(ii)}$$

From eq. (i)

$$\ln A = \ln k + \frac{E_a}{RT} \quad \dots \text{(iii)}$$

44.(AD) Large slope corresponds to large activation energy. Thus  $E_{a1}$  will be greater than  $E_{a2}$  and D, E will be the favorable products.

45.(AB)  $t_{3/4} = 2 \times t_{1/2}$

$$t_{15/16} = 4 \times t_{1/2}$$

These are standard relation of first order reaction.

46.(AC)  $\Delta H = E_f - E_b = (60 - 40) = + 20 \text{ kJ}$

$\Delta H = + \text{ve}$ ; hence it is an endothermic reaction.

47.(BD) Rate of zero order reactions do not depend on the concentration of reactant.

In the equation,  $k = Ae^{-E_a/RT}$

$$k = A \text{ when } E_a = 0$$

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303} \times \frac{1}{RT}$$

$$y = c + mx$$

When  $\log k$  is plotted against  $\frac{1}{T}$ , we get straight line with negative slope.

48.  $[A-p, q] \rightarrow [B-s] \rightarrow [C-r] \rightarrow [D-r]$

Half - life of a reaction depends on initial concentration or initial pressure of the reactant as

$$t_{1/2} \propto \frac{1}{a^{n-1}} \quad \text{or} \quad t_{1/2} \propto \frac{1}{p^{n-1}}$$

## INTEGER TYPE :

49.(2) According to rate law,  $\frac{dc}{dt} = k[A_2]^x[B_2]^y$

$$\frac{0.04}{0.08} = \frac{k[0.2]^x[0.2]^y}{k[0.2]^x[0.4]^y}$$

$$\left[\frac{1}{2}\right] = \left[\frac{1}{2}\right]^y \Rightarrow y = 1$$

$$\frac{0.04}{0.08} = \frac{k[0.1]^x[0.4]^y}{k[0.2]^x[0.4]^y}$$

$$\left[\frac{1}{2}\right] = \left[\frac{1}{2}\right]^x \Rightarrow x = 1$$

$$\text{Order} = 1 + 1 = 2$$

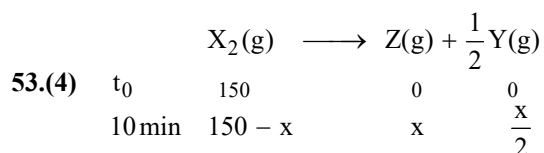
$$50.(9) \quad \frac{t_{1/8}}{t_{1/10}} = \frac{\frac{2.303}{k} \log \frac{1}{1/8}}{\frac{2.303}{k} \log \frac{1}{1/10}}$$

$$\frac{t_{1/8}}{t_{1/10}} \times 10 = \left[ \frac{\log 8}{\log 10} \right] \times 10 = 9$$

$$51.(3) \quad \frac{t_2}{t_1} = \left( \frac{a_1}{a_2} \right)^{n-1}$$

$$16 = \left( \frac{1}{1/4} \right)^{n-1} \quad \therefore \quad n = 3, \text{ i. e., third order reaction.}$$

$$52.(4) \quad t_{99.9\%} = 10 \times t_{50\%} = 10 \times 0.4 = 4 \text{ min}$$



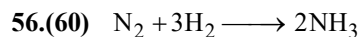
$$\text{Total pressure of reaction after 10 min} = 150 - x + x + \frac{x}{2} = 170 \Rightarrow x = 40$$

$$\text{thus rate of disappearance of } X_2 = \frac{-d[X_2]}{dt} = \frac{40}{10} = 4 \text{ mm / min.}$$

$$54.(9) \quad \frac{k_2}{k_1} = 2^n \text{ where } n = \frac{\text{change in temp}}{10}$$

$$\Rightarrow n = \frac{115 - 25}{10} = \frac{90}{10} = 9$$

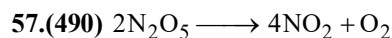
55.(2)  $K = [A][B]^2$ , if 'A' is taken in excess, then change in concentration of [A] wont effect rate of reaction. So, reaction rate will only depend upon concentration of B & hence reaction will be 2<sup>nd</sup> order reaction.



$$\text{rate} = -\frac{d[N_2]}{dt} = -\frac{1}{3} \frac{d[H_2]}{dt} = \frac{1}{2} \frac{d[NH_3]}{dt}$$

$$\frac{d[\text{NH}_3]}{dt} = 40 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\Rightarrow \frac{d[\text{H}_2]}{dt} = \frac{3}{2}(40 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}) = 60 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$



$$\text{Rate constant} = k = 3.38 \times 10^{-5} \text{ s}^{-1}$$

$$\text{Order of reaction} = 1 \Rightarrow \text{Rate} \propto [\text{N}_2\text{O}_5]^1$$

$$\text{Rate} = k[\text{N}_2\text{O}_5] \quad t = 10 \text{ min} = 600 \text{ s}$$

$$A_t = A_0 e^{-kt} \Rightarrow A_t = 500 e^{-3.38 \times 10^{-5} \times 600} \Rightarrow A_t = 490 \text{ atm}$$



$$\text{At } t = 0 \quad 1\text{M} \quad 0$$

$$\text{After 1h} \quad (1-x)\text{M} \quad x\text{M}$$

$$\text{Rate of reaction after 1h} = k(1-x) \quad \dots\dots\dots (i)$$

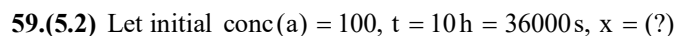
For the calculation of  $1-x$ , first order kinetic equation is used as follows :

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]_t}$$

$$\text{or} \quad 4.5 \times 10^{-3} \text{ min}^{-1} = \frac{2.303}{1 \times 60} \log_{10} \frac{1}{(1-x)} \quad \text{or} \quad \log \frac{1}{(1-x)} = \frac{4.5 \times 10^{-3} \times 60}{2.303} = 0.1172$$

$$\therefore \frac{1}{1-x} = 1.310 \quad \therefore (1-x) = 0.7633 \text{ M}$$

$$\text{Rate of reaction after 1h} = 4.5 \times 10^{-3} \times 0.7633 = 3.435 \times 10^{-3} \text{ M min}^{-1}$$

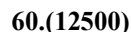


$$k = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

$$1.5 \times 10^{-6} = \frac{2.303}{3.6 \times 10^4} \log \frac{100}{(100-x)}$$

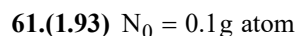
$$\therefore x = 5.2$$

Thus, initial concentration changed into product is 5.2%.



$$\text{For a second order reaction} \quad \frac{1}{A_t} = \frac{1}{A_0} + kt$$

$$\Rightarrow \frac{1}{0.5} = \frac{1}{1} + 8.0 \times 10^{-5} \times t \Rightarrow \frac{1}{8 \times 10^{-5}} = t \Rightarrow t = 12500 \text{ min}$$



$$t = 10 \text{ days and } t_{1/2} = 5 \text{ days}$$

$$\lambda = \frac{2.303}{t} \log \frac{N_0}{N}$$

$$\frac{0.693}{5} = \frac{2.303}{10} \log \frac{0.1}{N}$$

$$\therefore \text{Amount left after 10 days} = 0.0250 \text{ g atom}$$

Similarly if  $t = 11$  days

$$\frac{0.693}{5} = \frac{2.303}{11} \log \frac{0.1}{N}$$

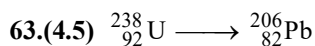
$\therefore$  Amount left after 11 days = 0.0218g atom

$\therefore$  Amount decayed in 11th day = 0.0250 – 0.0218

$$= 3.2 \times 10^{-3} \text{ g atoms}$$

$$= 3.2 \times 6.023 \times 10^{23} \times 10^{-3} \text{ atoms} = 1.93 \times 10^{21} \text{ atoms}$$

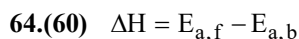
62.(75)



The rock contains equal number of U and Pb atoms

$\Rightarrow$  U is half decayed

$\therefore$  Age of rock =  $t_{1/2} = 4.5 \times 10^9$  years



$$-10 \text{ kJ mol}^{-1} = 50 \text{ kJ mol}^{-1} - E_{a,b}$$

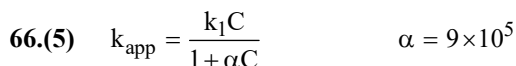
$$\Rightarrow E_{a,b} = 60 \text{ kJ mol}^{-1}$$

65.(0.693)

$$\frac{\left(-\frac{dN}{dt}\right)_1}{-\left(\frac{dN}{dt}\right)_2} = \frac{1}{1/10} = 10 \Rightarrow \frac{N_0}{N_t} = 10$$

$$k = \frac{2.303}{t} \log \left( \frac{N_0}{N_t} \right) \Rightarrow k = \frac{2.303}{2.303} \log(10) \Rightarrow k = 1$$

$$\Rightarrow t_{1/2} = \frac{0.693}{k} = 0.693 \text{ s}$$



Limiting value of  $k'_{app}$  when  $C \rightarrow \infty$

$$\Rightarrow \alpha C \gg 1 \Rightarrow 1 + \alpha C \approx \alpha C$$

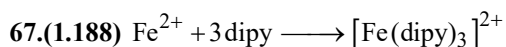
$$k_{app} = \frac{k_1 C}{\alpha C} = \frac{k_1}{\alpha}$$

To calculate the values of C for which  $k_{app}$  has 90% of  $k'_{app}$

$$k_{app} = \frac{90}{100} k'_{app} = \frac{k_1 C}{1 + \alpha C}$$

$$\Rightarrow \frac{9}{10} \cdot \frac{k_1}{\alpha} = \frac{k_1 C}{1 + \alpha C} \Rightarrow 9(1 + \alpha C) = 10 \alpha C \Rightarrow 9 + 9 \alpha C = 10 \alpha C$$

$$\Rightarrow 1 \alpha C = 9 \Rightarrow C = \frac{9}{9 \times 10^5} = 10^{-5} \text{ mol L}^{-1}$$



$$r_f = 1.45 \times 10^{13} [\text{Fe}^{2+}][\text{dipy}]^3$$

$$r_b = [\text{Fe}(\text{dipy})_3]^{2+} (1.22 \times 10^{-4})$$

At equilibrium,  $r_f = r_b$

$$\Rightarrow 1.45 \times 10^{13} [\text{Fe}^{2+}] [\text{dipy}]^3 = 1.22 \times 10^{-4} [\text{Fe}(\text{dipy})_3]^{2+}$$

$$\Rightarrow \frac{[\text{Fe}(\text{dipy})_3]^{2+}}{[\text{Fe}^{2+}] [\text{dipy}]^3} = \frac{1.45 \times 10^{13}}{1.22 \times 10^{-4}} \Rightarrow K_{(\text{stability constant})} = 1.188 \times 10^{17}$$

$$68.(70.8) \quad Q_{10} = \frac{k_{37^\circ\text{C}}}{k_{27^\circ\text{C}}} = \frac{A e^{-E_a/RT'}}{A e^{-E_a/RT}} = 2.5$$

$$\ln 2.5 = -\frac{E_a}{R} \left( \frac{1}{310} - \frac{1}{300} \right)$$

$$\Rightarrow E_a = \frac{0.9163 \times 8.314 \times 300 \times 310}{10} \Rightarrow E_a = 70.8 \text{ kJ mol}^{-1}$$

$$69.(6) \quad -\frac{d[\text{H}^+]}{dt} = 1 \times 10^7 \text{ mol L}^{-1} \text{ s}^{-1}$$

A drop  $\equiv 0.05 \text{ mL}$  solution contains  $3 \times 10^{-6} \text{ mol H}^+$

$$\Rightarrow \text{Concentration of } \text{H}^+ \text{ in drop} = \frac{3 \times 10^{-6} \text{ mol} \times 1000}{0.05 \text{ mL}} = 0.06 \text{ mol L}^{-1}$$

$1 \times 10^7 \text{ mol L}^{-1}$  will disappear in 1 second

$$\Rightarrow 0.06 \text{ mol L}^{-1} \text{ will disappear in } \frac{1}{10^7} \times 0.06 = 6 \times 10^{-9} \text{ sec}$$

70.(2.30) By observing reading 1 and 2, the reaction is first order w. r. t. B. Also by 3 and 4, the reaction is found to be 2nd order w.r.t. A.

$$\Rightarrow \text{rate} = k[\text{A}]^2 [\text{B}]^1$$

$$\Rightarrow 1.15 \times 10^{-4} = k[0.50]^2 (0.02)^1$$

$$\Rightarrow k = 230 \times 10^{-4} = 2.30 \times 10^{-2} \text{ L}^2 \text{ mol}^{-2} \text{ min}^{-1}$$

$$71.(48) \quad t_{99\%} = 32 \text{ min}$$

$$k = \frac{2.303}{32} \log \left( \frac{100}{1} \right)$$

$$k = 0.144 \text{ min}^{-1}$$

$$k_{99.9\%} = \frac{2.303}{0.144} \log \left( \frac{100}{0.1} \right) = 48 \text{ min}$$

$$72.(0.75) \quad \text{Rate} = k[\text{A}]^{1/2} [\text{B}]^{1/4} [\text{C}]^0$$

$$\text{Overall order} = \frac{1}{2} + \frac{1}{4} + 0 = \frac{3}{4}$$

$$73.(1) \quad \text{A} \longrightarrow \text{B} \quad \text{rate} = k[\text{A}]$$

$$t = 0 \quad 0.8 \quad -$$

$$t = 1 \text{ h} \quad 0.2 \quad 0.6$$

$$k = \frac{2.303}{1} \log \left( \frac{0.8}{0.2} \right) \quad k = 1.387 \text{ h}^{-1}$$

If  $A_0 = 0.9 \text{ mol}$  and  $B_t = 0.675 \text{ mol}$

$$\Rightarrow A_t = 0.9 - 0.675 = 0.225$$

$$t = \frac{2.303}{1.387} \log \left( \frac{0.9}{0.225} \right) \Rightarrow t = 1 \text{ h}$$

74.(7.33)  $k_f = 1.1 \times 10^{-2}$

$$k_b = 1.5 \times 10^{-3}$$

$$\text{Equilibrium constant } k_{eq} = \frac{k_f}{k_b} = \frac{1.1 \times 10^{-2}}{1.5 \times 10^{-3}} = 7.33$$

75.(3.2)  $E_a = 0$

$$k = 3.2 \times 10^6 \text{ s}^{-1} \text{ at } 300 \text{ K}$$

$$k_{(\text{at } 310 \text{ K})} = ?$$

$$\Rightarrow k = A e^0 \Rightarrow k = A$$

At 310 K,

$$k = (3.2 \times 10^6) e^{-E_a/RT} \Rightarrow k = 3.2 \times 10^6 \text{ s}^{-1}$$



$$\text{rate}(r) = k[A]^2$$

On doubling the concentration of A

$$r' = k[2A]^2 = 4k[A]^2 \Rightarrow r' = 4r$$



## Solutions - Advanced Problem Package | Chemistry

### Colligative Properties of Solution

- 1.(C) Solution with positive deviation from Raoult's law forms low boiling azeotrope.
- 2.(A)  $K_2SO_4$  has highest van't Hoff factor, therefore, aqueous solution of  $K_2SO_4$  has lowest freezing point.
- 3.(B)  $2KI(aq) + HgI_2(s) \longrightarrow K_2[HgI_4](aq)$   
 Number of solute particles decreases, therefore, freezing point is raised.
- 4.(C)  $\frac{\Delta P}{P_0} = X_B \approx \frac{n_B}{n_A}$   
 $\frac{3000 - 2985}{3000} = \frac{5/m_B}{100/18} \quad ; \quad m_B = 180$
- 5.(B) Increase in temperature has no effect on mass  
 $\therefore$  Molality does not change nor does % w/w  
 However, volume increases and  $\therefore$  Molarity decreases.
- 6.(A)  $HX \rightleftharpoons H^+ + X^-$   
 $t = 0 \quad 1 \quad 0 \quad 0$   
 $t_{eq} \quad (1-0.2) \text{ } 0.2 \text{ m}, \quad 0.04 \text{ m}, \quad 0.04 \text{ m},$   
 Total molality =  $0.16 \text{ m} + 0.04 + 0.04$   
 $= 0.24 \text{ m}$   
 $\Delta T = k_f \times m = 1.86 \times 0.24 = 0.45$   
 Freezing point  $-0.45^\circ\text{C}$
- 7.(A) Freezing point is the temp. at which vapour pressure of liquid solvent is equal to vapour pressure of solid solvent  
 So, equilibrium exists between  
 $Solvent(l) \rightleftharpoons solvent(s)$
- 8.(A)  $\pi(Na_2SO_4) \rightleftharpoons \pi(\text{glucose})$   
 $i \times 0.004 \times RT = 0.01 \times RT$   
 $i = 2.5$   
 $\alpha = \frac{i-1}{n-1} = \frac{2.5-1}{3-1} = 0.75$
- 9.(B)  $\alpha = \frac{i-1}{n-1} = \frac{2.74-1}{3-1} = \frac{1.74}{2} = 0.87$
- 10.(A)  $\Delta T = i \times K_f \times \frac{w_B \times 1000}{M_B \times w_A}$   
 $2 = i \times 1.72 \times \frac{20 \times 1000}{172 \times 50} \quad ; \quad i = 0.5$
- 11.(C) For isotonic solutions,  
 $\pi_1 = \pi_2$   
 $C_1 RT = C_2 RT$   
 or  $C_1 = C_2$  (same molar concentration)

$$12.(B) \quad \alpha = \frac{i-1}{n-1}$$

Here,  $n$  = number of ions formed =  $x + y$

$$\therefore \quad \alpha = \frac{i-1}{x+y-1}$$

13.(C) Greater is the molar concentration of solution, more is the osmotic pressure.

$$\therefore \quad P_2(\text{urea}) > P_1(\text{glucose}) > P_3(\text{sucrose})$$

14.(B) Due to increase in the inter-molecular force, the rate of evaporation is lowered. Thus, the solution of acetone and chloroform will give negative deviation.

$$15.(B) \quad \Delta T = K_b \times \frac{w_B \times 1000}{M_B \times w_A} = K_b \times \frac{Y \times 1000}{M \times 250} = \frac{4K_b Y}{M}$$

$$16.(C) \quad \frac{\Delta P}{P_0} = X_B \approx \frac{n_B}{n_A} = \frac{w_B \times M_A}{M_B \times w_A}$$

$$\left( \frac{w_B \times M_A}{M_B \times w_A} \right)_{\text{glucose}} = \left( \frac{w_B \times w_A}{M_B \times w_A} \right)_{\text{urea}}$$

$$\frac{w_B \times 18}{180 \times 100} = \frac{1 \times 18}{60 \times 50} \quad ; \quad w_B = 6 \text{ g}$$

$$17.(C) \quad \Delta T = K_f \times \frac{w_B \times 1000}{M_B \times w_A}$$

$$9.3 = 1.86 \times \frac{50 \times 1000}{62 \times w_A}$$

$$w_A = 161.29 \text{ gm (water)}$$

$$\text{Amount of ice} = 200 - 161.29 = 38.71 \text{ g}$$

$$18.(A) \quad P_{\text{Total}} \times y_A = P_A$$

$$\frac{1}{P_{\text{Total}}} = y_A \times \frac{1}{P_A}$$

When  $\frac{1}{P_{\text{Total}}}$  is plotted against  $y_A$ , we get straight line passing through origin having

Slope of  $\frac{1}{P_A}$ .

19.(A) Urea and glucose do not dissociate in aqueous medium, hence their solution has lowest depression in freezing point.

$$20.(C) \quad \text{When } X_A = 1, P = P_A^0 \quad \therefore \quad P_A^0 = 254 - 119 = 135 \text{ torr}$$

$$\text{When } X_A = 0, P = P_B^0 \quad \therefore \quad P_B^0 = 254 \text{ torr}$$

$$21.(A) \quad i = \frac{\text{Normal molar mass}}{\text{Observed molar mass}} = \frac{164}{65.6} = 2.5$$

$$\alpha = \frac{i-1}{n-1}, \quad n = 3 \text{ (number of ions)}$$

$$= \frac{2.5-1}{3-1} = \frac{1.5}{2} = 0.75 \quad \therefore \text{Percentage ionization of } MX_2 \text{ will be } 75\%$$

22.(B)  $X_A = \frac{1}{3}, X_B = \frac{2}{3}$

$$P = P_A^0 X_A + P_B^0 X_B = 150 \times \frac{1}{3} + 240 \times \frac{2}{3} = 50 + 160 = 210 \text{ mm}$$

$P_{\text{exp.}} < P_{\text{calculated}}$  ;  $\therefore$  There is negative deviation from Raoult's law.

23.(C) If 2 liquids are mixed the entropy of mixture is always more than entropy of components.

$$\Delta S_{\text{mix}} > 0$$

$$\Delta G_{\text{mix}} < 0$$

$$\Delta H_{\text{mix}} = 0 \text{ (for ideal solution)}$$

$\therefore$  Interaction between molecules in components are equal to interaction between molecules in mixture

24.(A) A : Benzene B : Toluene

$$P = P_A + P_B$$

$$P = P_A^0 X_A + P_B^0 X_B = 75 \times \frac{1}{2} + 22 \times \frac{1}{2} = 37.5 + 11 = 48.5$$

$$\text{Mole fraction of benzene in vapour, } Y_A = \frac{P_A}{P} = \frac{37.5}{48} = 0.78$$

Similarly, mole fraction of toluene in vapour,  $Y_B = 0.22$

$\therefore$  The vapour will contain higher percentage of benzene.

25.(B) 
$$\frac{(\Delta T)_P}{(\Delta T)_Q} = \frac{K_b \times \left( \frac{w_B \times 1000}{m_B \times w_A} \right)_P}{K_b \times \left( \frac{w_B \times 1000}{m_B \times w_A} \right)_Q}$$

$$\frac{0.2}{0.4} = \frac{\left( \frac{0.1 \times 1000}{100 \times 100} \right)_P}{\left( \frac{0.1 \times 1000}{100 \times m_B} \right)_Q}$$

$$\frac{1}{2} = \frac{(m_B)_Q}{100}$$

$$m_B = 50 \quad \therefore (m_B)_P : (m_B)_Q = 100 : 50 = 2 : 1$$

26.(A) For X,  $i = 2$

For Y,  $i = 0.5$  as Y dimerizes in solution

$\therefore$  Elevation in boiling point  $\Delta T_b = iK_b m$

$\Delta T_b$  for X  $>$   $\Delta T_b$  for Y

Depression in freezing point  $\Delta T_f = iK_f m$

$\Delta T_f$  for X  $>$   $\Delta T_f$  for Y

Osmotic pressure,  $\pi = iCRT$

$\therefore \pi$  for X  $>$   $\pi$  for Y

Relative lowering of vapour pressure,  $\frac{\Delta P}{P} = iX_B$  since value of  $i$  is not same,  $\therefore \frac{\Delta P}{P}$  will not be same.

|        |   |       |       |       |
|--------|---|-------|-------|-------|
| 27.(B) | X | 0.010 | 0.015 | 0.020 |
|        | P | 82    | 122   | 166   |

$$\Rightarrow K_H = \frac{P}{X} \quad 8200 \quad 8133.33 \quad 8300$$

So, average  $K_H = 8211 \text{ kPa}$

Also, graph of  $P$  v/s  $x$  is a straight line which cannot pass through origin.

28.(B) For  $\text{NaCl} \rightarrow i = 2$

$\text{BaCl}_2 \rightarrow i = 3$

$\therefore C_{\text{eff}}$  for  $\text{NaCl} = 0.1 \times 2 = 0.2 \text{ M}$ .

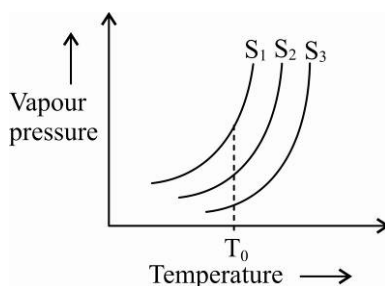
$C_{\text{eff}}$  for  $\text{BaCl}_2 = 0.05 \times 3 = 0.15 \text{ M}$

$\Rightarrow \text{NaCl(aq)} \text{ solution } S_1 \text{ is more concentrated than } S_2.$

So,  $S_1$  solution is hypotonic while  $S_2$  is hypertonic and osmosis will take place from  $S_2$  to  $S_1$ .

29.(B) At a particular temperature, more the concentration of solute in the solution, lesser will be the vapour pressure.

As vapour of  $S_1 > S_2 > S_3$



$\therefore$  concentration order is  $S_3 > S_2 > S_1$

30.(A) According to Dalton's law of partial pressures, total vapour pressure = sum of partial vapour pressure of all components

$\therefore$  OP = Partial vapour pressure of B at composition O

OQ = Partial vapour pressure of A at composition O

$\therefore$  OR = Total vapour pressure at composition O.

So,  $\text{OR} = \text{OP} + \text{OQ}$

31.(B)  $p = K_H x$  Henry's law

At a particular partial pressure P,

$$k_H \propto \frac{1}{x}$$

$\therefore$  lower the  $x$  (solubility), higher the value of  $k_H$ .

$\Rightarrow \text{O}_2$  has the highest solubility and

$\therefore$  has the lowest  $k_H$

So, we have  $\text{O}_2 < \text{N}_2 < \text{He}$

32.(B) For a particular gas, increase in temperature decreases the solubility of gas in the solution. So,  $T_1 > T_2 > T_3 > T_4$

33.(C) 
$$\alpha = \frac{1-i}{1-1/n} = \frac{1-i}{1-1/2}$$

$$\frac{\alpha}{2} = 1-i \quad ; \quad i = 1 - \frac{\alpha}{2}$$

$$34.(B) \quad \alpha = \frac{1-i}{1-\frac{1}{n}} = \frac{1-i}{1-\frac{1}{3}} \Rightarrow \frac{2}{3} \alpha = 1-i \quad \text{For} \quad \alpha=1, i=\frac{1}{3}$$

35.(C) Non-electrolytes neither dissociate nor associate in water  $\therefore$   $i$  remains equal to 1.

$$36.(C) \quad \alpha = \frac{i-1}{n-1}, \quad \alpha = 0.6, \quad n = 5 \quad \therefore \quad 0.6 = \frac{i-1}{5-1}$$

$$i = 0.6 \times 4 + 1 = 3.4$$

37.(A)  $\text{CH}_3\text{OH}$  is the most volatile out of the other

$$38.(C) \quad \Delta T_f = K_f m$$

$$\text{Molality}(m) \propto \frac{1}{\text{Molar mass}}$$

$\therefore$  For 124 g, glycerol will have lower molality than glycol.

$$39.(B) \quad \Delta T_f = K_f m$$

$$= 1.86 \times \frac{620}{62 \times 4} = 4.65$$

$\therefore$  Freezing point decreases by 4.65.

$$\text{Freezing point} = -4.65^\circ\text{C}$$

Let amount of water left after freezing be  $x$  kg

$\therefore$  If depression =  $6^\circ\text{C}$

$$6 = k_f \times \frac{10}{x} \Rightarrow 6 = \frac{18.6}{x} \Rightarrow x = 3.1 \text{ kg}$$

Amount of water freezing =  $9 \text{ kg} = 900 \text{ g}$

40.(B) The work of anti-freeze is to decrease the freezing point. Hence forth, more will be the depression in freezing point, the better will be the anti-freeze property.

$$\therefore \quad \Delta T_g \propto \frac{1}{M_B} \quad \text{where, } M_B = \text{molecular wt. of solute}$$

As order of molecular wt. is Glycerol > Glycol > Methanol

Hence, the order of usage / economy is Methanol > Glycol > Glycerol

41.(ABC) If solvent and solute react each other then solution becomes non-ideal and relations of colligative properties do not give accurate molecular mass of solute.

42.(ABD) Azeotropes are non-ideal solutions, hence they do not obey Raoult's law.

43.(ABC) In case of solutions showing negative derivation from Raoult's law, the A – B interactions are stronger than that between A – A and B – B.

$\therefore$  Stronger bonds are formed.

As a result,  $\Delta H_{\text{mix}} < 0$  and  $\Delta V_{\text{mix}} < 0$

$$\text{Also, } p_T < p_A^0 x_A + p_B^0 x_B$$

44.(AB)  $(\text{H}_2\text{O} + \text{C}_2\text{H}_5\text{OH})$  and  $(\text{C}_6\text{H}_6 + \text{C}_2\text{H}_5\text{OH})$  show positive deviation from Raoult's law while  $(\text{H}_2\text{O} + \text{HCl})$  and  $[\text{CHCl}_3 + (\text{CH}_3)_2\text{CO}]$  show negative deviation from Raoult's law.

45.  $[\text{A-q}, r] \rightarrow [\text{B-q}, r] \rightarrow [\text{C-p}, r] \rightarrow [\text{D-s}]$

Elevation in boiling point and depression in freezing point are both colligative properties.

$$\Delta T_b = k_b m \quad \text{and} \quad \Delta T_f = k_f m$$

Osmotic pressure is also a colligative property

46.(69.6) Given that :

$$p^\circ = 640 \text{ mm Hg}, p_s = 600 \text{ mm Hg} \text{ and } w = 2.175 \text{ g}, W = 39.0 \text{ also } M = 78$$

$$\frac{p^\circ - p_s}{p^\circ} = \frac{w \times M}{m \times W} \text{ therefore } \frac{640 - 600}{640} = \frac{2.175 \times 78}{m \times 39} \text{ or } m = 69.6$$

47.(46.9)  $n_{\text{H}_2\text{O}} = \frac{500}{18} = 27.78 \text{ mol}, n_{(\text{glucose})} = \frac{20}{180} = 0.11 \text{ mol}$

$$X_{\text{H}_2\text{O}} = \frac{n_{\text{H}_2\text{O}}}{n_{\text{H}_2\text{O}} + n_{(\text{glucose})}} = \frac{27.78}{27.78 + 0.11} = \frac{27.78}{27.89} = 0.996$$

According to Raoult's law,

$$\text{Vapour pressure of solution} = P_{\text{H}_2\text{O}} = P_{\text{H}_2\text{O}}^\circ X_{\text{H}_2\text{O}} = 47.1 \times 0.996 = 46.9 \text{ torr}$$

$$\text{Lowering of vapour pressure} = P_{\text{H}_2\text{O}}^\circ - P_{\text{H}_2\text{O}} = 47.1 - 46.9 = 0.2 \text{ torr}$$

48.(3.35) Given :  $P_A^\circ = 59.2 \text{ torr}, P_A = 51.3 \text{ torr}$

$$\text{By RLVP} \Rightarrow X_{\text{solute}} = \frac{7.9}{59.2} = 0.1334$$

$$\therefore (\text{molality}) m = \frac{X_B}{X_A} \times \frac{1000}{M_{\text{solvent}}} = \frac{X_B}{(1 - X_B)} \times \frac{1000}{M_{\text{solvent}}} \quad (\because X_A + X_B = 1)$$

$$m = \frac{0.1334}{0.8666} \times \frac{1000}{46} = 3.346 \text{ molal}$$

49.(100.25) Urea solution,  $\Delta T_b = K_b m$

$$0.25 = K_b \cdot \frac{1 \text{ g}}{60 \text{ g mol}^{-1}} \times \frac{1}{1}$$

$$\Rightarrow K_b = (0.25 \times 60) \text{ k kg mol}^{-1}$$

Glucose solution

$$\Delta T_b = K_b m = (0.25 \times 60) \times \frac{3}{180} \times \frac{1}{1} = 0.25$$

Boiling point of solution =  $100.25^\circ\text{C}$

50.(4.48) Concentration =  $\frac{68.4 \text{ g}}{342 \text{ g mol}^{-1}} \times \frac{1}{1} = 0.2 \text{ M}$

$$\pi = CRT = 0.2 \text{ mol L}^{-1} \times 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1} \times 273 \text{ K}$$

$$\pi = 4.48 \text{ atm}$$

51.(0.478) Pentane : hexane mole ratio = 1 : 4 (1n solution phase)

$$P_{\text{pentane}}^\circ = 440 \text{ mm Hg}$$

$$P_{\text{hexane}}^\circ = 120 \text{ mm Hg}$$

$$y_A = \frac{P_A^\circ x_A}{P_T}$$

$$P_T = P_A^\circ x_A + P_B^\circ x_B = 440 \left( \frac{1}{5} \right) + 120 \left( \frac{4}{5} \right) = 184 \text{ mm Hg}$$

$$y_{\text{pentane}} = \frac{440 \times \frac{1}{5}}{184} = 0.478$$

52.(84.86°C) Using the result :

$$\Delta T_b = K_b \left( \frac{\frac{g_B}{M_B}}{g_A} \times 1000 \right) = 2.57 \left( \frac{\frac{5}{180}}{\frac{15}{15}} \times 1000 \right) = 4.76^\circ\text{C}$$

$$T_b = T_b^0 + \Delta T_b \quad [T_b^0 = \text{Boiling point of pure solvent}]$$

$$= 80.1 + 4.76 = 84.86^\circ\text{C}$$

53.(2736 gm)  $\Delta T_f = 3.72^\circ\text{C}$ ,  $M_{\text{sugar}} = 342$

$$\Delta T_f = K_f \left( \frac{\frac{g_B}{M_B}}{g_A} \times 1000 \right) \Rightarrow g_B = \frac{\Delta T_f g_A M_B}{1000 K_f}$$

$$\Rightarrow g_B = \frac{3.72 \times 4 \times 10^3 \times 342}{1000 \times 1.86} = 2736 \text{ gm}$$

54.(83.5%)  $i$  for  $\text{AgNO}_3 = \frac{\text{Normal mol. wt.}}{\text{Observed mol. wt.}} = 1 + \alpha \Rightarrow \alpha = \frac{170}{92.64} - 1 = 0.835 = 83.5\%$

55.(2.24 mg) Solubility of oxygen  $C = k \times P$

$$= 1.4 \times 10^{-3} \text{ mol L}^{-1} \text{ atm}^{-1} \times 0.5 \text{ atm}$$

$$= 0.7 \times 10^{-3} \text{ mol L}^{-1}$$

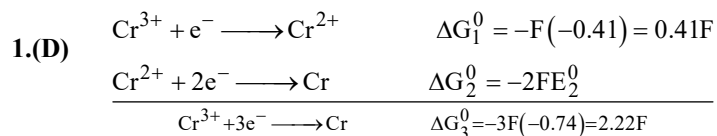
$$\Rightarrow 1 \text{ L of solution contains} = 0.7 \times 10^{-3} \text{ moles O}_2$$

$$\Rightarrow 0.1 \text{ L of solution contains} = 0.7 \times 10^{-4} \text{ moles O}_2$$

$$= (0.7 \times 10^{-4} \times 32) \text{ g O}_2 = 2.24 \text{ mg O}_2$$

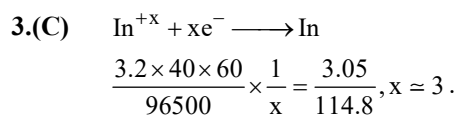
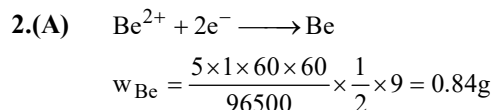
## Solutions - Advanced Problem Package | Chemistry

### Electrochemistry



$$\Delta G_1^0 + \Delta G_2^0 = \Delta G_3^0$$

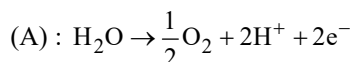
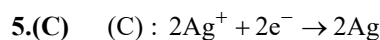
$$-1.81 F = 2FE_2^0 \quad ; \quad E_2^0 = -0.905$$



Equivalents of Cr = Equivalents of Ag

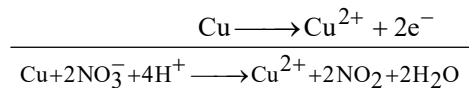
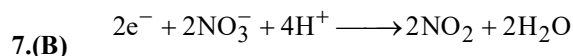
$$\frac{W_{\text{Cr}}}{52} \times 3 = \frac{0.54}{108} \times 1$$

$$W_{\text{Cr}} = 0.0866\text{g}.$$



6.(D)  $E = E^0 - \frac{RT}{nF} \ln \frac{[\text{Cr}^{3+}]}{[\text{Ag}^+]^3} = 1.5 - \frac{0.059}{3} \log \frac{0.1}{(0.1)^3} = 1.5 - 0.039 = 1.461$

$$\Delta G = -nFE = -3 \times 96500 \times 1.461 = -422959\text{J} = -422.96\text{KJ}.$$



$$E = \left( E^0_{\text{NO}_3^-/\text{NO}_2} - E^0_{\text{Cu}^{2+}/\text{Cu}} \right) - \frac{0.059}{2} \log \frac{[\text{Cu}^{2+}](P_{\text{NO}_2})^2}{[\text{NO}_3^-]^2 [\text{H}^+]^4}$$

$$E = (0.79 - 0.34) - \frac{0.059}{2} \log \frac{0.1 \times 10^{-6}}{[\text{HNO}_3]^4}$$

$$0 = 0.45 - \frac{0.059}{2} \log \frac{10^{-7}}{[\text{HNO}_3]^4}$$

$$15.2 = \log \frac{10^{-7}}{[\text{HNO}_3]^4} \Rightarrow 1.58 \times 10^{15} = \frac{10^{-7}}{[\text{HNO}_3]^4} \Rightarrow [\text{HNO}_3] = 2.82 \times 10^{-6}\text{M}.$$



8.(C)  $\Lambda_{\text{eq}} = \Lambda_{\text{m}} / \text{Total cationic or anionic change in 1 mole of electrolyte}$

$$\Lambda_{\text{eq}} = \frac{150}{3} = 50 \text{ S cm}^2 \text{ eq}^{-1}.$$

9.(D)  $\text{Pt} | \text{H}_2 | \text{HA}(\text{cM}) || \text{H}^+(1\text{M}) | \text{H}_2 | \text{Pt}$

Net Reaction:  $\text{H}_c^+ \longrightarrow \text{H}_A^+$

$$E_{\text{cell}} = \frac{0.059}{1} \log \frac{[\text{H}_c^+]}{[\text{H}_A^+]} = 0.059 \log \frac{1}{[\text{H}_A^+]}$$

To get maximum  $E_{\text{cell}}$ , the  $[\text{H}_A^+]$  must be least.

10.(D)  $\Delta S = nF \left( \frac{dE}{dT} \right)$

$$\Delta G = -nFE = \Delta H - T \times nF \left( \frac{dE}{dT} \right)$$

$$\Delta H = nF \left[ T \left( \frac{dE}{dT} \right) - E \right] = 1 \times 965000 \left[ \left\{ 298 \times (-5.5 \times 10^{-4}) \right\} - 0.265 \right]$$

$$\Delta H = -41388.85 \text{ J} = -41.38 \text{ kJ}$$

11.(B)  $k_{\text{soln}} = k_{\text{AgCl}} + k_{\text{water}}$

$$\therefore k_{\text{AgCl}} = k_{\text{soln}} - k_{\text{water}} = 1.26 \times 10^{-6} = k_{\text{Ag}^+} + k_{\text{Cl}^-}$$

$$1.26 \times 10^{-6} = \frac{\lambda_{\text{Ag}^+} [\text{Ag}^+]}{1000} + \frac{\lambda_{\text{Cl}^-} [\text{Cl}^-]}{1000}$$

$$1.26 \times 10^{-3} = (53.9 + 72.1) [\text{Ag}^+]$$

$$[\text{Ag}^+] = \frac{1.26 \times 10^{-3}}{126} = 10^{-5} \text{ moles / L}$$

Or  $1.435 \times 10^{-3} \text{ g / L}$ .

12.(D) The conducting ability of an electrolyte is measured using its equivalent conductivity.

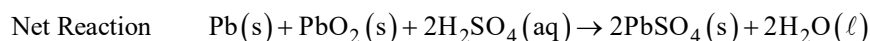
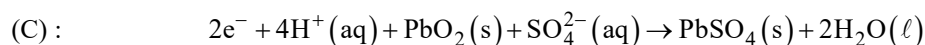
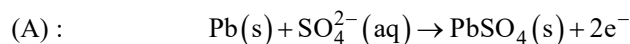
$$\begin{array}{cccc} \text{AgCl} & \text{BaSO}_4 & \text{AlPO}_4 & \text{Ba}_3(\text{PO}_4)_2 \\ \Lambda_{\text{eq}} = \frac{200}{1} = 200 & \frac{300}{2} = 150 & \frac{500}{3} = 166.6 & \frac{700}{6} = 116.6 \end{array}$$

The one with highest equivalent conductivity is the most conducting.

13.(C)  $\alpha = \frac{\Lambda_{\text{m}}^c}{\Lambda_{\text{m}}^\infty} = \frac{5.5}{275} = 0.02$

$$K_a = \frac{C \alpha^2}{1 - \alpha} = \frac{0.1(0.02)^2}{0.98} = 4.08 \times 10^{-5}$$

14.(C) The reactions during discharging is



15.(A) First conductance decreases due to replacement of  $H^+$  by  $Na^+$  till the equivalence point and then increases due to presence of free  $OH^-$  ions.

16.(B) Potential of the cell will be same as that of the  $Fe^{3+}/Fe^{2+}$  couple.

$$E_{\text{cell}} = E_{Fe^{3+}/Fe^{2+}} = E_{Fe^{3+}/Fe^{2+}}^{\circ} - \frac{0.059}{1} \log \frac{[Fe^{2+}]}{[Fe^{3+}]} \Rightarrow E_{\text{cell}} = 0.8 - \frac{0.059}{1} \log \frac{\frac{2}{3}}{\frac{1}{3}} \Rightarrow E_{\text{cell}} = 0.782V$$

17.(D) At equivalence point,  $[Fe^{2+}]$  and  $[Fe^{3+}]$  will be equal.

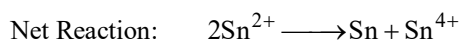
$$\therefore E_{\text{cell}} = E_{Fe^{3+}/Fe^{2+}} = E_{Fe^{3+}/Fe^{2+}}^{\circ} = 0.8V$$

18.(B)  $E_{\text{cell}} = E_{Ce^{4+}/Ce^{3+}}^{\circ} - E_{Fe^{3+}/Fe^{2+}}^{\circ} - \frac{0.059}{1} \log K$

$$0 = (1.64 - 0.8) - 0.059 \log K$$

$$\log K = \frac{0.84}{0.059}, K = 10^{0.84/0.059}$$

19.(BD)

$$\begin{array}{l} 2e^- + Sn^{2+} \longrightarrow Sn \\ Sn^{2+} \longrightarrow Sn^{4+} + 2e^- \end{array}$$


$$E_{\text{cell}}^{\circ} = E_{Sn^{2+}/Sn}^{\circ} - E_{Sn^{4+}/Sn^{2+}}^{\circ} = -0.14 - (-0.13) = -0.01$$

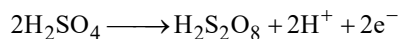
$Sn^{2+}$  is stable and does not undergo disproportionation.

20.(CD)  $\Lambda_{\text{AcOH}}^{\infty} = (91 + 426.2 - 126.5) = 390.7$

$$\Lambda_{\text{HNO}_3}^{\infty} = (426.2 + 145) - 149.9 = 421.3$$

21.(ABC)(C) :  $2H^+ + 2e^- \longrightarrow H_2$

(A) :  $2OH^- \longrightarrow \frac{1}{2}O_2 + H_2O + 2e^-$



Equivalents of  $H_2$  formed = Equivalents of  $O_2$  formed + Equivalents of  $H_2S_2O_8$  formed.

22.(ABD)

(A)  $H_C^+ \longrightarrow H_A^+$

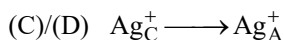
$$E_{\text{cell}} = \frac{0.059}{1} \log \frac{[H_C^+]}{[H_A^+]}$$

The reaction would be spontaneous when  $[H_C^+] > [H_A^+]$ .

(B)  $H_2(A) \longrightarrow H_2(C)$

$$E_{\text{cell}} = \frac{0.059}{2} \log \frac{(P_{H_2})_A}{(P_{H_2})_C}$$

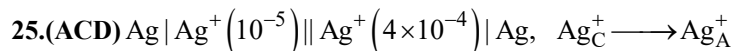
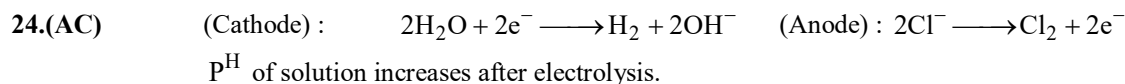
This cell reaction is spontaneous when  $(P_{H_2})_A > (P_{H_2})_C$ .



$$E_{\text{cell}} = \frac{0.059}{1} \log \frac{[\text{Ag}_C^+]}{[\text{Ag}_A^+]}$$

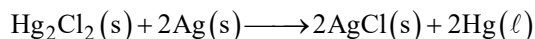
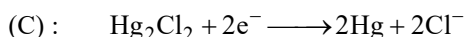
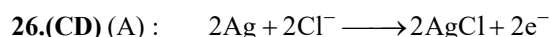
The cell reaction is spontaneous when  $[\text{Ag}_C^+] > [\text{Ag}_A^+]$ .

23.(ACD)  $\Delta G^\circ = -RT \ln K_{\text{eq}}$ .

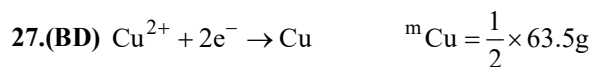


$$E_{\text{cell}} = \frac{0.059}{1} \log \frac{4 \times 10^{-4}}{1 \times 10^{-5}} = 0.059 \times 1.6 = 0.0944$$

It is an electrolyte concentration cell and the reaction occurring in it is spontaneous.

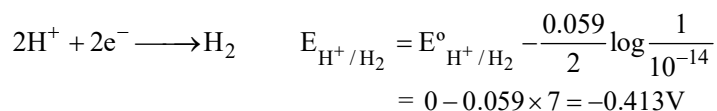
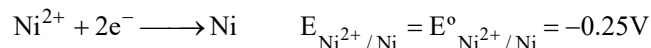


$$E_{\text{cell}} = E^\circ_{\text{Cl}^-/\text{Hg}_2\text{Cl}_2/\text{Hg}} - E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}}$$

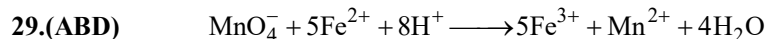


28.(BC) At anode,  $\text{H}_2\text{O}$  gets oxidized in preference to  $\text{SO}_4^{2-}$  to form  $\text{O}_2$  gas.

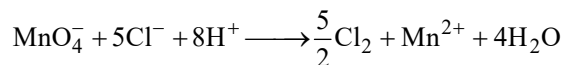
At cathode, the possible reactions are



$\therefore$   $\text{Ni}^{2+}$  will be reduced in preference to  $\text{H}^+$ .



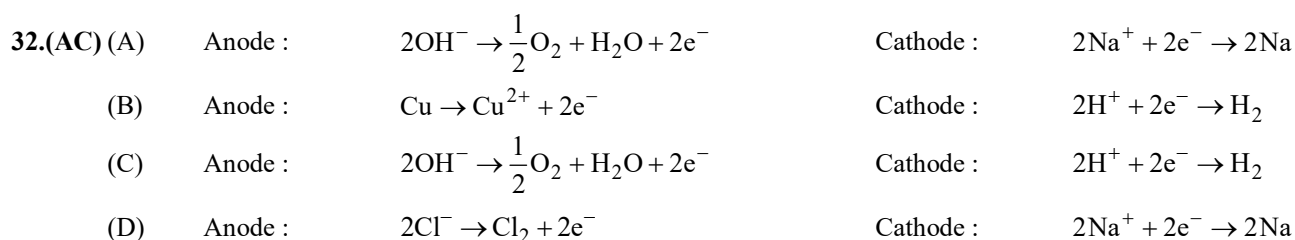
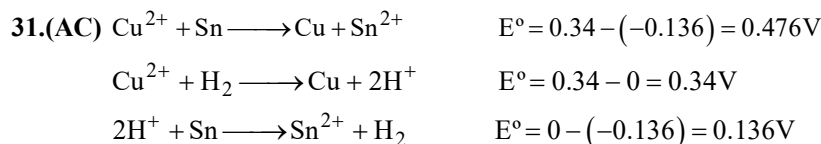
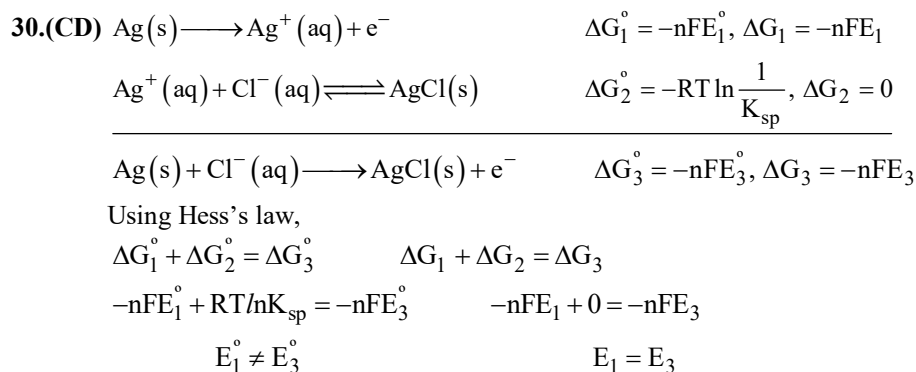
$$E^\circ = 1.5 - 0.77 = 0.73\text{V}$$



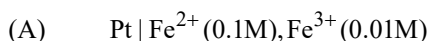
$$E^\circ = 1.5 - 1.4 = 0.1\text{V}$$

Equivalents of  $\text{MnO}_4^-$  used = Equivalent of  $\text{Fe}^{2+}$  + Equivalent of  $\text{Cl}^-$ .

Estimation of  $\text{Fe}^{2+}$  can not be done unless and until amount of  $\text{Cl}^-$  is known using  $\text{MnO}_4^-$  in aqueous  $\text{HCl}$ .

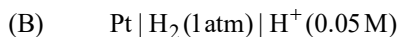


33.(ACD) Redox half cell.

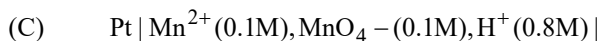


Represents conversion of  $\text{Fe}^{2+}$  to  $\text{Fe}^{3+}$  if half cell behaves as anode.

$\text{Fe}^{3+}$  to  $\text{Fe}^{2+} \rightarrow$  if half cell behaves as cathode.

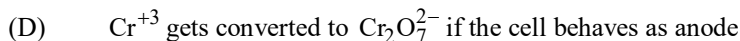


Represent anode (oxidation half cell, not redox half cell)



$\text{Mn}^{2+}$  converts to  $\text{MnO}_4^-$  if half cell forms anode.

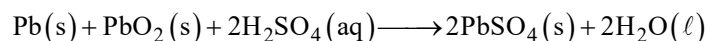
$\text{MnO}_4^-$  convert to  $\text{Mn}^{2+}$  if half cell forms cathode.



$\text{Cr}_2\text{O}_7^{2-}$  gets converted to  $\text{Cr}^{+3}$  if the cell behaves as cathode

34.(ABD) A more electronegative halogen displaces a less electronegative halogen from its solution.

35.(ABC) Discharging reactions:

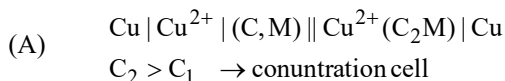


Reactions during charging is exactly reverse of the above.

With the lowering of temp, the viscosity of electrolyte increases which leads to difficulty in movement of ions.

36.(AC) A spontaneous electrolyte concentration cell will have concentration of ion at cathode, greater than that of anode.

37. [A-p, s] → [B-s] → [C-p, q, s] → [D-q, r, s]

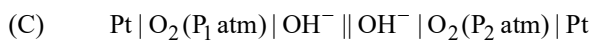


In A concentration cell anode and cathode are made up of same metal metal ion but diff concentration of ion in both halves.

It represents an electrochemical cell.



Represents Galvanic cell

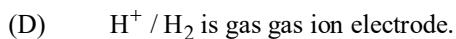


→ Represents concentration cell

∴ Partial pressure of  $\text{O}_2$  in both the halves are different

→ It also represents gas-gas ion electrode ( $\text{O}_2$  gas,  $\text{OH}^-$  ion)

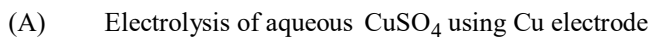
→ Represents Galvanic cell.



→ pH measurement can be done from  $\text{H}^+$  concentration.

→ represents Galvanic cell

38. [A-q, s] → [B-p, r, s, t] → [C-r, s, t] → [D-p, s]



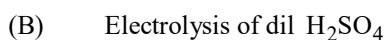
→ In electrolytic cell, positive electrode is Anode and negative electrode is cathode.

At Cathode :-  $\text{Cu}^{2+}$  gets deposited into Cu

At Anode :- Cu forms  $\text{Cu}^{2+}$

∴  $[\text{SO}_4^{2-}]$  remains constant

$[\text{Cu}^{2+}]$  remains constant

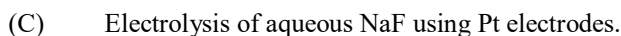


At cathode : -  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

+ve cathode:- Anode

-ve cathode : - Cathode.

At anode : -  $\text{O}_2$  is liberated

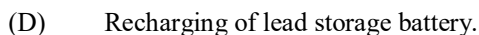


Cathode  $\text{H}_2\text{O}$  gets reduced to  $\text{H}_2$

In electrolytic all – cathode –ve charged

Anode → +ve charged

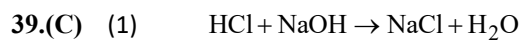
Anode -  $\text{H}_2\text{O}$  gets oxidized to  $\text{O}_2$



→ It behaves as electrolytic cell during recharging

∴ cathode → –ve charged electrode

Anode → +ve charged electrode

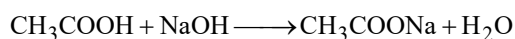
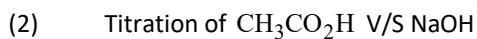
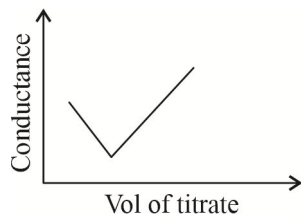


On titration conc of  $\text{H}^+$  ion decreases

$\therefore$  conductance decreases. (since  $\text{H}^+$  converts to  $\text{H}_2\text{O}$ )

When titration is complete, conductance becomes minimum.

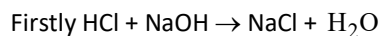
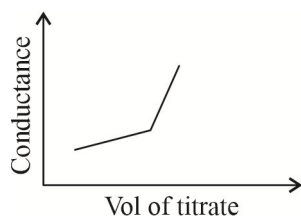
On further addition of  $\text{OH}^-$ , conduction increases



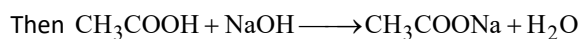
$\therefore$  Conc. of ions increases with titration when titration is complete and even then  $\text{NaOH}$  is added

$[\text{OH}^-] \uparrow$  es. And

$\therefore$  Conductance increases

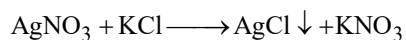
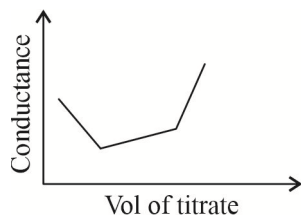


Conduction decreases



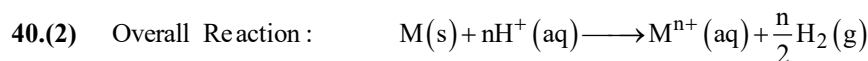
Buffer is formed

No. of ion increases  $\therefore$  Conduction increase slowly.



$\therefore$   $\text{AgCl}$  is ppt.

Conductance decreases & when ppt is complete on further addition of  $\text{KCl}$ , conductance again starts increasing.



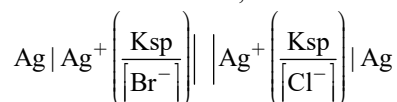
$$E_{\text{cell}} = E^{\circ}_{H^+/H_2} - E^{\circ}_{M^{n+}/M} - \frac{0.059}{n} \log \frac{[M^{n+}](P_{H_2})^{\frac{n}{2}}}{[H^+]^n}$$

$$0.81 = 0 - (-0.76) - \frac{0.059}{n} \log 0.02 \quad ; \quad 0.81 - 0.76 = -\frac{0.059}{n} \log 0.02$$

$$0.05 = \frac{0.059 \times 1.7}{n}$$

$$n = 2$$

41.(2) The cell is reduced to,



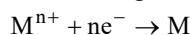
The overall reaction is :  $Ag_c^+ \longrightarrow Ag_A^+$

$$\therefore E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.059}{1} \log \frac{[Ag_A^+]}{[Ag_c^+]}$$

$$0.118 = \frac{0.059}{1} \log \frac{K_{sp}(AgCl)}{[Cl^-]} \times \frac{[Br^-]}{K_{sp}(AgBr)} \quad ; \quad 2 = \log \frac{1 \times 10^{-10} \times [Br^-]}{5 \times 10^{-13} \times [Cl^-]}$$

$$2 = \log 200 \frac{[Br^-]}{[Cl^-]} \quad ; \quad \therefore \frac{[Cl^-]}{[Br^-]} = 2.$$

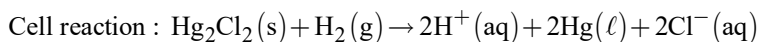
42.(4) Let the charge on metal cation be  $n+$ . The reduction half reaction would be



The quantity of electricity required to deposit 106.4 g metal =  $n$  mole of electron =  $n \times 96500$  C .

$$\therefore 2.977 \text{ g metal will be deposited by } = \frac{n \times 96500 \times 2.977}{106.4} = n \times 2700 \text{ C of electricity}$$

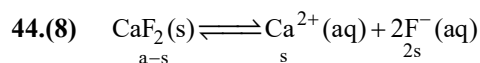
$$\text{The electricity actually passed} = 4 \times 45 \times 60 \text{ C} \quad \therefore \quad n = \frac{4 \times 45 \times 60}{2700} = 4.$$



$$E_{\text{cell}} = E^{\circ}_{\text{calomel}} - E^{\circ}_{H^+/H_2} - \frac{0.06}{2} \log \frac{[H^+]^2 [Cl^-]^2}{P_{H_2}}$$

$$\therefore \quad 0.68 = 0.28 - 0 - \frac{0.06}{2} \log \frac{[H^+]^2 1^2}{1} \quad ; \quad 0.48 = 0.06 \text{ pH}$$

$$\text{pH} = \frac{0.48}{0.06} = 8$$



$$K_{sp} = [Ca^{2+}] [F^-]^2 = s(2s)^2 = 4s^3$$

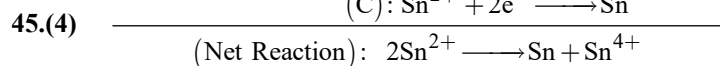
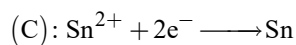
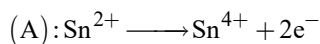
$$K_{\text{solution}} - K_{\text{water}} = K_{CaF_2} = K_{Ca^{2+}} + K_{F^-}$$

$$(5.2 \times 10^{-5} - 0.2 \times 10^{-5}) = 5 \times 10^{-5} = \frac{[\text{Ca}^{2+}]}{1000} \approx_{\text{Ca}^{2+}} + \frac{[\text{F}^-]}{1000} \approx_{\text{F}^-}$$

$$5 \times 10^{-5} = \frac{[\text{Ca}^{2+}]}{1000} (\approx_{\text{Ca}^{2+}} + 2 \approx_{\text{F}^-}) = \frac{[\text{Ca}^{2+}]}{1000} (120 + 130)$$

$$5 \times 10^{-5} = \frac{[\text{Ca}^{2+}]}{1000} \times 250 \quad ; \quad [\text{Ca}^{2+}] = 2 \times 10^{-4} = s$$

$$K_{sp} = 4(s)^3 = 4(2 \times 10^{-4})^3 = 4 \times 8 \times 10^{-12} \quad ; \quad \therefore x = 8$$



$$\text{No. of moles of } e^- \text{ passed} = \frac{96.5 \times 100 \times 60}{96500} = 6$$

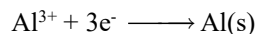
$$\text{Moles of } \text{Sn}^{2+} \text{ consumed} = 6$$

$$\text{Initial moles of } \text{Sn}^{2+} = 8$$

$$\text{Moles of } \text{Sn}^{2+} \text{ left} = 2$$

$$[\text{Sn}^{2+}]_{\text{left}} = 4M.$$

$$46.(9) \quad \text{Number of moles of } e^- \text{ passed} = \frac{9650 \times 100}{96500} = 10$$



$$\text{Moles of Al produced} = \frac{10}{3}$$

$$\text{Mass of Al produced} = \frac{10}{3} \times 27 = 90 \text{ g}$$

$$\text{No of cans of Al that can be made} = 9$$

47. (-1.66 V)

$$E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}$$

$$E^\circ_{\text{cell}} = -E^\circ_{\text{Al}^{3+}/\text{Al}} + E^\circ_{\text{Ag}^+/\text{Ag}}$$

Reduction
Reduction  
potential of Al
potential of Ag

$$\Rightarrow 2.46 = -E^\circ_{\text{Al}^{3+}/\text{Al}} + 0.80 \text{ V}$$

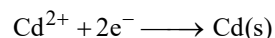
$$\Rightarrow E^\circ_{\text{Al}^{3+}/\text{Al}} = -1.66 \text{ V}$$

48. (18.99 g)

$$\text{Power} = 100 \text{ watt}$$

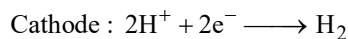
$$\text{Voltage} = 110 \text{ V}$$

$$\text{Current } I = \frac{100}{110} = 0.909 \text{ Ampere}$$



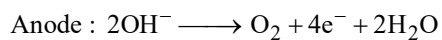
$$W = zIt = \frac{112}{2 \times 96500} \times 0.909 \times 10 \times 60 \times 60 = 18.99 \text{ g}$$



**49.(193 coulombs)**

$$\Rightarrow 2\text{F} \equiv 1 \text{ mole of } \text{H}_2 \equiv 22400 \text{ mL}$$

$$\text{or } 1\text{F} \equiv 0.5 \text{ mole of } \text{H}_2 \equiv 11200 \text{ mL}$$



$$\Rightarrow 4\text{F} \equiv 1 \text{ mole of } \text{O}_2 \equiv 22400 \text{ mL}$$

$$\text{or } 1\text{F} \equiv 0.25 \text{ mole of } \text{O}_2 \equiv 5600 \text{ mL}$$

From two electrode reactions, it clear that hydrogen and oxygen are evolved in the mole ratio of 2 : 1, hence their volumes will also be in the same ratio.

$$\Rightarrow \text{volume of } \text{H}_2 = \frac{2}{3}(33.6) = 22.4 \text{ mL}$$

$$\text{As } 2\text{F} \equiv 22400 \text{ mL } \text{H}_2$$

$$\Rightarrow 0.002 \text{ F of charge is passed through the electrolytic cell.}$$

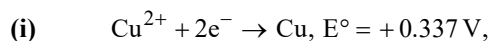
$$\Rightarrow \text{Amount of electricity} = 0.002 \times 96500 \text{ C} = 193.0 \text{ Coulombs}$$

**50.(0.521V)**

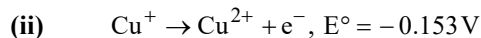
From given data,

$$\therefore E_{\text{Cu}^{2+}/\text{Cu}^+}^\circ = 0.153 \text{ V}$$

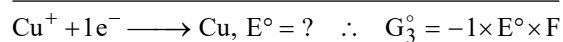
$$\therefore E_{\text{Cu}^+/\text{Cu}^{2+}}^\circ = -0.153 \text{ V}$$



$$\therefore \Delta G_1^\circ = -2 \times 0.337 \times \text{F} = -0.674 \text{ F}$$



$$\therefore \Delta G_2^\circ = -1 \times \text{F}(-0.153)$$



$$\Delta G_3^\circ = \Delta G_2^\circ + \Delta G_1^\circ$$

$$-E^\circ \times \text{F} = -0.674 \text{ F} + 0.153 \text{ F}$$

$$\text{or } -E^\circ \text{F} = -0.521 \text{ F}$$

$$\therefore E^\circ = 0.521 \text{ V}$$

## Solutions - Advanced Problem Package | Chemistry

### The Solid State

- 1.(B) Packing efficiency =  $\frac{\text{Volume occupied by particles}}{\text{Volume of unit cell}}$
- 2.(C) Relation between edge length and radius of atom in simple cubic is  $a = 2r$ , body centred cubic is  $\sqrt{3}a = 4r$  and face centred cubic is  $\sqrt{2}a = 4r$ .
- 3.(C)
- 4.(B)  $O_2$  molecule is non polar and at very low temperature it crystallises into molecular crystal having  $O_2$  molecule at lattice points. The type of interaction in this solid will be induce dipole induce dipole (or) London forces.
- 5.(D) Position of B represents octahedral void, whose coordination number is 6.
- 6.(A) This kind of planes have closest arrangement of atoms.
- 7.(A) In FCC, octahedral and tetrahedral voids are located on body diagonal at a distance of  $\frac{\sqrt{3}a}{4}$ .
- 8.(B) For the given formula, 18 copper ions present with 10 sulphide ions. Let  $(18 - x) Cu^+$  and  $x Cu^{2+}$  are present. Using charge balance,  $(18 - x) + 2x = 20$ ,  $x = 2$ , % of  $Cu^{2+} = 11.11$ .
- 9.(A) Shortest inter ionic distance is  $\sqrt{3}a / 2$ .
- 10.(B) Replacement of every  $2Na^+$  ions by one  $Sr^{2+}$  ion produces one cation vacancy.
- 11.(C) Occupying CCP lattice with A atoms, there are 4A per unit cell while occupying 50% tetrahedral voids by B, there are 4B per unit cell and coordination number of A with respect to B is 4.
- 12.(B) On heating oxygen gets evaporated, leaving excess zinc metal whose electrons can be excited by absorbing visible light (Learn as a fact & Read NCERT)
- 13.(A)  $\frac{\rho_{BCC}}{\rho_{FCC}} = \frac{Z_{BCC}}{Z_{FCC}}$
- 14.(C) In FCC arrangement,  $4O^{2-}$ ,  $\frac{1}{5}$  tetrahedral voids i.e.  $\frac{8}{5}X^{2+}$  and  $\frac{1}{2}$  octahedral voids i.e.  $2Y^{3+}$  are present. Thus the formula is  $X_4Y_5O_{10}$ .
- 15.(D)  $x\left(4 - \frac{1}{4}\right)y(4-1)z(8-2) = x_5y_4x_8$
- 16.(B) With respect to corner position, there are 12 equidistant face centre positions exist.
- 17.(C) There is no direct contact of tetrahedral and octahedral voids.
- 18.(A) Tetrahedral void is located over body diagonal at a distance of  $\frac{\sqrt{3}a}{4}$  from corner. We can see it by cutting a cube into 8 sub cubes. Distance between centres of two sub cubes = distance between two nearest tetrahedral voids in FCC (As centre of sub cube represents a tetrahedral void).
- 19.(C)
- 20.(A) Volume = Area of basal plane  $\times$  height =  $6\sqrt{3} r^2 \times 2\left[2r\sqrt{\frac{2}{3}}\right] = 24\sqrt{2} r^3$

21.(D)

|                                         |                                        |
|-----------------------------------------|----------------------------------------|
| H. C. P.                                | C. C. P                                |
| One-unit cell have 12 tetrahedral voids | One-unit cell have 8 tetrahedral voids |

In close packing of “N” identical spheres there will be “N” octahedral voids and “2N” tetrahedral voids. For C.C.P arrangement N = 4 and for H.C.P arrangement N = 6.

22.(C) Distance between two layers is equal to height of tetrahedron

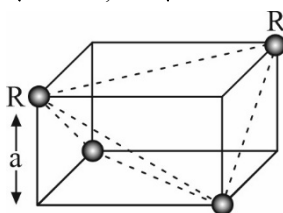
23.(B)  $\text{density} = \frac{ZM}{N_A V}$

24.(D) In Zinc blende type of structure  $S^{2-}$  occupy CCP arrangement while  $Zn^{2+}$  present in half of tetrahedral voids.

25.(C) B is present in tetrahedral void, whose coordination number is 4.

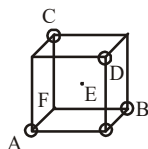
26.(C) Closest packed lattice of four atoms resembles tetrahedral arrangement and these four atoms are assumed to present at four corners of a imaginary cube.

27.(C) Face diagonal =  $a\sqrt{2} = 2R$ ;  $a = \sqrt{2}R$



Body diagonal =  $\sqrt{3}a = \sqrt{6}R$ .

28.(B)



Length AB =  $\sqrt{2}x = 2R$

Length DF =  $\sqrt{3}x = \sqrt{6}R$

Length DE =  $\frac{\sqrt{3}x}{2} = R + r_t$

$$r_t = \left( \sqrt{\frac{3}{2}} - 1 \right) R$$

29.(ACD)

| Crystal systems | Inter axial angles $[\alpha, \beta, \gamma]$      |
|-----------------|---------------------------------------------------|
| Cubic           | $\alpha = \beta = \gamma = 90^\circ$              |
| Hexagonal       | $\alpha = \beta = 90^\circ, \gamma = 120^\circ$   |
| Monoclinic      | $\alpha = \gamma = 90^\circ, \beta \neq 90^\circ$ |
| Triclinic       | $\alpha \neq \beta \neq \gamma \neq 90^\circ$     |

30.(ABC) In a close pack arrangement within a layer the voids are trigonal and the voids created by adjacent arrangement of layer's are tetrahedral and octahedral.

31.(ACD)

(A) In  $Cs^+Cl^-$  structure a cation is surrounded by 8 anions and an anion is surrounded by 8 cations then coordination number of the ions is 8.

(B) In a B.C.C arrangement of identical spheres C.N. of a sphere = 8

(C) In a NaCl type structure.

$$a = 2[r^+ + r^-]$$

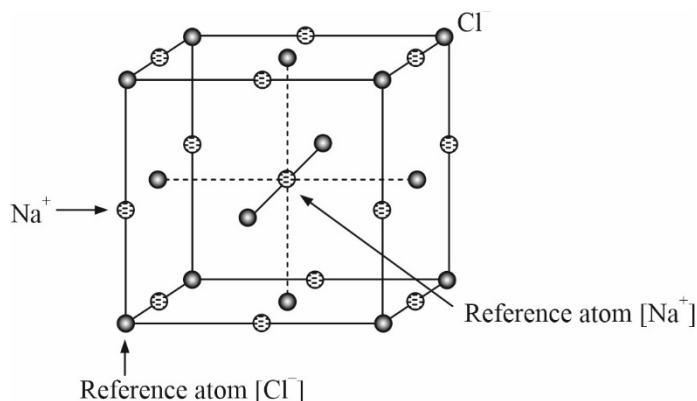
Here  $r^+ = 95 \text{ pm}$

$$r^- = 181 \text{ pm} \quad \therefore \quad a = 2[95 + 181] = 552 \text{ pm}$$

(D) Ionic crystals are formed by close packing of ions and in close packing sharing of atoms by adjacent unit cell is a common feature.

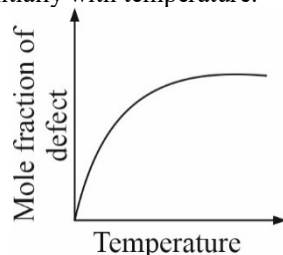
- 32(BD) (B) In H.C.P structure effective number of atom = 6  
 Number of tetrahedral voids = 12  
 $\therefore$  Number of tetrahedral voids/atom = 2.
- (D) In antifluorite structure ( $\text{Li}_2\text{O}$ ,  $\text{Rb}_2\text{S}$  etc).  
 Cation occupy tetrahedral voids and anions form a C.C.P lattice.

33.(ABC)



It can be visualise that a  $\text{Na}^+$  is surrounded by 6  $\text{Cl}^-$  ions and a  $\text{Cl}^-$  ion is surrounded by 6  $\text{Na}^+$ .  
 The next nearest neighbour of  $\text{Na}^+$  ion will be 12  $\text{Na}^+$ .

- 34.(ABCD) (A) Schottky defect is usually favoured by a very small difference in size of cation and anion.
- (B) Each defect has its energy barrier (E) and the fraction of defects is proportional to the Boltzmann factor  $[e^{-E/kt}]$ . Hence mole fraction of Frenkel defect increases exponentially with temperature.



- (C) Trapping of an electron in a lattice is the cause of F-centre.
- (D) Usually ionic solids having F-centre act as a semiconductor.

35.(ABC)

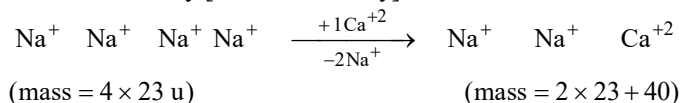
Unpaired electron doesn't exist in-case of schottky defect.

- (D) Usually ionic solids having F-centre act as a semiconductor.

36.(AD)

Doping of solids  $\text{NaCl}$  with  $\text{CaCl}_2$  cause.

Replacement of some  $\text{Na}^+$  ions by  $\text{Ca}^{+2}$ . As charge of the dopant is greater than charge of the source hence it will create metal deficiency [cationic vacancy] as well as decrease in density of the host solid.



Hence doping will cause decrease in mass of the host lattice.

37.(ABCD)

- (A)  $\text{CsCl}$  structure can show Schottky defect due to comparable sizes of ions.

- (B)  $\text{ZnS}$  structure can show Frenkel defect due to smaller radius ratio  $\left( \frac{r_{\text{Zn}^{2+}}}{r_{\text{S}^{2-}}} \right)$

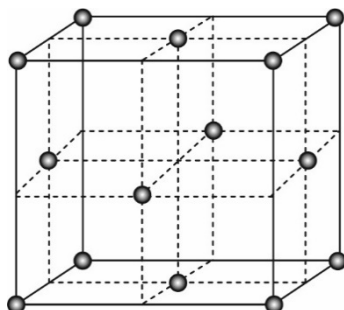
- (C) Both H.C.P and C-C.P arrangements of identical spheres have C.N = 12.

- (D) Increase in pressure over solids can forcefully change arrangement of atoms into more compact arrangement with higher coordination number.

38.(ABCD)

| Crystal system        | Inter axial angles                      |
|-----------------------|-----------------------------------------|
| Cubic                 | $\alpha = \beta = \gamma = 90^\circ$    |
| Tetragonal            | $\alpha = \beta = \gamma = 90^\circ$    |
| Orthorhombic          | $\alpha = \beta = \gamma = 90^\circ$    |
| Rhombohedral/Trigonal | $\alpha = \beta = \gamma \neq 90^\circ$ |

39.(ACD)



Out of these 8 small cubes only 4 are having carbon atoms. The centres of these small cubes are tetrahedral voids.

In diamond.

- 4 out of 8 tetrahedral voids are occupied
- Packing efficiency of diamond unit cell = 34%
- All octahedral voids are vacant.
- C.N. of a carbon atom = 4 in this unit cell.

40.(C) (1) NaCl type crystal.

$$\left[ \begin{array}{l} \text{CN. of cation} = 6 \\ \text{CN. of anion} = 6 \\ a = 2[r_c + r_a] \\ \text{Anion forming C.C.P lattice} \end{array} \right]$$

(2) CsCl type crystal

$$\left[ \begin{array}{l} \text{C.N. of cation or anion} = 8 \\ a\sqrt{3} = 2[r_c + r_a] \\ \text{Both ions forming simple cubic lattice} \end{array} \right]$$

(3) ZnS type crystal

$$\left[ \begin{array}{l} \text{CN. of ions} = 4 \\ a\sqrt{3} = 4[r_c + r_a] \\ \text{Anion forming C.C.P lattice (Zinc blende)} \end{array} \right]$$

(4) CaF<sub>2</sub> type crystal

$$\left[ \begin{array}{l} \text{CN. of cation} = 8 \\ \text{CN of anion} = 4 \\ a\sqrt{3} = 4[r_c + r_a] \\ \text{[cation forming C.C.P lattice]} \end{array} \right]$$

41(A)

(Set – I)

(Set – II)

(1) Cubic

(p)

All angles are equal

(R)

Have maximum elements of symmetry

(2) Ortho rhombic

(P)

All angles are equal

(3) Monoclinic

(S)

Contain only two Bravais lattices [Primitive and End centred].

(4) Hexagonal

(Q)

Only two sides have equal length

42.(A) In simple cubic system face of the unit cell has Q type of atomic arrangement

In body centred cubic system, diagonal plane of symmetry has R type of atomic arrangement. In face centred cubic system, diagonal plane of symmetry has S type of atomic arrangement.

In hexagonal close packing system, basal plane has P type atomic arrangement.

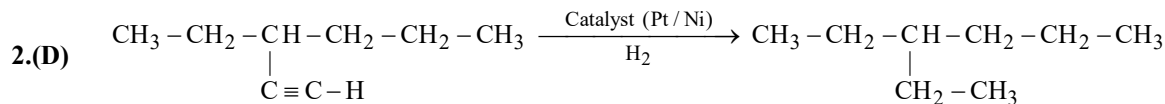
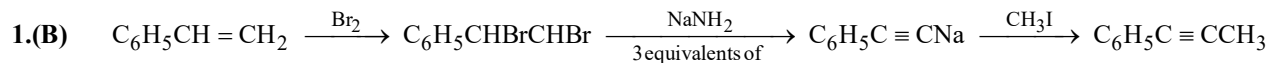
- 43.(4) This kind of arrangement is observed in F.C.C. system.  
Rank of the unit cell i.e. effective no. of atoms per unit cell is equal to 4.
- 44.(2) In body centred cubic crystal, the nearest distance between two atoms is  $\frac{\sqrt{3}a}{2}$ . Its effective no. of atoms per unit cell is 2.
- 45.(4) Cubic, tetragonal, orthorhombic, rhombohedral.
- 46.(2) The given arrangement follows face centred cubic system which contain 8 tetrahedral voids and 4 octahedral voids per unit cell.
- 47.(4) Body diagonal length is  $\sqrt{3}a$  while the nearest distance between octahedral and tetrahedral voids is  $\frac{\sqrt{3}a}{4}$ .
- 48.(3) The given axis is 4 fold axis. Removal of body centred  $\text{Na}^+$  will give 3  $\text{Na}^+$  ions per unit cell.
- 49.(7) Formula of the compound is  $\text{Zn Al}_2 \text{O}_4$ .
- 50.(4)  $\text{Ni}_{10.98} \text{O}_1$  contain 98 Nickel ions with total charge of 200 units.  

$$3x + (98 - x) 2 = 200$$

$$x = 4$$
- 51.(0) None of the face of a truncated octahedron is octahedral face.
- 52.(3) In ABC packing, that is in F.C.C. system, distance between two successive layers of 'A' is  $\sqrt{3}$  times of lattice parameter.
- 53.(5) Density =  $\frac{ZM}{N_A V}$ ,  $Z = 4$ ,  $M = 6.023y \text{ amu / formula unit} = y \times 10^{-26} \text{ kg / formula unit}$ ,  $V = a^3 = (2y^{1/3})^3 \text{ nm}^3$
- 54.(8) Formula of the compound is  $\text{A}_4\text{B}_3\text{C}$ .

## Solutions - Advanced Problem Package | Chemistry

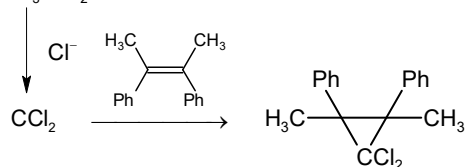
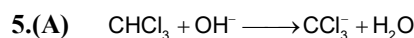
### IOC & Hydrocarbons



In catalytic hydrogenation double bonds and triple bonds are reduced.

3.(C)  $\text{sp}$  hybridized carbons are more electro negative than  $\text{sp}^2$ -hybridized carbons so alkynes are more acidic than alkenes.

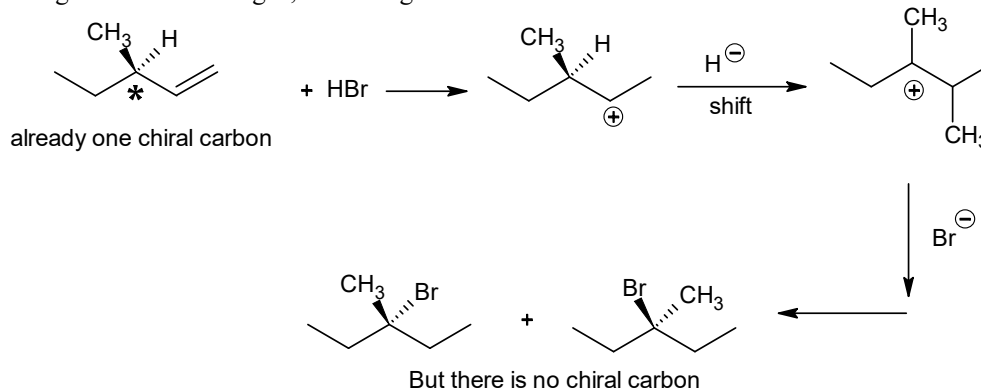
4.(A)  $\text{Br}_2$  gives Anti addition.



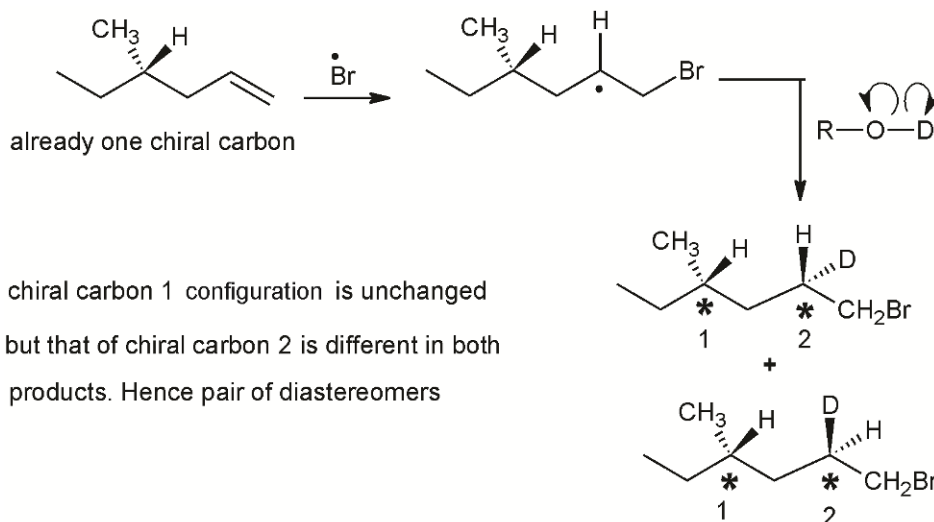
6.(C) Reaction I :

Chiral carbon 1 configuration is unchanged, but configuration are different at chiral carbon 2. Thus diastereomers.

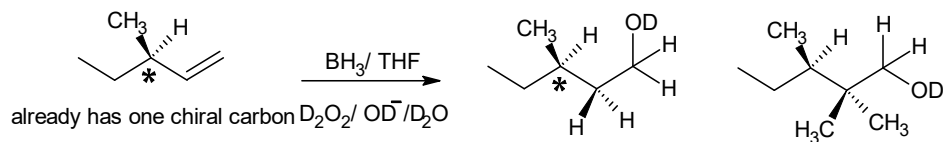
Reaction II :



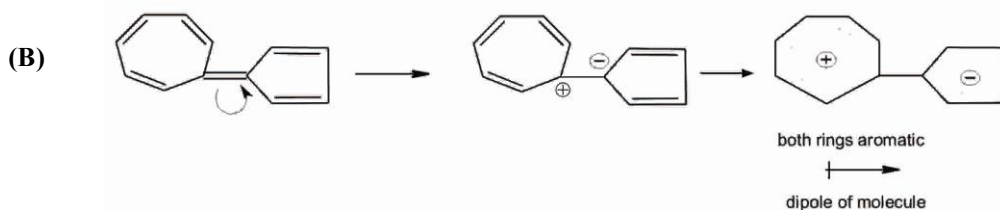
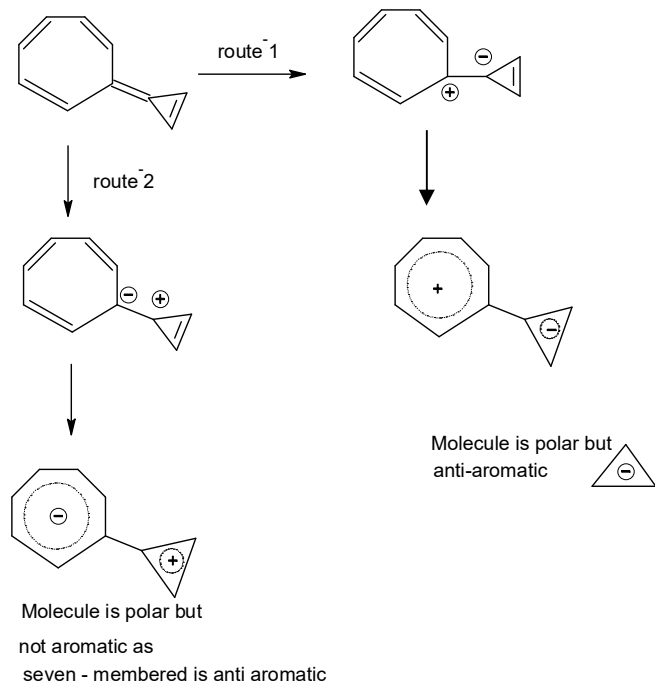
Reaction III :



Reaction IV :



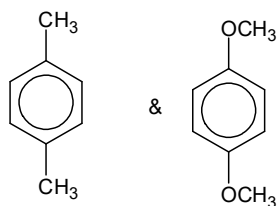
7.(B) (A)



(C) is non-polar in nature.

(D) is non-aromatic in nature because of non-polar tub-like structure.

8.(C)





- 9.(A) A meso compound must have at least 1 chiral centre and then it must have at least one element of symmetry.
- 10.(C) Benzene ring activated for electrophilic aromatic substitution can easily undergo Friedel Craft's alkylation reaction.

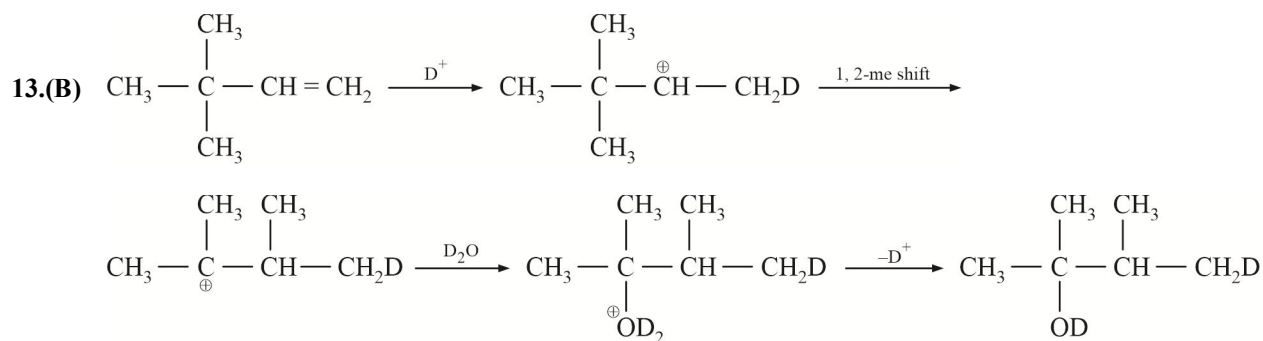
$$11.(D) \quad \% \text{ enantiomeric excess} = \frac{\text{measured specific rotation of mixture}}{\text{specific rotation of pure enantiomer}} \times 100 = \frac{11.55}{23.1} \times 100 = 50 \%$$

or  $11.55 = 23.1 \times a + (1-a)(-23.1)$  [Where 'a' is the fraction of dextro form.]

$\Rightarrow a = 0.75$ , so dextro form is 75% and 25% form is laevo form

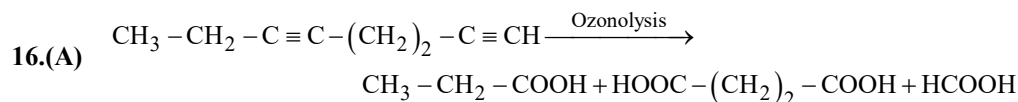
It means 50% is the enantiomeric excess of dextro or (+) form

- 12.(C) Extended conjugation.



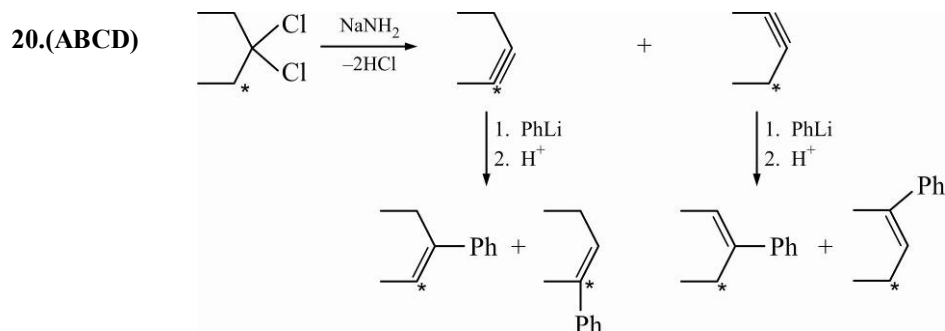
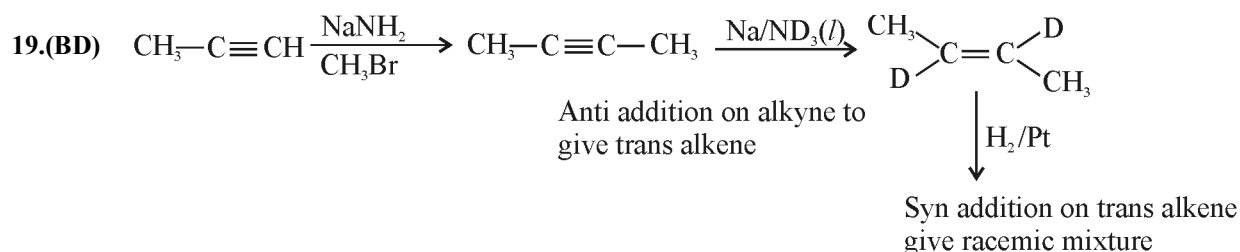
- 14.(C) Hint: meso product is formed

- 15.(B) Hydroboration-oxidation



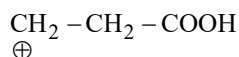
- 17.(ABC) In option (A) elimination product is the minor product, in option (D) first product is the major product because E2 elimination is carried out by anti-planar arrangement.

- 18.(AC) In option (B) isopropyl-Benzene is formed.



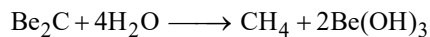
21.(BD) In first reaction addition takes place via Markonikov's rule.

In second reaction addition takes place via anti- Markonikov's rule forming more stable intermediate

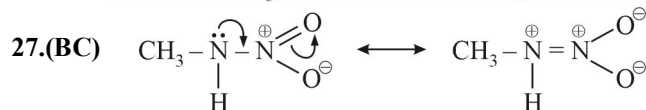
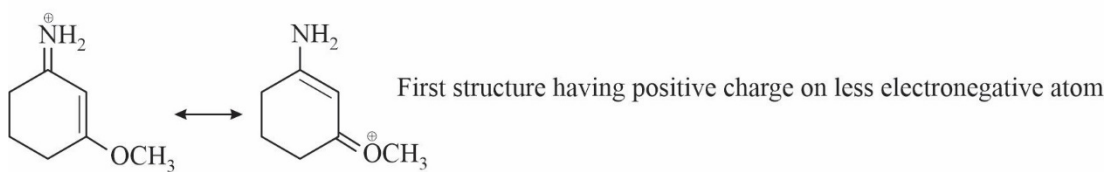
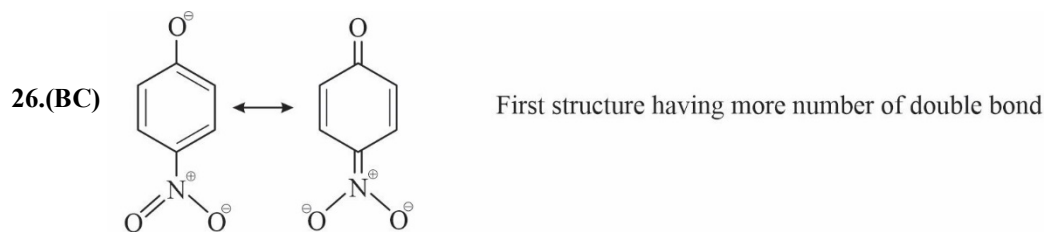
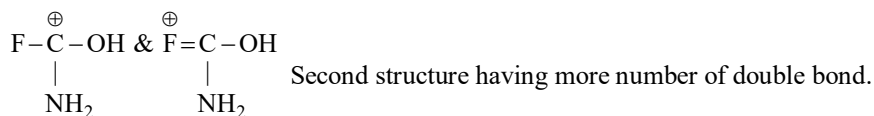
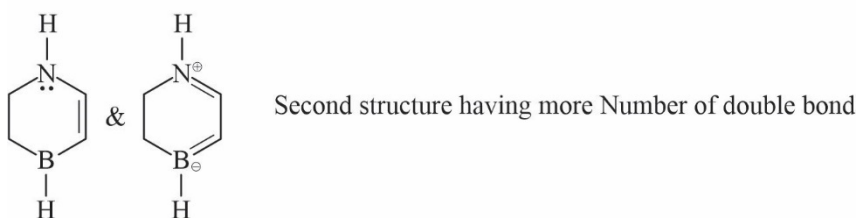
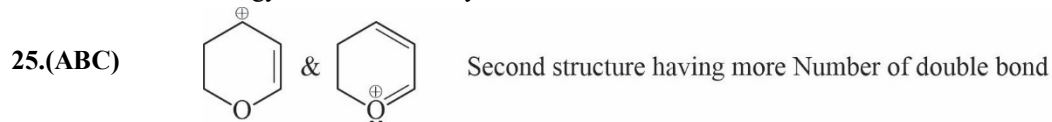


22.(BD) The structure must have three active H-atoms; red P and HI cannot reduce  $-\text{C} \equiv \text{C}-$

23.(AD)  $\text{Al}_4\text{C}_3 + 12\text{H}_2\text{O} \longrightarrow 3\text{CH}_4 + 4\text{Al}(\text{OH})_3$

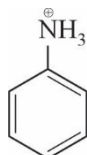


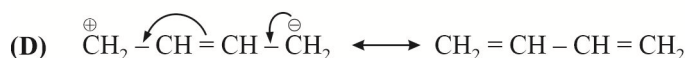
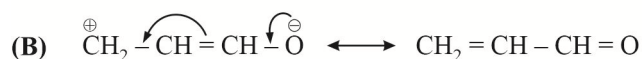
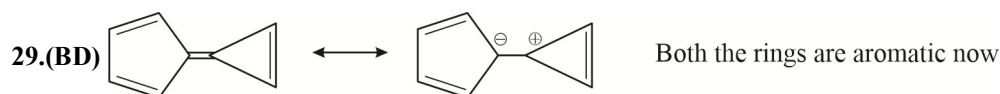
24.(ABCD) It decreases the energy of system by delocalisation. Resonating structure are hypothetical structure having decrease energy than resonance hybrid.



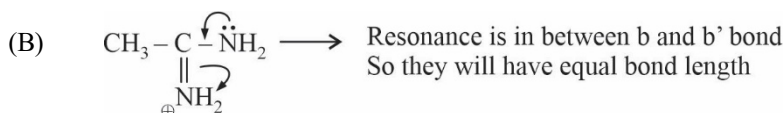
28.(C) It can not show -M effect Because nitrogen cannot form five bonds

It can not show +M effect because nitrogen neither having lone pair nor -ve charge



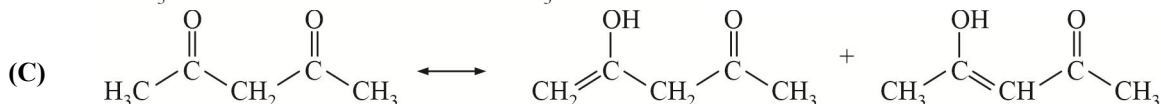
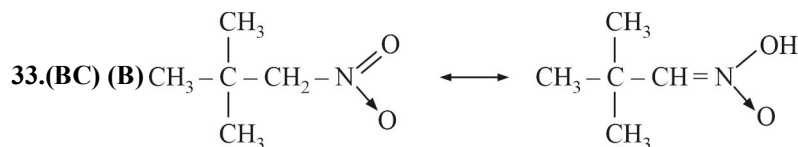


31.(ABD) (A)  $a' > a \rightarrow$  Because of resonance single bond acquire some double bond character. Hence, the bond length decreases



(D)  $d > d' \rightarrow$  Because of resonance, double bond acquire some single bond character. Here, the bond length along d will increase

32.(BCD) Equivalent contributing structures make resonance very important because of equal stability  
Contributing structures are hypothetical and less stable than resonance hybrid.



34.(ABC)

(A) More electronegative atom having -ve charge

(B) More electronegative atom having -ve charge

(C) Molecule having more No. of double bond

35.(ACD)

It is free radical allylic substitution reaction. For allylic bromination NBS,  $\text{Br}_2 / h\nu$  and  $\text{Br}_2 / \Delta$  can be used.

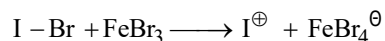
36.(ABC)

(A) Free radical bromination is highly regioselective.  $3^\circ\text{H}$  atom is easily substituted by Br atom.

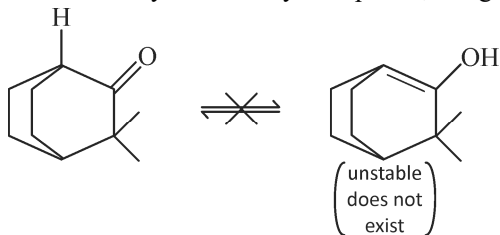
(B) It is allylic bromination

(C) It is free radical benzylic bromination of side chain.

(D) It is electrophilic substitution reaction. In this reaction iodination of benzene takes place



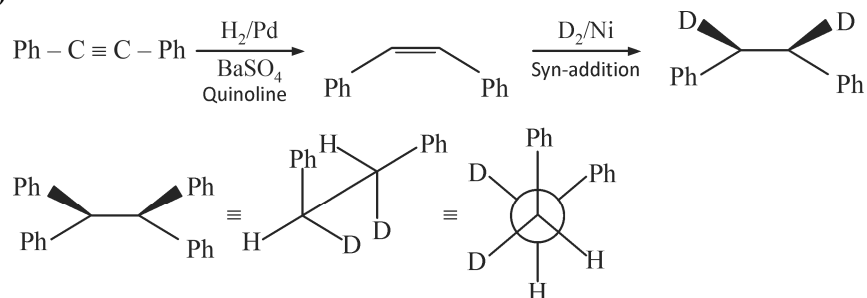
37.(AB) In case of bicyclic carbonyl compound, bridge head position can't be planar hence may not exhibit tautomerism



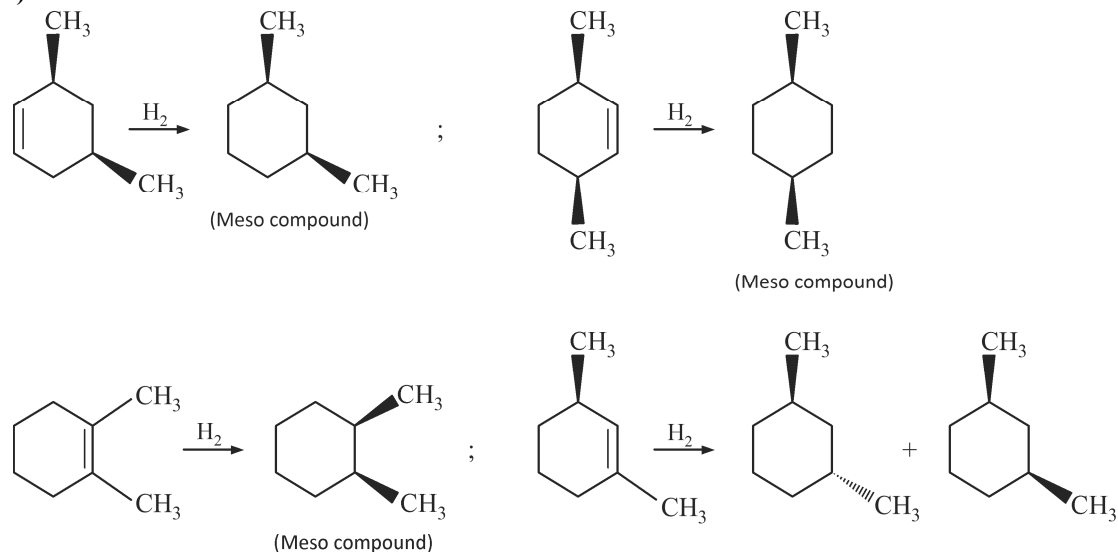
38.(ABCD)

All are suitable catalyst for partial reduction of alkyne to alkene. Rosenmund catalyst is  $\text{Pd} - \text{CaCO}_3$  poisoned with sulphur and quinoline.

39.(BC)



40.(ABC)

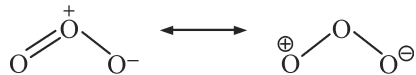


41.(ABC)

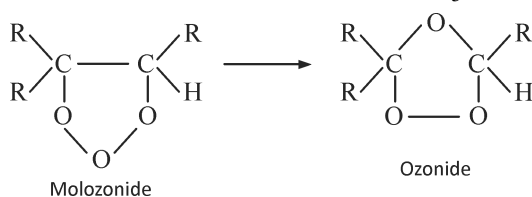
Cold  $\text{alk} \cdot \text{KMnO}_4$  and  $\text{OsO}_4$  &  $\text{Na}_2\text{CO}_3$  are used for syn hydroxylation of alkene.

42.(ABCD)

(A) Ozone is 1, 3-dipolar ion and acts as both electrophile as well as nucleophile

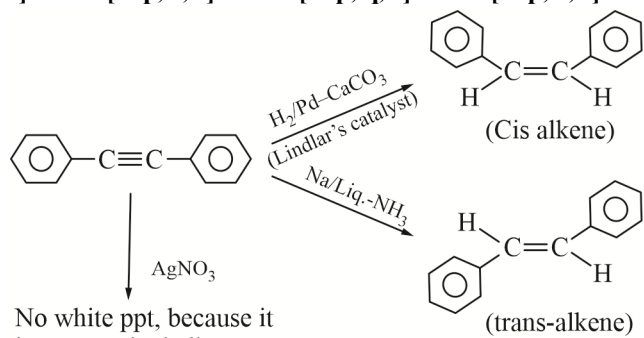


(B) Alkene is nucleophilic substrate hence  $\text{O}_3$  interact with alkene as electrophile



43. [A-p, q, s] [B-p, r, s] [C-p, q, s] [D-p, r, s]

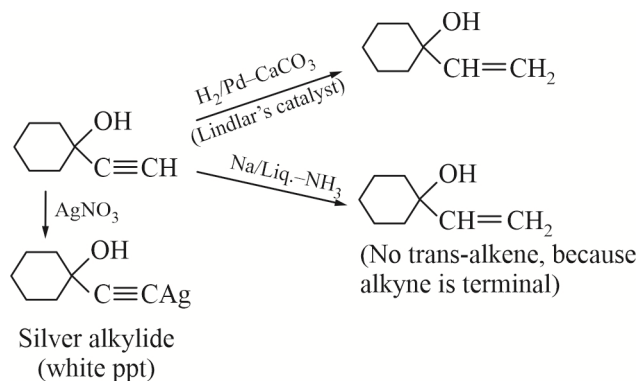
(A)



$$D.B.E = \frac{2C + 2 + N - X - H}{2} = \frac{2(14) + 2 - 10}{2} = 10$$

So,  $A \rightarrow p, q, s$ 

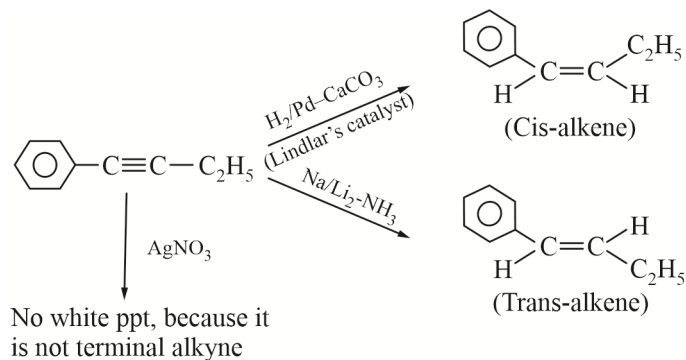
(B)



$$D.B.E = \frac{2C + 2 + N - X - H}{2} = \frac{2(8) + 2 - 12}{2} = 3$$

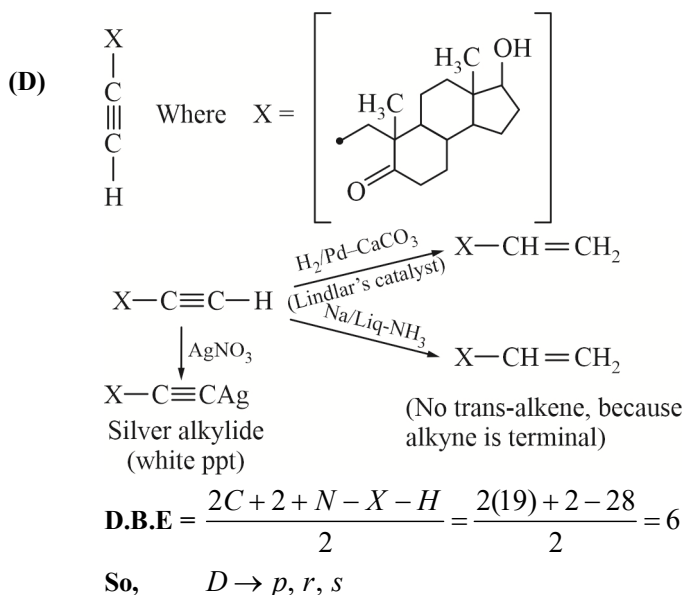
So,  $B \rightarrow p, r, s$ 

(C)

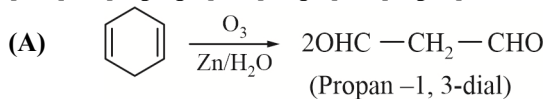


$$D.B.E = \frac{2C + 2 + N - X - H}{2} = \frac{2(10) + 2 - 10}{2} = 6$$

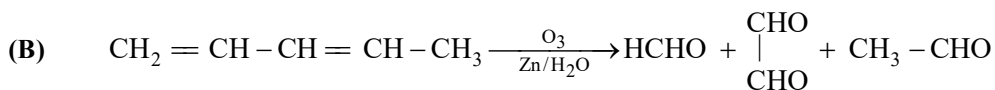
So,  $C \rightarrow p, q, s$



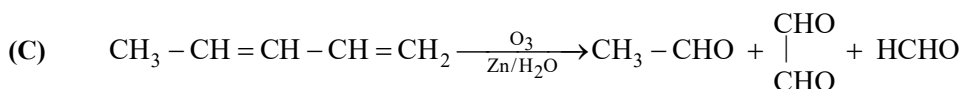
44. [A-r]  $\rightarrow$  [B-p, q, s]  $\rightarrow$  [C-q, s]  $\rightarrow$  [D-p, s]



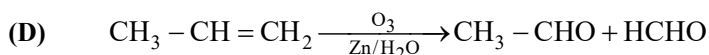
**So,**  $A \rightarrow r$



**So,**  $B \rightarrow p, q, s$



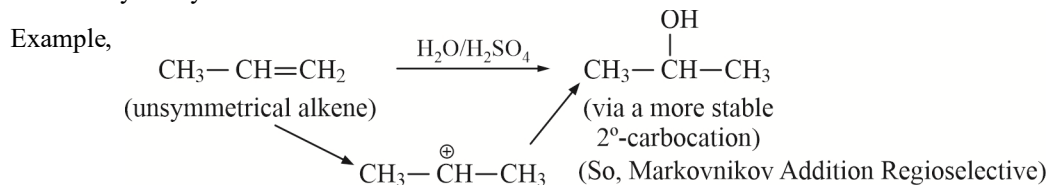
**So,**  $C \rightarrow p, q, s$



**So,**  $D \rightarrow p, s$

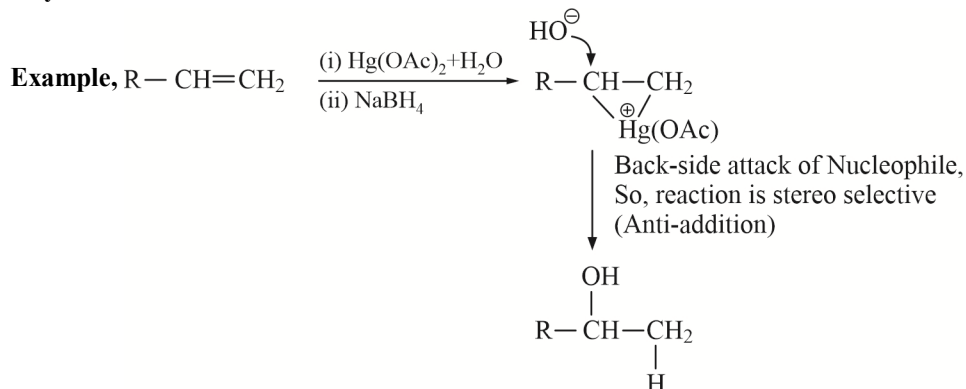
45. [A-p, s]  $\rightarrow$  [B-p, s, r]  $\rightarrow$  [C-q, s]  $\rightarrow$  [D-r, s]

(A) Acid catalyzed hydration of alkenes

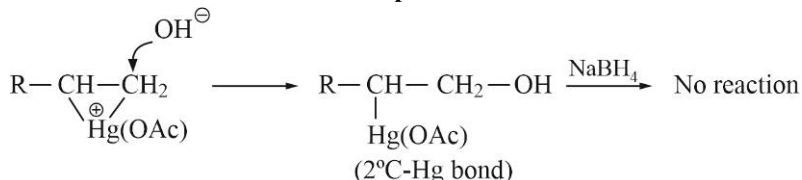


- No stereoselectivity, because carbocation is planar, and attack of nucleophile on carbocation can be from both above the plane as well as below the plane.

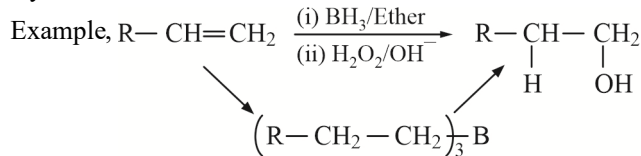
**So,**  $A \rightarrow p, s$

**(B) Oxymercuration-Demercuration of Alkenes**

- The reaction is also Regioselective, because only  $1^\circ C-Hg$  bond can be reduced by  $NaBH_4$ . So, Markovnikov Addition takes place.



So, B  $\rightarrow$  p, r, s

**(C) Hydroboration-Oxidation of Alkenes**

- The reaction is Regioselective, because Boron always attached to sterically less hindered carbon. So, Anti-Markovnikov Addition takes place.
- The reaction is stereoselective, because it proceed through cyclic transition state [So, syn Addition]

So, C  $\rightarrow$  q, r, s

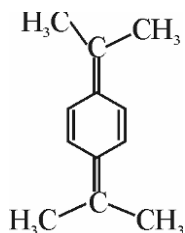
**(D) Dehydration of Alcohols to Alkenes**

For  $2^\circ$  and  $3^\circ$  Alcohols, elimination in acidic medium takes place through E1 mechanism, where carbocation may undergo rearrangement, and a Saytzeff product is always formed. So, regioselectivity is there.

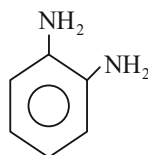
For  $1^\circ$  alcohols, elimination in acidic medium takes place through E2 mechanism, where principle of anti-peri planarity is followed i.e. Anti-elimination. So, stereoselectivity is there.

So, D  $\rightarrow$  r, s

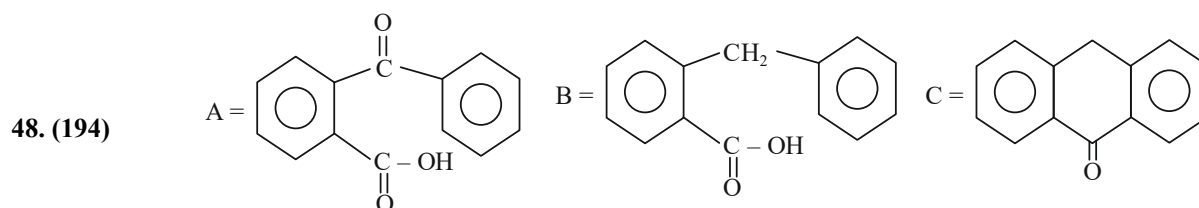
46.(4) Compound A is



47. (4) Product (P) is

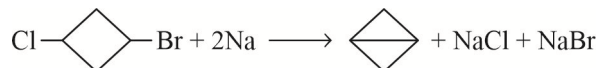


$$\text{Molecular weight} = M = 108 \quad \therefore \frac{M}{27} = \frac{108}{27} = 4$$



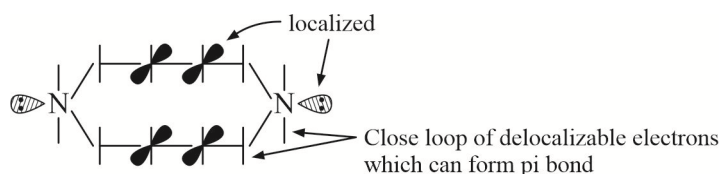
Molecular weight = M = 194

49. (5) It is intramolecular wurtz reaction.

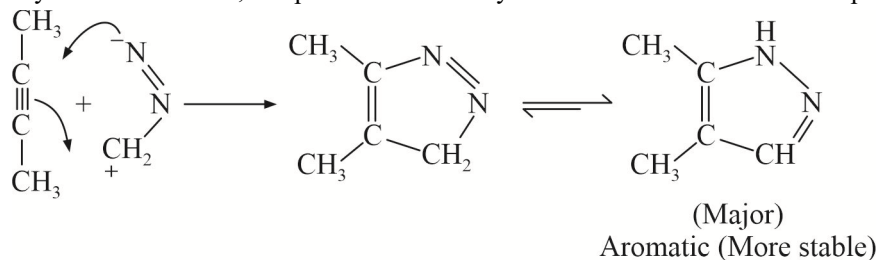


50. (10) Given compound obey Huckel's rule hence it possess a close loop of ten delocalizable pi electrons.

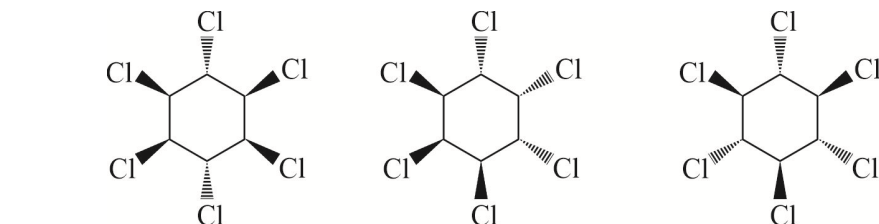
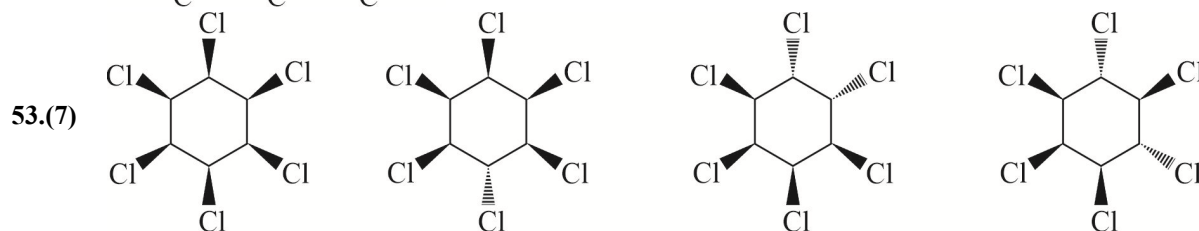
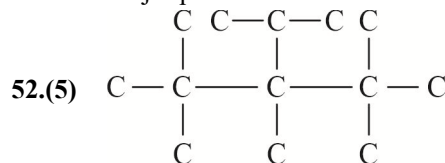
$$x\pi = 10\pi \Rightarrow x = 10$$



51. (6) Diazomethylene also acts as 1, 3-dipolar ion and it may add across carbon-carbon multiple bond.



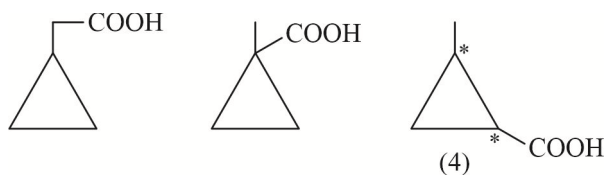
Major product is aromatic and it possess six delocalizable pi electrons.



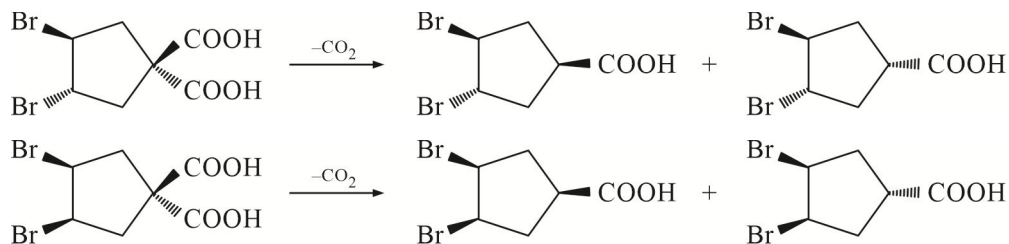
54. (6) (2, 3, 4, 5, 6, 8)



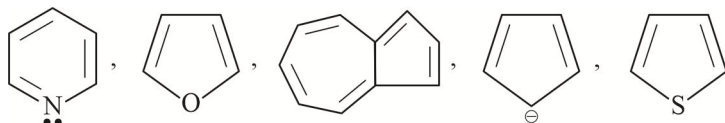
55.(6)



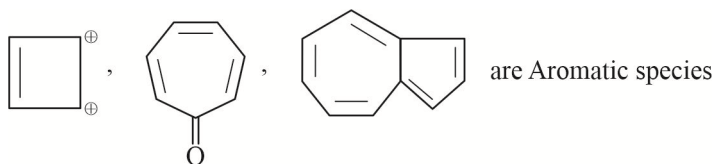
56.(4)



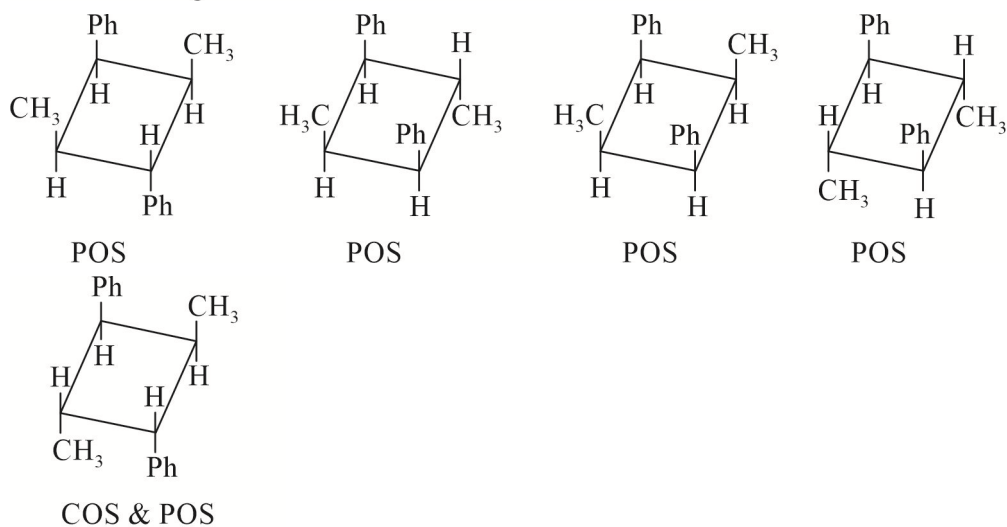
57.(5)



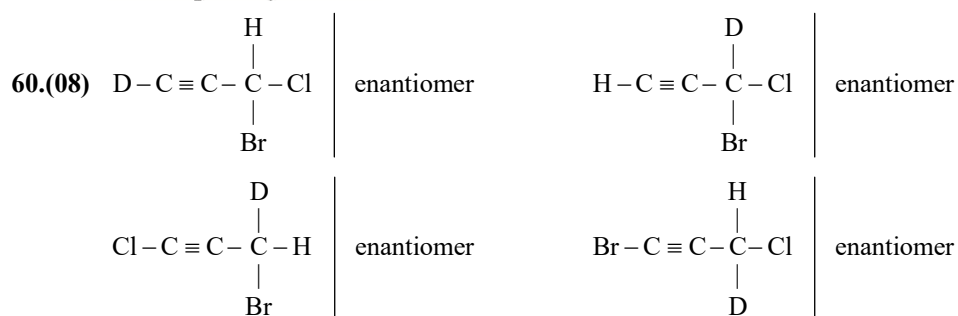
58.(3)

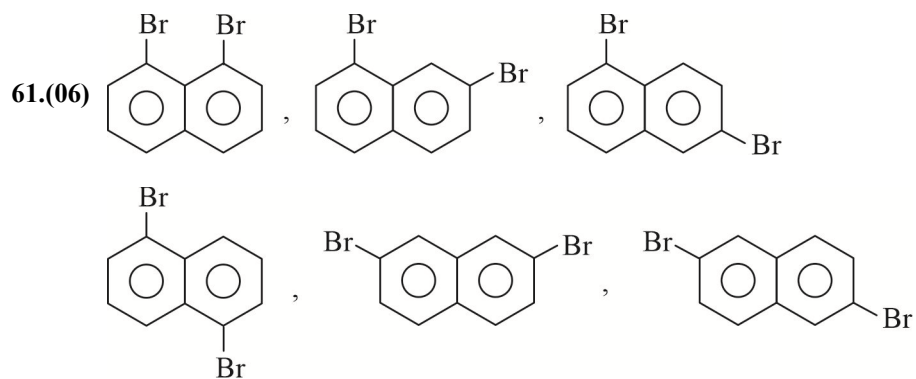


59.(05)



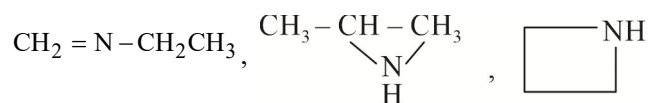
05 all optically inactive





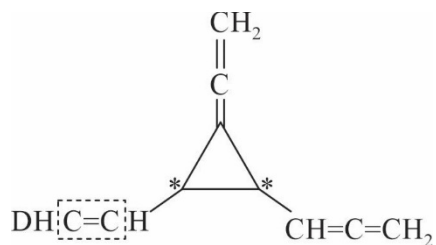
62.(5) (iii), (iv), (vi), (vii), (viii)

63.(3)  $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{NH}$ ,  $\text{CH}_3 - \text{CH} = \text{CH} - \text{NH}_2$ ,  $\text{CH}_3 - \text{CH} = \text{N} - \text{CH}_3$



64.(8) The molecule contains 3 stereocenters

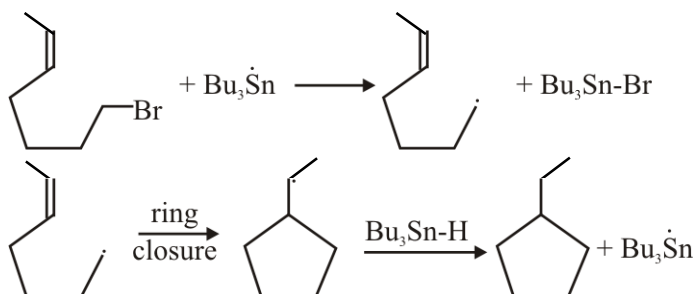
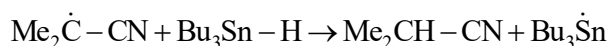
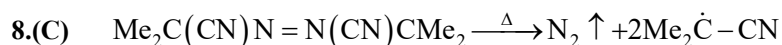
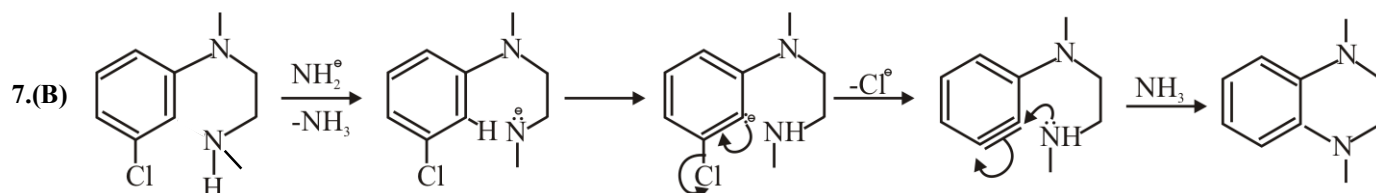
$\therefore$  Total stereoisomers =  $2^3 = 8$



## Solutions - Advanced Problem Package | Chemistry

## Organic Halides &amp; Organic Concepts

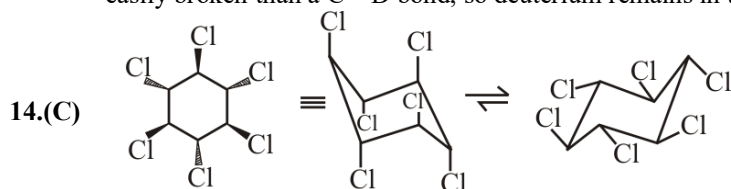
- 1.(D) 2° allylic halides undergo S<sub>N</sub>1 reaction faster than simple 2° halides. 3° bridged halides undergo S<sub>N</sub>1 reaction too slowly because of difficulty in forming a planar sp<sup>2</sup> hybridized carbon at bridge head.
- 2.(B) Cl atom is substituted by PhS<sup>-</sup> and inversion of configuration takes place as reaction is S<sub>N</sub>2.
- 3.(A) The unhybridized p-orbital in the S<sub>N</sub>2 transition state will be capable of delocalization of e<sup>-</sup> density into the benzene ring and then to the carbonyl group. 1° benzyl halide would react faster than a 2° benzyl halide.
- 4.(B) 2° benzyl halide which gives the most stable carbocation after loss of leaving group, will undergo S<sub>N</sub>1 reaction fastest.
- 5.(C) In presence of a strong base like NaOEt in EtOH and heating, the elimination would be preferred by E2 route as the substrate is 2°.
- 6.(B) Rate of those S<sub>N</sub>2 reactions can be increased by increasing the polarity of the solvent when the reactants are uncharged.



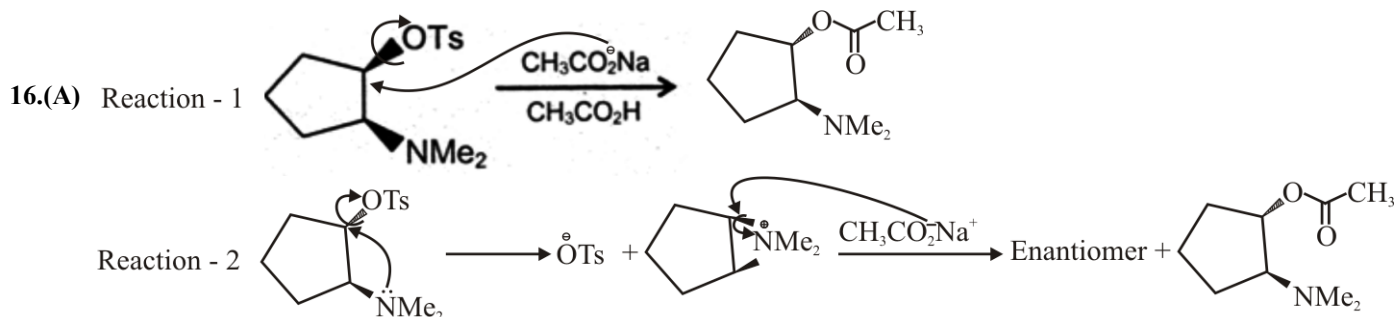
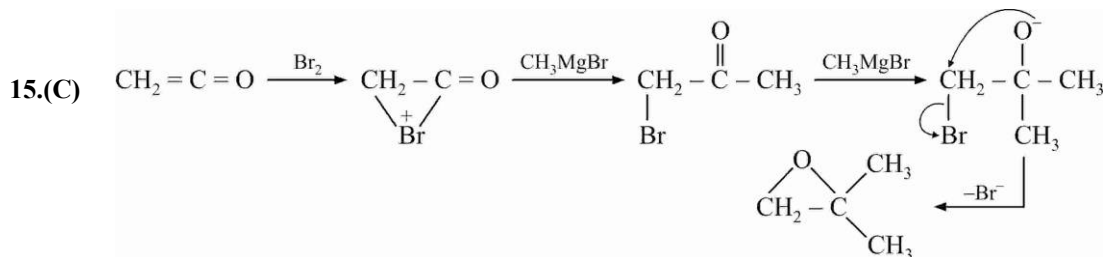
- 9.(B) Since H<sub>2</sub>O is a far better ion-solvating agent than ethanol, so H<sub>2</sub>O would be the preferred nucleophile to give product as alcohol.
- 10.(D)

- 11.(B) α - Halo ketones are more reactive than simple 2° halides towards S<sub>N</sub>2 attack.
- 12.(B) In presence of dilute OH<sup>-</sup>, the reaction proceeds via neighbouring group participation to form product with retention of configuration.

- 13.(A) A 1° halide cannot undergo elimination by E<sub>1</sub> pathway, so E<sub>2</sub> elimination route is preferred. C – H bond is more easily broken than a C – D bond, so deuterium remains in the product.

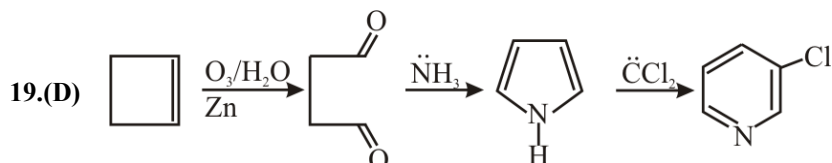


Since, H and Cl are not at anti positions, so elimination by E<sub>2</sub> route is least favoured.



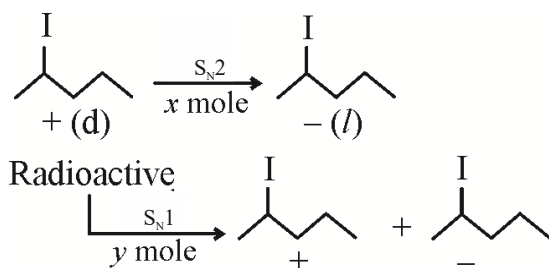
- 17.(D) Br and H atoms are in anti-position in compound 3 to undergo E<sub>2</sub> elimination giving Saytzeff's product. In compound 1, it undergoes elimination to give Hofmann product. ∴ Order of reactivity is 3 > 1 > 2.

- 18.(C) Hofmann's product can be obtained by bulky and sterically crowded bases.



- 20.(B) α - hydrogen of carbonyl compounds is more acidic than β - hydrogen, so elimination becomes rapid.

- 21.(D)



Loss in Radio activity = x

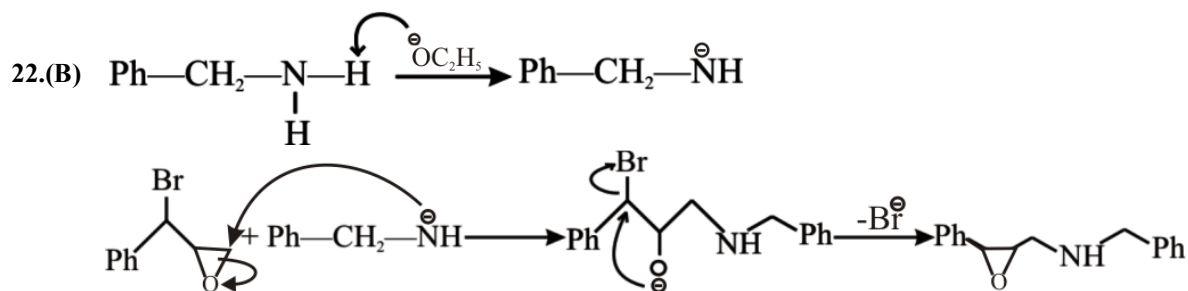
Loss in Radio activity = y

Loss in optical activity = 2x

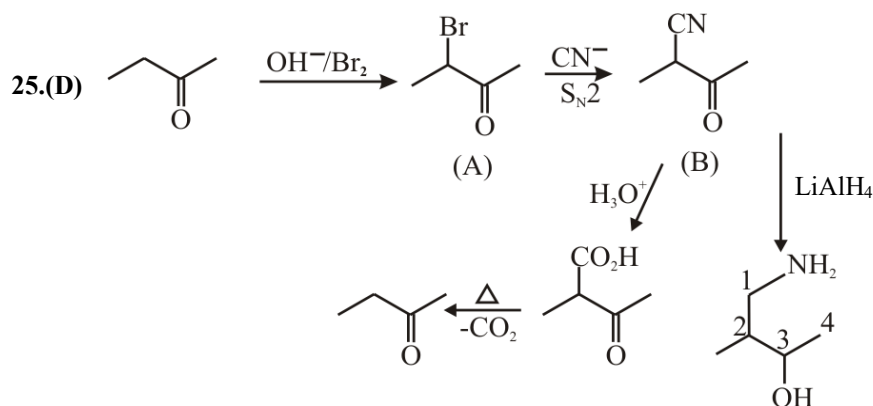
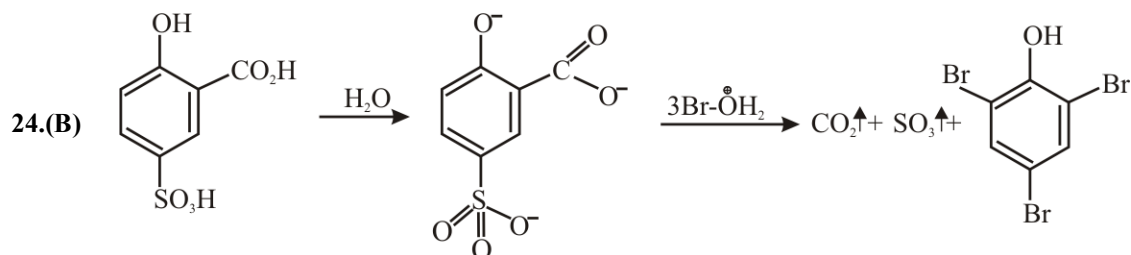
Loss in optical activity = y

Total loss in Radio activity = x + y

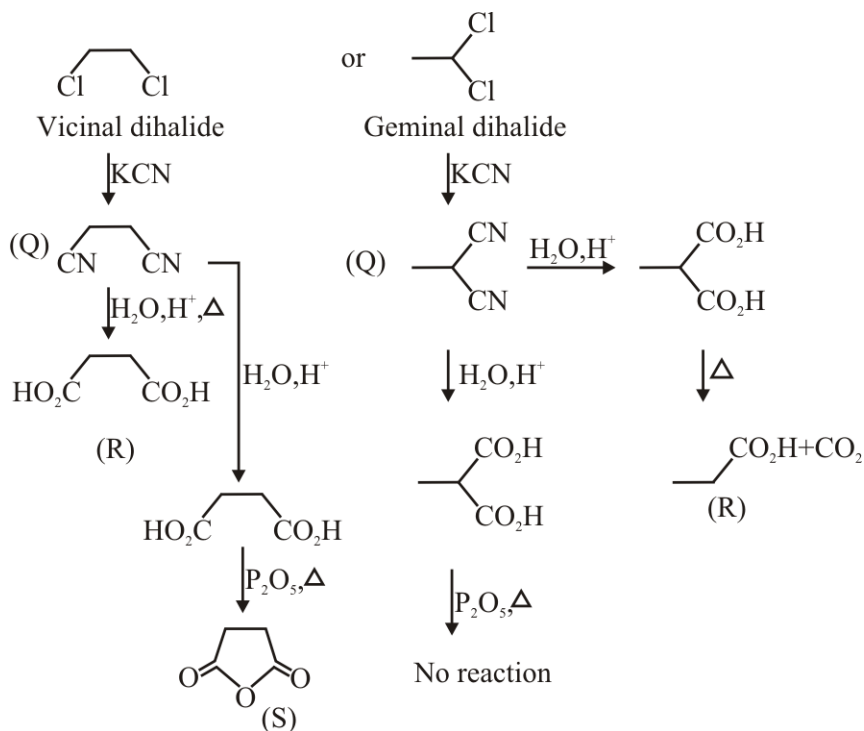
$$(2x + y) = 1.80(x + y) = \frac{x}{y} = 4 \quad ; \quad \% S_N2 = \frac{x}{x + y} \times 100 = \frac{4}{5} \times 100 = 80\%$$



23.(A) Compound (P) has H and Br in axial trans locations while compound (Q) has no such arrangement.



26.(ACD) (P) can be



27.(ABC)  $\text{Ag}^+$  takes out the halide ion and speeds up the formation of carbocation.  $\alpha$  - haloketones undergo reaction by  $\text{S}_{\text{N}}2$  mechanism only but as carbocation is destabilized by electron withdrawing carbonyl group, it does not react by  $\text{S}_{\text{N}}1$  mechanism. Acidic solution having  $\text{H}^+$  helps in abstracting  $\text{F}^-$  ion from t-BuF, so its reaction becomes faster in acidic solution than in  $\text{H}_2\text{O}$ .  $\text{S}_{\text{N}}2$  reaction on allylic substrates may undergo rearrangements.

28.(BC) The substrate is a  $2^\circ$  halide, so it can react by  $\text{S}_{\text{N}}1$  and  $\text{S}_{\text{N}}2$  pathways both.  $\text{SCN}^-$  is an ambidentate nucleophile with S atom as soft base and N atom as hard base. Hard base undergoes reaction via  $\text{S}_{\text{N}}1$  pathway as carbocation is a hard acid while soft bases prefer to go via  $\text{S}_{\text{N}}2$  route.

29.(AC) Rate of Reaction = Rate due to  $\text{S}_{\text{N}}2$  + Rate due to  $\text{S}_{\text{N}}1$

$$= 3.2 \times 10^{-5} [2\text{-Bromobutane}] + 1.5 \times 10^{-6} [2\text{-bromobutane}] = [2\text{-bromobutane}] 3.35 \times 10^{-5}$$

$$\therefore \% \text{ of reaction occurring by } \text{S}_{\text{N}}1 \text{ mechanism} = \frac{1.5 \times 10^{-6} [2\text{-Bromobutane}]}{3.35 \times 10^{-5} [2\text{-bromobutane}]} \times 100 = 4.4\%$$

$$\therefore \% \text{ of reaction occurring by } \text{S}_{\text{N}}2 \text{ mechanism} = 95.4\%$$

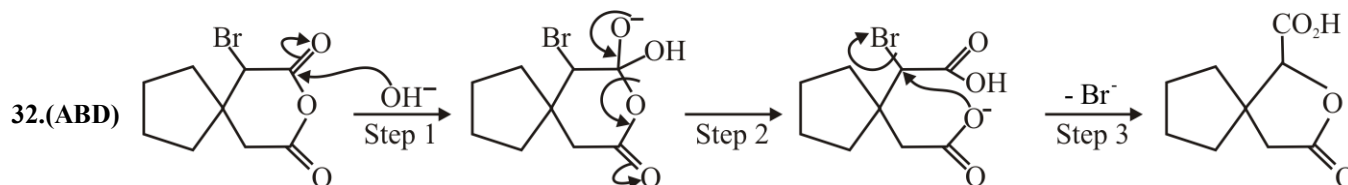
$$\% \text{ of racemization in the reaction} = 4.4\%$$

$$\% \text{ of inversion product in the reaction} = 95.4 + 2.2 = 97.6\%.$$

30.(ABCD) Peroxide effect is shown by HBr only and other hydrogen halides even in the presence of peroxides follow polar mechanism.

31.(CD) Alcoholic  $\text{AgNO}_3$  will give immediate white ppt. of  $\text{AgCl}$  with  $\text{CH}_2 = \text{CH} - \text{CH}_2\text{Cl}$  while  $\text{CH}_3\text{CH}_2\text{CH}_2 - \text{Cl}$  will give only on warming.

$\text{CH}_2 = \text{CH} - \text{CH}_2\text{Cl}$  will decolorize cold aq.  $\text{KMnO}_4$  while there is no decolorization by  $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ .

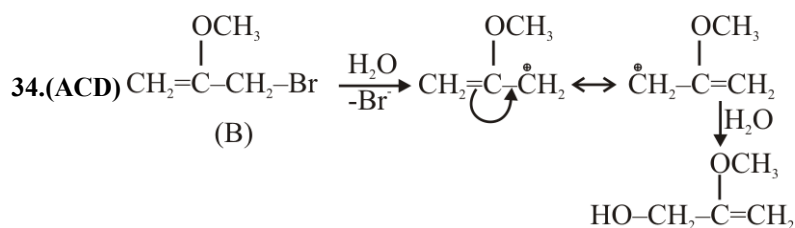
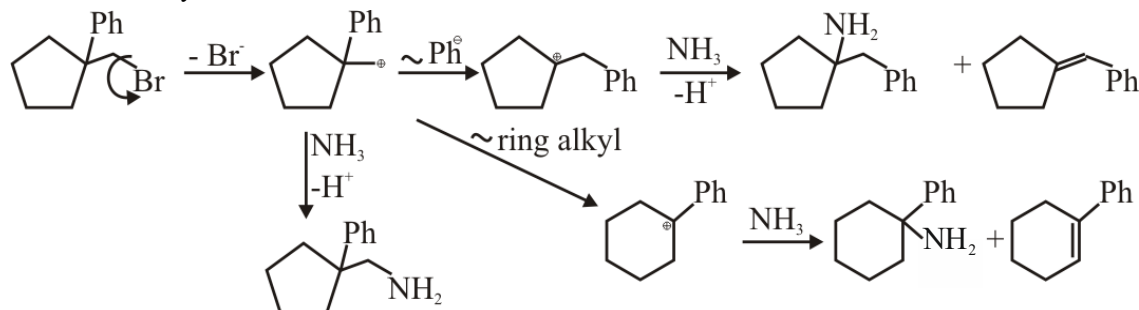


Step 1 and Step 2 involves nucleophilic

addition on acyl carbon followed by elimination of  $\text{CO}_2^-$  group but overall reaction is referred as nucleophilic substitution at acyl carbon.

Step 3 involves nucleophilic substitution at alkyl carbon.

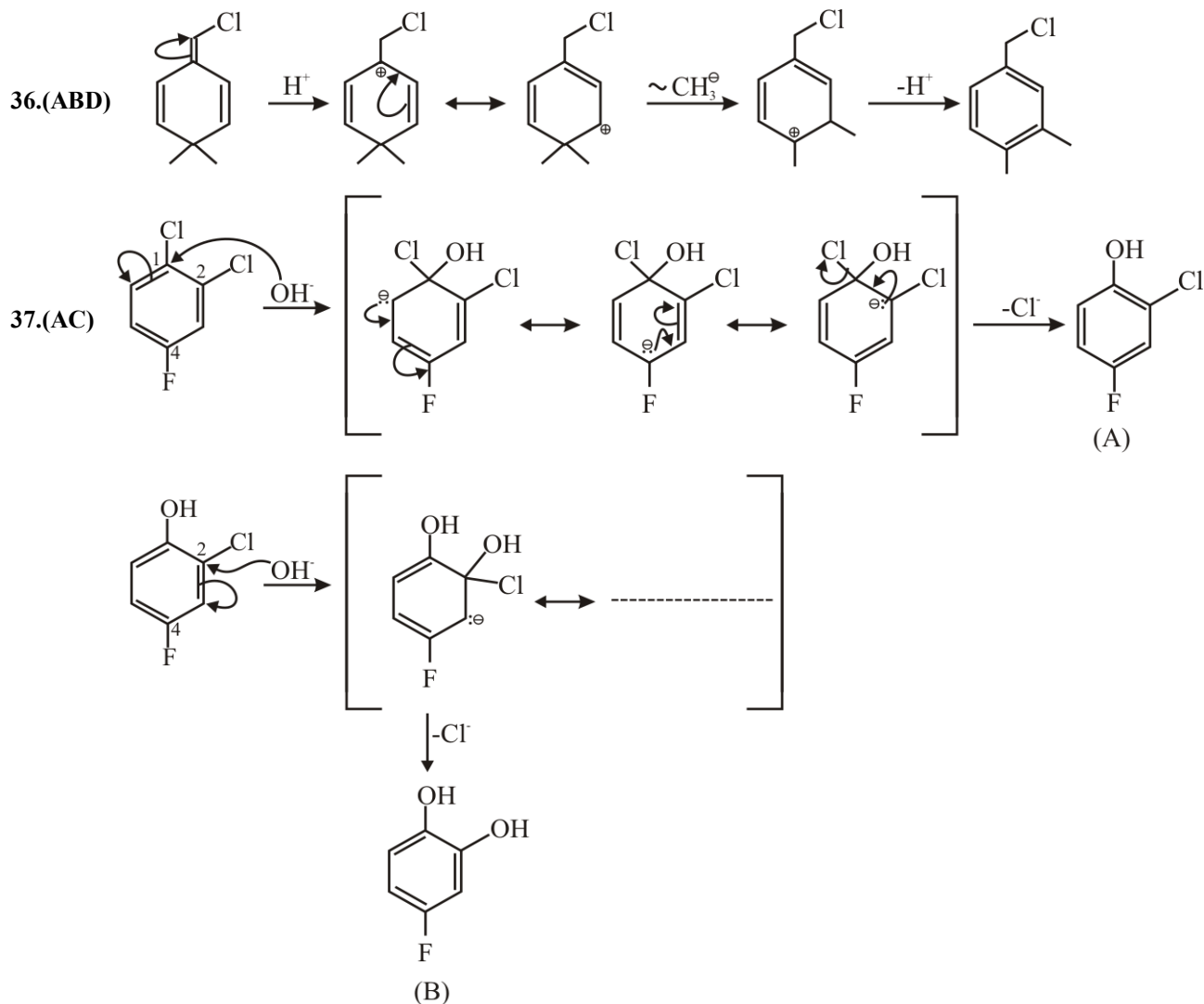
33.(ABD) As the substrate has 2  $\beta$ -substitutes, so it can not react by  $\text{S}_{\text{N}}2$  mode and  $\text{NH}_3$  being a polar solvent, the reaction occurs by  $\text{S}_{\text{N}}1$  mode.



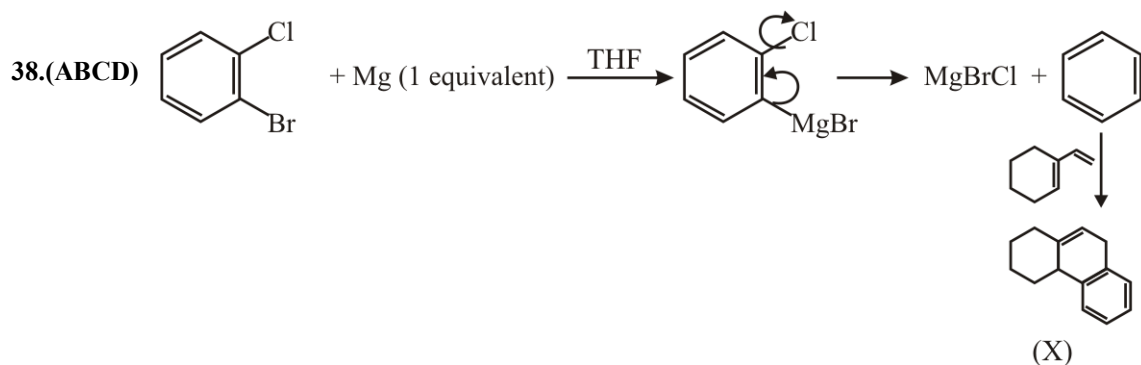
-I effect of  $\text{OCH}_3$  will disperse the  $e^-$  density in the transition state of compound (B) more than that of compound (A) in  $\text{S}_{\text{N}}2$  reaction.

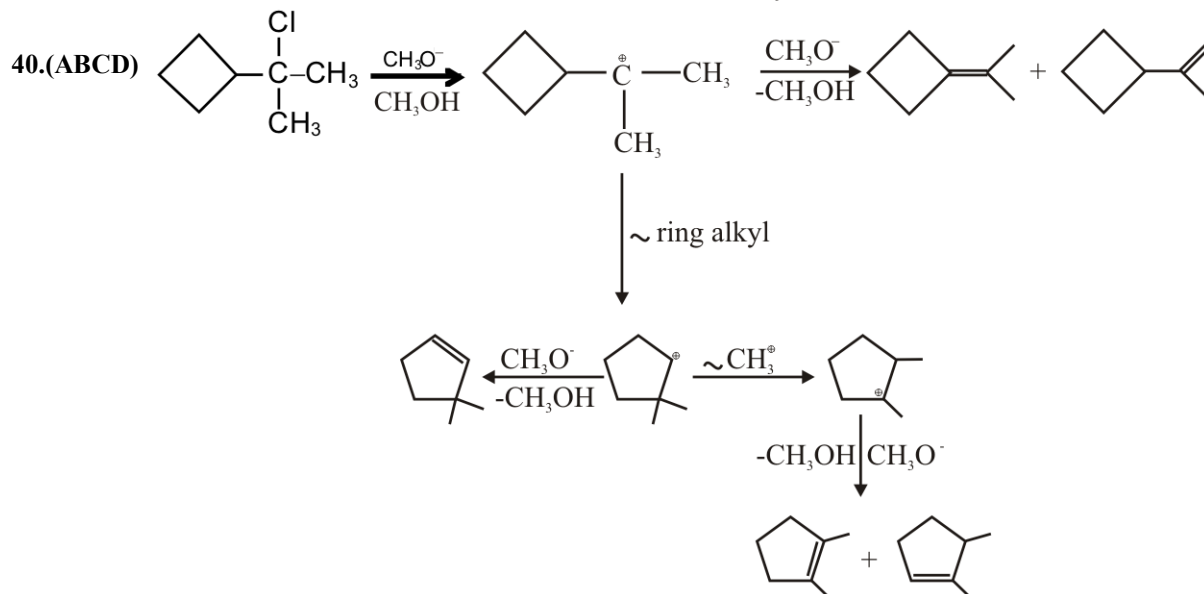
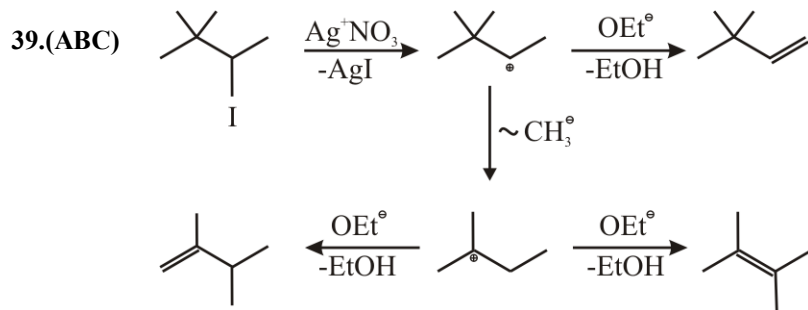
Compound (A) gives rearranged and non-rearranged product when reacted by via  $\text{S}_{\text{N}}1$  as well as  $\text{S}_{\text{N}}2$  pathways.

35.(ABC)  $\text{CH}_3\text{CH}_2\text{O}^-$  is a strong nucleophile, so it prefers  $\text{S}_{\text{N}}2$  attack in reaction (A).  $\text{NH}_3$  is a strong nucleophile, so it reacts by  $\text{S}_{\text{N}}2$  mode in reaction (B).  $\text{CH}_3\text{O}^-$  is a strong nucleophile, so undergoes  $\text{S}_{\text{N}}2$  reaction in (C).  $\text{CH}_3\text{OH}$  is a weak nucleophile.

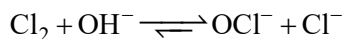
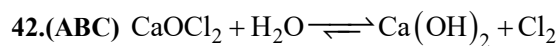
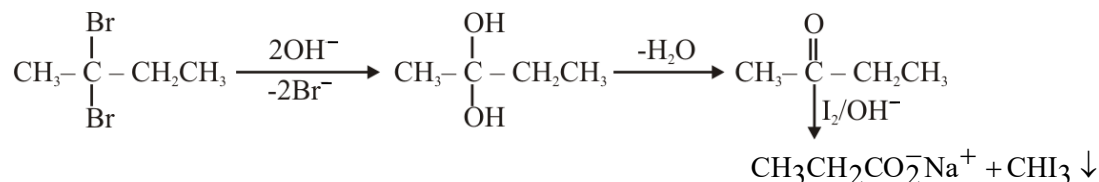


-I effect of OH group in (A) makes Cl at 2 more electrophilic than F at position 4.





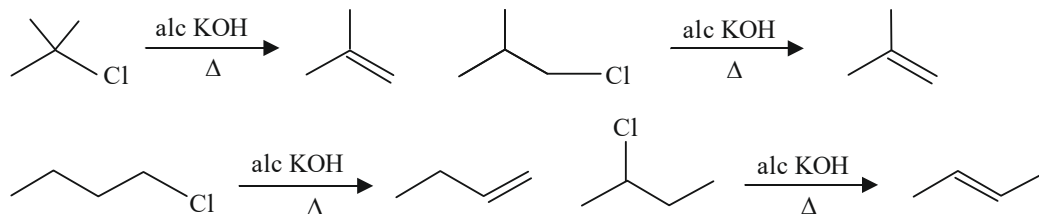
41.(ABD) Compounds containing  $\text{CH}_3-\text{C}(=\text{O})-$  unit or  $\text{CH}_3-\text{CH}(\text{OH})-$  unit or any compound that can be converted into these units under the reaction condition gives iodoform test.



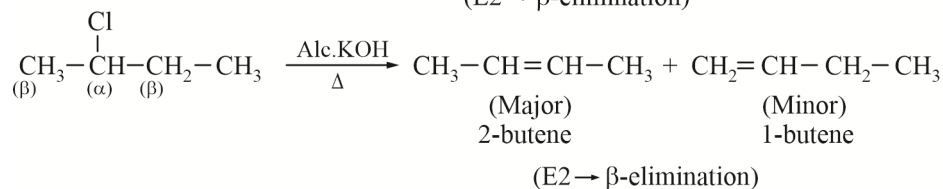
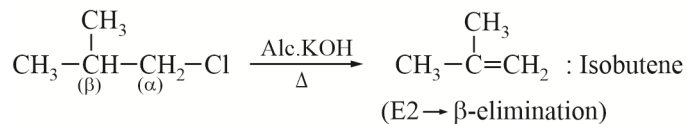
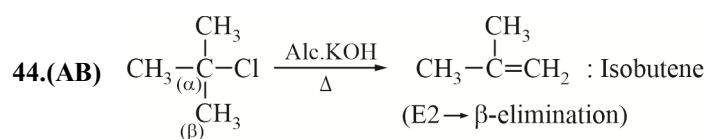
Bleaching powder provides  $\text{Cl}_2$  (Chlorinating agent),  $\text{OCl}^-$  (oxidizing agent) and  $\text{OH}^-$  (hydrolyzing agent).

43.(ABD) With  $\text{AgCN}$ , the major product is alkyl isocyanide.

With  $\text{NaCN}$ , the major product is alkyl cyanide. With  $\text{NaNO}_2$ , the major product is alkyl nitrite and with  $\text{AgNO}_2$ , it is nitro alkane.



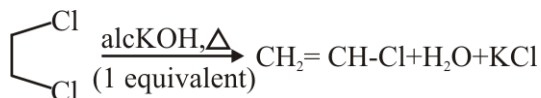
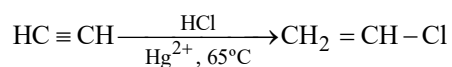
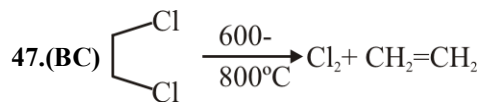
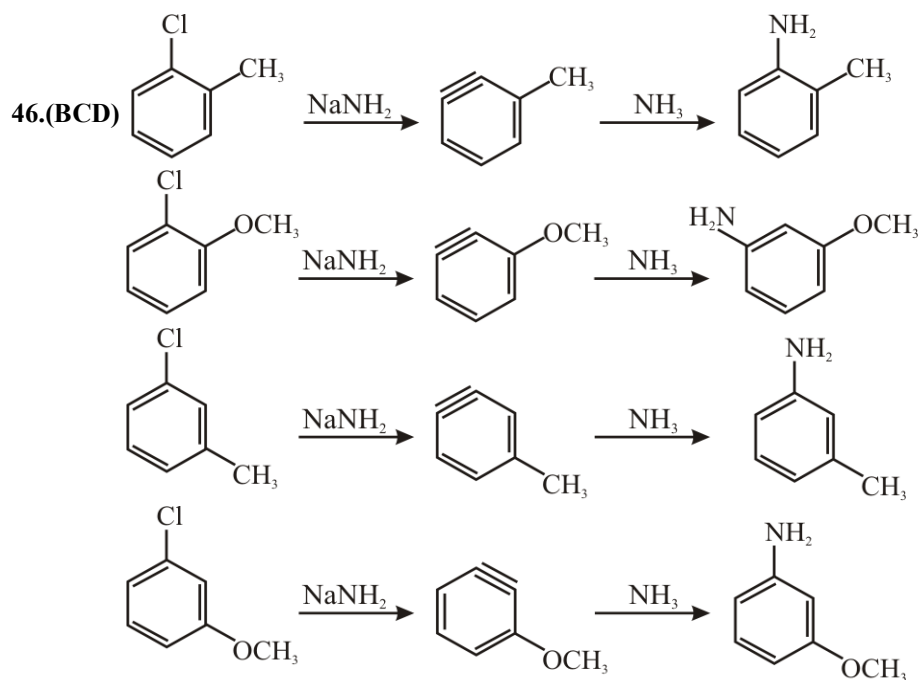


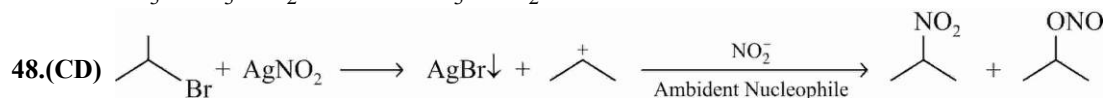
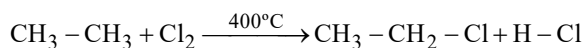


45.(BC) Reactivity order (a)  $1^\circ > 2^\circ > 3^\circ$  R in  $\text{RCO}_2\text{Ag}$ .

(b)  $\text{Br}_2 > \text{Cl}_2$

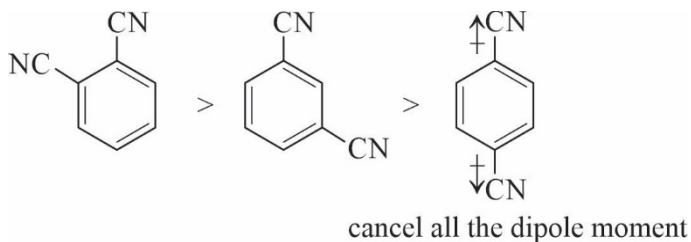
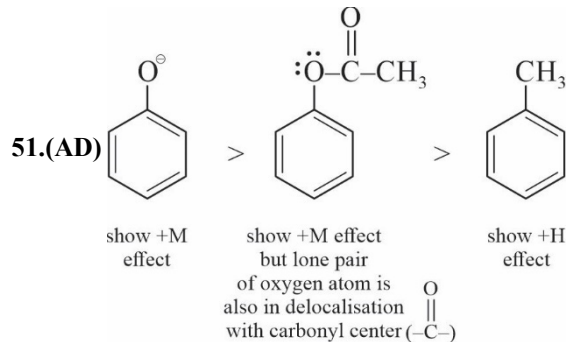
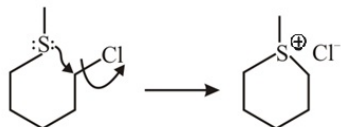
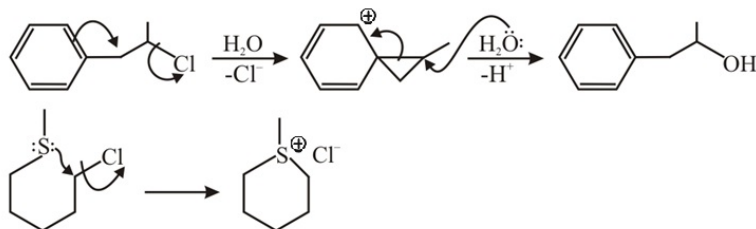
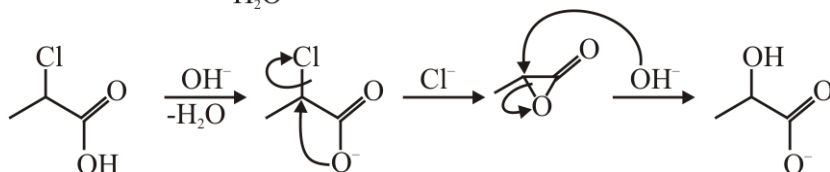
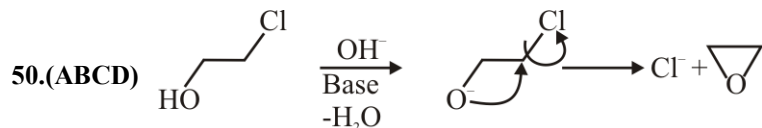
In case  $\text{X} = \text{I}$  Simmonini reaction happens :  $2\text{R}-\text{COOAg} \xrightarrow[\text{CCl}_4]{\text{I}_2} \text{RCOOR} + 2\text{AgI} + \text{CO}_2$





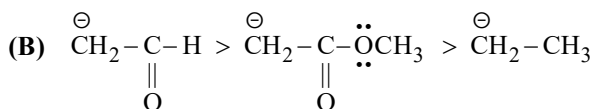
49.(AB) Rate of  $\text{S}_{\text{N}}1$  reaction

- (a)  $\text{R} - \text{I} > \text{R} - \text{Br} > \text{R} - \text{Cl} > \text{R} - \text{F}$   
 (b)  $\text{H}_2\text{O} > \text{CH}_3\text{OH} > \text{CH}_3\text{SH}$  (decreasing polarity of solvent)  
 (c) Greater the steric crowding in the substrate, greater is the rate (Steric acceleration)



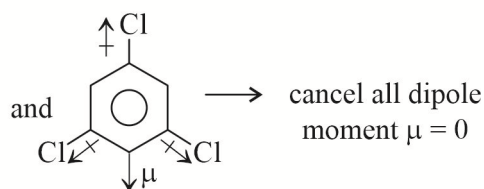
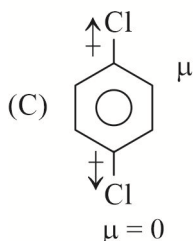
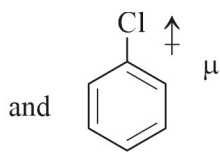
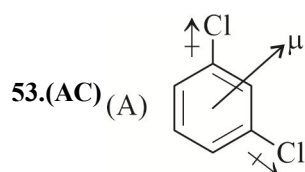
52.(ABCD)

(A)  $\text{II} > \text{III} > \text{I} \rightarrow$  More stronger is the  $\bar{\text{e}}$  withdrawing group, more stable is the  $-\text{ve}$  charge.

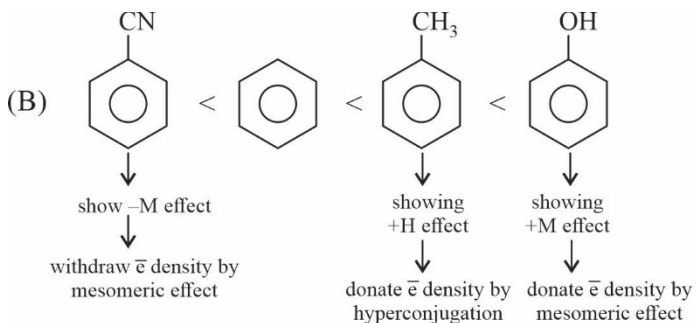
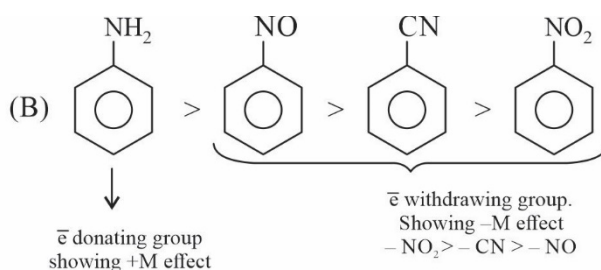
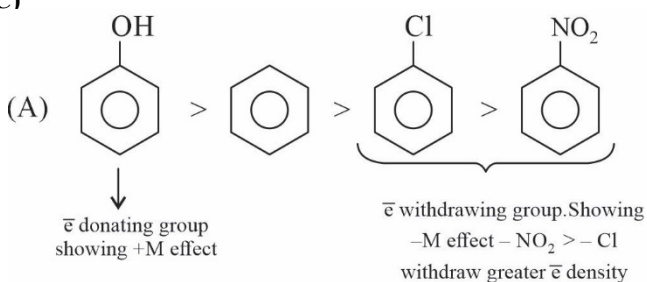


(C)  $\text{I} > \text{II} > \text{III} \rightarrow$  More is the conjugation, more is the stability of  $-\text{ve}$  charge

(D)  $\ominus\text{CH}_3 > \ominus\text{CD}_3$  More +I effect of D destabilize carbanion.

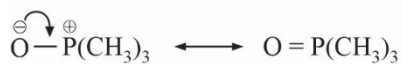


54.(ABC)



→ Mesomeric effect is stronger than Hyperconjugation

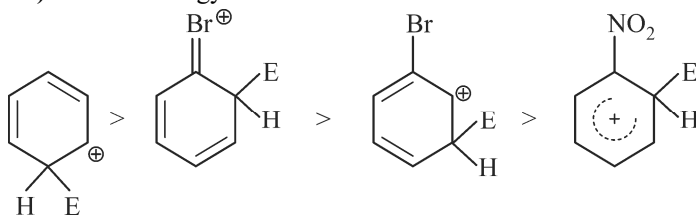
55.(BCD)  $\ominus\text{O} \Rightarrow \text{P}(\text{CH}_3)_2$  donating of -ve charge into vacant orbital of phosphorus



$\ominus\text{O} \Rightarrow \text{B}(\text{CH}_3)_2$  donating of -ve charge into vacant orbital of Boron

56. (BD) It is free radical substitution reaction. Intermediate of this reaction is free radical. It is planar intermediate having unpaired electron in p-orbital.

57. (ABCD) Order of energy of intermediate arenium ion is:



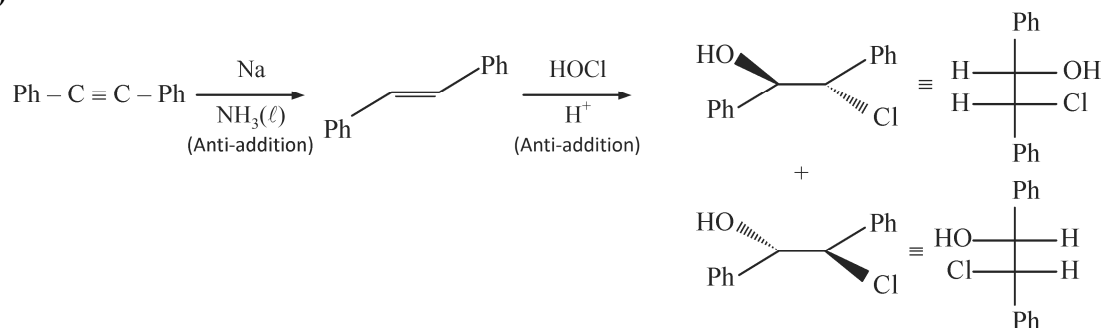
Br and NO<sub>2</sub> are electron withdrawing groups and presence of electron withdrawing group decreases the stability of arenium ion.

**58.(BD)** Strongly deactivated substrates are not suitable for F.C. acylation. Phenolic compounds and aniline are also not suitable for F.C. reaction because these form complex with Lewis acid which is used as catalyst in F.C. reaction.

**59.(BCD)**

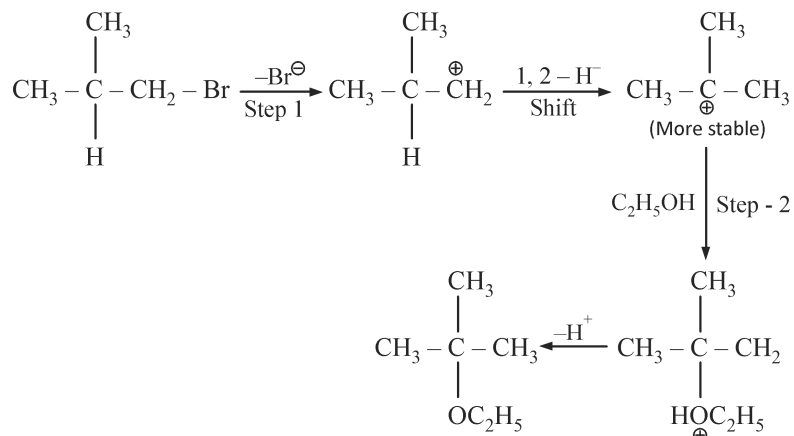
Left hand side phenyl ring of   $\text{O}-\text{CH}_2-\text{Ph}$  and   $\text{NH}-\text{COPh}$  is activated for electrophilic aromatic substitution reaction.

**60.(BC)**



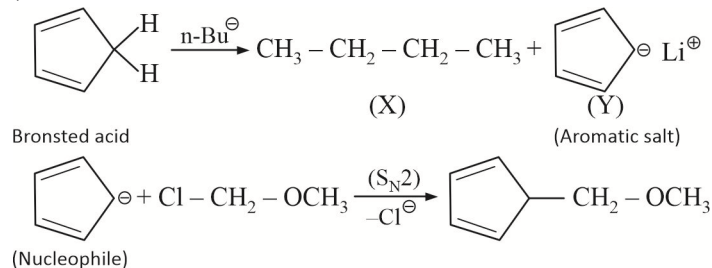
**61.(ABC)**

It is solvolysis reaction and proceed by  $S_N1$  reaction mechanism.



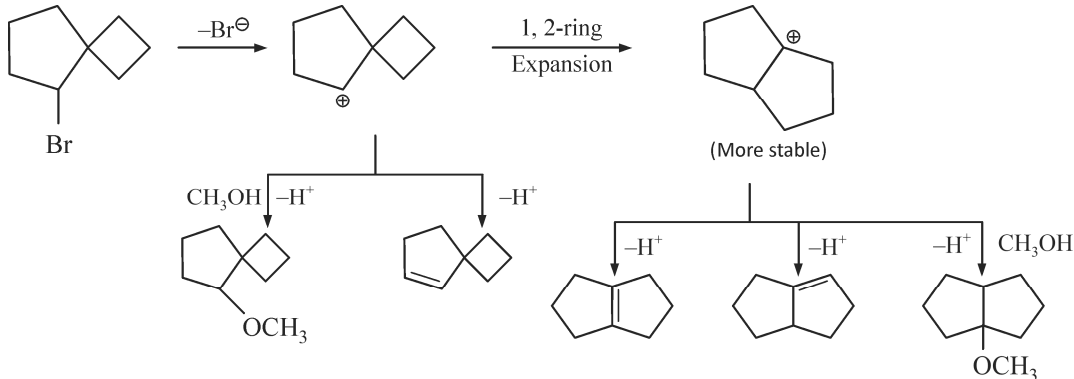
For  $S_N1$ , rate of reaction depends upon conc. of only alkyl halide.

**62.(ABC)**



63.(ABCD)

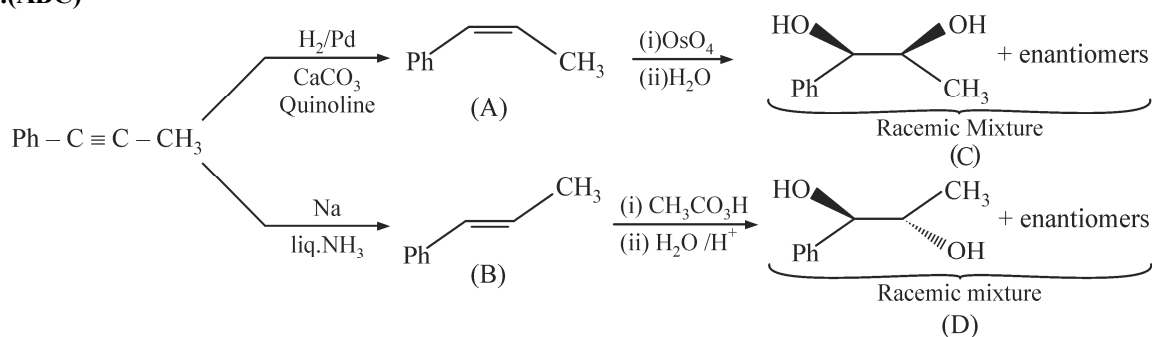
It is solvolysis reaction, takes place by formation of intermediate carbocation. Minor product is also formed through E1 elimination reaction mechanism



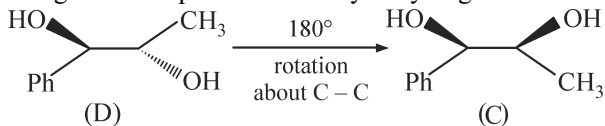
64.(ABCD)

Primary organic halide undergoes nucleophilic substitution by  $S_N2$  reaction mechanism.

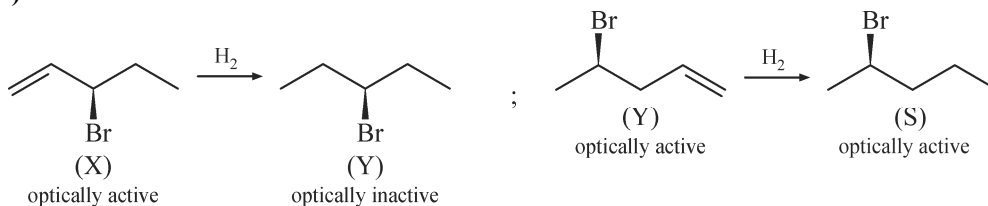
65.(ABC)

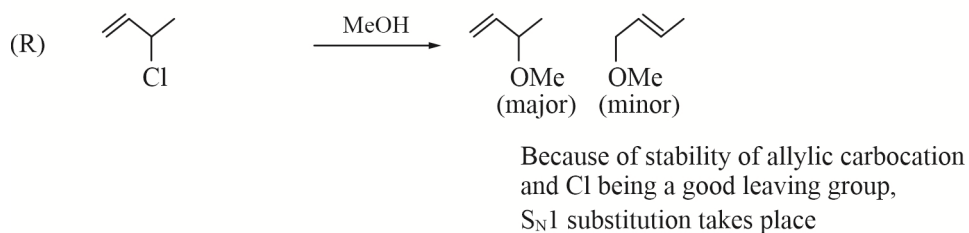
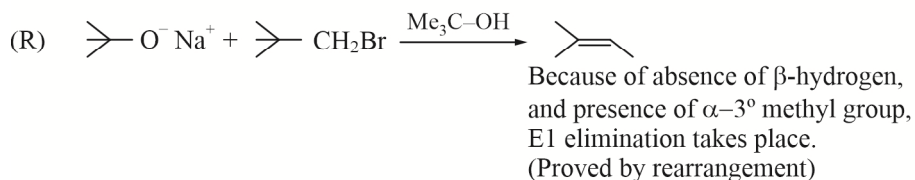
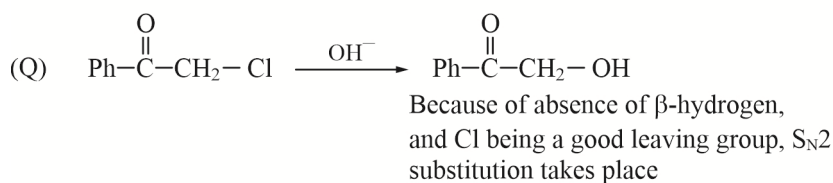
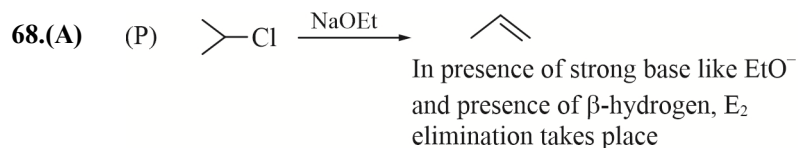
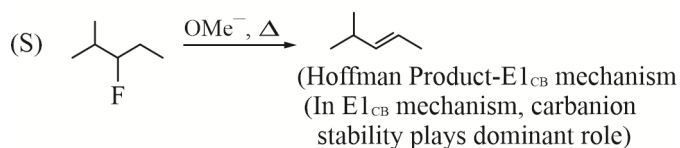
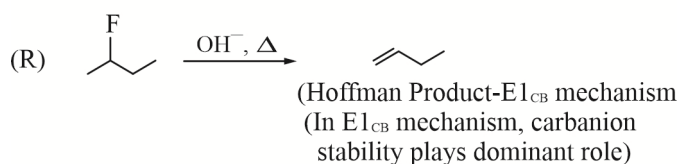
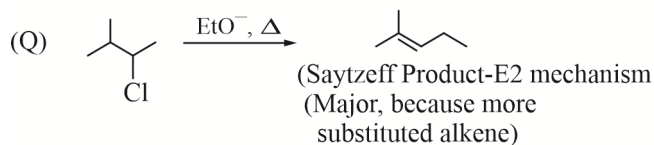
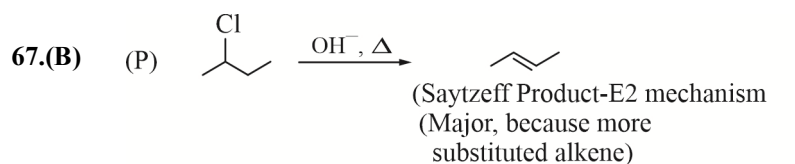


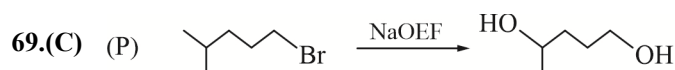
Product (C) and (D) are identical and can't be separated by fractional distillation. (A) and (B) are diastereomers and gives same product on catalytic hydrogenation.



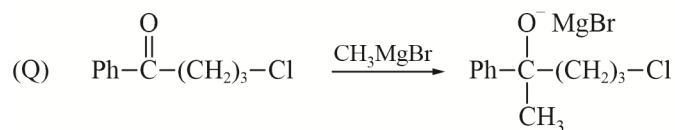
66.(ABD)



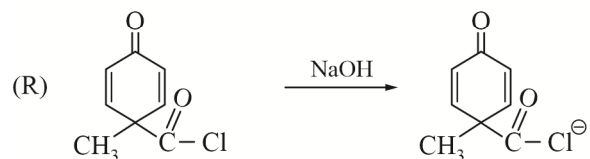




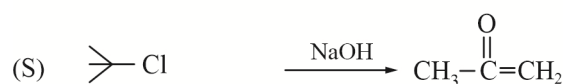
Because of 1°-halide group,  
Nucleophilic substitution through  $S_N2$   
mechanism take place



Nucleophilic addition of  
Grignard reagent across a  
carbonyl group

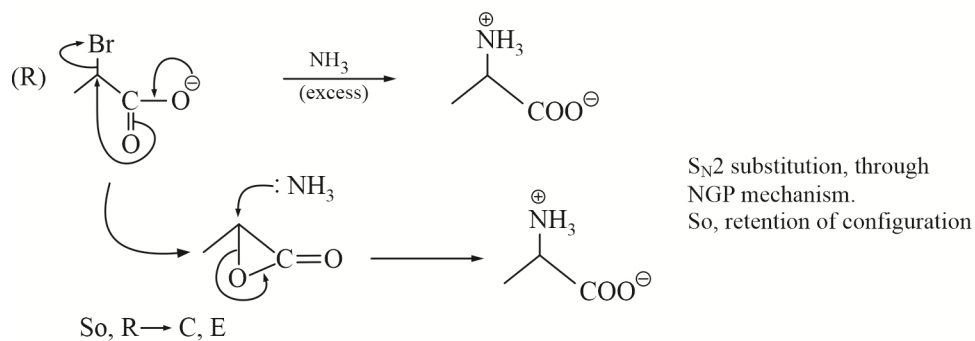
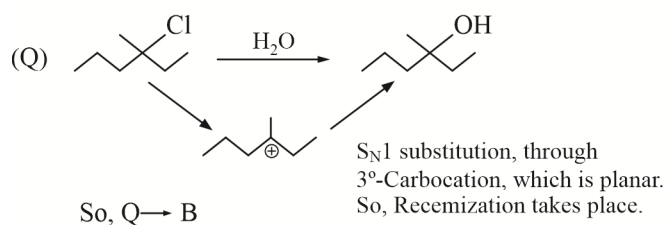
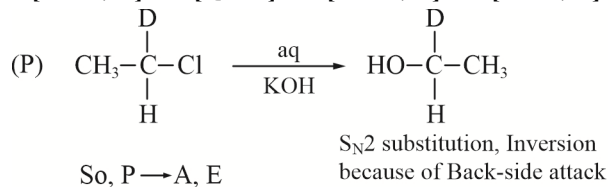


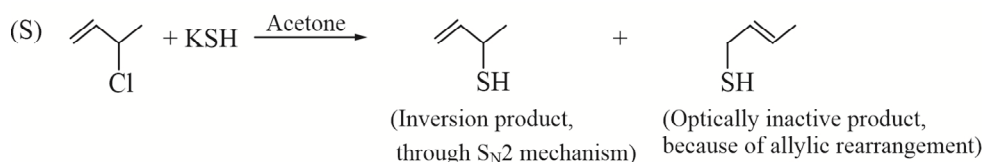
Nucleophilic acyl substitution of  
acid chloride, in presence of base



Because of 3° halide group,  
Cl is a good leaving group,  $E1$   
elimination takes place via a 3°-carbocation

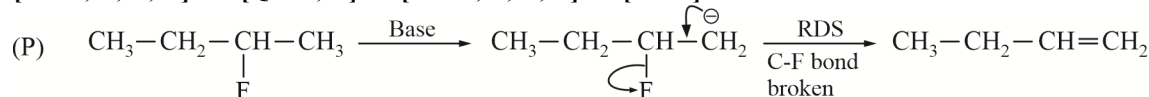
70. [P - A, E]  $\rightarrow$  [Q - B]  $\rightarrow$  [R - C, E]  $\rightarrow$  [S - D, E]



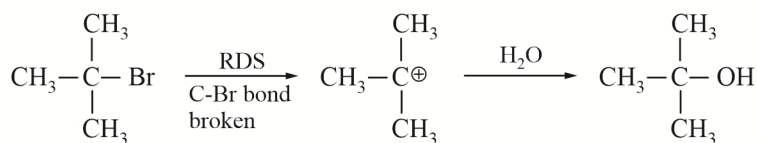


So, S → D, E

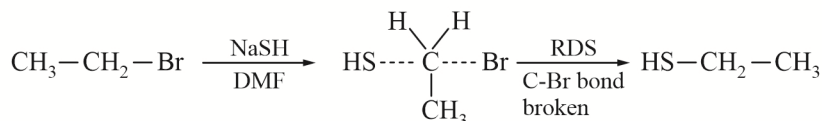
71. [P - A, B, C, D] → [Q - C, D] → [R - A, B, C, E] → [S - A]



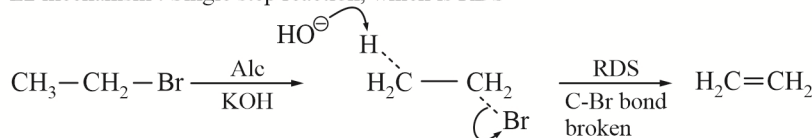
S<sub>N</sub>1 mechanism :



S<sub>N</sub>2 mechanism : Single step reaction, which is RDS



E2 mechanism : Single step reaction, which is RDS



So, P → A, B, C, D

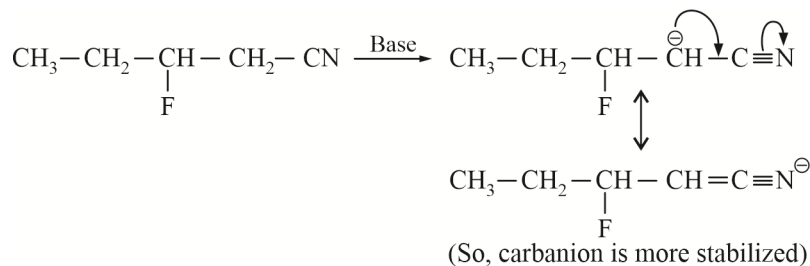
(Q) Due to EWG at β-position, in case of E2 mechanism, β-H becomes more acidic and in case of S<sub>N</sub>2 mechanism, it makes α-C more electrophilic. So, Q → C, D

(R)  $\frac{K_H}{K_D} = 1$ , when C-H bond is not breaking in the RDS step. In such cases, kinetic isotope effect is absent.

In E1<sub>CB</sub> mechanism, S<sub>N</sub>2 mechanism → no. C-H bond breaks in RDS step, but in E2 mechanism α C-H bond breaks in RDS step.

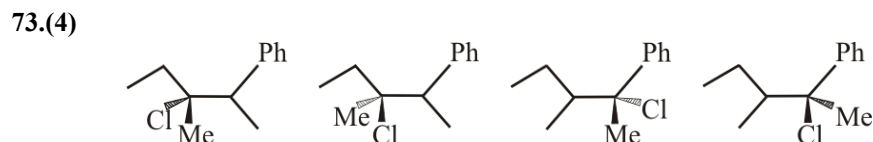
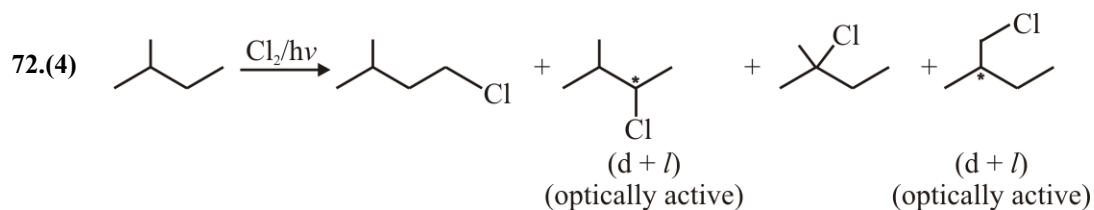
In S<sub>N</sub>1 mechanism no C-H bond breaking in RDS step. So,  $\frac{K_H}{K_D} \approx 1$  So, R → A, B, C, E

(S) In E1<sub>CB</sub> mechanism,



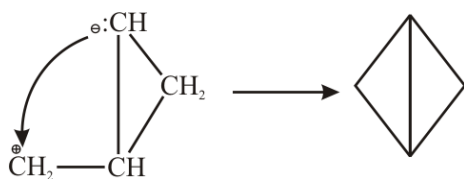
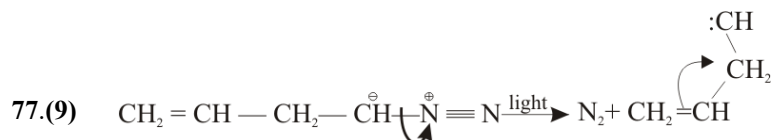
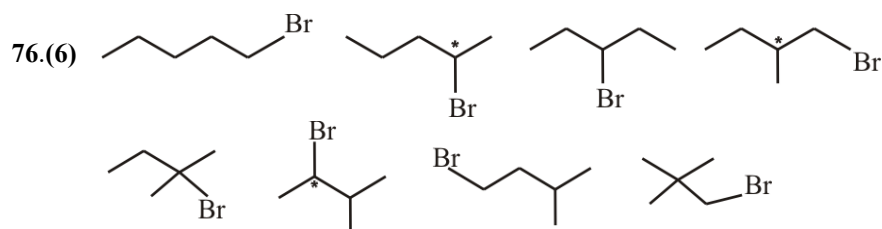
So, formation of carbanion in step - I is highly favored, which favors product formation in E1<sub>CB</sub> mechanism





74.(9) Compounds 1, 2, 5, 6, 8, 9, 10, 11 and 12 will undergo reaction by  $S_N2$  pathway.  
 $1^\circ$  allylic,  $1^\circ$  benzylic halides and  $\alpha$ -halo ketones undergo reaction by  $S_N2$  mode.

75.(6) Aryl halides with more electron withdrawing groups at ortho and para positions are more susceptible to bimolecular  $S_NAr$  reaction. Therefore, compound 6 is most reactive.

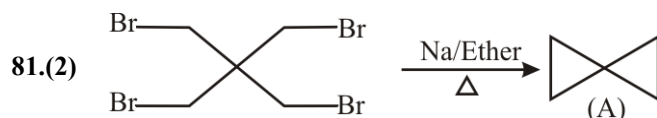
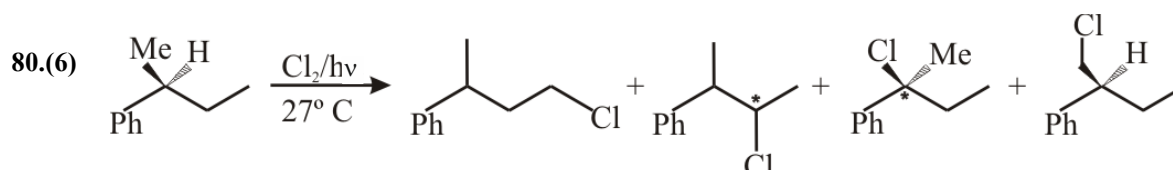


Molar mass = 54

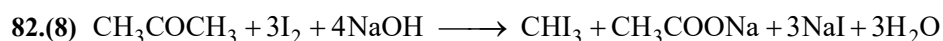
$$\therefore x + y = 5 + 4 = 9$$

78.(6) Compounds 5, 8, 9 form same products by  $S_N1$  and  $S_N2$  mechanism while compound 11 reacts by  $S_N2$  mechanism only, 6, 7 react by  $S_N1$  only.

79.(5) In reaction no. 1, 2, 5, 8, and 10, the rate of reaction increases with increase in concentration of nucleophile.

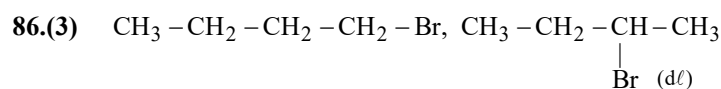
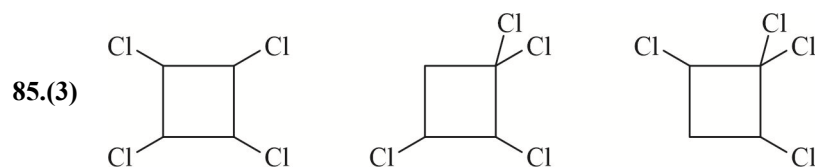
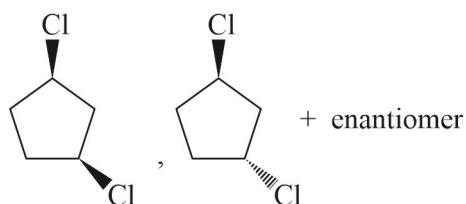
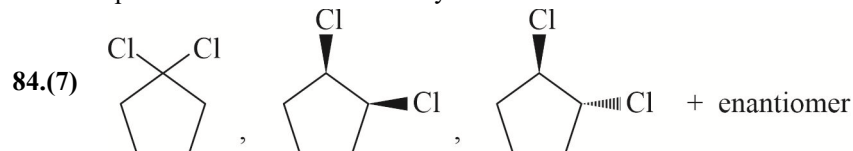


The product has 2 rings, so degree of unsaturation in (A) is 2.

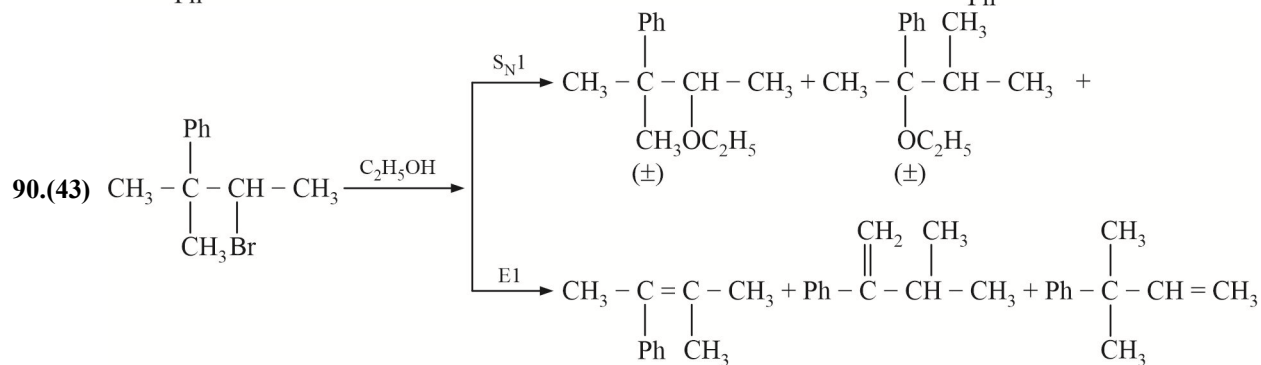
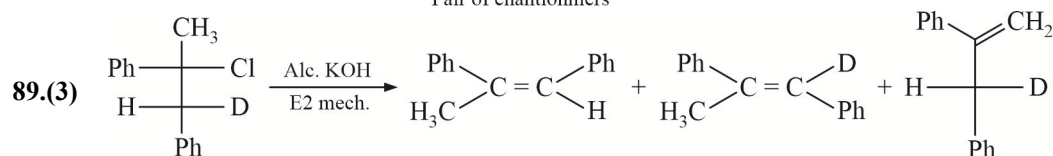
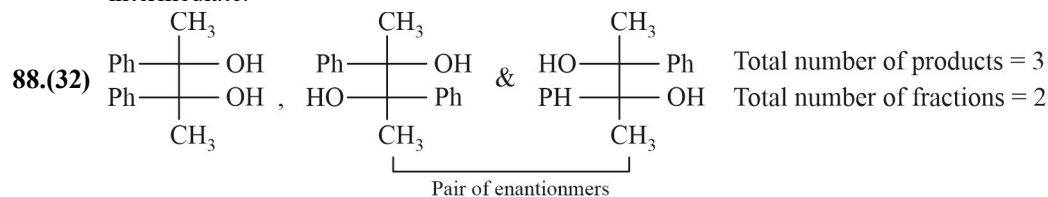




Principal chain of ester contains only one carbon atom.



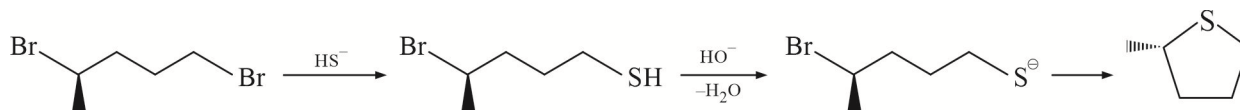
87.(9) Intermediate is tert-butyl carbocation  $(\text{CH}_3)_3\text{C}^+$  and total nine hyperconjugative H-atoms are present in this intermediate.



91.(31)

92.(8)

93.(31.37%)



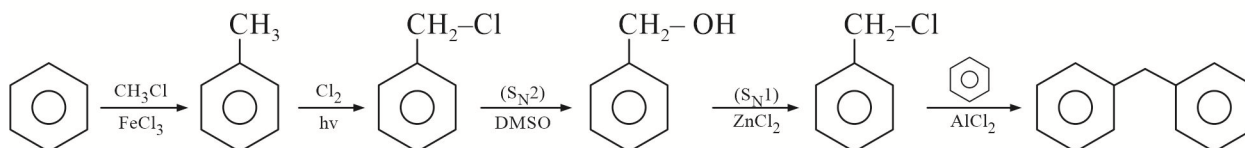
94.(8) All these reactions are correct with respect to product and mechanism of reaction.

95.(4) 1, 4, 7, 9 proceed by  $S_N1$  pathway.

96.(8) All are correct.

97.(8) All are correct.

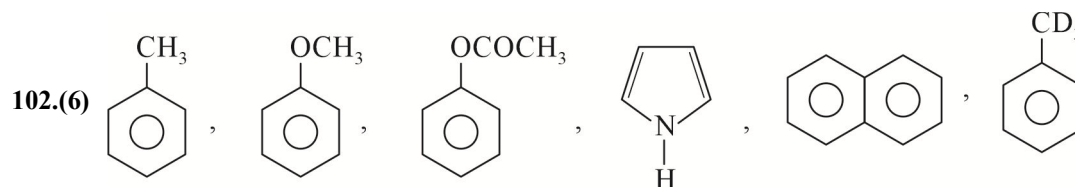
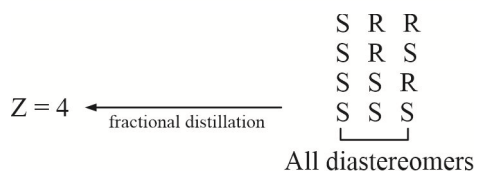
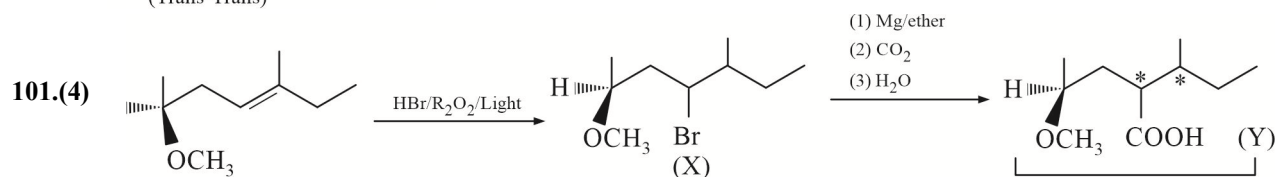
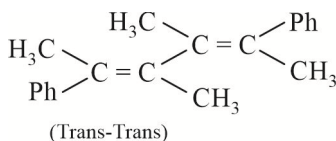
98.(2)



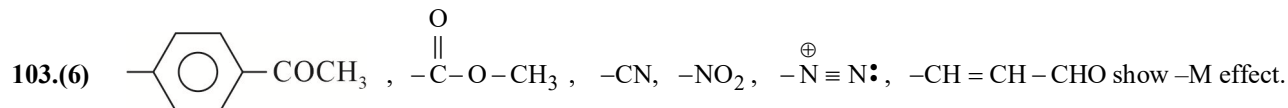
99.(3) Rearrangement of carbon skeleton of substrate is possible when carbocation intermediate is formed. This is possible in –

Electrophilic addition on alkenes,  $S_N1$ , E1

100.(1) Only one product (single stereoisomers)



are more reactive than benzene towards nitration.

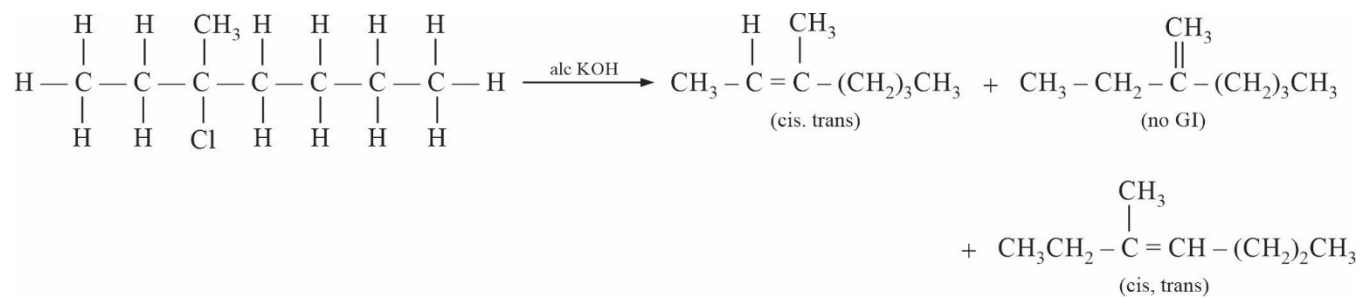


104.(3)

105.(5) 5(i, iii, iv, v, vii)

106.(4) Presence of electron releasing group activate alkene for electrophilic addition reaction.

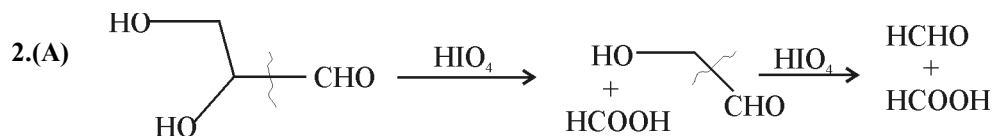
107.(5)



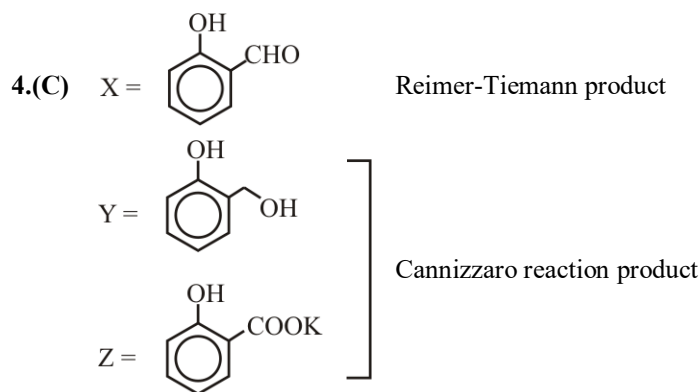
## Solutions - Advanced Problem Package | Chemistry

## Oxygen Containing Organic Compounds-I

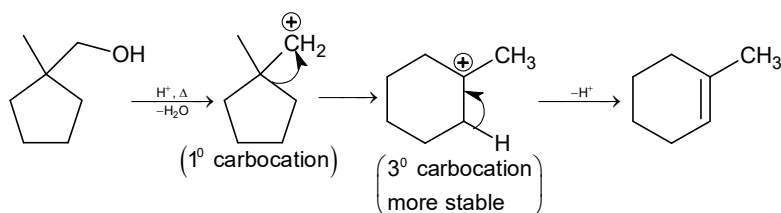
- 1.(A) Conjugated isomer is more stable than the non-conjugated; out of these trans isomer is more stable. Hence option (a) is the major product.



- 3.(D) At every step with one mole  $\text{HIO}_4$  carry out cleavage of carbon-carbon bond with one step up oxidation.

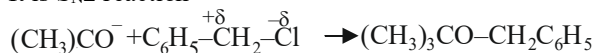


- 5.(C)

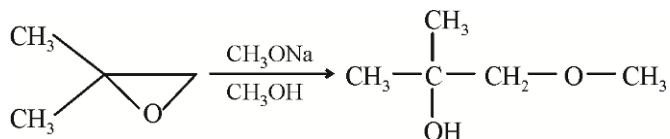


- 6.(C) Here the oxygen containing ring act as a  $-\text{OR}$  group. It activates the benzene ring thereby the positions ortho and para w.r.t. O are more electron rich than the position meta w.r.t. O. Hence both ortho & para w.r.t. O would be attacked by electrophile.

- 7.(B) It is  $\text{S}_{\text{N}}2$  reaction

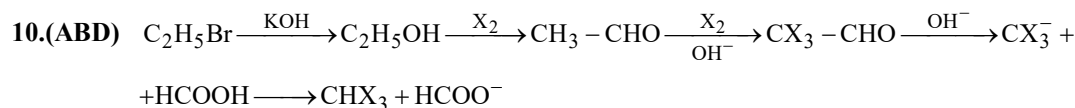


- 8.(A)



In base catalyzed ring opening nucleophile attacks on less substituted carbon.

- 9.(D) Iodoform reaction.

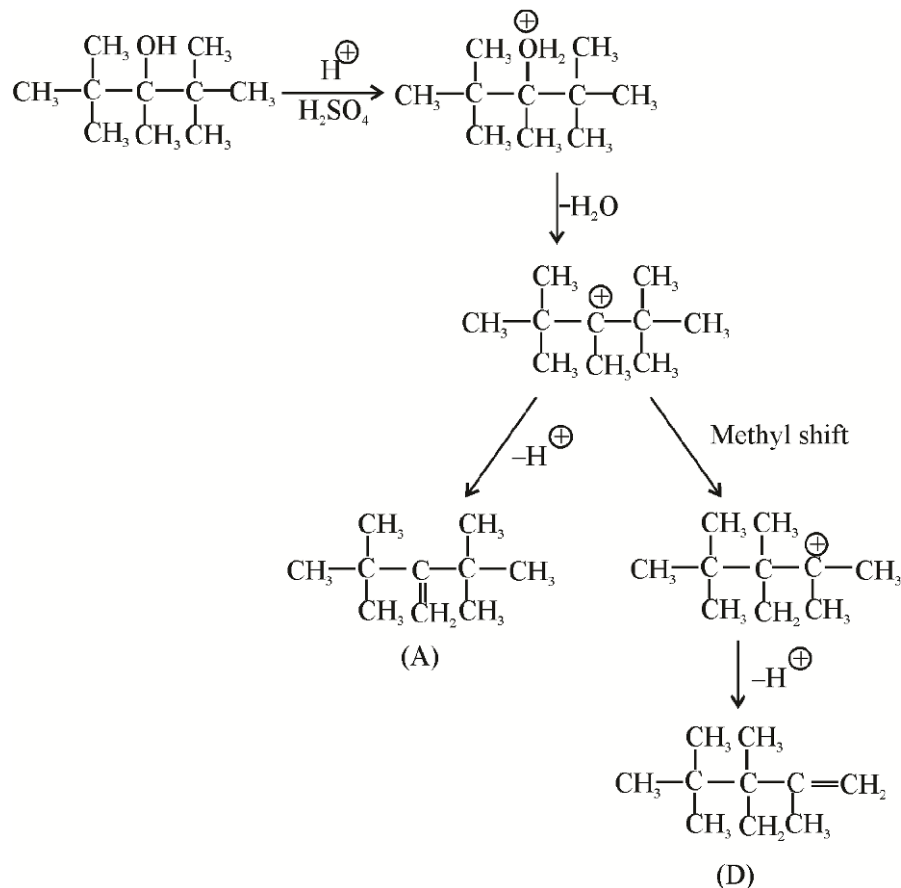


A, B, D  $\longrightarrow$  also gives haloform reaction

11.(AC) Tert. alkyl halides and aryl halides do not undergo Williamson synthesis

12.(ABC) Aryl halides do not undergo nucleophilic substitution.

13.(AD)



14.(ABCD) All compounds with  $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-$  group give haloform test.

15.(CD) Alkyl halides give elimination reaction

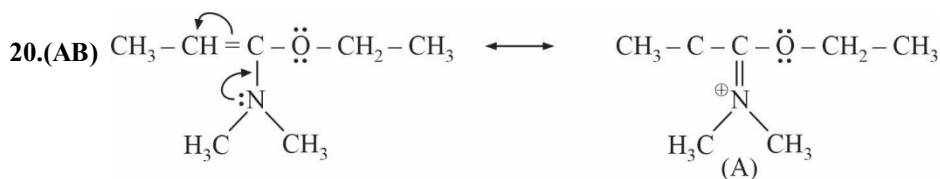
16.(ABC)  $\text{FeCl}_3$  solution is used to test phenol.

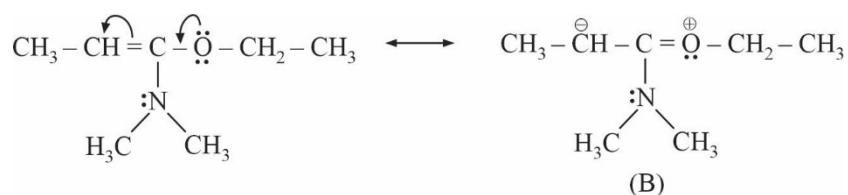
17.(AB) In option (C) and (D) ethene is formed as the major product.

18.(CD) (A) and (B) carry out syn-hydroxylation, (C) and (D) carry out anti hydroxylation.

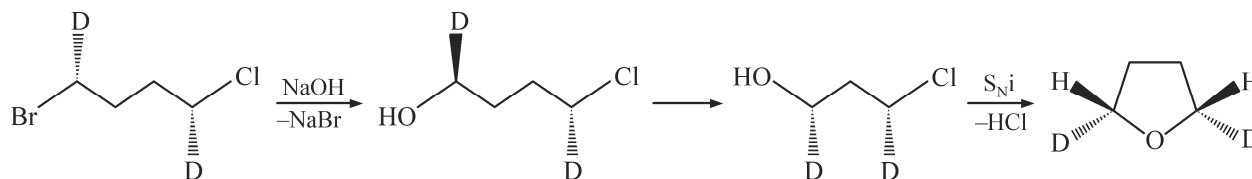
19.(AB) (C) forms  $\text{CH}_3-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$

(D) forms  $\text{CH}_3-\underset{\text{OH}}{\text{CH}}-\text{CH}_3$

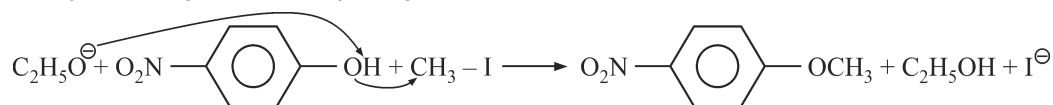




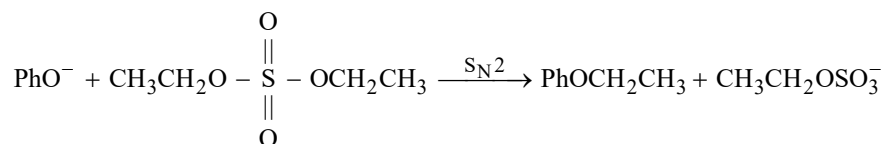
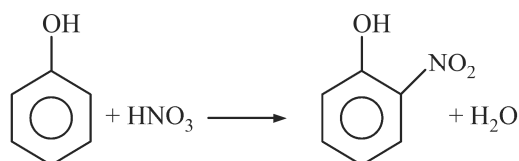
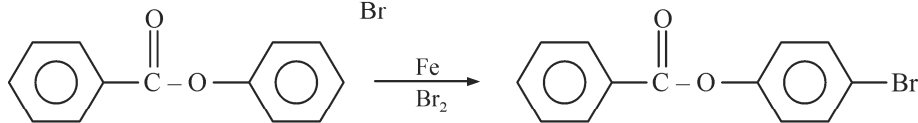
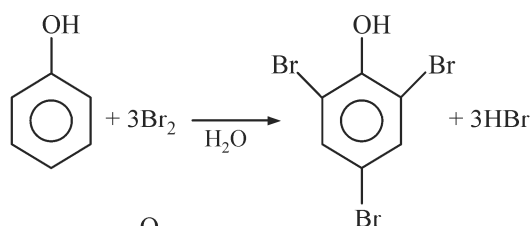
- 21.(BC) Bromide ion is better leaving group than chloride ion. Compound X is formed by  $\text{S}_{\text{N}}2$  reaction mechanism while Y is formed by intramolecular  $\text{S}_{\text{N}}2$  reaction mechanism.



- 22.(AB)  $\text{C}_2\text{H}_5\text{O}^\ominus + \text{CH}_3\text{I} \longrightarrow \text{C}_2\text{H}_5\text{OCH}_3 + \text{I}^\ominus$



- 23.(ABD)

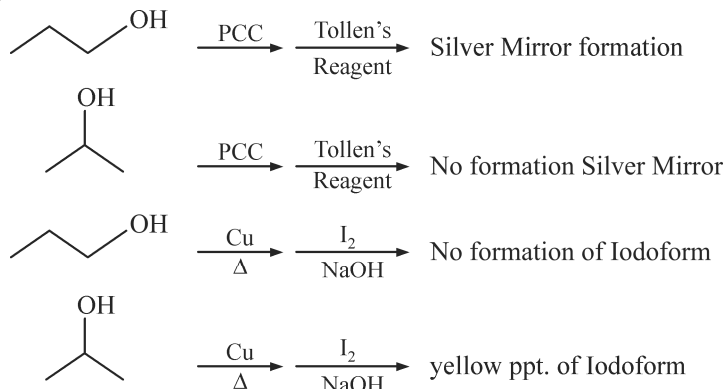


- 24.(ABCD)

- (A) Electrophilic attack of  $\text{D}^+$  on strongly activated benzene
- (B) It is diazocoupling reaction and it is an example of electrophilic substitution reaction.
- (C) Desulphonation is also an electrophilic substitution reaction. In this reaction attacking electrophile is  $\text{H}^+$  ion and leaving group is  $\text{SO}_3$
- (D)  $\text{CO}_2$  acts as electrophile. Kolbe – Schmidt reaction of sodium phenoxide with  $\text{CO}_2$  is an electrophilic substitution reaction.

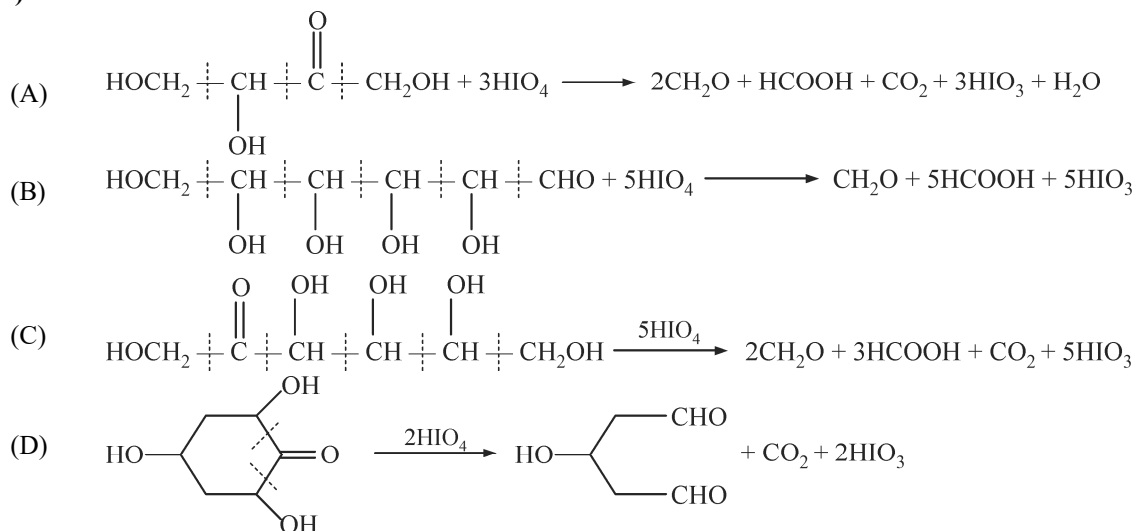
25.(AD) X and Y are monosubstituted benzene containing chiral carbon atom. Also contain primary or secondary alcoholic OH group.

26.(BC)

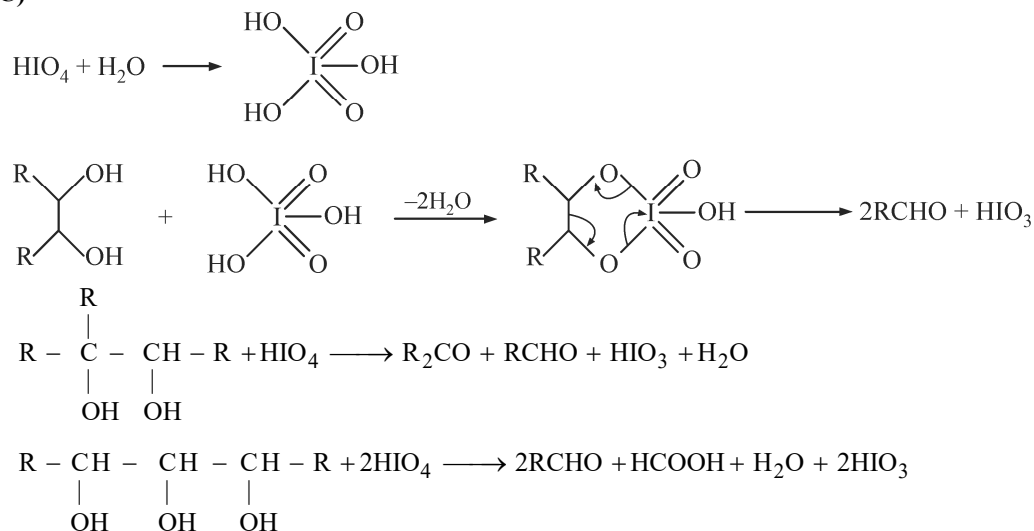


Both 1-propanol and 2-propanol are oxidized by  $\text{KMnO}_4$ . Reaction with conc.  $\text{H}_2\text{SO}_4$  form 1-propene and further there is no reaction of propene with Fehling's solution.

27.(ACD)

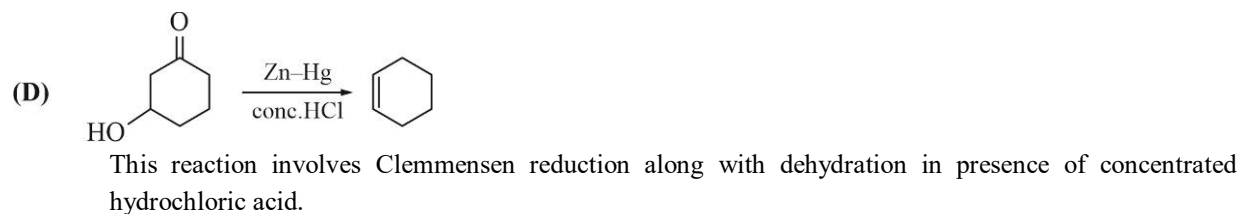
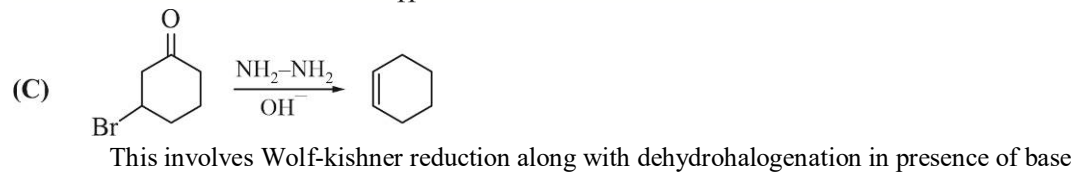
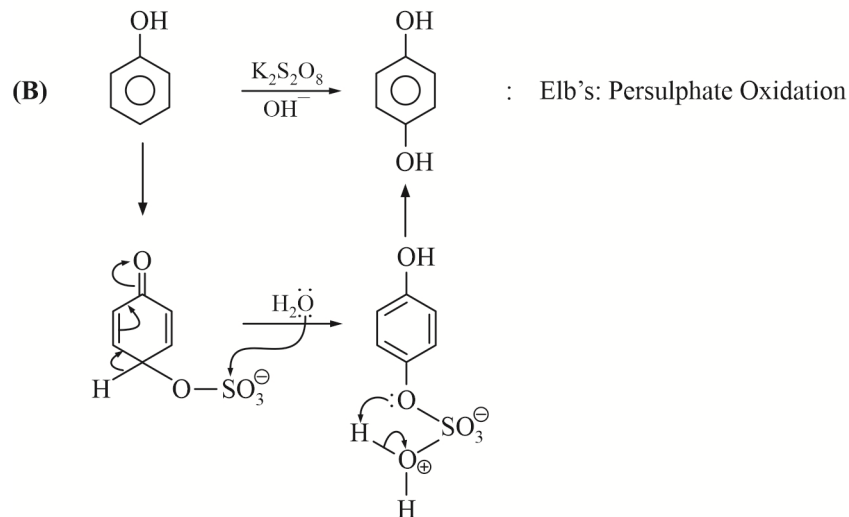
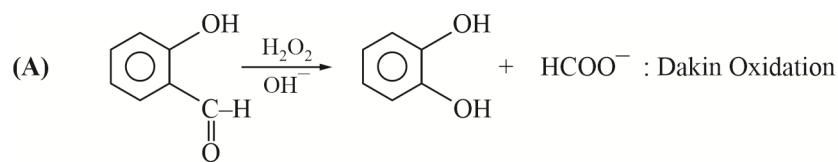


28.(ABC)

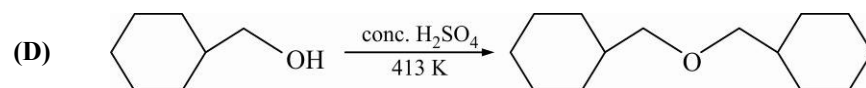
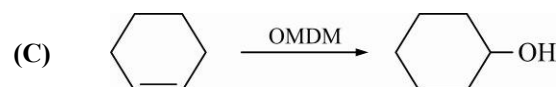
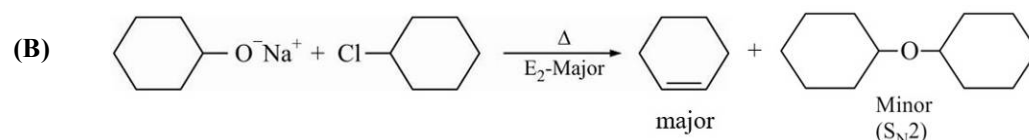
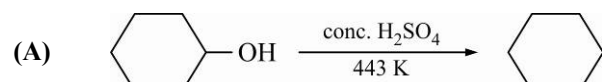




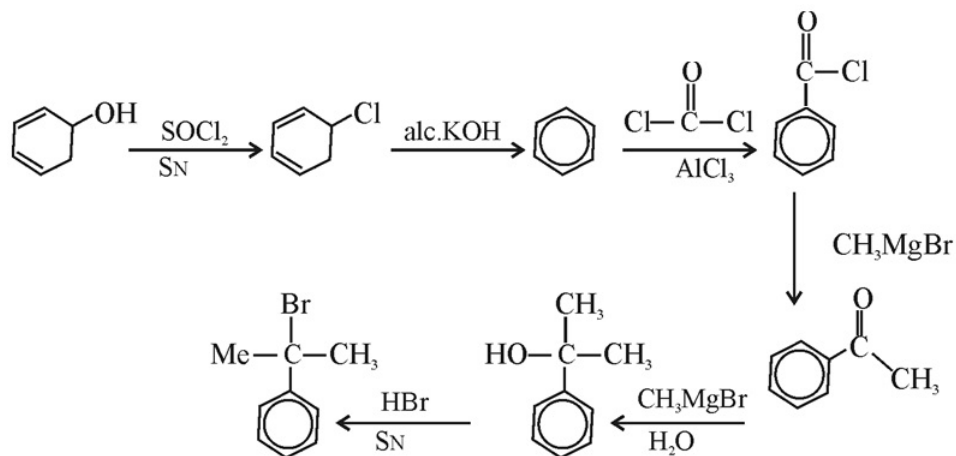
29. [A-p] → [B-r] → [C-s] → [D-q]



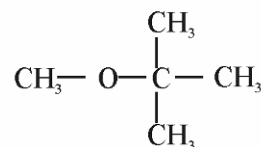
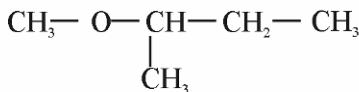
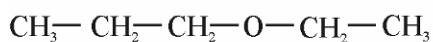
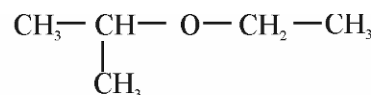
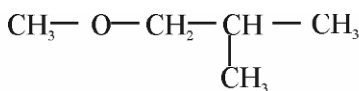
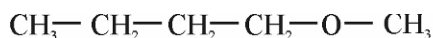
30. [A-r] → [B-r, q] → [C-p] → [D-s]



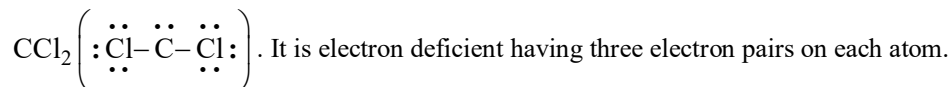
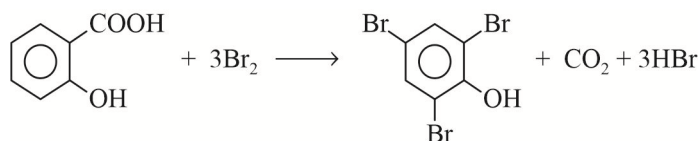
31.(2)



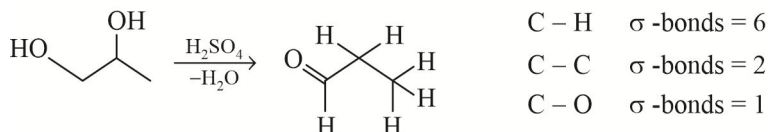
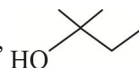
32.(6)



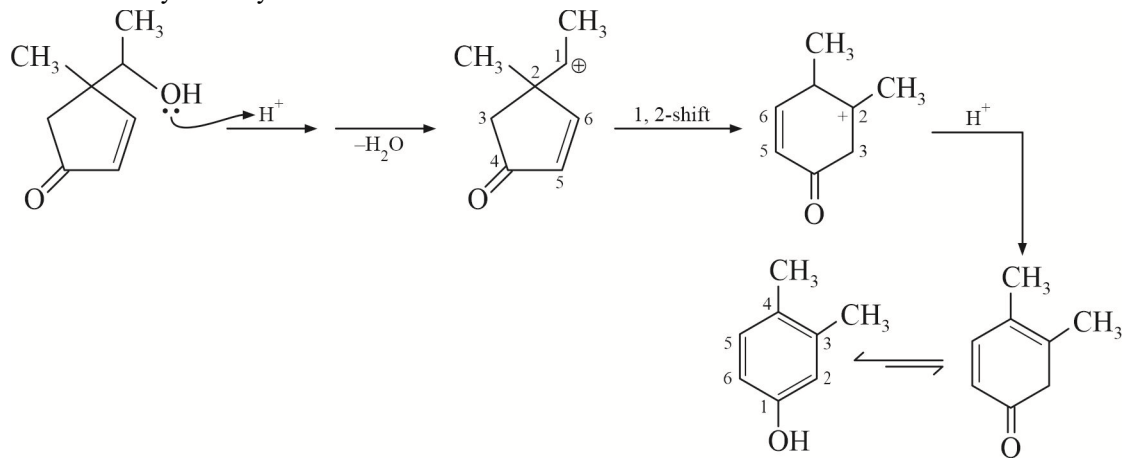
33.(9) It is Reimer-Tiemann reaction and attacking reactive species of this reaction is dichloro carbene,

34.(9) Salicylic acid on reaction with  $\text{Br}_2$  water undergo bromination cum decarboxylation. Product [P] is 2, 4, 6-tribromophenol.Mole of  $\text{Br}_2$  used =  $3 \times$  moles of salicylic acid taken.

35.(9) It is pinacol-pinacolone type rearrangement

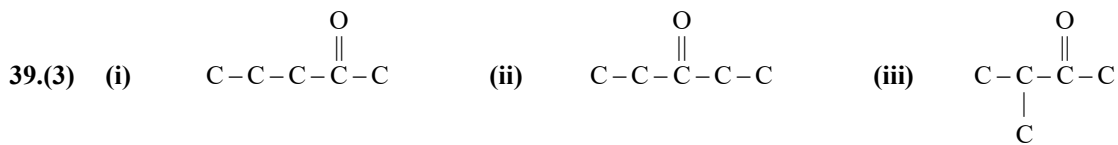
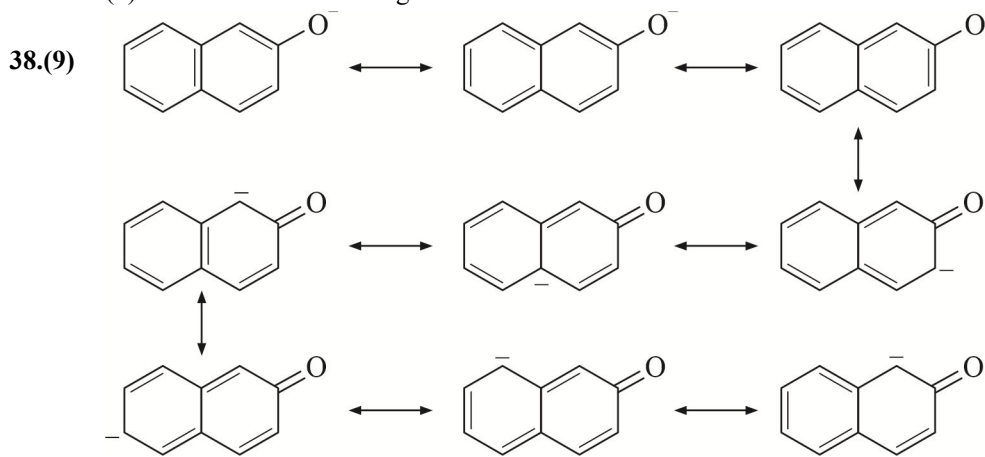
36.(1) PCC can oxidise only primary and secondary alcohols. It can't oxidise tertiary alcohol. Only one tertiary alcohol is possible for molecular formula  $\text{C}_5\text{H}_{12}\text{O}$  i.e.,

**37.(8)** It is acid catalyzed dehydration of alcohol.

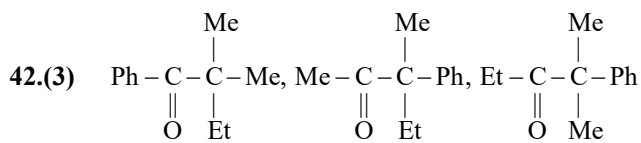
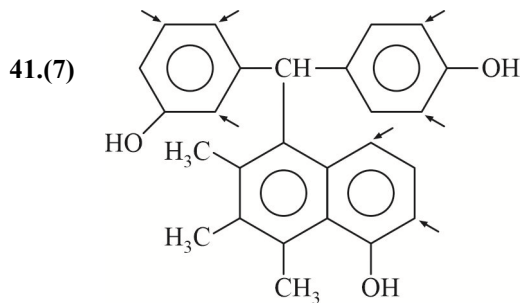


**Note:**

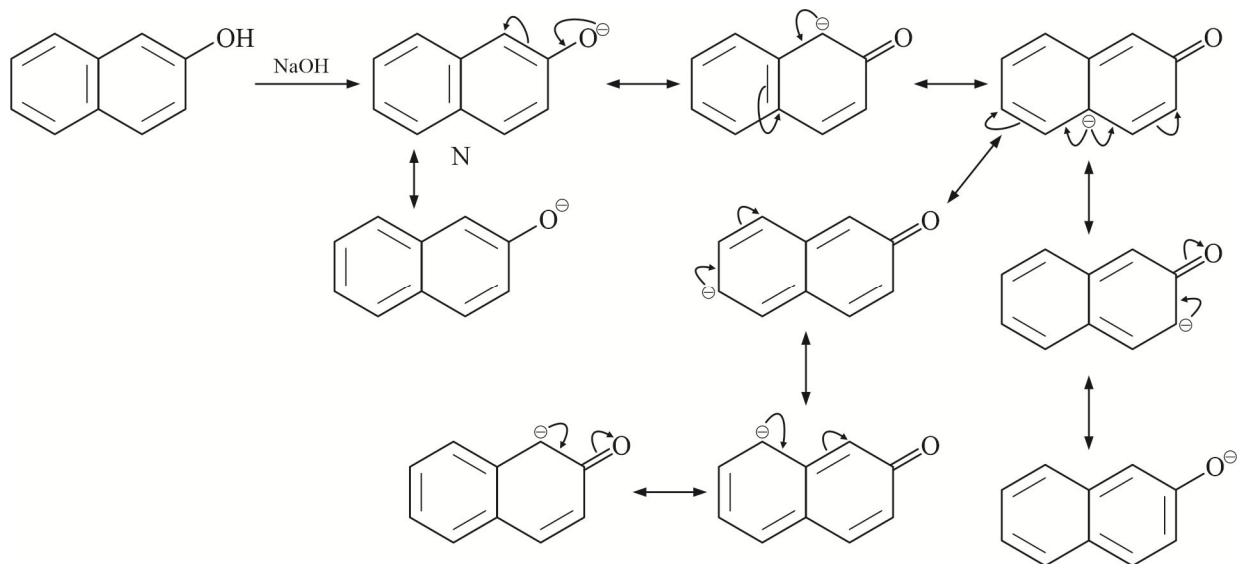
- (1) Migratory aptitude of vinyl group is more than that of methyl.
- (2) Enolic form having aromatic character is more stable than keto form.



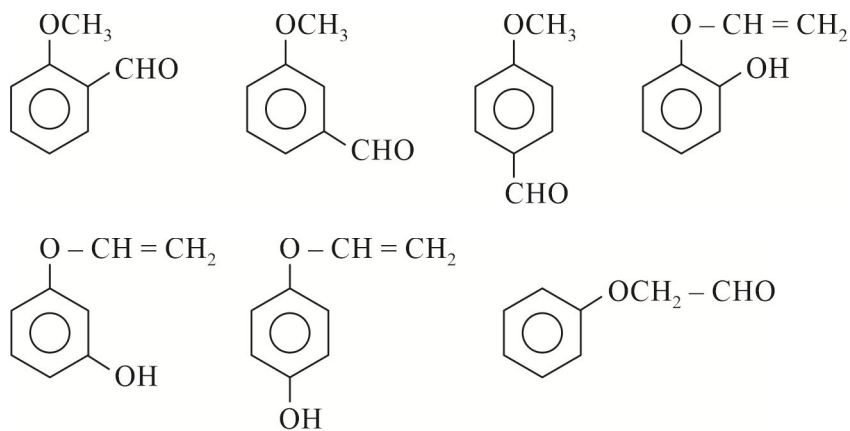
**40.(2)** (p & v)



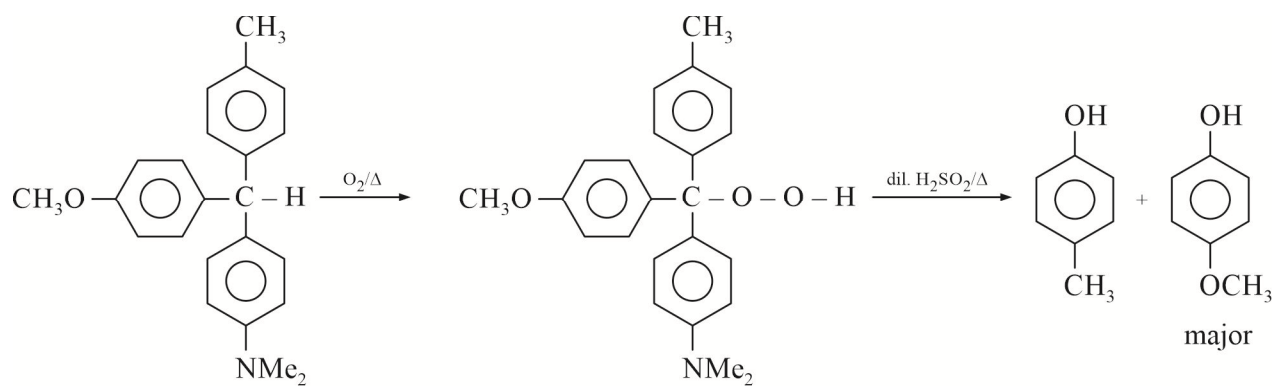
43.(9)

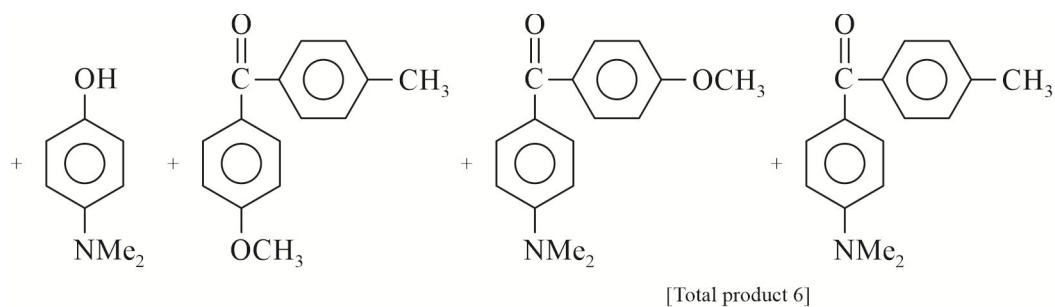


44.(07)



45.(06)

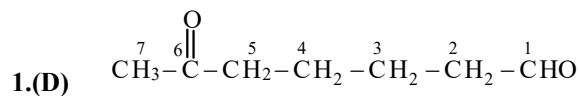




- 46.(05)  $\text{R}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}$  produces red coloration with neutral  $\text{FeCl}_3$  (tested in inorganic Qual. Analysis as well) while phenolic  $^-\text{OH}$  group produces violet coloration with neutral  $\text{FeCl}_3$

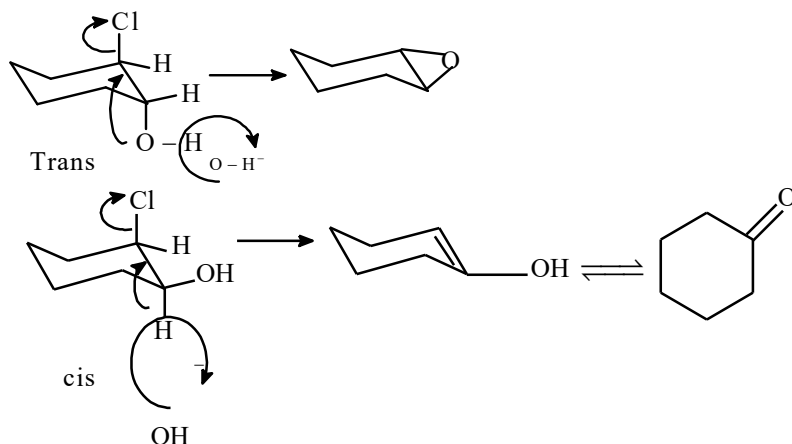
## Solutions - Advanced Problem Package | Chemistry

## Oxygen Containing Organic Compounds-II



During addition of NaOH, firstly carbanion is generated from the said compound, and that will attack  $-\text{C}=\text{O}$  group internally. Carbanion from C-2 can attack carbonyl group at C-6 to give option (A). Similarly carbanion from C-7 and C-5 can attack carbonyl group of C-1 to give option (B) and (C) respectively. So (D) cannot be formed.

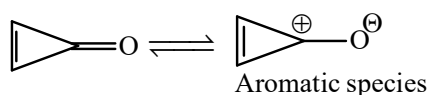
2.(B)



3.(D) Use stoichiometry of reaction.

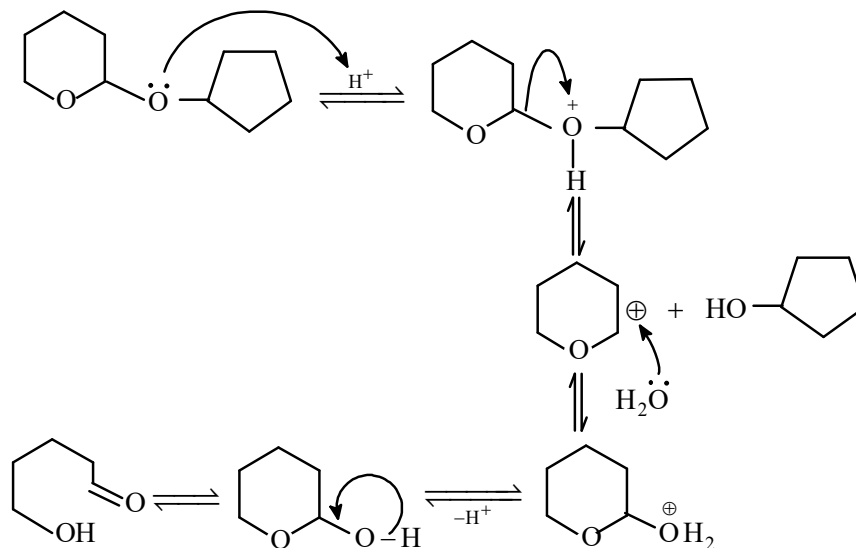
4.(B) Sterically hindered ketonic group cannot react with glycol.

5.(A) Since (A) exist as aromatic species so would have highest dipole moment.

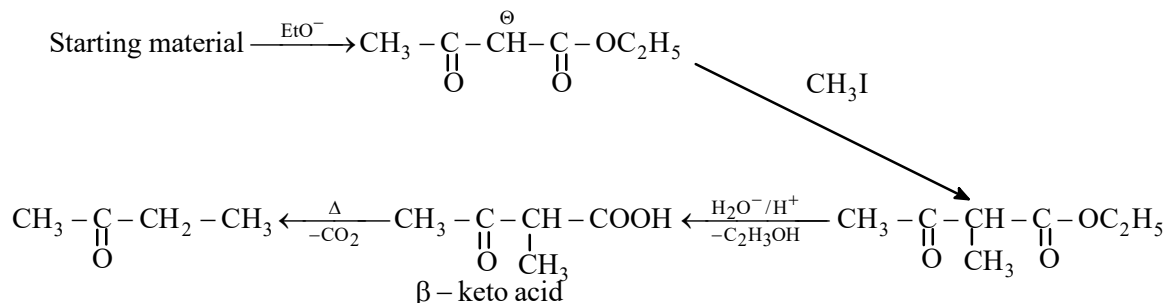


6.(A) Hydrolysis of (I) will give  $\text{CH}_2=\overset{\text{OH}}{\text{C}}-\text{CH}_3$  which on tautomerism give  $\text{CH}_3\text{COCH}_3$  whereas hydrolysis of (II) will give  $\text{CH}_3-\text{CH}_2-\text{CHO}$ . Acetone and propanal can be distinguished by Fehling, Benedict and Tollen's reagent whereas Brady's reagent react with both.

7.(C) Starting compound is acetal which on hydrolysis will give aldehyde or ketone



8.(A)



9.(B)  $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$ , gives iodoform test and hydrazine test. In dil.alkaline medium, it would first make

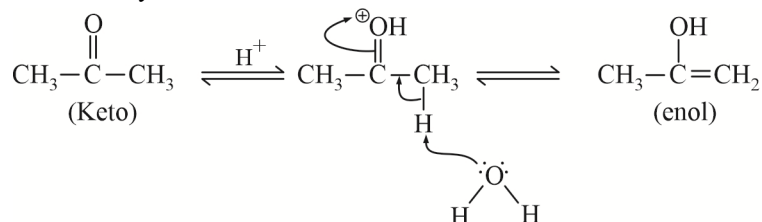
carbanion which on abstraction of proton will generate racemic mixture.

10.(B) Except (D), all compounds are optically active as they have one chiral carbon in them. (A) is ketone so won't give tollens test. Out of (B) and (C), (C) would undergo racemisation as  $\alpha$ -H is attached to chiral carbon, which deprotonates first and then on protonation it results in racemisation. So compound in bottle (Y) is B.

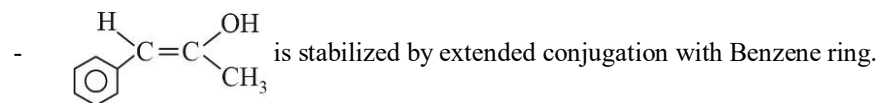
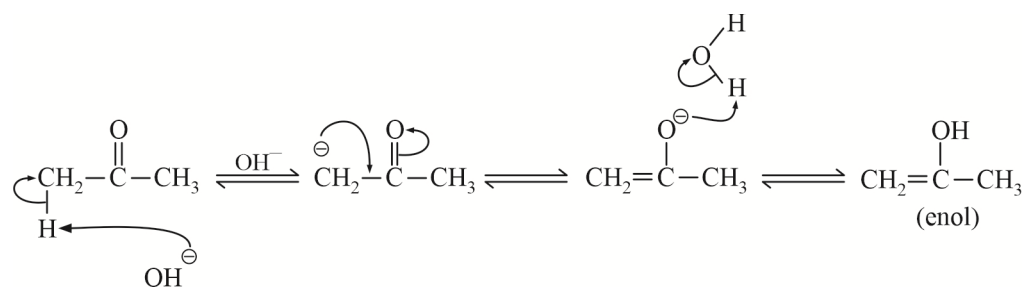
11.(ABCD) All are correct Statements.

Generally enol is less stable than keto because  $\text{C}=\text{O}$  bond energy is 799 kJ/mol, as against  $\text{C}=\text{C}$  bond energy is 622 kJ/mol. So,  $\text{C}=\text{O}$  bond is stronger than  $\text{C}=\text{C}$  bond.

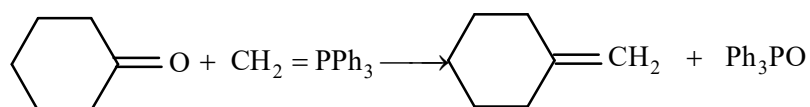
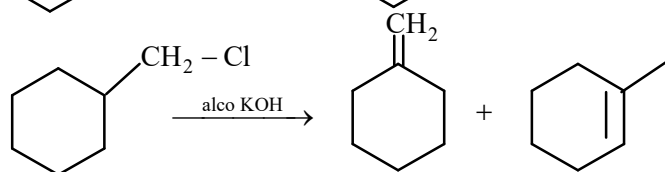
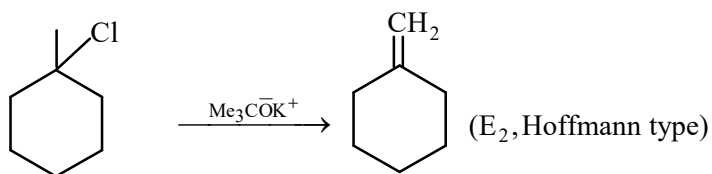
- Acid catalyzed tautomerism:



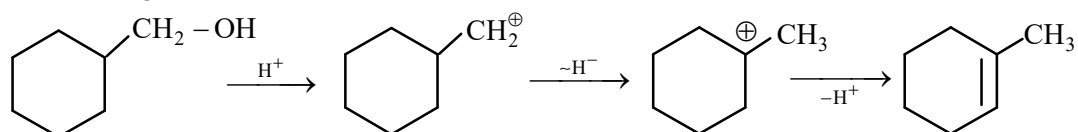
- Base catalyzed tautomerism:



## 12.(ABC)

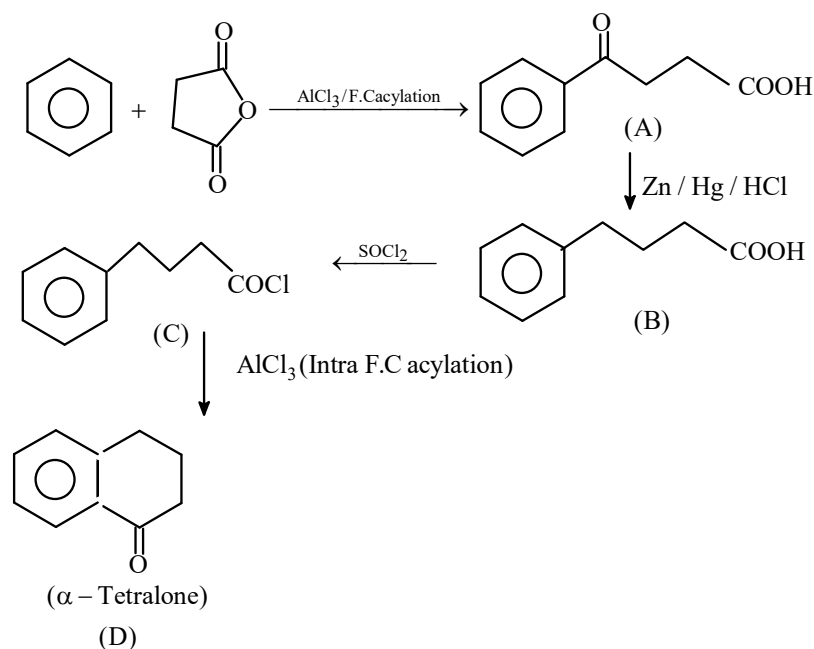


It is a wittig reaction





13.(ABCD)

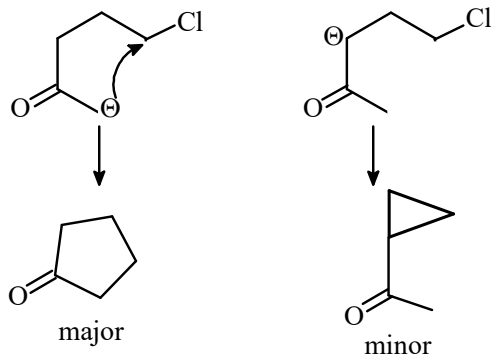


14.(ABC)

(A)  $\text{CH}_3\text{COCH}_2\text{CH}_3$  in basic medium undergo kinetically controlled enolisation whereas in acidic medium it undergo thermodynamically controlled enolisation.

(B) Perkin condensation

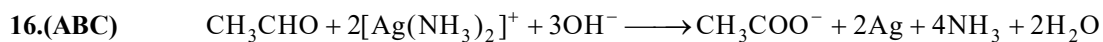
(C)



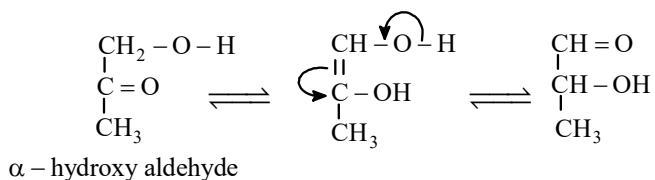
15.(BCD)

All are correct statements:

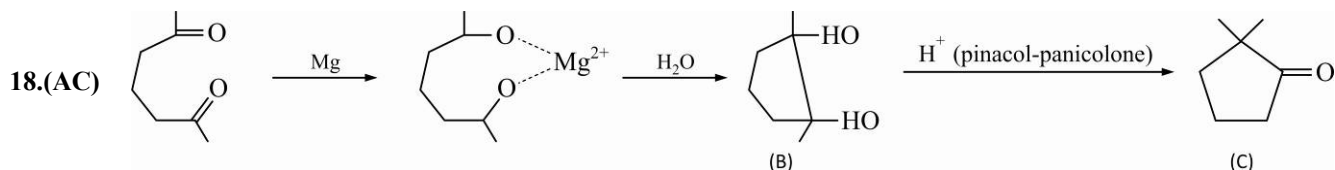
- C-Mg bond is 35% Ionic, whereas C-Cu bond is 21% Ionic and C-Cd bond is 18% Ionic
- The reactivity of  $\text{EtMgBr}$  (Grignard Reagent) is more than  $\text{Et}_2\text{Cd}$  and  $\text{Et}_2\text{CuLi}$  (Gilman Reagent)
- $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \xrightarrow{\text{EtMgBr}} \text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Et}$  : Nucleophilic Addition Elimination Reaction because Cl is a good leaving group.
- $\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Et} \xrightarrow{\text{EtMgBr}} \text{Me}-\overset{\text{OMgBr}}{\underset{\text{Et}}{\text{C}}}-\text{Et}$  Nucleophilic Addition only because, alkyl is not good leaving group
- Ketones further react with  $\text{RMgX}$  to give 3° alcohols because they do not contain suitable leaving groups. So, they undergo Nucleophilic Addition rather than Nucleophilic substitution. [Although, it is true that ketones are more reactive to nucleophile than acid chlorides]



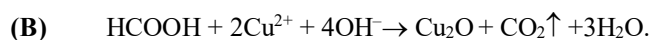
$\alpha$ -hydroxy aldehyde undergo Tollen reagent test, as it rearranges to aldehyde as shown below:



- 17.(AC) Carbonyl group of (A) is more electrophilic so  $\text{OH}^-$  will attack (A) rather than carbonyl group of (B). Therefore (A) would be hydride ion donar.

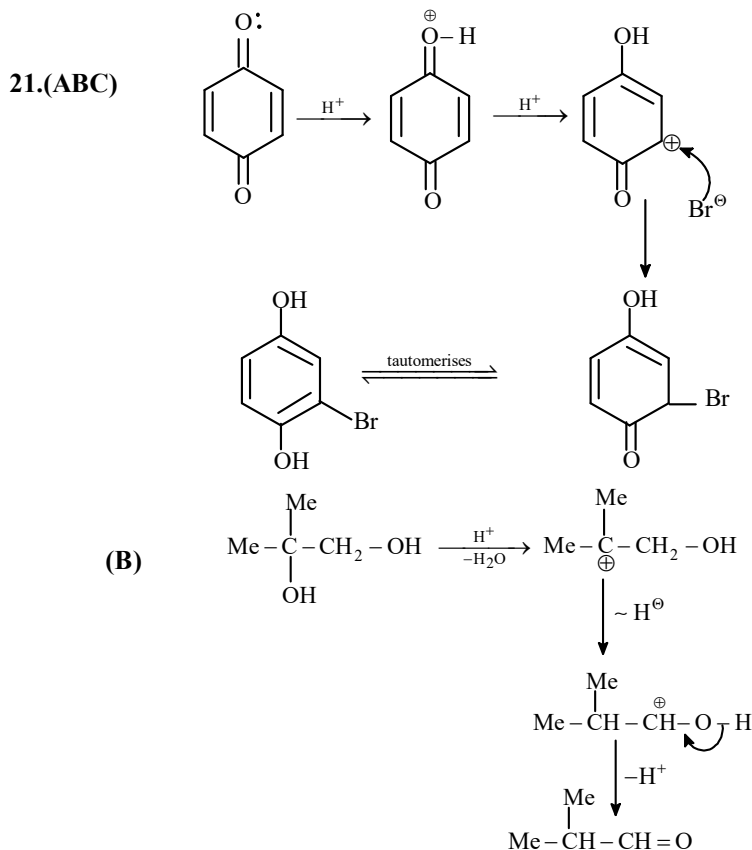


- 19.(ABCD) (A) Iodoform Reaction

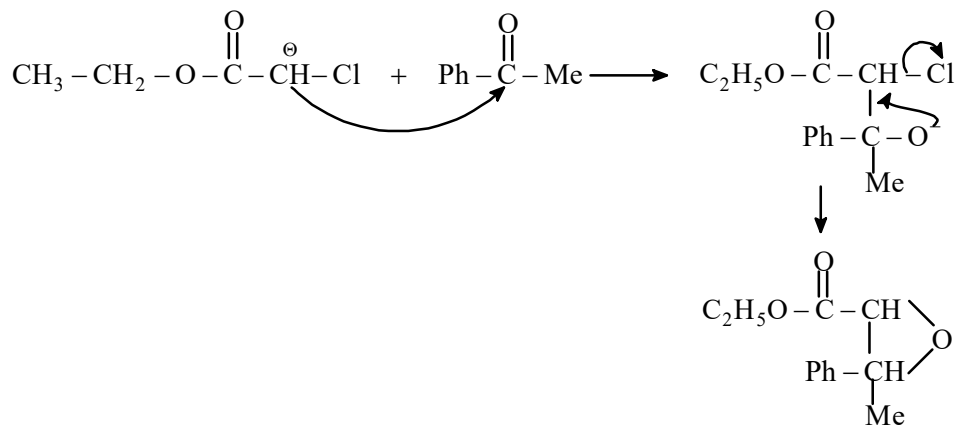


(C) and (D):  $\text{SeO}_2$  convert methylene or methyl group adjacent to carbonyl group to carbonyl group and this reagent may be used for oxidation of allylic carbon to carbonyl group.

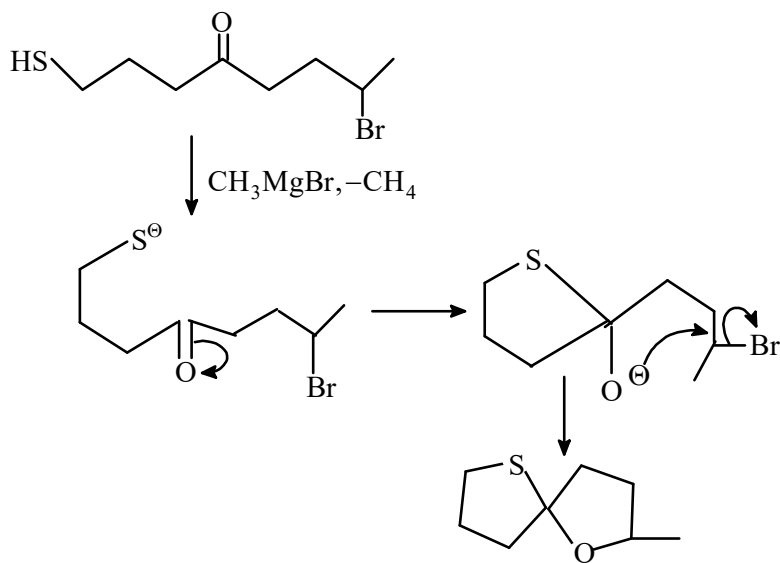
- 20.(CD) Aldehyde give test with Tollen's reagent where as ketones wont. DNP test is given by both aldehydes and ketones  $\text{CH}_3\text{CO}^-$  give iodoform test.

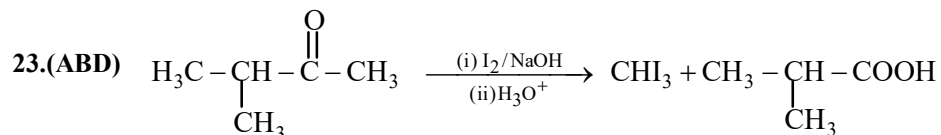


(C)

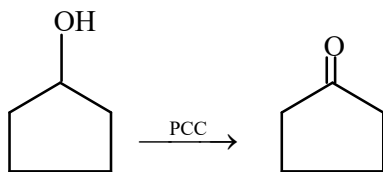
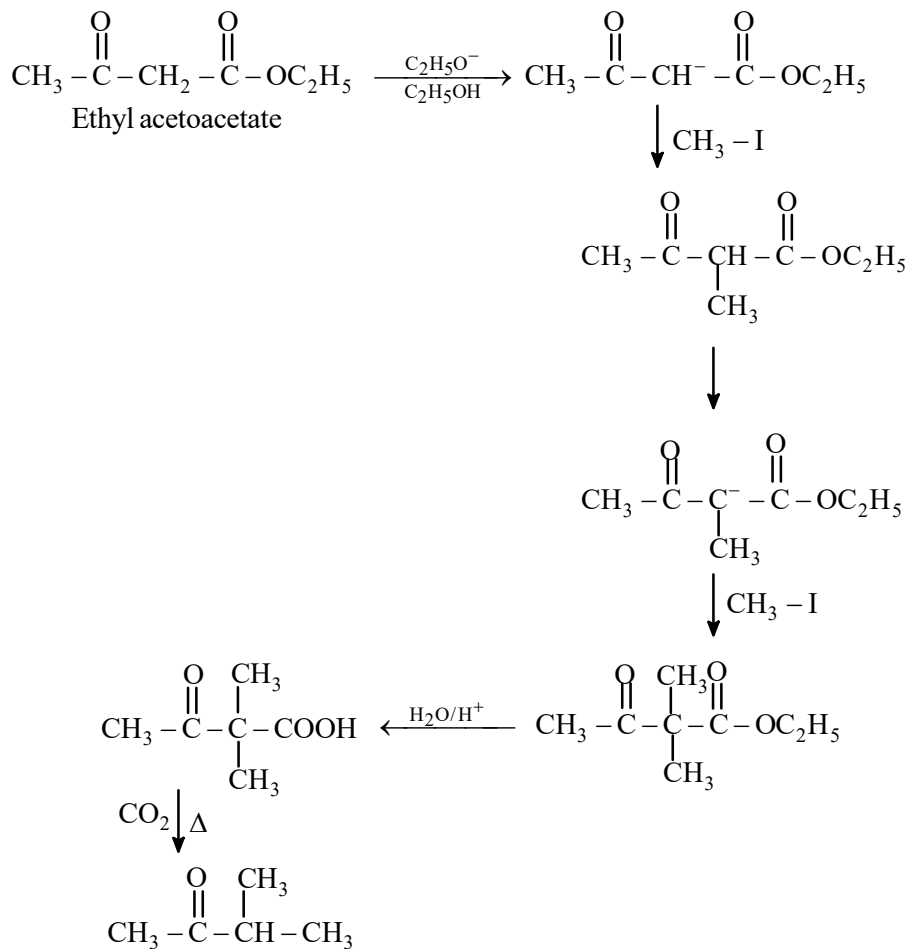


**22.(BCD)** It is Claisen condensation in which ester having  $\alpha$ -H undergo self condensation in presence of strong base

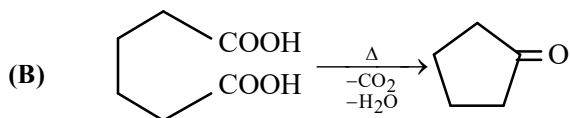




(C) option is incorrect

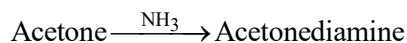
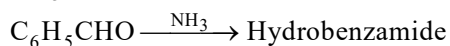
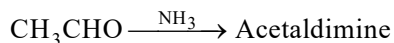
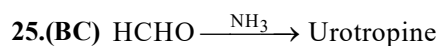


24.(AB) (A)

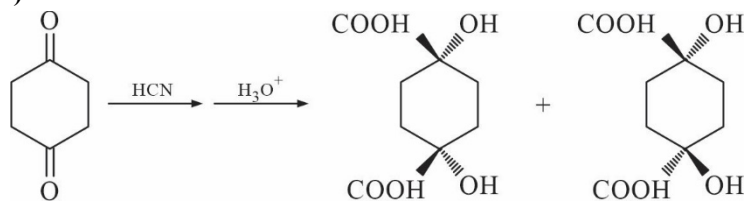


(C) It is Rosenmund's Reduction which give Pentanal

(D) It will give cyclopentylmethanol

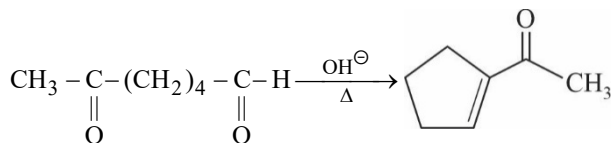


26.(ABD)

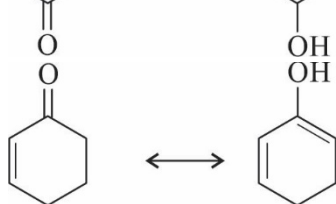
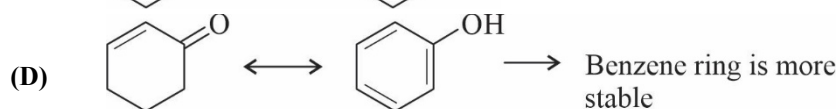
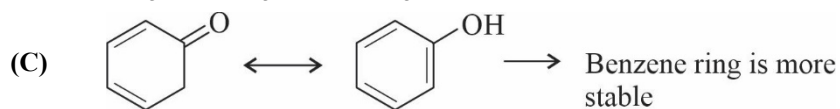
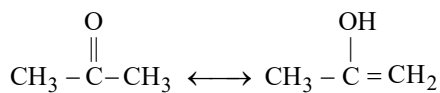
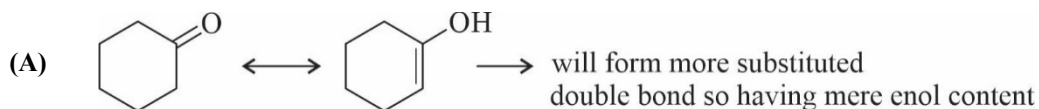


The products are optically inactive due to presence of plane of symmetry and G.I. (diastereomers) of the molecules exist.

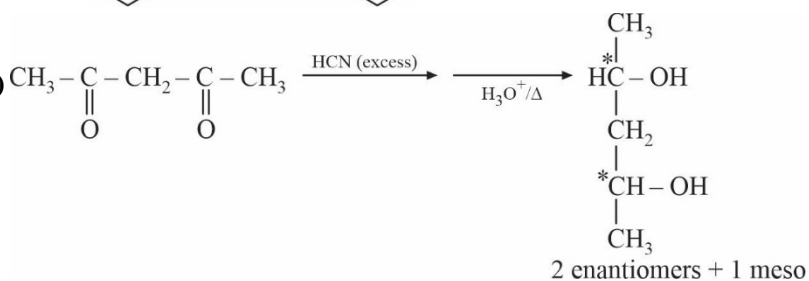
27.(ABC)



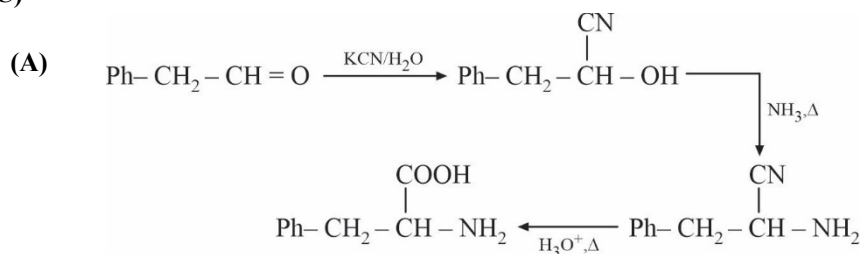
28.(ACD)

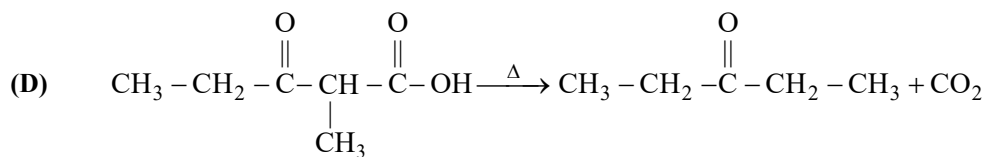
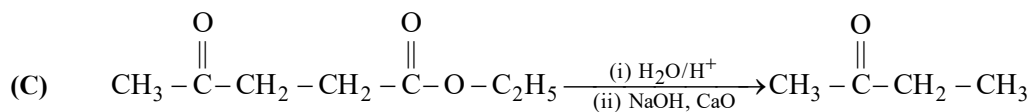
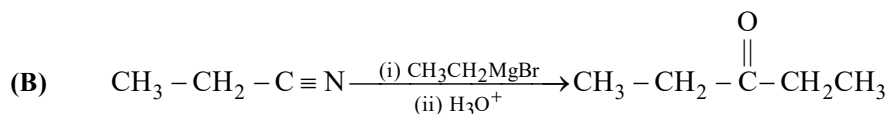
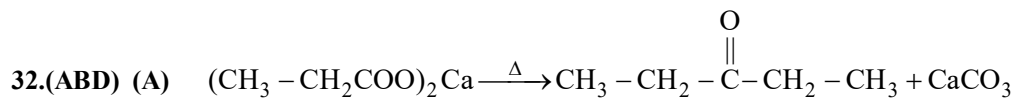
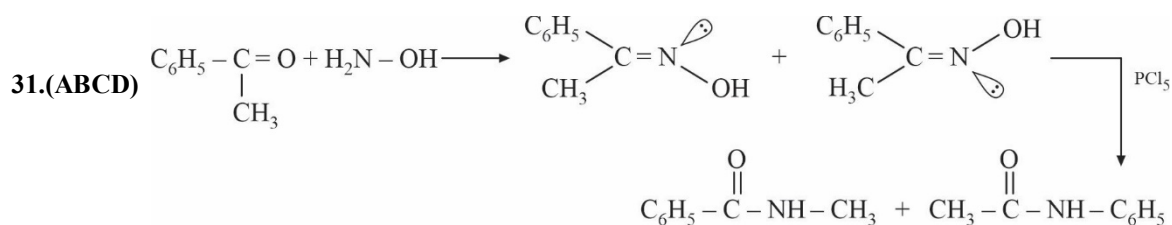
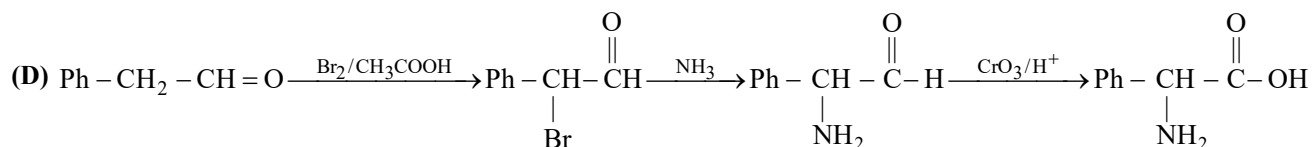
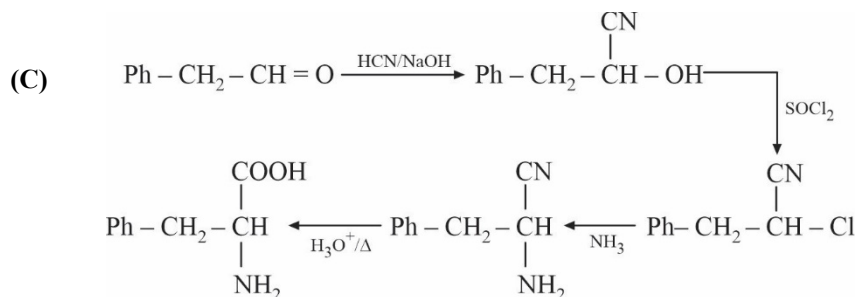
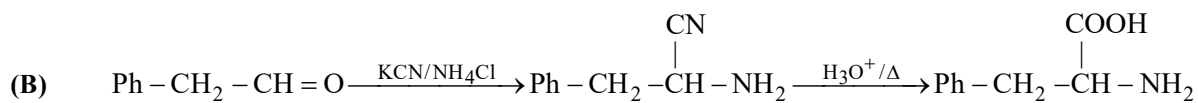


29.(BCD)

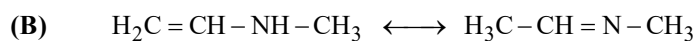
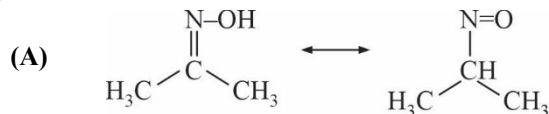


30.(ABC)

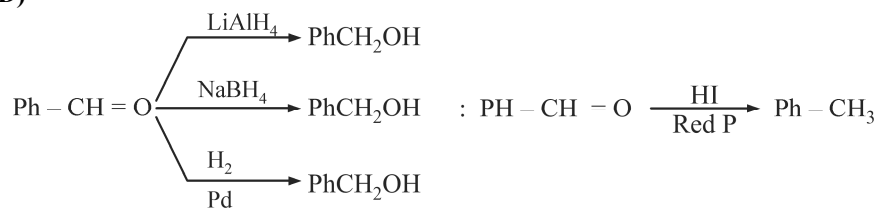




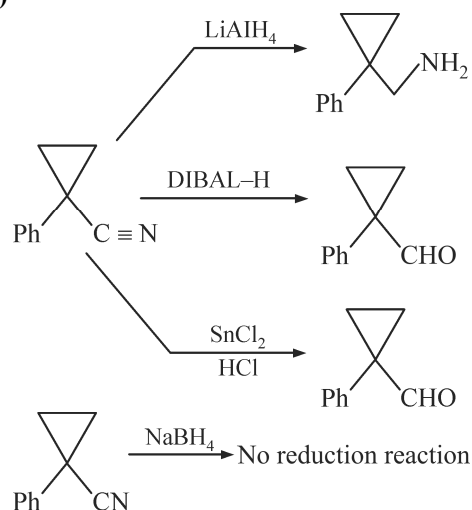
33.(AB)



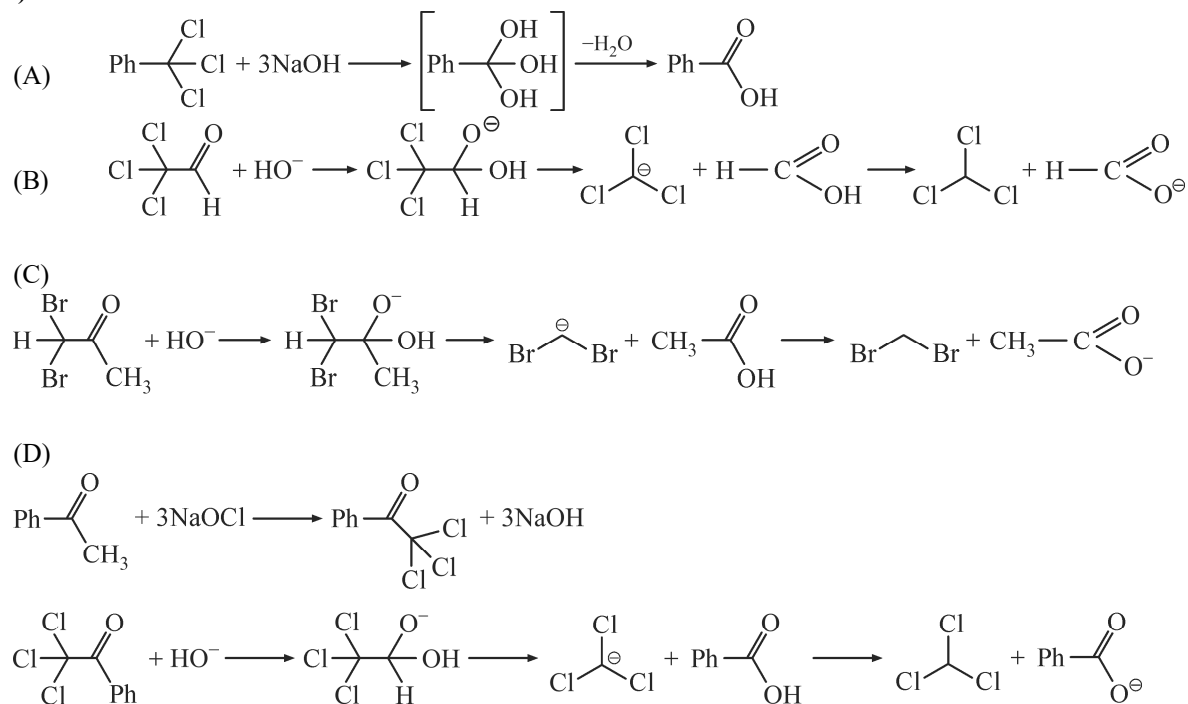
34.(ABD)



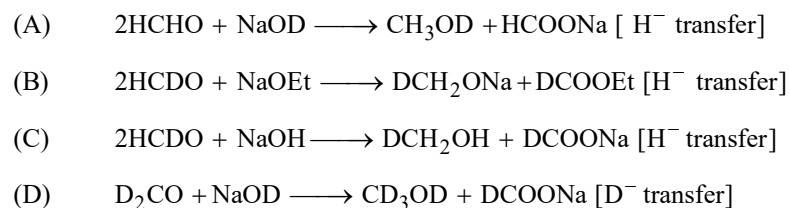
35.(CD)



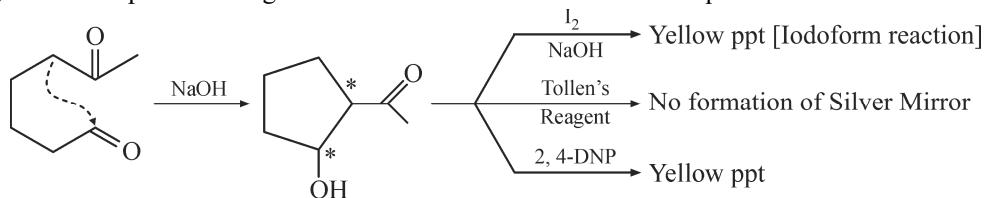
36.(ACD)



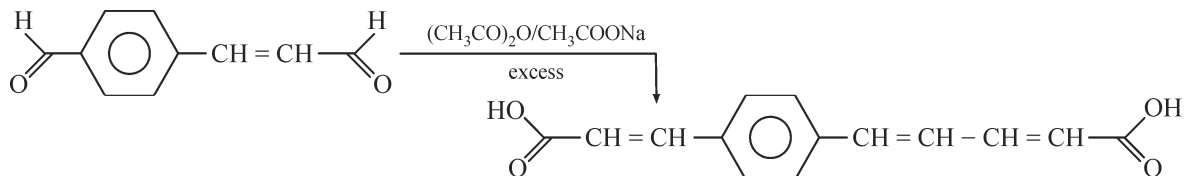
37.(ABCD)



38.(AC) Given compound undergoes intramolecular aldol reaction with aq. NaOH

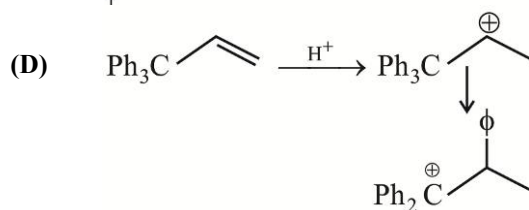
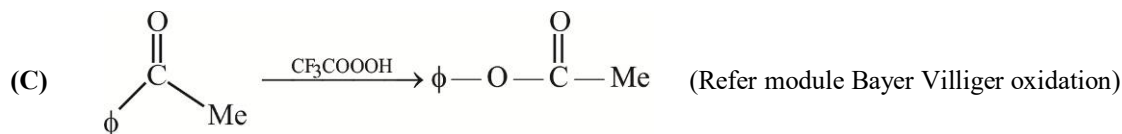
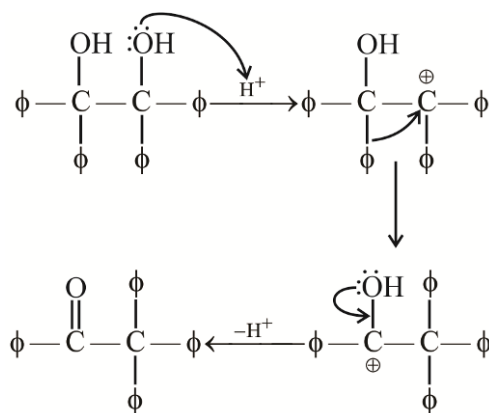
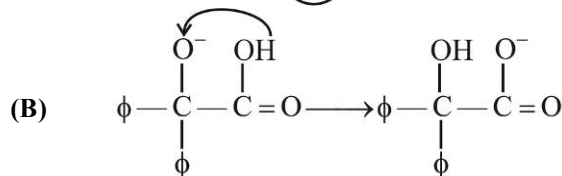
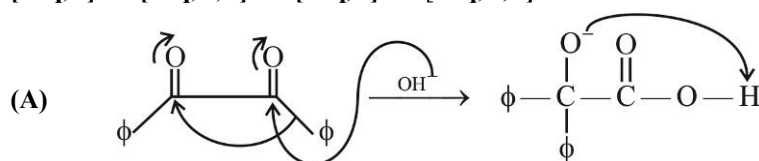


39.(B) Given reaction is Perkin reaction

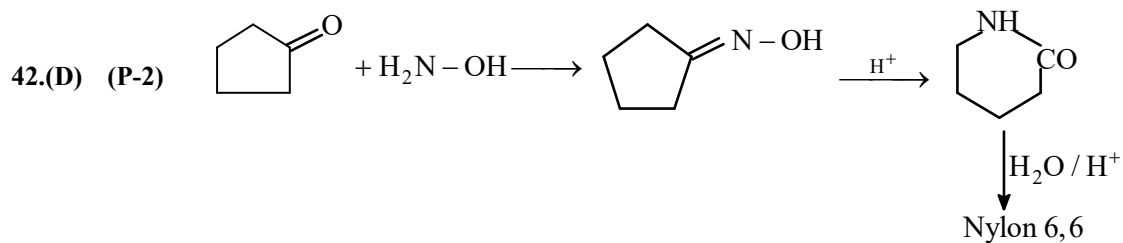


40.(AB) DIBAL-H selectively reduces ester and cyanide group to aldehyde group.

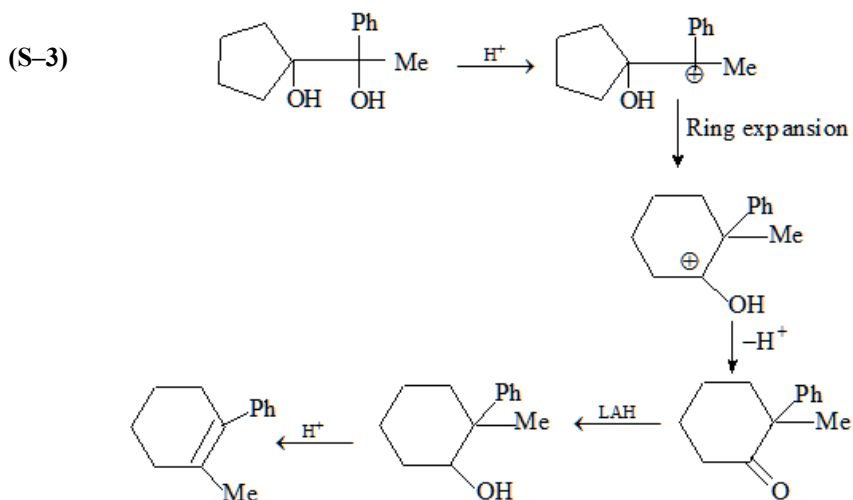
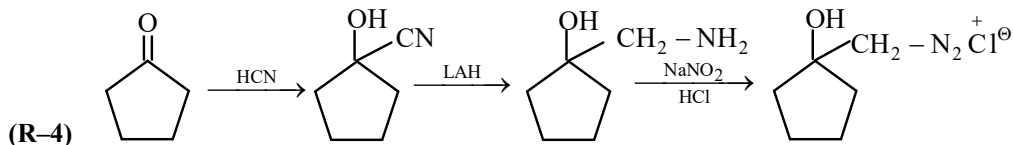
41. [A-q, s] → [B-q, r, s] → [C-q, s] → [D-q, r, s]



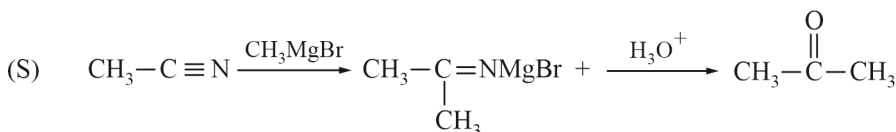
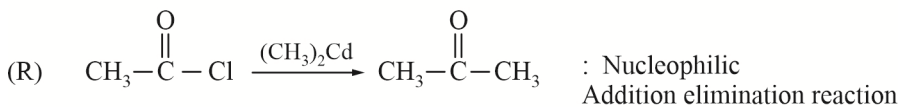
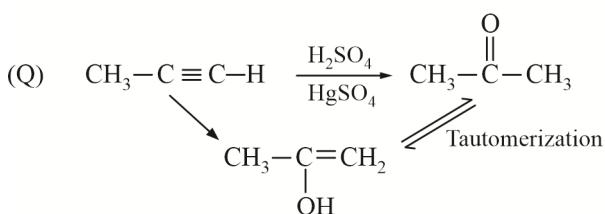
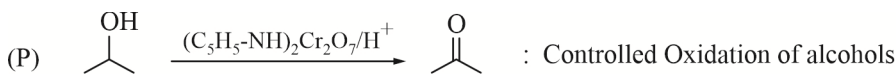




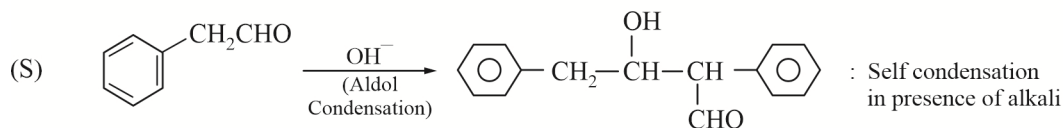
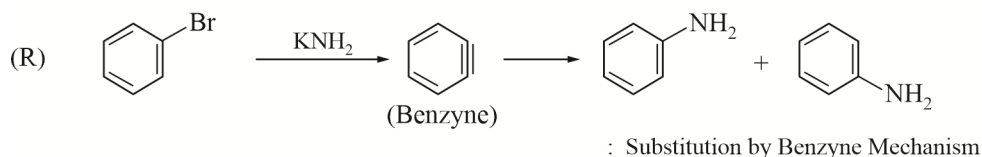
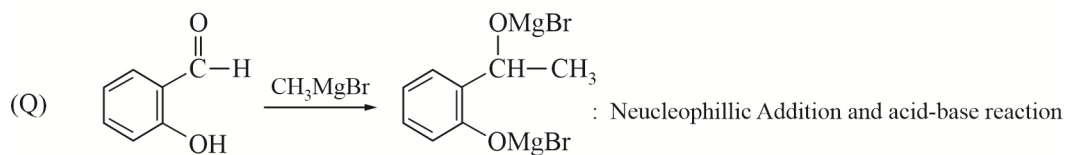
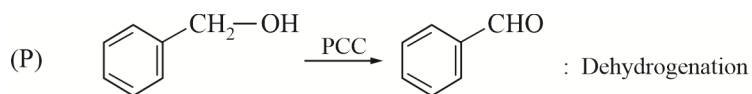
(Q-1) It is crossed intra aldol condensation



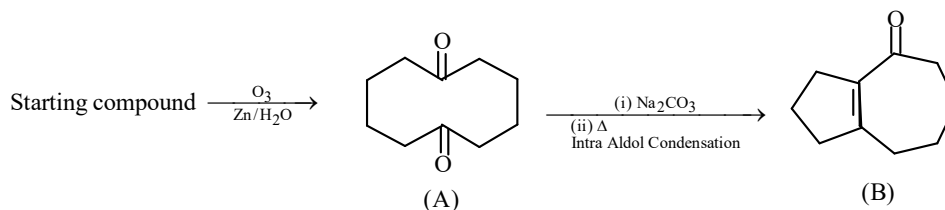
43.(C)



44.(C)

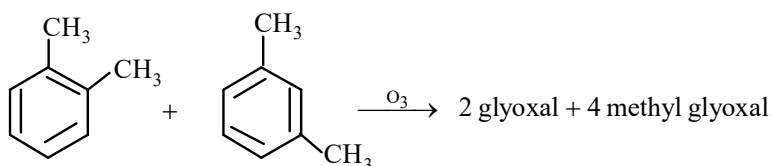


45.(2)

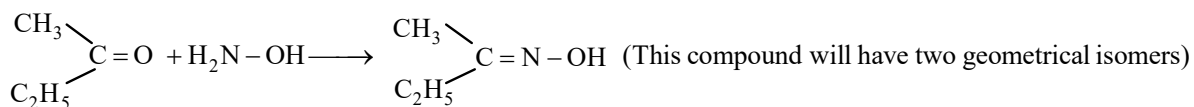
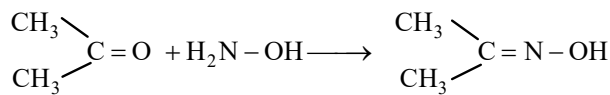


$$\frac{\text{DU in (B)}}{\pi - \text{bonds in (B)}} = \frac{4}{2} = 2.$$

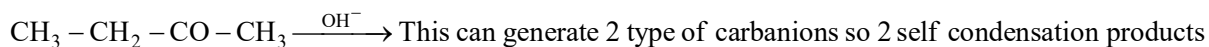
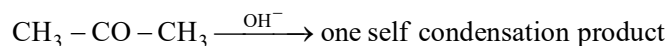
46.(8)



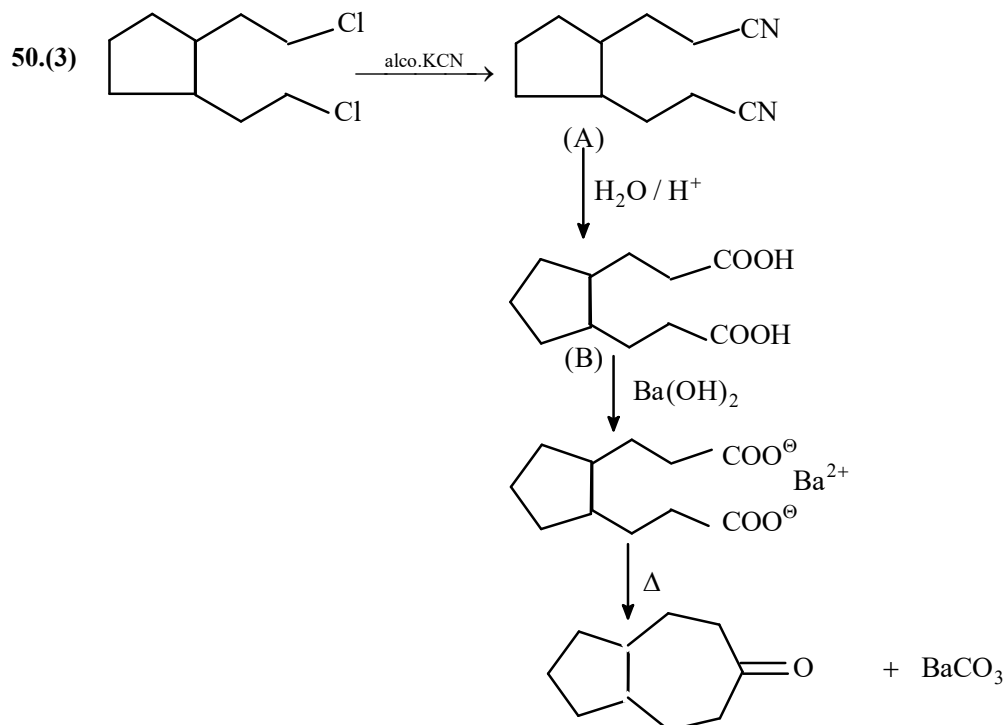
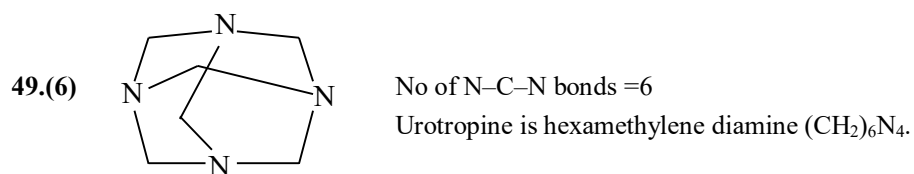
47.(3)



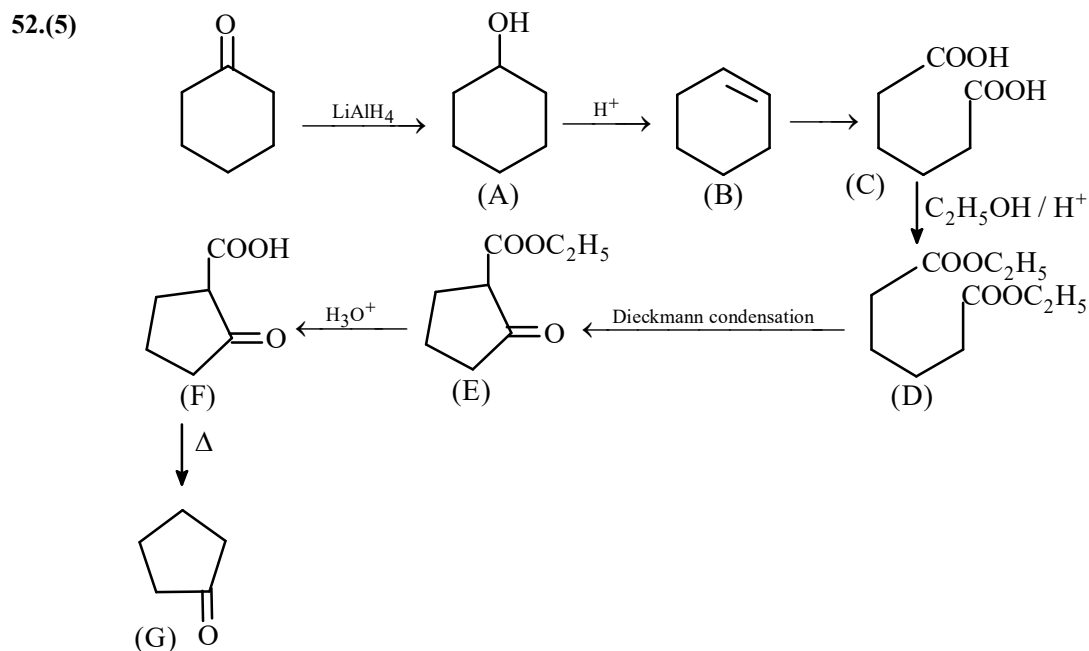
Total compound formed = 3



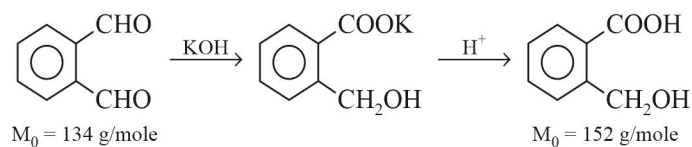
48.(6) Crossed condensation product will be 3. Total products = 6



51.(8) Iodoform test is generally given by those compounds which have or can generate  $\text{CH}_3\text{-CO-}$  group in them.

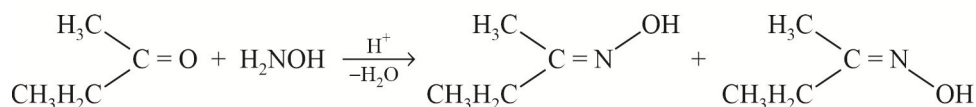


53.(9) It is intramolecular cannizzaro reaction. In this reaction molecular mass of reactant increases by 18 units.



$$\frac{152 - 134}{2} = \frac{18}{2} = 9$$

- 54.(2) Carbonyl compounds on reaction with hydroxyl amine produce oximes. Oximes may show geometrical isomerism.

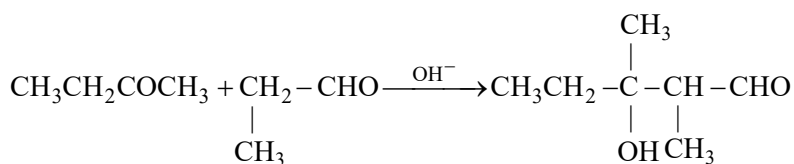
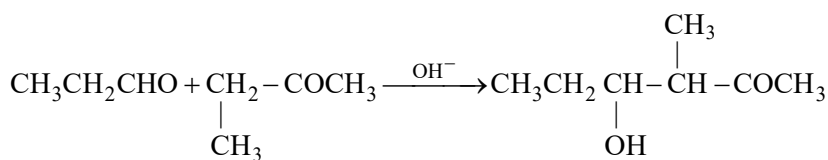
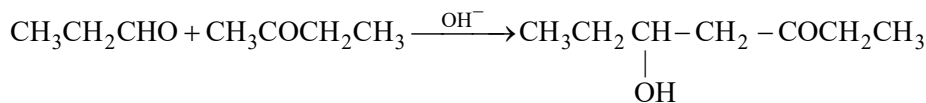
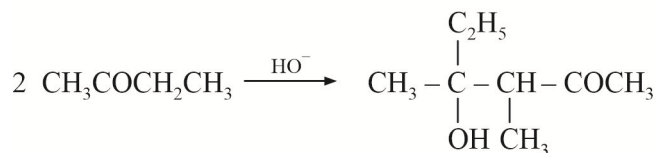
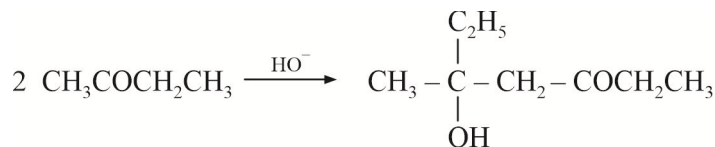
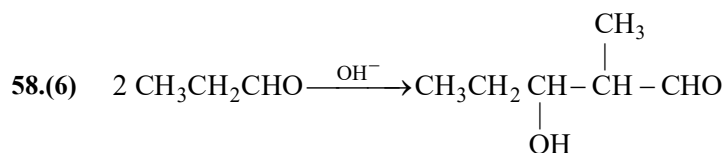


- 55.(3) Disproportionation reaction of formaldehyde with KOH is Cannizzaro reaction. Cannizzaro reaction follows third order kinetics.

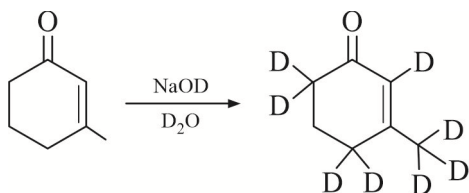
$$r = K[\text{HCHO}]^2[\text{KOH}]$$

- 56.(9) Except (v) all other reactions are used to produce benzaldehyde. Vicinal disubstituted alkene on reaction with  $\text{O}_3/\text{H}_2\text{O}$  form carboxylic acid because it is oxidative ozonolysis reaction.

- 57.(9) Except (viii) all other reaction product is an aromatic compound.

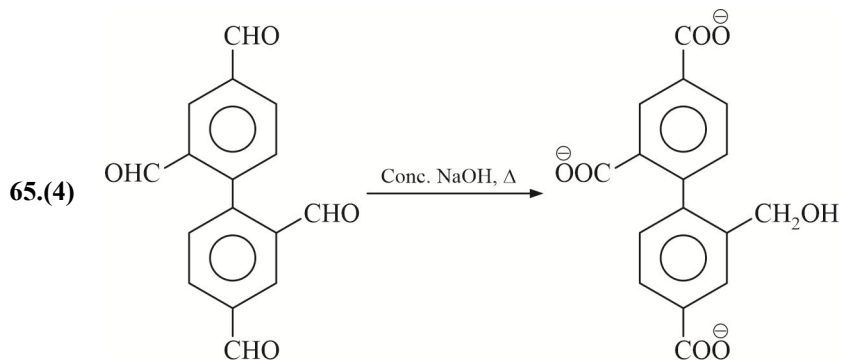
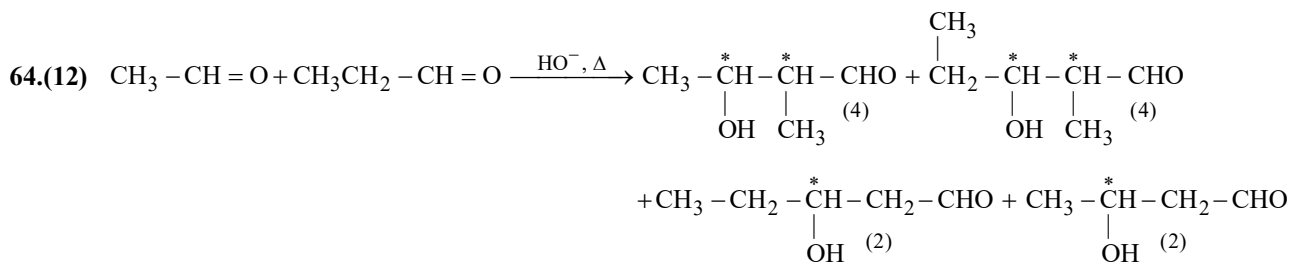
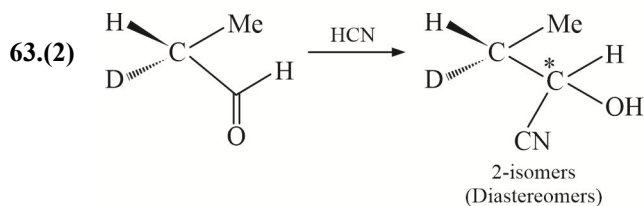
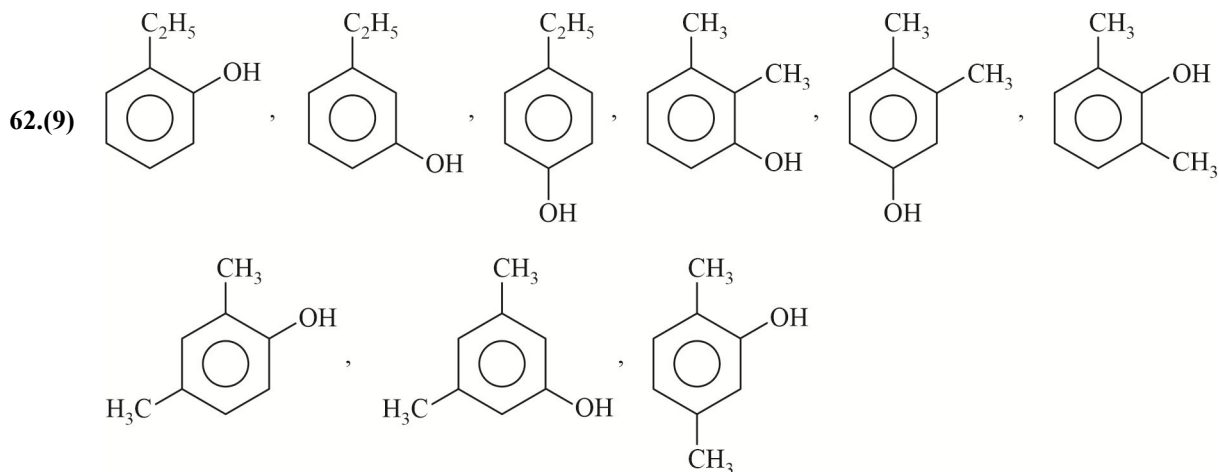
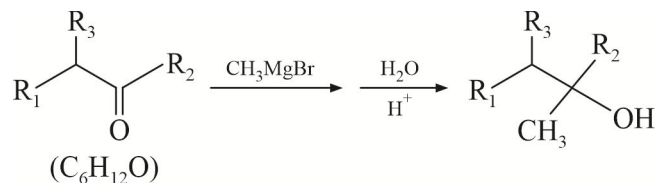


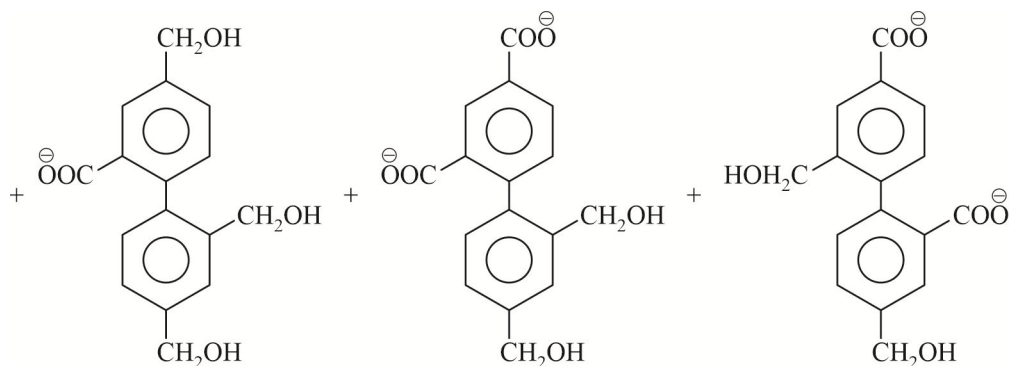
- 59.(8) All  $\alpha$  and  $\gamma$  H atoms (in conjugation with  $\text{C}=\text{O}$  group) will be exchanged with D atoms.



60.(5) Electron withdrawing group increases rate of nucleophilic addition reaction and aldehyde is more reactive than ketone.

61.(0) Only racemic mixture or single stereo isomer is formed. Ketone is  $C_6H_{12}O$ .

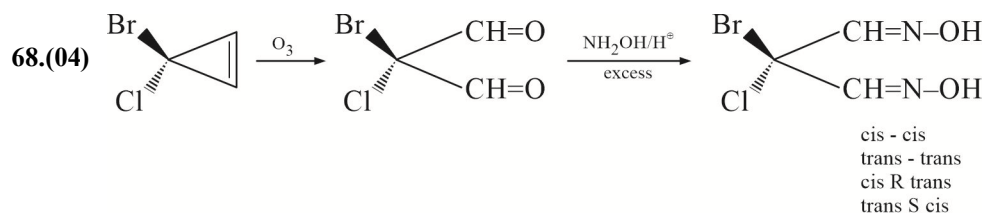




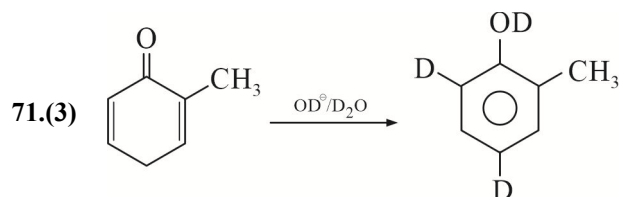
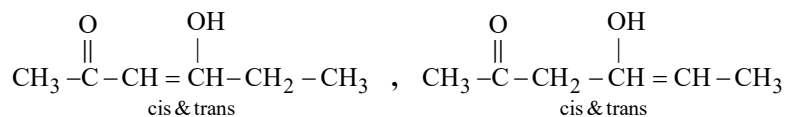
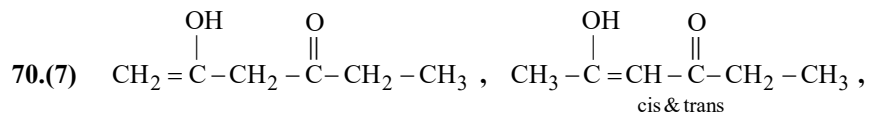
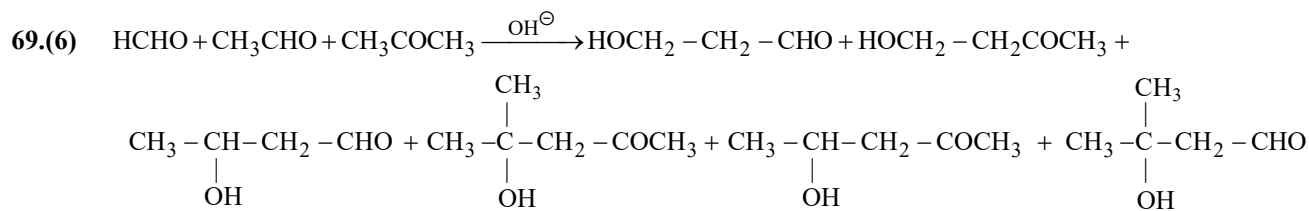
66.(3) 4, 5, 6

67.(12) X has 3 stereocentres with similar ends. Hence, 6 stereoisomers.  $Y = \text{CH}_3 - \underset{\text{D}}{\text{CH}} - \text{CHO}$  has only one stereocentre,

hence 2 stereoisomers.  $Z = \text{CH}_3 - \underset{\text{D}}{\text{CH}} - \text{CH} = \text{NOH}$  has 2 stereocentre, hence 4 stereoisomers.



Total isomeric product = 4



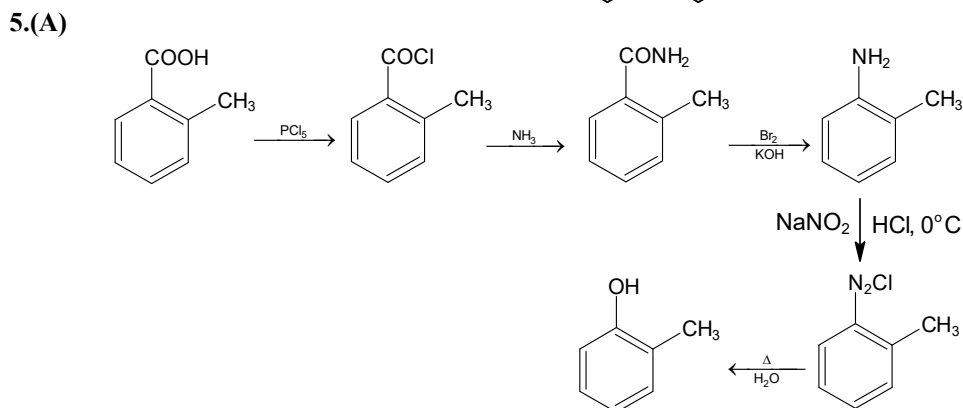
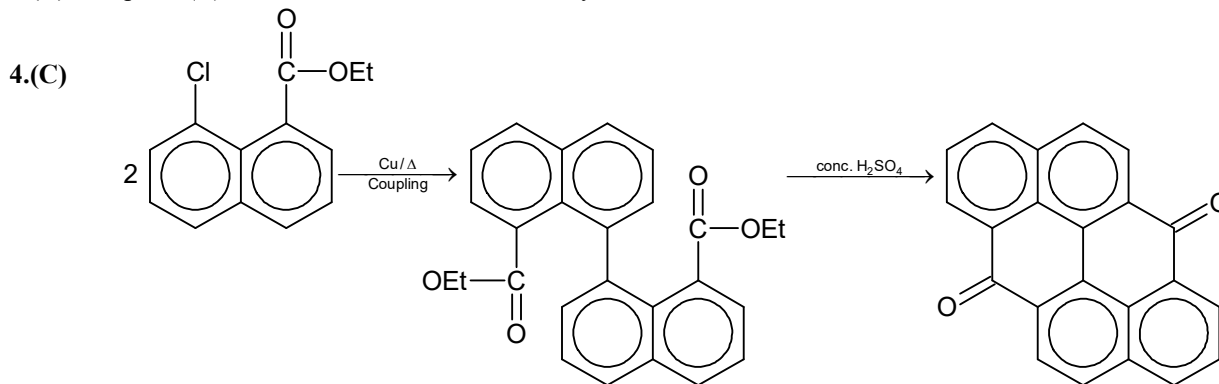
## Solutions - Advanced Problem Package | Chemistry

### Oxygen Containing Organic Compound-III

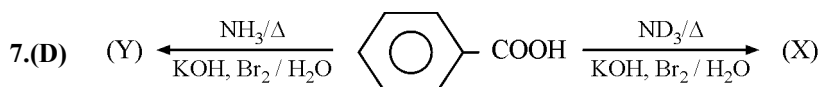
1.(B) Elimination of CO takes place on heating  $\alpha$ -keto acids.

2.(B) Elimination of CO<sub>2</sub> takes place on heating  $\beta$ -keto acids.

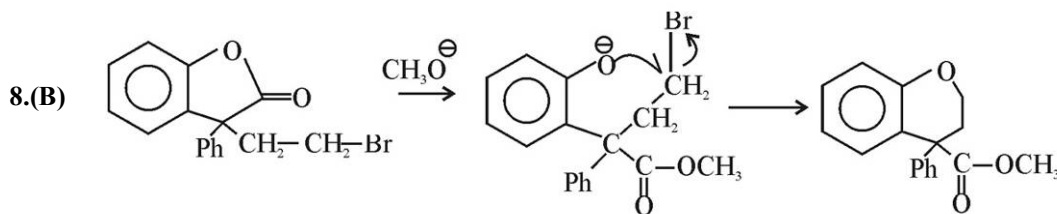
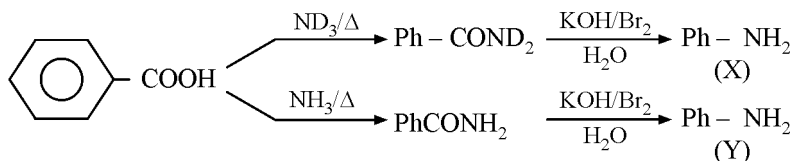
3.(C) Option (C) is chiral and it does not decarboxylate.

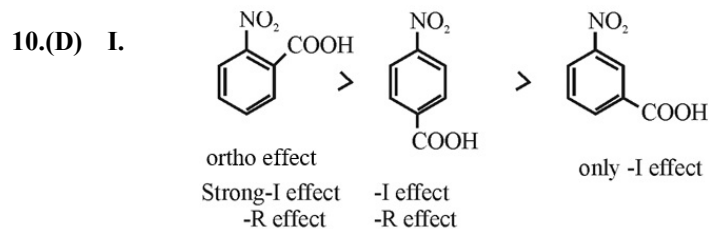
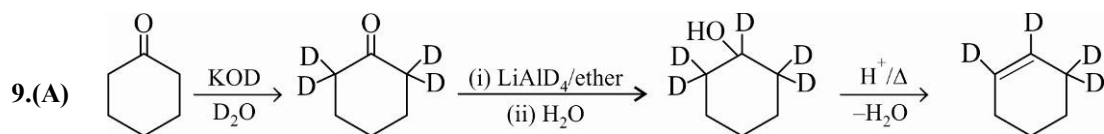


6.(B) Hoffmann Bromamide reaction takes place towards left end and Bromoform reaction takes place towards right end.



Both (X) and (Y) will be c1ccccc1C(=O)N





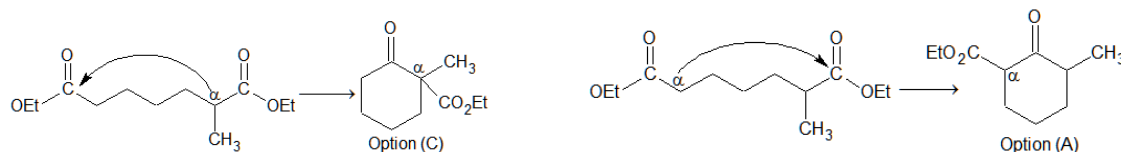
II. Higher the acidity of an acid lower the basicity of its conjugate base.

III. Higher the stability of an alkene lower the heat of hydrogenation.

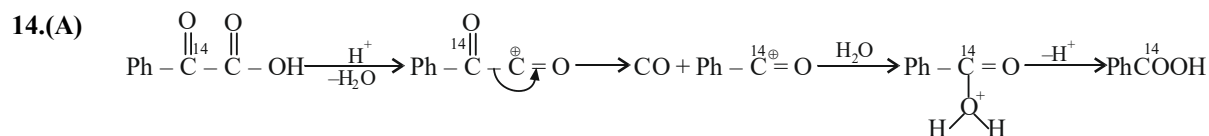
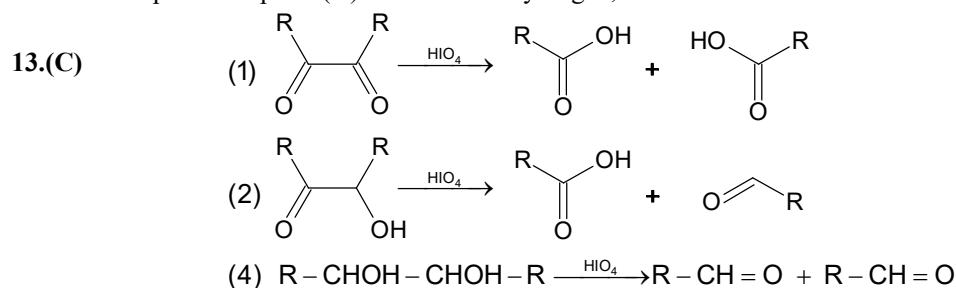
IV.  $\text{Ph}-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{C}-\text{COOH}$  is the  $\beta$ -keto acid so it would be having maximum ease for decarboxylation.

11.(B) This is an example of Cross – Cannizzaro reaction. The molecule first attacked is converted to sodium salt of benzoic acid in a Cannizzaro reaction. The presence of  $-\text{NO}_2$  group activates the  $-\text{CHO}$  group towards nucleophilic attack. Therefore nitrobenzaldehyde is preferentially attacked.

12.(A) The two probabilities are:



Option (C) lacks an acidic  $\alpha$ -hydrogen, which is required to drive equilibrium of Dieckmann's reaction towards completion. Option (A) has an acidic hydrogen, so it is the correct answer.



15.(ABCD)

(A)  $\text{R}-\text{C}\equiv\text{N} \xrightarrow[\text{(ii) H}_3^+\text{O}]{\text{(i) Sn+HCl}} \text{Product (Stephen Reaction)}$

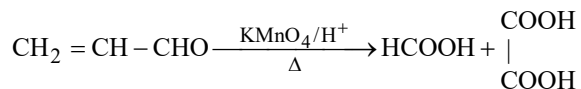
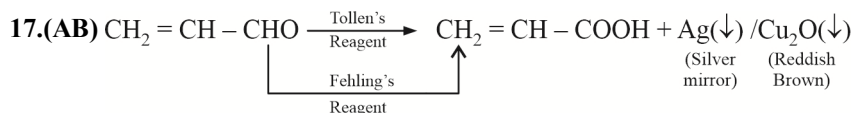
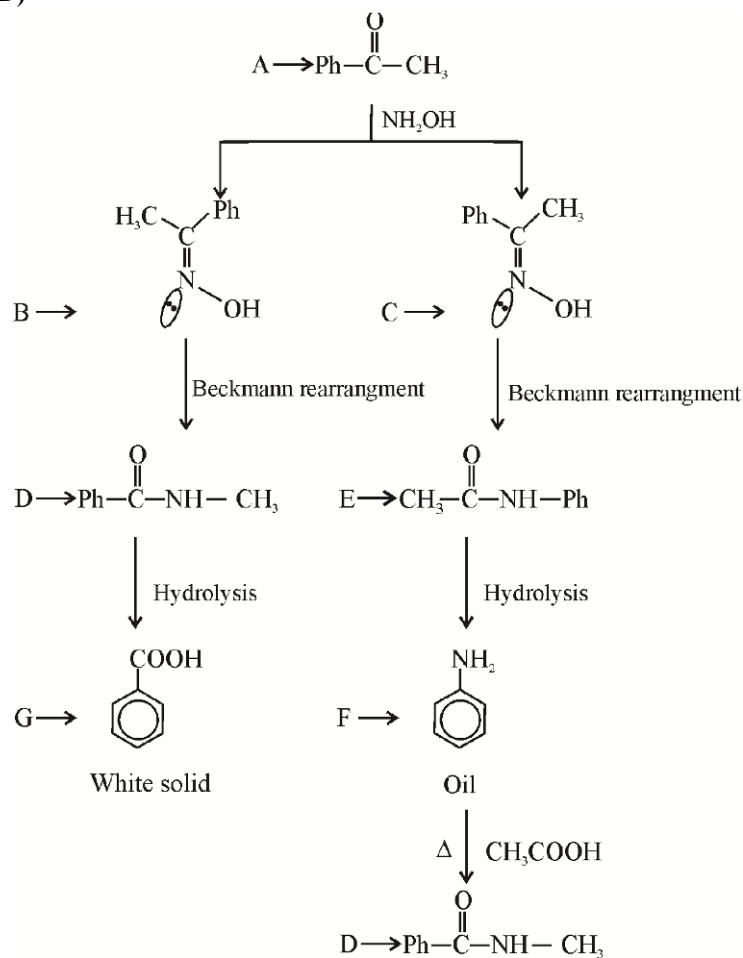
(B)  $\text{RCH}_2\text{OH} \xrightarrow[\text{CH}_2\text{Cl}_2]{\text{PCC}} \text{Product R-CHO}$

(C)  $\text{RCOCl} \xrightarrow{\text{LiAlH}(\text{OBu})_3} \text{Product R-CHO}$  due to bulky nature of  $\text{LiAlH}(\text{OBu})_3$   
RCHO further does not react to form alcohol

(D)  $\text{RCOCl} + \text{H}_2 \xrightarrow[\text{S or Quinoline}]{\text{Pd-BaSO}_4} \text{Product (Rosenmund reaction)}$  all above reaction are used to form aldehyde only. Hence the (A, B, C, D)



16.(BCD)



In (C), MPV reaction cannot be used for this purpose, as it requires a secondary alcohol compound, and also it perform reduction of aldehyde, not oxidation. So, A and B are correct.

18.(ABC) A is isobutyric ester.

19.(BCD) The above reaction is Hoffmann Bromamide reaction.

20.(AC) (B) is long chain and (D) is trans isomer

21.(BC) Baeyer Villiger followed by hydrolysis of ester

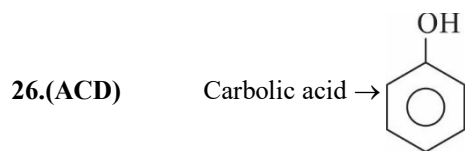
22.(ABCD)  $\beta$ -keto acid, gem-dicarboxylic acids,  $\alpha$ -Nitro-carboxylic acid,  $\beta$ ,  $\gamma$ -unsaturated carboxylic acid on heating decarboxylate to give CO<sub>2</sub>

23.(AC) Esterification in (A) and amide formation in (C)

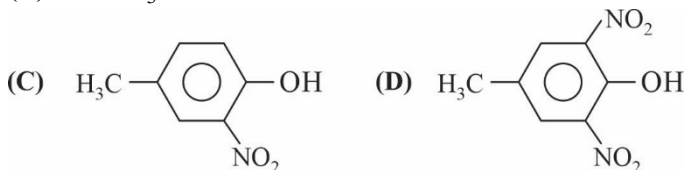
24.(ABC) Oxidation of side chain gives carboxylic acid. In option (D) O-atom attached cannot be oxidised to -COOH.

25.(ACD)(A)  $\bar{e}$  withdrawing group increases the acidic strength ( $-\text{NO}_2 > -\text{F}$ )

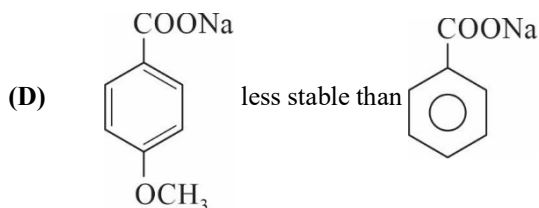
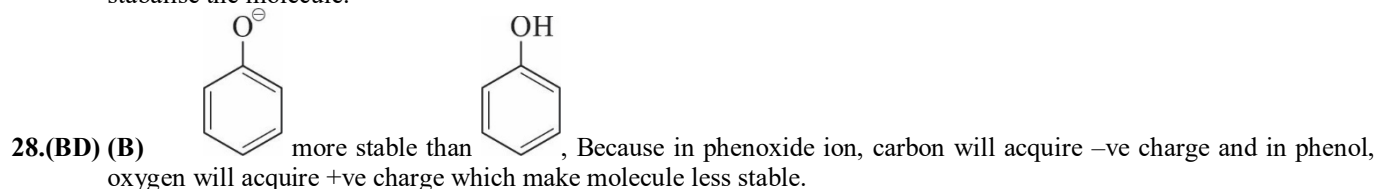
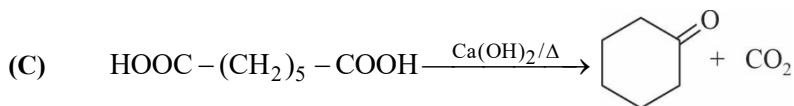
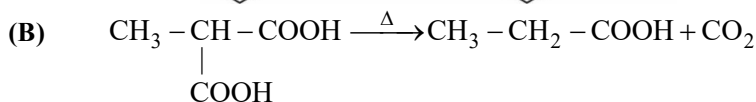
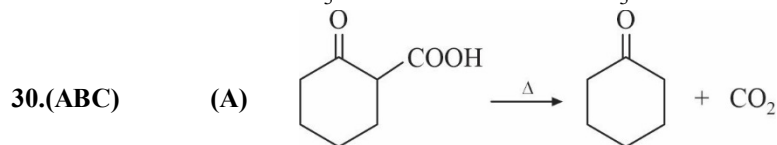
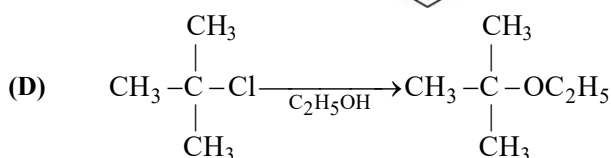
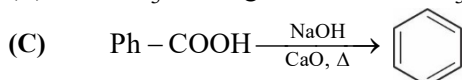
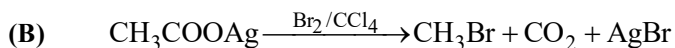
$\bar{e}$  donating group decreases the acidic strength.

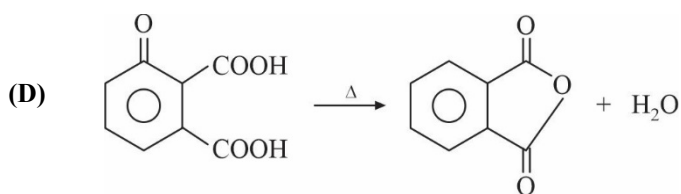
(C)  $(-\text{NO}_2 > -\text{F}) \rightarrow -\text{I}$  effect(D)  $-\text{CH}_2\text{CH}_3$  show  $+\text{I}$  effect

This is less acidic than

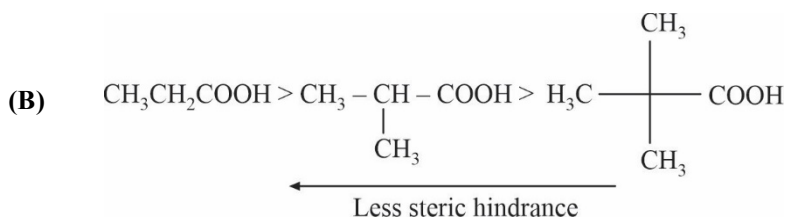
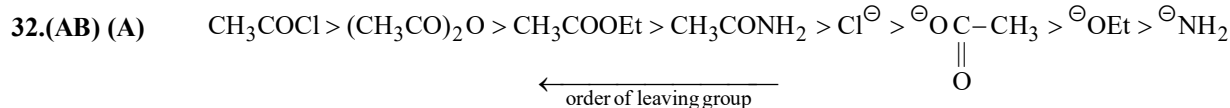
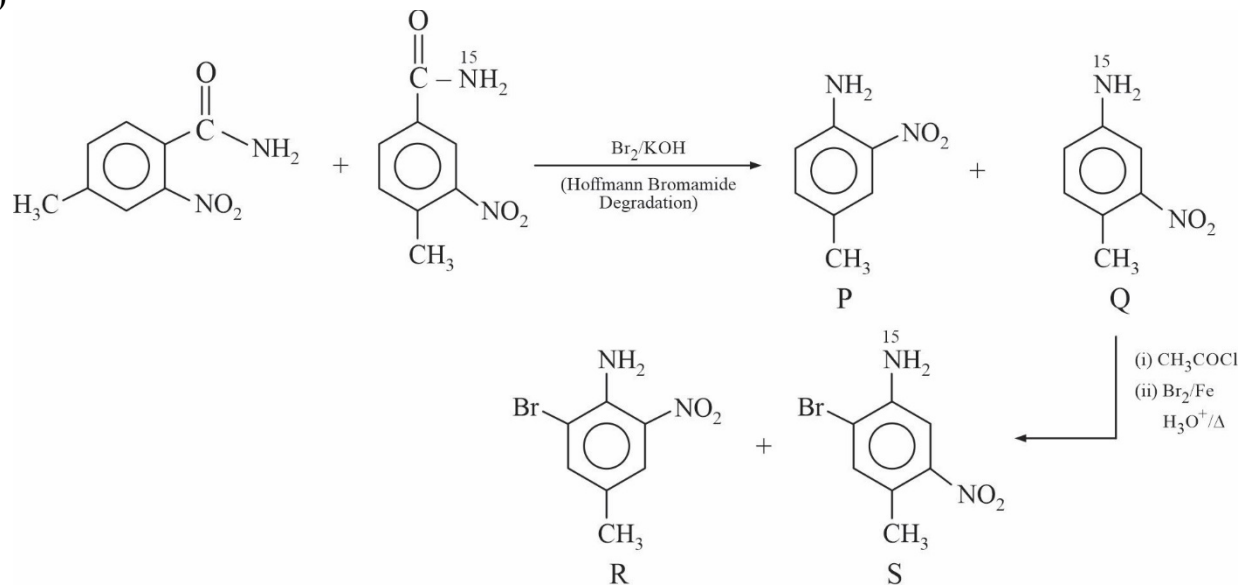
(A)  $\text{CH}_3\text{COOH}$  $\bar{e}$  withdrawing group increases the acidic strength

27.(BC) It is ascorbic acid and  $\text{H}_\text{b}$  is more acidic because after deprotonation,  $\text{O}^-$  will show delocalization which will stabilise the molecule.

Because  $-\text{OCH}_3$  group shows  $+\text{M}$  effect which will destabilise the  $-\text{COO}$  group.



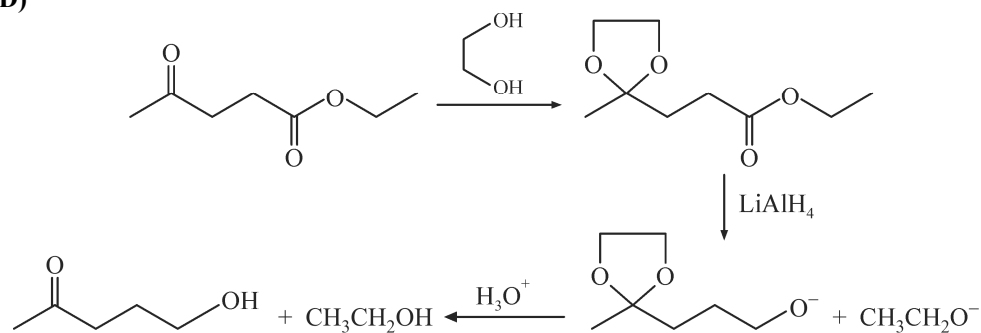
31.(BC)



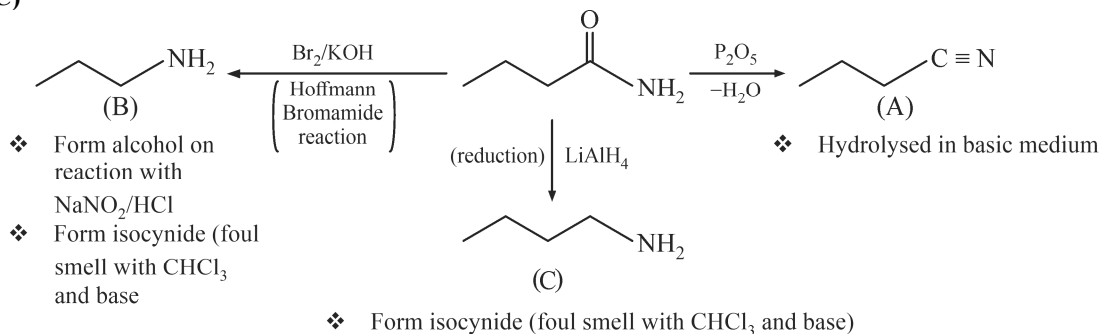
33.(ABCD)

- (A)  $\text{LiAlH}_4 \Rightarrow$  Reduce Ester, Acid, Amide, Ketone & Nitro  
 (B)  $\text{NaBH}_4 \Rightarrow$  Reduce only Ketone  
 (C)  $\text{Na} / \text{C}_2\text{H}_5\text{OH} \Rightarrow$  Reduce Ester and Ketone  
 (D) MPV reduction is only for Ketone

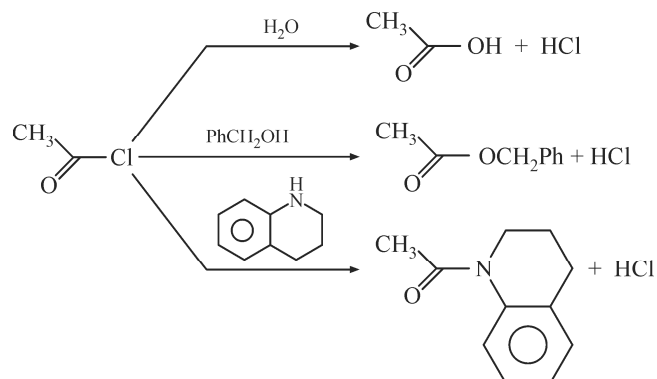
34.(ACD)



35.(ABC)

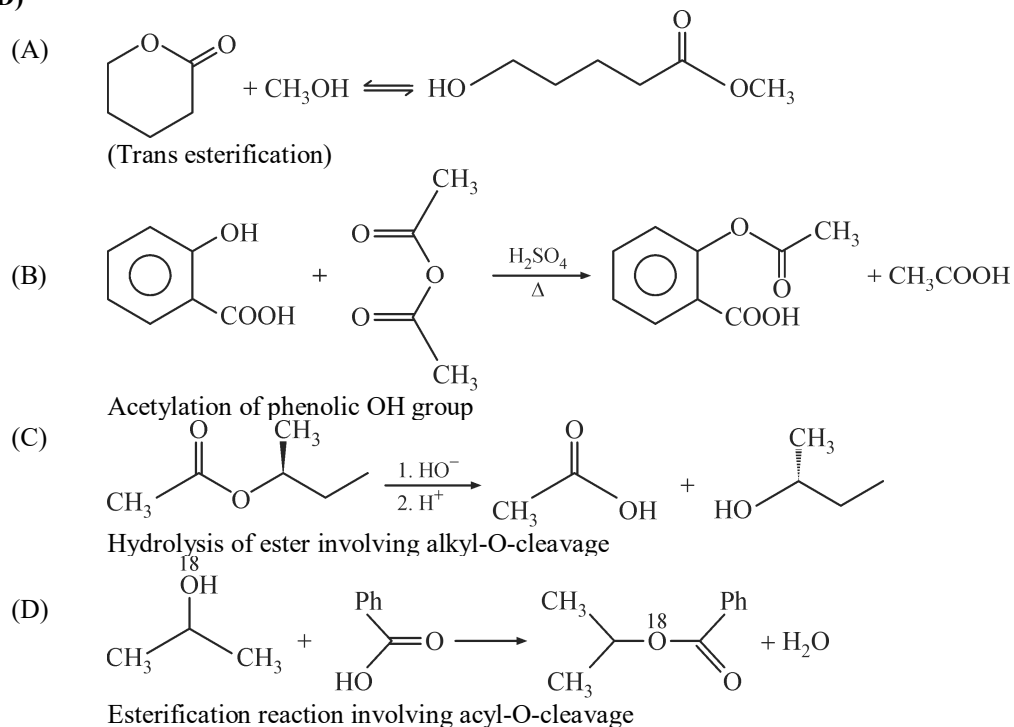


36.(C)



No reaction with 3° amine

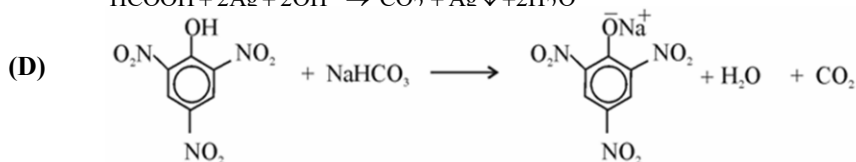
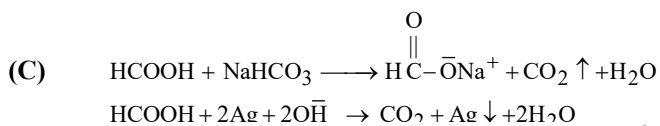
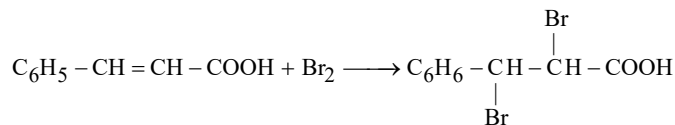
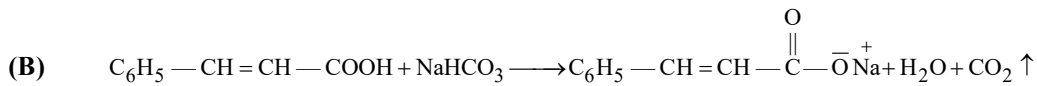
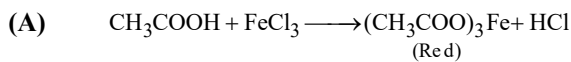
37.(ABD)



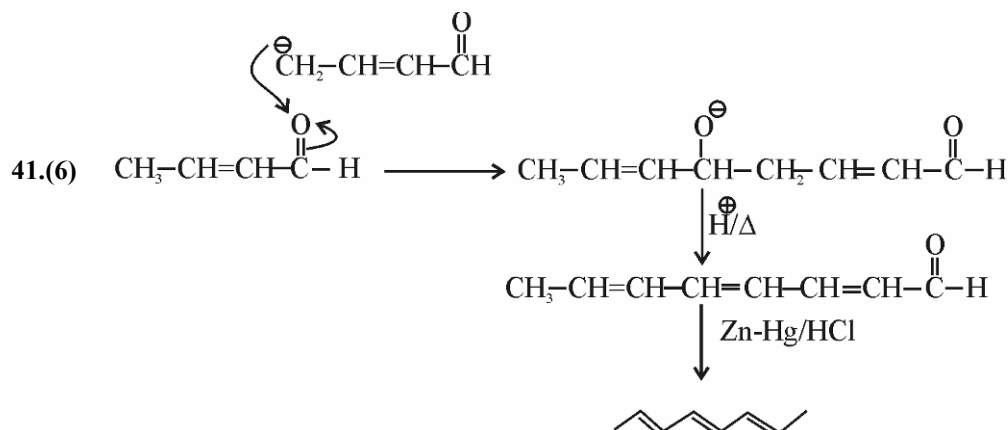
38.(ABCD)

$\text{LiAlH}_4$  can reduce ketone, cyanide, ester and anhydride groups.  $\text{NaBH}_4$  can reduced only ketone.

39. [A → p, r; B → p, q; C → p, s; D → p, r]

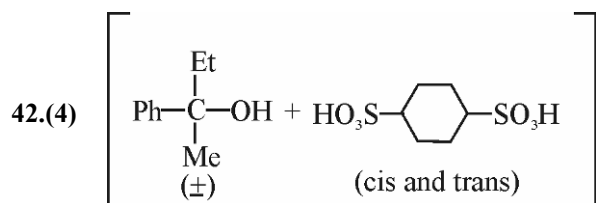


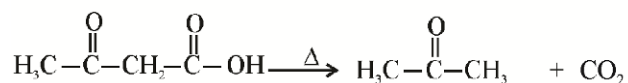
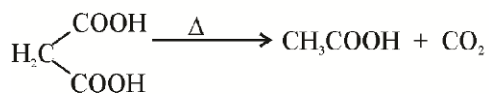
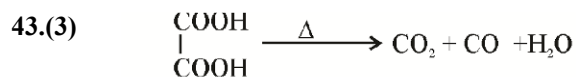
40. [A → r; B → s; C → q; D → p]



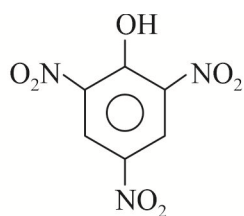
No. of stereo isomers are

cis, cis, cis  
 cis, cis, trans  
 cis, trans, cis  
 cis, trans, trans  
 trans, cis, trans  
 trans, trans, trans

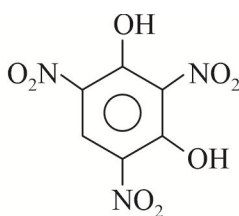




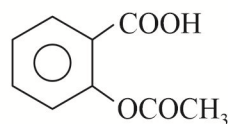
44.(4) Aspirin, Penicillin, Valeric acid and Anthranilic acid contain carboxylic acid functional group.



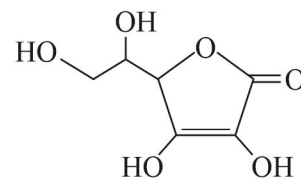
Picric acid



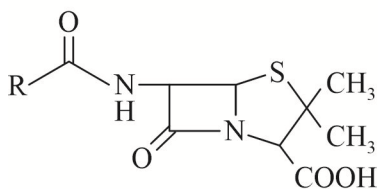
Styphnic acid



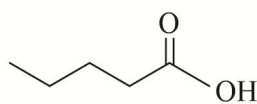
Aspirin



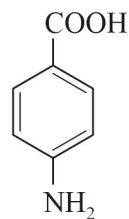
Ascorbic acid



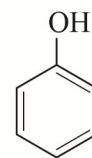
Penicillin



Valeric acid



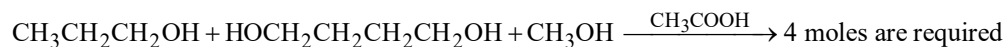
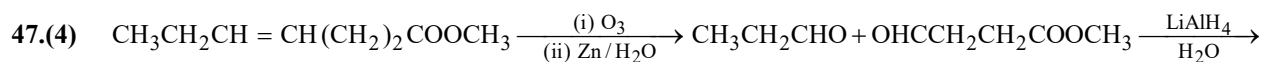
Anthranilic acid



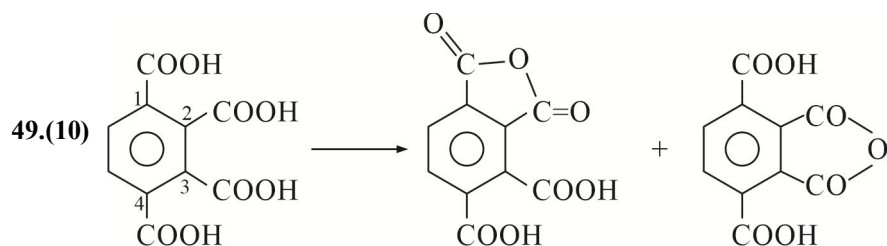
Carboic acid

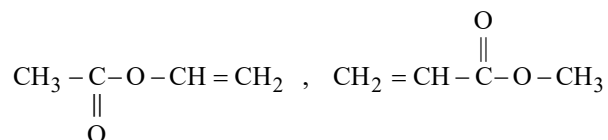
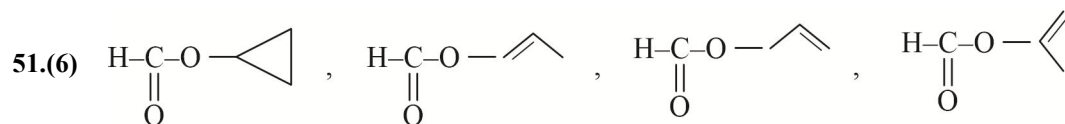
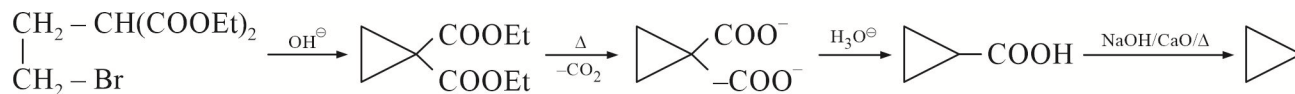
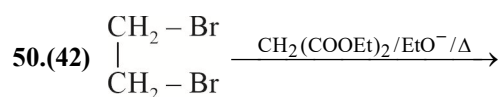
45.(22)

46.(21)

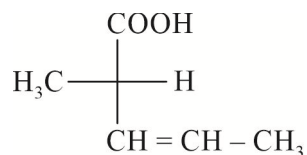


48.(4) 5, 1, 3, 8 only.

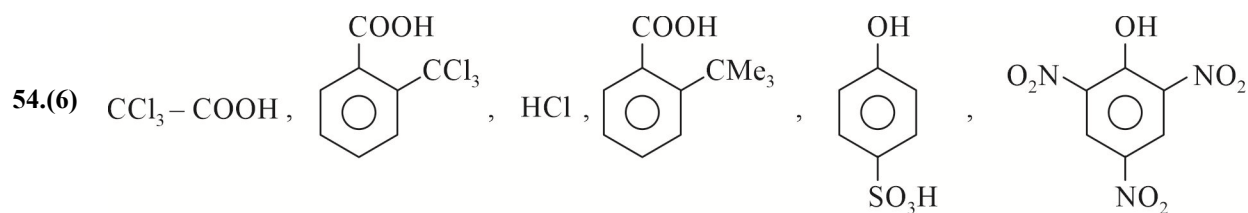
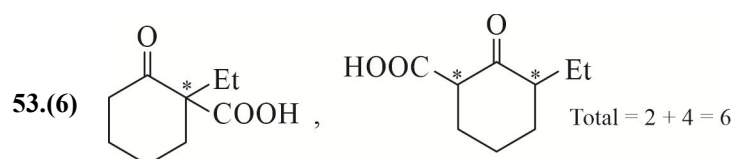




52.(6) The smallest optically active alkenoic acid which can also show geometrical isomerism can be written as



Hence its molecular formula is  $\text{C}_6\text{H}_{10}\text{O}_2$ .

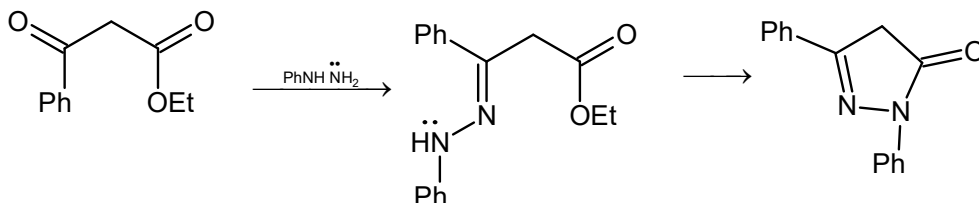


## Solutions - Advanced Problem Package | Chemistry

### Nitrogen Containing Organic Compound

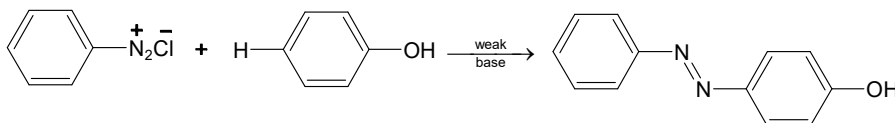
1.(B) Diazotization coupling takes place at most activated position of ring.

2.(B)

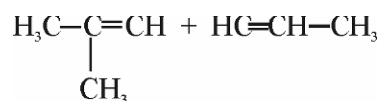


3.(C) Beckmann's rearrangement

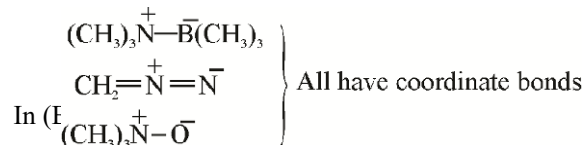
4.(B)



5.(C) Product mixture:

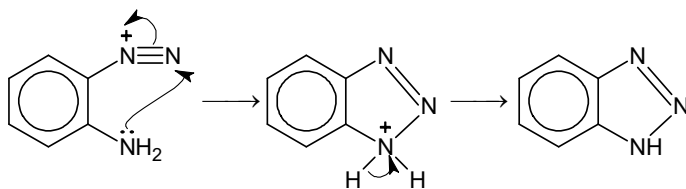


6.(B)

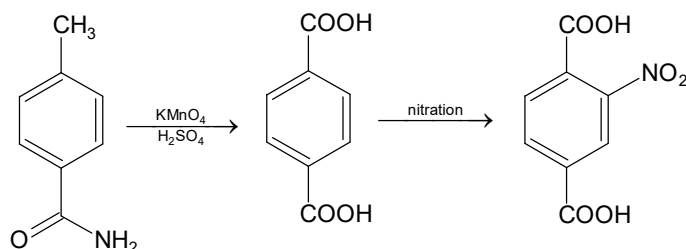


7.(D)  $\text{N}^2$  is pyrrole and is least basic due to conjugation while  $\text{N}^3$  is most basic due to being amine and +I effect.

8.(C)

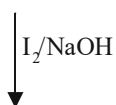
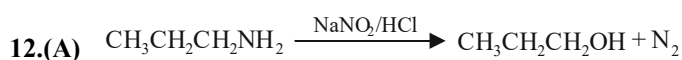
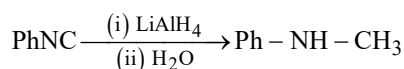
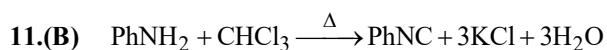
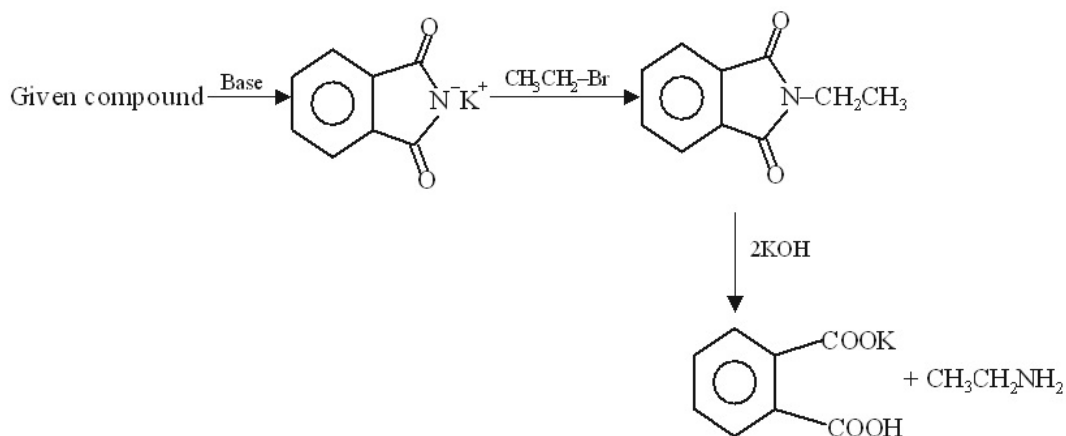


9.(A)

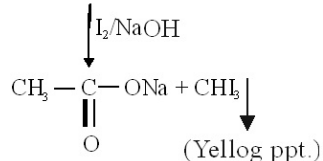
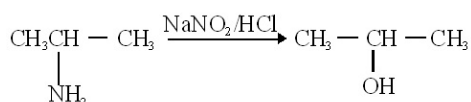




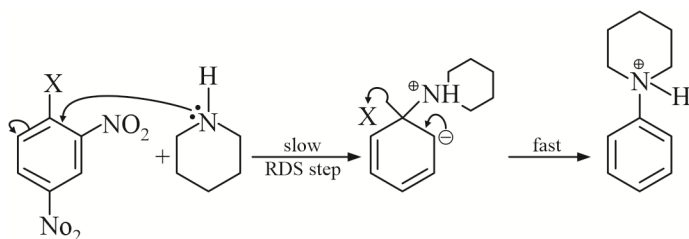
10.(A)



No iodoform test



13.(BCD) This is a nucleophilic substitution reaction on aromatic compound, whose EWG  $-\text{NO}_2$  at ortho and para position increases the rate of reaction, by increasing the electrophilicity of the ring.



Here, RDS step is step-I because here aromaticity is lost. Order of reactivity is  $\text{F} > \text{Cl} > \text{Br} > \text{I}$ , because exceptionally high electronegativity of F increase electrophilicity and so, the rate of nucleophilic attack is maximum with  $\text{X} = \text{F}$ . So, the correct choice are B, C, D.

14.(BCD) (A) on reduction form secondary amines

15.(AB) Amides are less basic than amines.

16.(BCD) Aliphatic  $1^\circ$  amine released  $\text{N}_2$  on treatment with  $\text{HNO}_2$ .

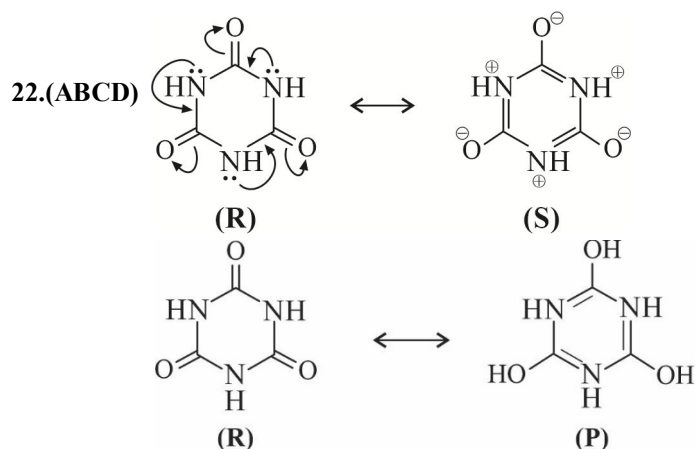
17.(CD) Substituted aryl amines do not give this reaction.



19.(BC) In reaction 1, elimination product is governed by Hoffmann's rule.  
In reaction 2, elimination product is governed by Saytzeff's rule.

20.(CD) N-nitroso amine is formed by 2°-amine.

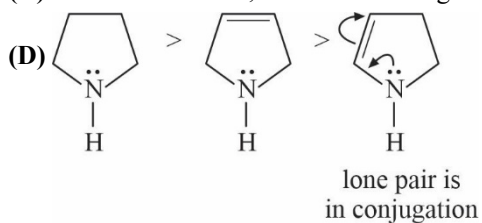
21.(ACD) Tollen's test is given by aldehydes.



Show both resonance and tautomerism

- 23.(BD) (B) (I = V)  $\rightarrow$  Kekule structure  
(II = IV)  $\rightarrow$  same structure  
(I = V) > (II = IV)  $\rightarrow$  Neutral molecule is more stable than charged molecule.  
(D) (II = IV) > III  $\rightarrow$  Opposite charge at less distance is more stable

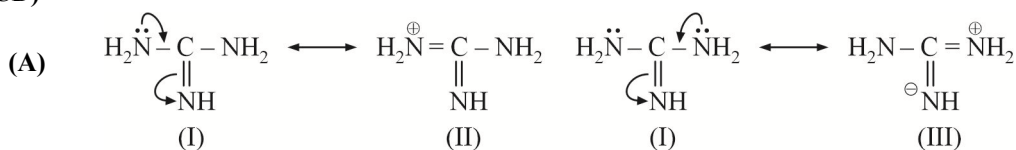
- 24.(BCD) (B) More +I effect, more basic strength,  
(C) less delocalization, more basic strength



25.(ABCD)

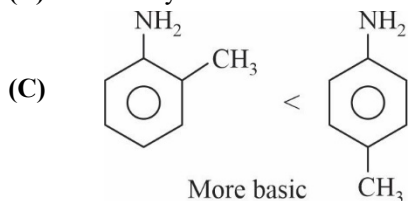
- (A) Both nitrogen of pyrimidine having same basic strength because both are  $\text{sp}^2$  hybridized and having No delocalization.  
(B) N-3 having no delocalization, So it's lone pair are more available for protonation.  
(C) Only N-9 is able to delocalize  
(D) All molecules are aromatic.

26.(ABCD)



Both II and III having equal contributing resonating structure.

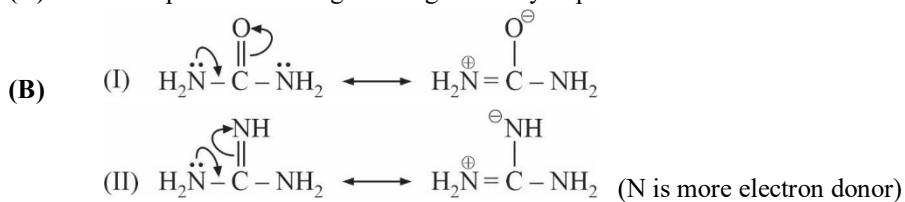
(B) Diethyl amine will have more H-bonding.



(D) Because of +I effect of methyl groups

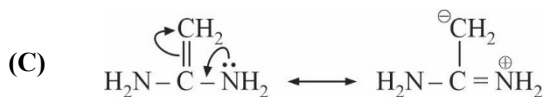
27.(ABCD)

(A) Lone pair on the bridged nitrogen is easy to protonate



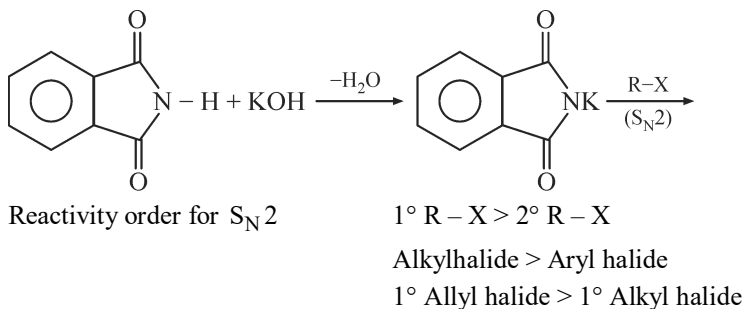
Secondary amine is more basic than primary amine

28.(BCD)

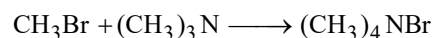
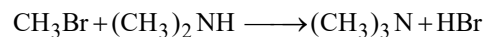
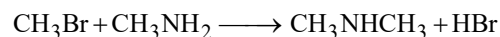
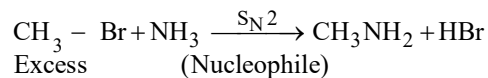
(B)  $\text{CH}_3\text{COOH}$  is more acidic than  $\text{Ph}-\text{OH}$ 

(D) Lone pair of Nitrogen is in conjugation. So it will not be available for protonation.

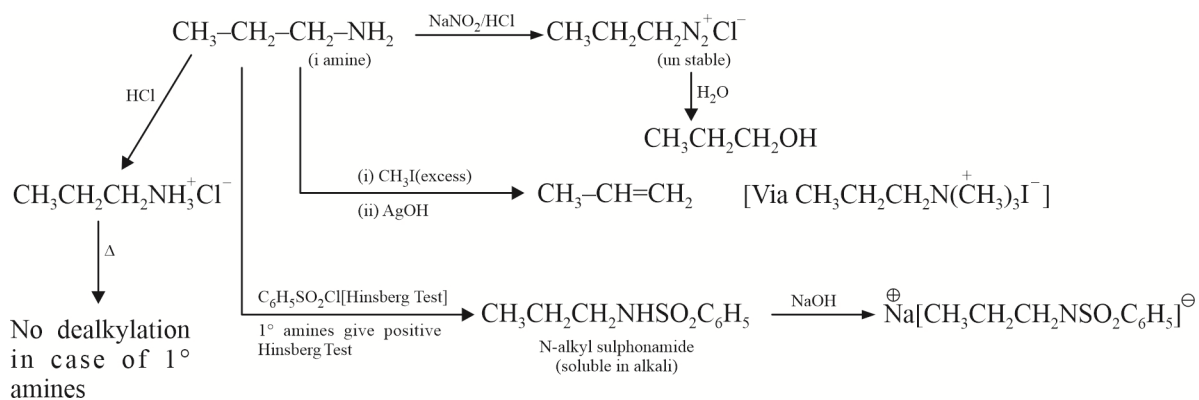
29.(CD)



30.(AB)

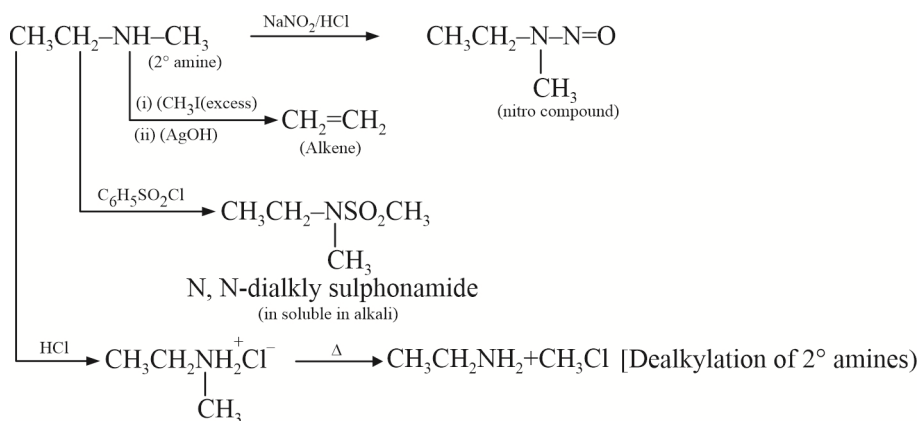


31.  $A \rightarrow r, s, t$ ;  $B \rightarrow p, r, s$ ;  $C \rightarrow s$ ;  $D \rightarrow q, t$   
(A)



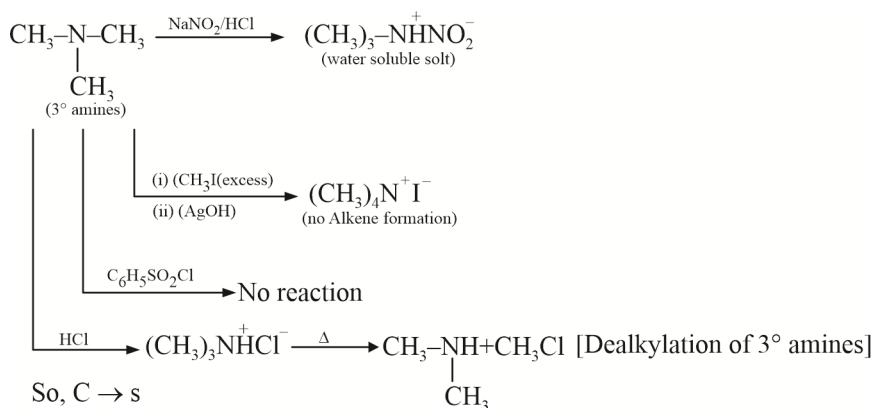
So,  $A \rightarrow r, t$ .

(B)



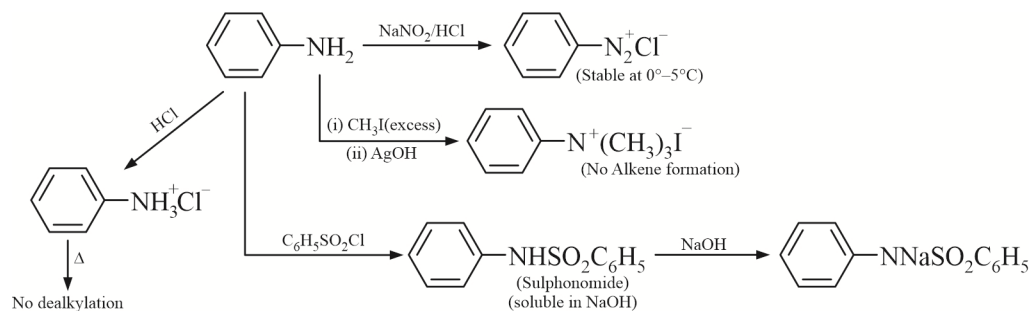
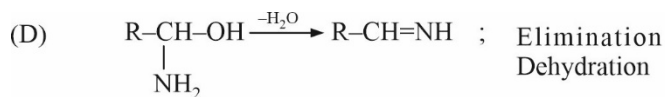
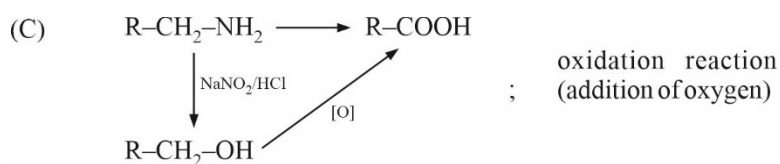
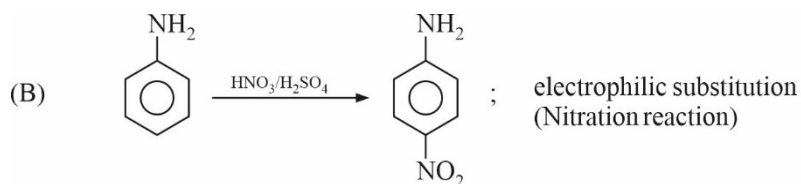
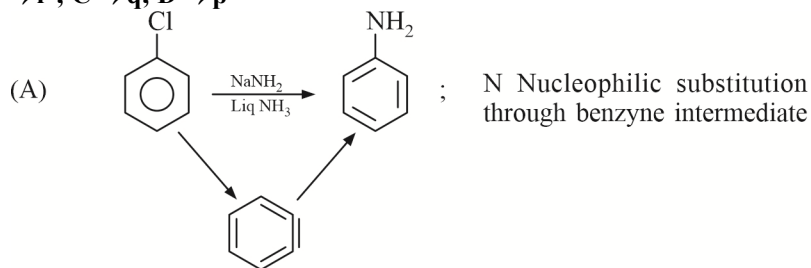
So,  $B \rightarrow p, r, s$

(C)

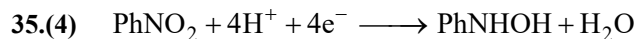
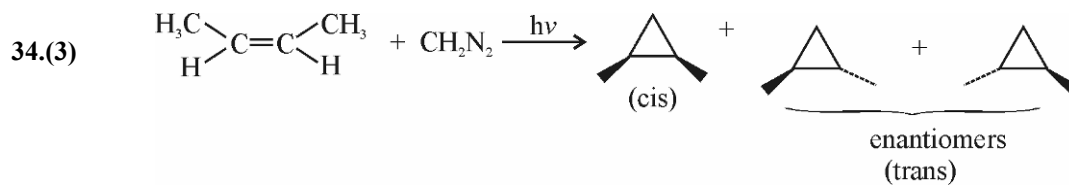


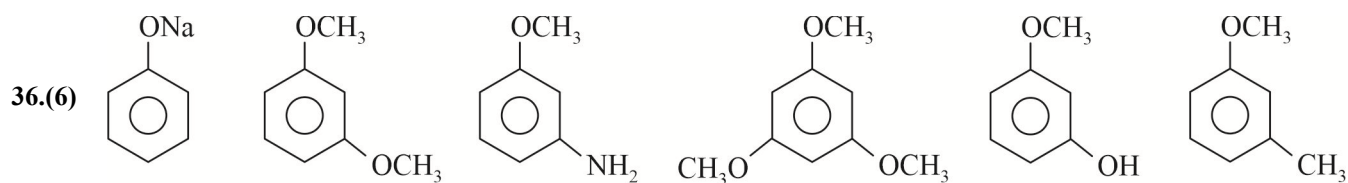
So,  $C \rightarrow s$

(D)

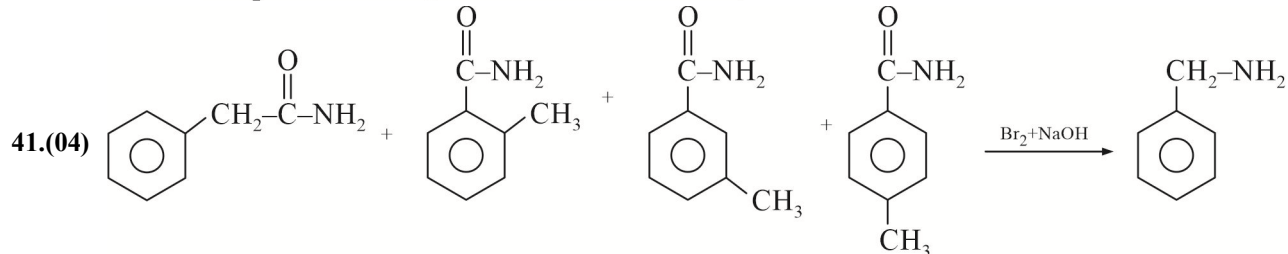
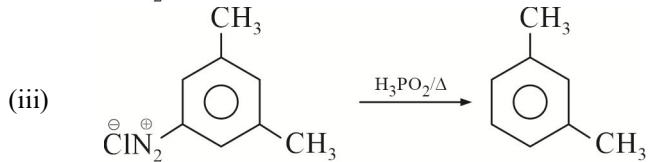
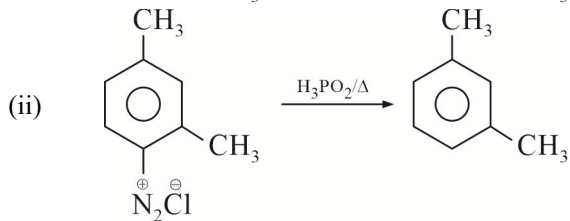
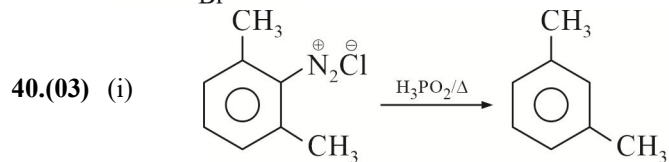
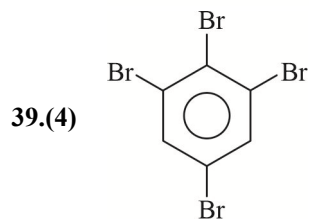
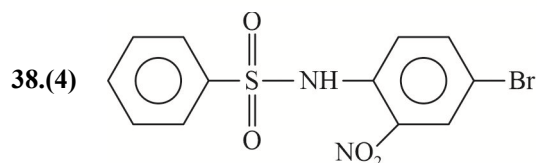
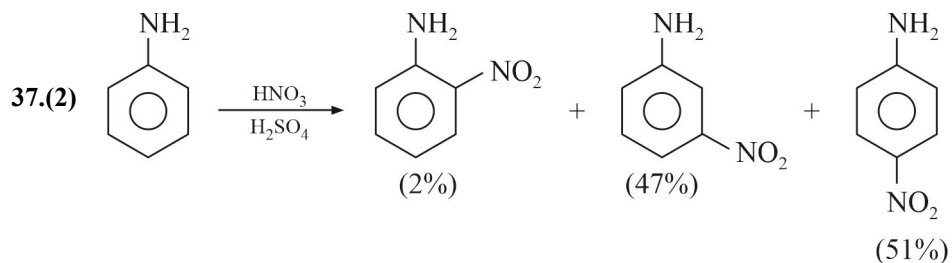
So, D  $\rightarrow$  q, t32. A  $\rightarrow$  s; B  $\rightarrow$  r; C  $\rightarrow$  q; D  $\rightarrow$  p

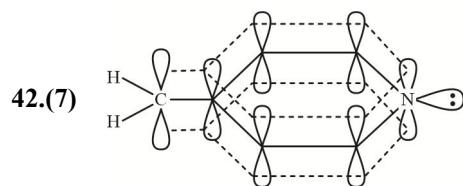
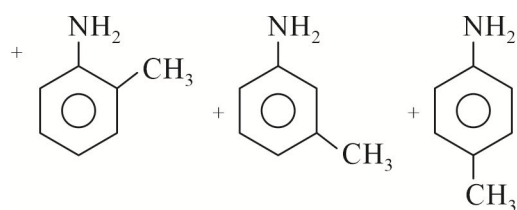
33.(3) It is Hofmann's Bromamide reaction and in this reaction cross over products are not obtained.



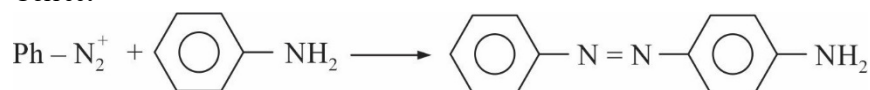


Presence of electron releasing group in anisole will make it more reactive toward coupling reaction with diazonium salt.

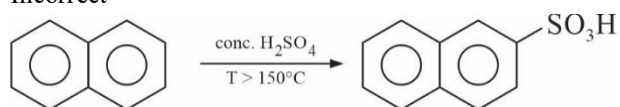




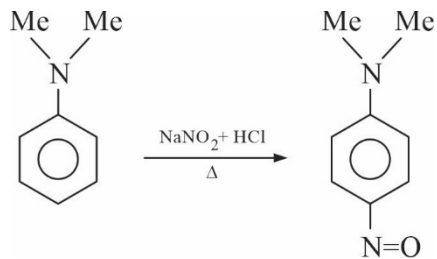
43.(2) (i) Correct



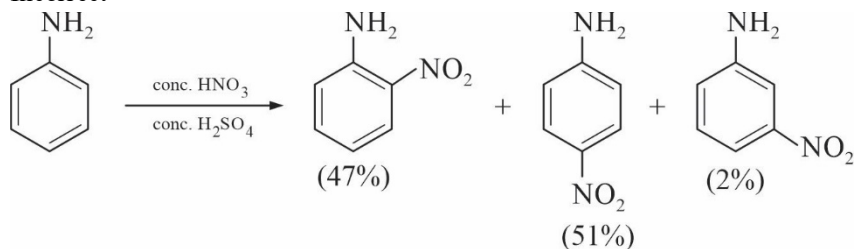
(ii) Incorrect



(iii) Correct



(iv) Incorrect



## Solutions - Advanced Problem Package | Chemistry

### Surface Chem, Biomolecules, Practical Organic Chem & Polymers

1.(B) Phenols give violet colour with  $\text{FeCl}_3$  solution while alcohols do not.



$$n_{\text{NH}_3} = 1 = n_{\text{RCONH}_2}$$

$$\therefore \text{Mo}(\text{amide}) = 59\text{g}$$

$$\therefore \text{Mo}(\text{acid}) = 59 - 15 + 16 = 60\text{g i.e. CH}_3\text{COOH}$$

3.(A)  $\%C \Rightarrow 38.7 \Rightarrow \frac{38.7}{12} = 3.225\text{ mol}$

$$\%H \Rightarrow 16.17 \Rightarrow \frac{16.17}{1} = 16.17\text{ mol}$$

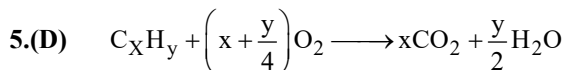
$$\%N \Rightarrow 45.17 \Rightarrow \frac{45.17}{14} = 3.225\text{ mol}$$

Divide by 3.225

$$C = 1, H = 5, N = 1$$

so, empirical formula is  $\text{CNH}_5$  i.e.  $\text{CH}_3\text{NH}_2$

4.(D) CAN is a test of  $-\text{OH}$  group and not  $-\text{COOH}$  group



$$\text{Moles of organic compound} = \frac{20}{12x + y}$$

$$\therefore \text{moles of H}_2\text{O} = \frac{10y}{12x + y} = \frac{10}{18} \Rightarrow 18y = 12x + y \quad \therefore 17y = 12x$$

$$\% H = \frac{y}{12x + y} \times 100 = \frac{\frac{12x}{7}}{12x \left[1 + \frac{1}{17}\right]} \times 100 = \frac{1}{18} \times 100 = 5.55\%$$

or Mass of Hydrogen ( $\text{H}_2$ ) in the sample =  $\frac{10}{18} \times 2\text{g}$  ( $\because$  after combustion all the hydrogen in the hydrocarbon will

$$\text{be present in the water formed}) \Rightarrow \% \text{ of hydrogen} = \frac{\frac{10}{18} \times 2}{20} \times 100 = 5.55\%$$

6.(C) Mass of sulphur =  $\frac{0.35}{233.43} \times 32\text{g} \Rightarrow \% \text{ sulphur} = \frac{0.35 \times 32}{233.43} \times 100 = 18.5\%$

7.(D) Mass of nitrogen =  $200 \times 10^{-3} \times 14\text{g} \Rightarrow \% \text{ of nitrogen} = \frac{(200 \times 1) \times 10^{-3} \times 14}{6} \times 100 = 46.67\%$

8.(A)  $-\text{OH}$  group below the plane of paper at C-1 in Haworth projection

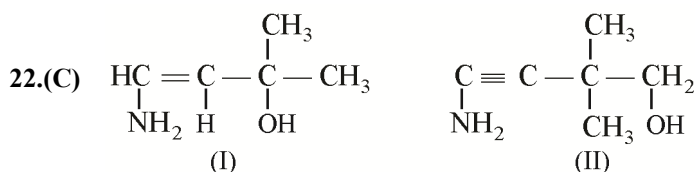
9.(C) Other three have aldehyde linkage.

10.(A) Mutarotation is an equilibrium phenomenon of  $\alpha$  - form, open chain &  $\beta$  - form

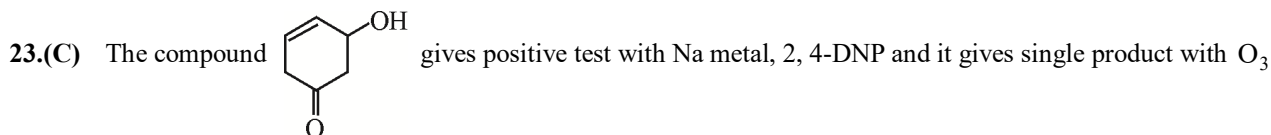
11.(C) Muta rotation result (Learn as a fact)



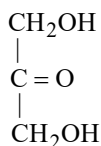
- 12.(D) Adding NaCl will not cause any change in the pH of the solution.
- 13.(D)  $[H^+] = \sqrt{K_a1 K_a2}$  as  $[(\text{conjugate acid})] = [\text{conjugate base}]$
- 14.(A) Silk is a natural product made from the fibres found in the cocoon of the mulberry silk worm. It is actually made up of long strands of protein sericin and fibroin which are polymers of amino acids.
- 15.(C) Vulcanised rubber contains linear chains having some branches.
- 16.(C)  $\text{CH}_3 - \text{CH}_2 - \text{NH}_2$  has only one functional group, hence it can't undergo self-polymerisation
- 17.(B) Polyethylene, PVC and polycrylonitrile (orlon) are addition polymers.
- 18.(B) Steric crowd makes  $\text{NaHSO}_3$  addition difficult around carbonyl group.
- 19.(C) Because C-I of both units are linked through glycoside linkage.
- 20.(D)  $\text{Fructose} \xrightleftharpoons{\text{OH}^-} \text{Mannose} \rightleftharpoons \text{Glucose}$  (Fact)
- 21.(C) Because C-I of both units are linked through Glycosidic linkage.

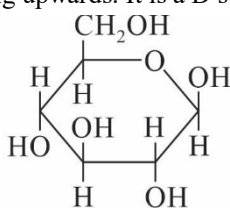
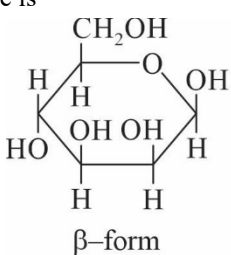
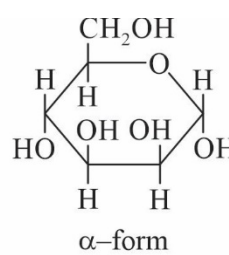


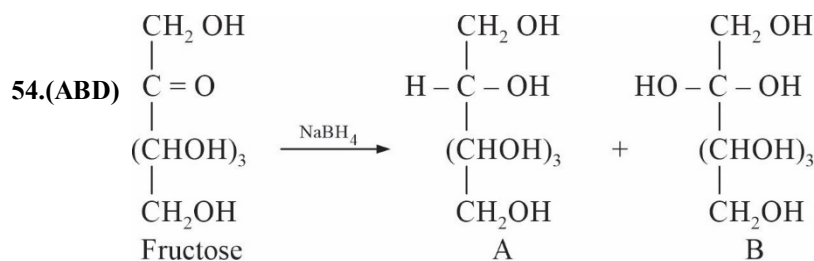
(I) gives immediately turbidity by Lucas reagent and (II) does not give turbidity appreciably.



- 24.(B) The compound  $\text{C}_{30}\text{H}_{60}\text{O}$  has one degree of unsaturation since it gives negative test with  $\text{Br}_2 / \text{H}_2\text{O}$ , 2, 4-DNP and Na metal it must be an epoxide.
- 25.(A) Sodium aluminium silicate (zeolite) acts as an ion exchange medium. Calcium and magnesium cations in hard water are exchanged for the sodium ions of the zeolite.
- 26.(B) Silver sol is an example of lyophobic colloid.
- 27.(A) When Freshly precipitated  $\text{Fe}(\text{OH})_3$  is shaken with little amount of dilute solution of  $\text{FeCl}_3$  electrolyte, peptization process takes place by converting the  $\text{Fe}(\text{OH})_3$  precipitate into positively charged colloidal solution of  $\text{Fe}(\text{OH})_3$ .
- 28.(C) Rubber plating and chrome tanning are based on electrophoresis.
- 29.(A) Gold number  $\propto \frac{1}{\text{protective power}}$
- 30.(C) Both absorption and viscosity are bulk phenomenon.
- 31.(D) Colloidal particle range  $\rightarrow 1\text{nm}$  to  $1000\text{nm}$ .
- 32.(B) An oligosaccharide is a carbohydrate whose molecules are composed of relatively small number of monosaccharide units (typically 3 to 10).
- 33.(C) Nucleic acid is deoxyribose.
- 34.(A) The simplest ketose is dihydroxyacetone or ketotriose which has only 3 carbon atoms and is the only one with no optical activity



- 35.(A) The given unsymmetrical molecule has 4 chiral carbon so it has  $2^4 = 16$  optical isomers.
- 36.(A) Ziegler-Natta polymerization is an example of addition polymerization.
- 37.(D) All three (A, B & C) correctly represents addition polymerization process.
- 38.(A)  $pK_{a3}$  value of side chain determines the nature of amino acid.
- 39.(C) For acidic amino acids  $PI = \frac{pK_{a1} + pK_{a3}}{2} = \frac{1.88 + 3.65}{2} = \frac{5.53}{2} = 2.77$
- 40.(B) For basic amino acids  $PI = \frac{pK_{a2} + pK_{a3}}{2} = \frac{8.95 + 10.53}{2} = \frac{19.48}{2} = 9.74$
- 41.(AC) (A)  $\Delta G = \Delta H - T\Delta S < 0$  as  $\Delta S < 0$  so  $\Delta H = -ve$   
 (B) Micelles formation will take place above  $T_K$  and above CMC  
 (C) Read NCERT  
 (D)  $Fe^{3+}$  ion will have greater flocculability power so smaller flocculating value.
- 42.(AC) By the condition of the question I & IV both compounds do not give iodoform test and I and II both give similar test.
- 43.(AB) Methylaniline is the only base in the mixture and benzoic acid is a much stronger acid than phenol.
- 44.(AC) Physisorption requires low temperature and approx zero activation energy.
- 45.(BC) The pressure decreases due to the adsorption of gas molecules on the surface of charcoal.
- 46.(AB) When positive & negative colloids are added in suitable amounts, mutual coagulation takes place.
- 47.(AC) Sulphur and gold sol  $\Rightarrow$  multimolecular colloid, egg albumin in water = macromolecule colloid, soap solution associated colloid
- 48.(AC) Blue color of sky is due to tyndall effect and ultra microscope is also based on the same principle of light scattering.
- 49.(ACD) The given structure is similar to pyran, so it is said to be in pyranose form. It is a  $\beta$ -anomer since the  $-OH$  at anomeric C is facing upwards. It is a D sugar.
- 50.(AC) D-Glucose is
- 
- So D-Manose is
- 
- 
- 51.(BC) Glucose and fructose are reducing sugars since they react with Tollen's Reagent and Fehling solution.
- 52.(AB) Starch and cellulose are polysaccharides of glucose. Sucrose is a disaccharide composed of glucose and fructose. Lactose is a disaccharide composed of glucose and galactose.
- 53.(BD) II and III are D sugars since the lowest chiral C has  $-OH$  group on the right side on the contrary. I has  $-OH$  on the left side and so it is a L-sugar.



A and B are diastereomers. They are C-2 epimers. They are both optically active hexahydroxy compounds.

55.(ABCD) All the statements are correct.

56.(ABC) Protein, nylon-6, 6 and nylon-6 are polyamide polymers since all of them have amide linkage.

57.(CD) Since here two different substances polymerise, copolymerization is said to have occurred. Also since a water molecule is lost in the process it is condensation polymerisation.

58.(BCD) Starch is a mixture of two different polysaccharides : Amylose (15 – 20%) and Amylopectin (80 – 85%). It involves the (C<sub>1</sub> – C<sub>4</sub>) α – glycosidic linkage between two α – D glucose units.

It involves branching by (C<sub>1</sub> – C<sub>6</sub>) glycosidic linkage.

59.(CD) α – D glucose undergoes mutarotation when dissolved in water to form β – D glucose.

Sucrose is a disaccharide composed of α – D – glucose and β – D – fructose.

60.(ACD) Neoprene is  $\left( \text{CH}_2 - \underset{\text{Cl}}{\underset{|}{\text{C}}} = \text{CH} - \text{CH}_2 \right)_n$

61.(ABC) At isoelectric point:  $\text{H}_3\text{N}^+ - \text{CH}_2 - \underset{\text{O}}{\underset{||}{\text{C}}} - \text{O}^-$

In acidic medium :  $\text{H}_3\text{N}^+ - \underset{\text{R}}{\underset{|}{\text{CH}}} - \text{COOH}$

In basic medium :  $\text{H}_2\text{N} - \underset{\text{R}}{\underset{|}{\text{CH}}} - \text{COO}^-$

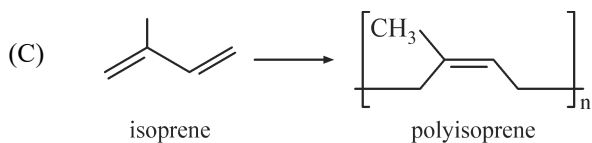
62.(ABCD) Anomers have different stereochemistry at C – 1 (anomeric carbon).

When pure α – D – glucopyranose is dissolved in water mutarotation occurs.

63.(AC)

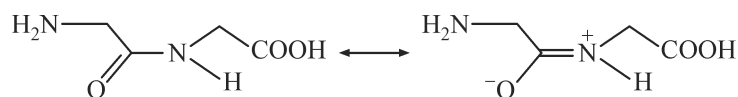
(A)  $n\text{HOOC} - (\text{CH}_2)_4 - \text{COOH} + n\text{H}_2\text{N} - (\text{CH}_2)_6 - \text{NH}_2 \longrightarrow \text{Nylon} - 6,6 + \text{H}_2\text{O}$

(B) Cellulose triacetate is semi synthetic polymer

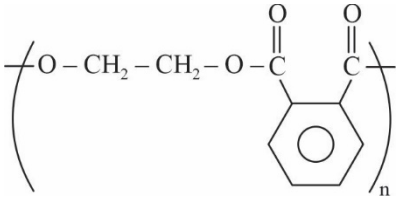
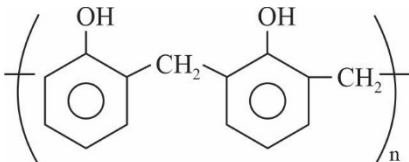


(D) Sucrose is disaccharide

64.(ABCD)



Peptide linkage is an amide linkage capable to form H-bond (hydrophilic in nature) and connects polypeptide chain in protein through H-bond.

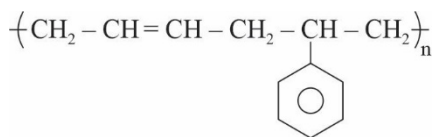
65. [A-r] → [B-s] → [C-p, q] → [D-r]  
 Kjeldahl's method and dumas method for nitrogen  
 Calius method for halogen and sulphur Leibig's method for carbon and hydrogen
66. [A-p, q, s] → [B-p, q, r] → [C-r] → [D-s]  
 CAN Test = for alcohol and phenol  
 Ester test = for carboxylic acid and alcohols  
 Oxidation test = for alcohols and aldehydes  
 Libermann's test = For 2° amines (Nitroso amines) and phenol
67. [A-p, r, s, t] → [B-p, s, t] → [C-p, s, t] → [D-q, r, s]  
 • Reducing sugar ⇒ Where the aromatic carbon has an OH group attached that can reduce other compounds  
 • Non-reducing sugar ⇒ where the aromatic carbon don't have an OH group attached so they can't reduce other compounds.  
 • Anomers ⇒ they are diastereoisomers of cyclic forms of sugar or similar molecules differing in configuration at anomeric carbon.  
 • Epimers ⇒ Two isomers with different configuration of atoms about one or several asymmetric carbon atoms present.  
 • Mutarotation ⇒ change in optical rotation because of the change in equilibrium between, anomers, when the corresponding stereocentres interconvert.
68. [A-q, r, s] → [B-r, s] → [C-p, r, s] → [D-p, s]  
 • Acidic amino acids ⇒ Aspartic acid, Glutamic acid  
 • Basic amino acids ⇒ Serine, Threonine, asparagine, Glutamic acid  
 • Optically active amino acids ⇒ All except glycine
69. [A-q, r] → [B-p, r] → [C-s] → [D-r]  
 (A) Nylon 6, 10  $\left( \text{C}(=\text{O})-(\text{CH}_2)_6-\text{C}(=\text{O})-\text{NH}-\text{CH}_2-\text{NH} \right)_n$   
 Monomers ⇒  $\text{HOOC}-(\text{CH}_2)_8-\text{COOH}$  and  $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$   
 (B) Glyptal ⇒  
  
 Monomers : Phthalic acid and glycol  
 (C) Teflon ⇒  $(\text{CF}_2-\text{CF}_2)_n$   
 Monomer ⇒  $\text{CF}_2 = \text{CF}_2$   
 (D) Bakelite ⇒  


70. [A-p, r] → [B-p, s] → [C-q, s] → [D-q, s]

(A) Polythene ⇒  $(\text{CH}_2 - \text{CH}_2)_n$

Monomer ⇒  $\text{CH}_2 = \text{CH}_2$

(B) Buna-S-rubber ⇒

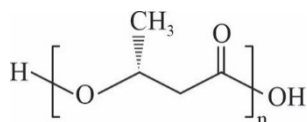


Monomer ⇒  $\text{H}_2\text{C} = \text{CH} - \text{C}_6\text{H}_5$  and  $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$

(C) Melamine – Formaldehyde resin

Monomer ⇒ Melamine and formaldehyde

(D) Poly-hydroxy butyrate-Co-β-Hydroxy valerate



71. [A-p, s] → [B-r] → [C-q] → [D-p, s]

(A)  $\text{Fe}(\text{OH})_3$  ⇒ Lyophobic colloid & hence irreversible

(B) Micelles ⇒ They are associated colloids

(C) Gelatin ⇒ It is a lyophilic colloid

(D) Arsenious sulphide ( $\text{As}_2\text{S}_3$ ) ⇒ lyophobic colloid and hence irreversible

72. [A-s] → [B-r] → [C-q] → [D-p]

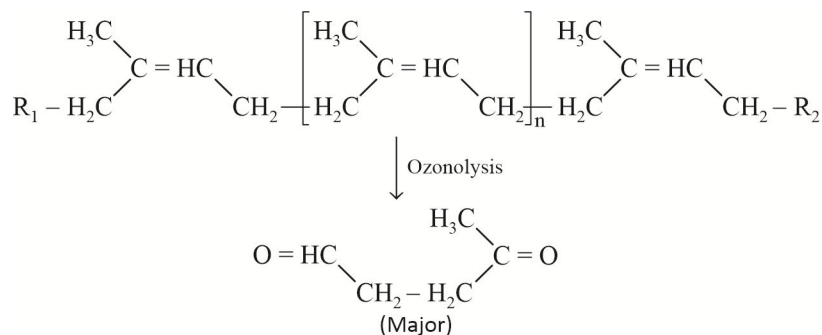
(A) Coagulation ⇒ It is done by addition of electrolyte

(B) Dialysis ⇒ It is a method of preparation of colloid

(C) Peptization ⇒ It is defined as the process of converting a precipitate into colloidal solution by shaking it with dispersion medium in the presence of small amount of electrolyte

(D) Tyndall effect : The scattering of light as the light beam passes through a colloid.

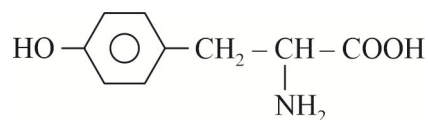
73.(5)



$\text{R}_1$  and  $\text{R}_2$  are chain terminating agents.

74.(8) Except (ix) all other observations are correct.

2-amino-3 (p-hydroxy phenyl) propanoic acid is



This is not formed on hydrolysis.

75.(8) Compounds (I), (IV), (V), (VII) and (IX) give positive haloform test. Compound (III), (VII) and (IX) gives positive Lucas reagent test,  $x = 5$ ,  $y = 3$  so,  $(5 + 3) = 8$ .

76.(6) Na Metal,  $\text{HCl} + \text{ZnCl}_2$ ,  $\text{FeCl}_3$ ,  $\text{NaOH} + \text{Phenolphthalein}$ ,  $\text{dil. KMnO}_4$ ,  $\text{Br}_2 / \text{H}_2\text{O}$  gives positive test with ascorbic acid.

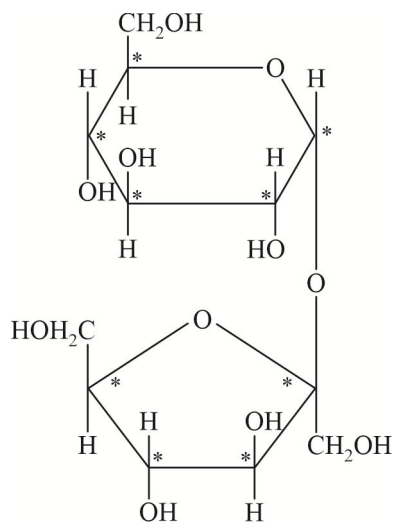
77.(5) 1, 2, 3, 6, 8

78.(8)

79.(9)  $\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$  (Glycine) +  $\text{H}_2\text{N}-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$

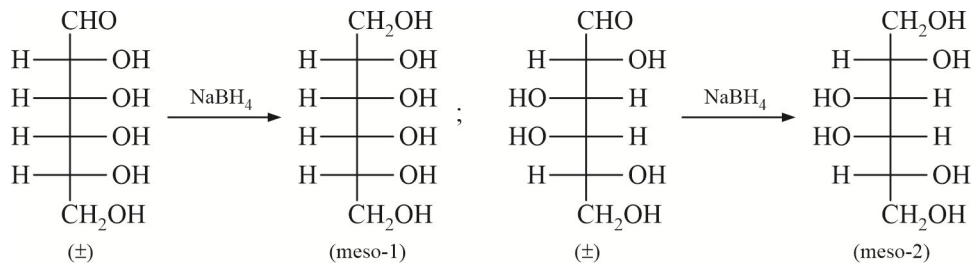
- |                   |                   |                   |                   |
|-------------------|-------------------|-------------------|-------------------|
| (1) Gly-Gly       | (2) Gly-ala(+)    | (3) Gly-ala(-)    | (4) Ala(+)-Gly    |
| (5) Ala(-)-Gly    | (6) Ala(+)-Ala(+) | (7) ala(+)-ala(-) | (8) ala(+)-ala(-) |
| (9) ala(-)-ala(+) |                   |                   |                   |

80.(9)



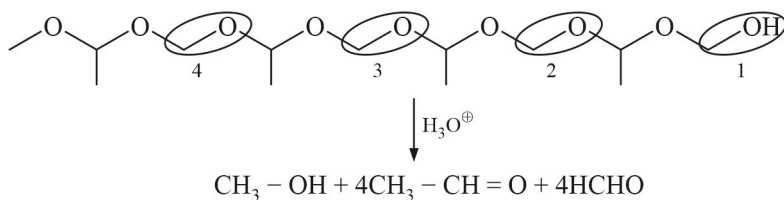
The star sign indicates chiral centre.

81.(4)



So in total 4 stereoisomers are reduced to meso products.

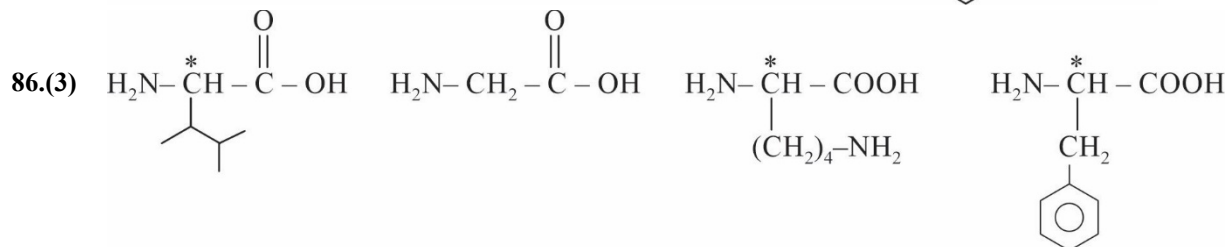
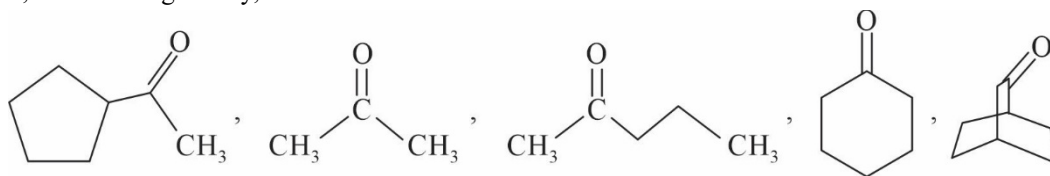
82.(4)



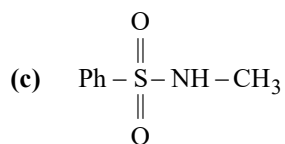
83.(4) (A) has no C-OH group,  $\alpha$  to the  $\text{CH}=\text{O}$  group. So it forms only phenylhydrazone with 1 mole of  $\text{PhNH NH}_2$ .  
(B) has (C-OH) group  $\alpha$  to the  $\text{CH}=\text{O}$  group, so it forms only osazone by reaction with 3 mole of  $\text{PhNH NH}_2$ .

84.(6) Neutral amino acid & acidic amino acid are negatively charged at  $\text{pH} = 7.0$   
Alanine, Cysteine, Glutamic acid, Glycine, Leucine, Aspartic acid

85.(5) 2, 4-DNP test given by,



87.(1) In case both N and S are present in an organic compound. Blood red color is obtained due to formation of sodium thiocyanate.



88.(2.5)

$$\frac{x}{m} = k \cdot P^{1/n} \text{ since } \log k = 0.699, \text{ hence } k = 5; \quad \text{Hence } k = 5$$

$$\text{Slope} = \frac{1}{n} = \tan 45^\circ = 1, \text{ thus, } \frac{x}{m} = 5 \times 0.5 = 2.5 \text{ g/g absorbent}$$

## Solutions - Advanced Problem Package | Chemistry

### Coordination Compounds

- 1.(B) Both  $\text{Cr}^{2+}$  and  $\text{Fe}^{2+}$  contain 4 unpaired electrons in each.
- 2.(B) According to Effective atomic number rule, EAN of Co is 36. Hence  $\text{Co}(\text{CO})_4$  attain stability either by reduction or by dimerization.  
 Effective atomic number of  $[\text{Co}(\text{CO})_4]^- = 27 - (-1) + 2 \times 4 = 36$   
 Effective atomic number of  $[\text{Co}_2(\text{CO})_8] = \frac{(27 \times 2) + (1 \times 2) + (2 \times 8)}{2} = 36$
- 3.(A)  $[\text{Pt}(\text{gly})_2]^{2+}$  is square planar complex and glycinate is bidentate ligand with different donor atoms
- 4.(B) (a)  $[\text{Cr}(\text{EDTA})]^{2-}$   
 Here Cr is in +2 oxidation state  
 $\text{EAN} = 24 - 2 + 12 = 34$
- (b)  $[\text{Co}(\text{en})_3]^{2+}$   
 $\text{EAN} = 27 - 2 + 12 = 37$
- (c)  $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$   
 $\text{EAN} = 26 - 3 + 12 = 35$
- (d)  $[\text{Ni}(\text{CN})_4]^{2-}$   
 $\text{EAN} = 28 - 2 + 8 = 34$
- 5.(B) Primary valency is equal to oxidation state while secondary valency is the number of electron pairs accepted.
- 6.(A) Number of unpaired electrons are based on oxidation state and crystal field theory.
- 7.(A) Basis on unpaired electron.
- 8.(A) Based on number of ions produced by ionization.
- 9.(C) Formula of given complex is  $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$  which ionizes to give one mole of  $\text{Cl}^-$  per mole of compound.
- 10.(A)
- |   |                                    | Conductance $\propto$ no. of ions                  |             |
|---|------------------------------------|----------------------------------------------------|-------------|
|   |                                    | Molecular formula                                  | No. of ions |
| A | $\text{PtCl}_4 \cdot 6\text{NH}_3$ | $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$            | 5           |
| B | $\text{PtCl}_4 \cdot 5\text{NH}_3$ | $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$   | 4           |
| C | $\text{PtCl}_4 \cdot 4\text{NH}_3$ | $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}_2$ | 3           |
- (A) Has maximum conductance
- 11.(B) No action of HCl suggests that all  $\text{NH}_3$  are present in coordination sphere.
- 12.(B)  $\Delta T_f = iK_f m$  ;  $0.0054 = i \times 1.8 \times 0.001$  ;  $i = 3$
- 13.(D) Statement 1 is incorrect  
 $[\text{Co}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}$  and  $[\text{Co}(\text{NH}_3)_5(\text{ONO})]\text{Cl}$  are linkage isomers  
 Statement 2 is also correct as both compounds give  $\text{Cl}^-$  in solution.



- 14.(A) Trans isomer has plane of symmetry.
- 15.(A)  $F^-$  is weak field ligand while  $NH_3$  acts as strong field ligand.
- 16.(A) Exchange of negatively charged ligand of coordination sphere and anion of ionization sphere produce ionization isomerism.
- 17.(C) The two forms are  $[Cr(NH_3)_4ClBr]Cl$  and  $[Cr(NH_3)_4Cl_2]Br$
- 18.(C) The given square planar complex is  $[M(NH_3)_2Cl_2](CrO_4)$ . Further A can react with  $Ag_2C_2O_4$ , indicates it could be cis form because the two similar monodentate ligands from cis positions can be replaced by one bidentate ligand.
- 19.(B) For 100% dissociation, Van't Hoff's factor(i) = no. of ions produced (n). Thus ratio is 5/7.

20.(C)  $[Cu(NH_3)_4]^{2+}$  is a square planar complex.

- 21.(D) I.  $[Fe(H_2O)_6]^{3+}$   
 $Fe^{3+} : 3d^5$   
 – It is high spin complex  
 – No. of unpaired  $e^- = 5$

- II.  $[Cr(NH_3)_6]^{3+}$   
 $Cr^{3+} : 3d^3$   
 – No. of unpaired  $e^- = 3$

- III.  $[CoCl_4]^{2-}$   
 $Co^{2+} : d^7$  complex  
 – High spin complex  
 – No. of unpaired  $e^- = 3$

22.(B)  $Cl^- = HCl = NaOH$



Thus 1 mol of complex will form n mol of HCl

1 mole of complex = n mol of HCl = n mole of NaOH

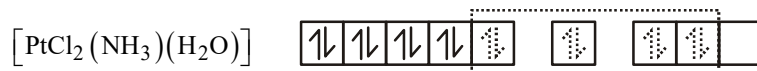
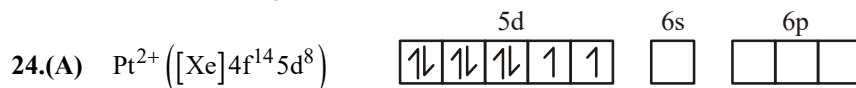
$$\text{Mole of complex} = \frac{0.319}{266.5} = 0.0012 ; \text{mole of NaOH used} = \frac{28.5 \times 0.125}{1000} = 0.0036$$

So 0.0012 mole of complex = 0.0036 mole of NaOH = 0.0036 mole of HCl

$$1 \text{ mole of complex} = \frac{0.0036}{0.0012} = 3 \text{ mole of HCl} \quad \therefore n = 3$$

So complex is  $[Cr(H_2O)_6]Cl_3$ .

- 23.(B) CFSE depends on the relative magnitude of crystal field splitting,  $\Delta_0$  and pairing energy and in turns  $\Delta_0$  depends upon the field produced by ligand and charge on the metal ion. The order of increasing crystal field strength is  $F^- < H_2O < NH_3 < CN^-$



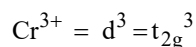
$dsp^2$  hybrid orbitals

Four pairs of electrons from four  $Cl^-$

$Ma_2bc$  have cis and trans isomers.

25.(B) Chelate complexes are more stable than non chelate complexes

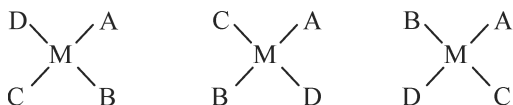
26.(D)  $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$  has the largest CFSE because of stable configuration



$$\text{CFSE} = (-0.4 \times 3)\Delta_0 = -1.2\Delta_0$$

27.(B) IUPAC Name : Potassium trioxalatoaluminate (III)

28.(C)  $\text{M}(\text{ABCD})$  have 3 geometrical isomers



29.(C)

|                                                    | No of ions |
|----------------------------------------------------|------------|
| $[\text{Pt}(\text{NH}_3)_4]\text{Cl}_4$            | 5          |
| $[\text{Pt}(\text{NH}_3)_2\text{Cl}_4]$            | 0          |
| $[\text{Pt}(\text{NH}_3)_5\text{Cl}]\text{Cl}_3$   | 4          |
| $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]\text{Cl}_2$ | 3          |

30.(C)  $\text{cis} [\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$  is Optically Active because of absence of symmetry element.

$\text{Trans} [\text{M}(\text{AA})_2\text{a}_2]$  is optically inactive due to presence of POS while  $\text{cis}[\text{M}(\text{AA})_2\text{a}_2]$  is Optically Active due to absence of Symmetry element.

31.(B)  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 \longrightarrow [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+} + 2\text{Cl}^-$

100 ml 0.1 M

= 10 mmol

– 20 mmol

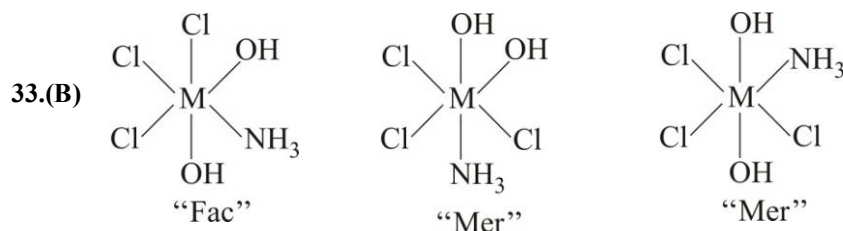
Moles of  $\text{AgCl}$  obtained = 0.02 mol

32.(A) (A) Is correct as Tetrahedral complexes do not show Geometrical Isomerism

(B) Is incorrect as square planar Complexes show G.I.

(C) Incorrect .  $\text{Ma}_3\text{b}$  will not shown G.I.

(D)  $[\text{Pt}(\text{gly})_2]$  show cis-trans isomerism



34.(C) 4.91 BM Indicates 4 unpaired  $e^-$ .

- Ion could be  $\text{Fe}^{2+}$

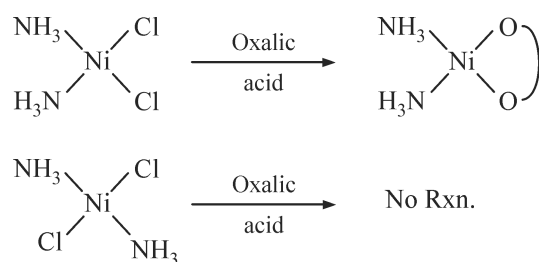
-  $\text{Fe}^{2+}$  has 4 unpaired electron. ( $3d^6$  configuration)

-  $\text{Fe}^{2+}$  with strong field ligand has 0 unpaired electron

35.(B)  $[\text{Ni}(\text{NH}_3)_4]^{2+} \xrightarrow{\text{HCl}} [\text{Ni}(\text{NH}_3)_2\text{Cl}_2]$

exist in 2 forms

Cis and Trans



36.(D)  $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$  shows Ionisation Isomerism because of presence of ion in co-ordination and Ionisation sphere.

37.(A) Magnitude of  $\Delta_0$  directly proportional to principal quantum number of d sub-shell.

38.(C) In octahedral field, ligands pointing towards lobes of  $d_{x^2-y^2}$  and  $d_{z^2}$  orbitals.

39.(C)  $\text{Ti}_{(\text{aq})}^{3+}$  coloured because of d-d transition while  $\text{Ti}^{4+}$  is colourless because of  $d^0$  configuration.

40.(D)  $[\text{Ni}(\text{PPh}_3)_2\text{Br}_2]$  is tetrahedral complex

#### Hybridisation

- |    |                                 |                |               |
|----|---------------------------------|----------------|---------------|
| a) | $[\text{Ni}(\text{CN})_4]^{2-}$ | $\text{dsp}^2$ | Square planar |
| b) | $[\text{Ni}(\text{CO})_4]$      | $\text{sp}^3$  | Tetrahedral   |
| c) | $[\text{NiCl}_4]^{2-}$          | $\text{sp}^3$  | tetrahedral   |

41.(A) (A)  $[\text{RhCl}(\text{CO})(\text{pph}_3)_2]$

$\text{Rh}^+ : 4d^8$

Hybridization :  $\text{dsp}^2$  (square planar)

(B)  $\text{K}_3[\text{Cu}(\text{CN})_4] : d^{10} \longrightarrow \text{sp}^3$

(C)  $\text{K}_2[\text{Zn}(\text{CN})_4] : d^{10} \longrightarrow \text{sp}^3$

(D)  $[\text{Ni}(\text{CO})_4] : d^{10} \longrightarrow \text{sp}^3$

42.(D)  $[\text{Pt}(\text{gly})_2]$  shows cis-trans isomerism

$[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NH}_2\text{CH}_3)]\text{Cl}$  shows cis-trans isomerism

43.(D)  $[\text{Ni}(\text{CO})_4] : 3d^8 4s^2 \xrightarrow{\text{SFL}} 3d^{10}$

- It is  $\text{sp}^3$  hybridized

- Tetrahedral

(D) is False because Metal-Carbon bond order increases due to synergic Bonding

44.(C)  $\text{Ni}(\text{CO})_4$   $\text{sp}^3$  - diamagnetic

$\text{Fe}(\text{CO})_5$   $\text{dsp}^3$  - diamagnetic

$\text{V}(\text{CO})_6$   $d^2\text{sp}^3$  - Paramagnetic

$\text{Cr}(\text{CO})_6$   $d^2\text{sp}^3$  - diamagnetic

45.(D) All statements are correct

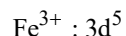
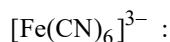
Metal carbon bond in metal carbonyls possess double bond character due to synergic bonding. As extent of synergic bonding increases, Metal-carbon bond order increases and C-O bond order decreases.

46.(ABD) EDTA is Polydentate (denticity = 6)

- Flexidentate (it can change its denticity if required)

- Chelating ligand as it form Ring with metal.

- $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$  is more stable than  $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$  due to more  $Z_{\text{eff}}$ .
- 47.(BCD)** - (B) is correct Octahedral complexes of  $\text{Ma}_3\text{b}_3$  show fac-mer isomer.
- $[\text{Mg}(\text{EDTA})]^{2-}$  - contains 5 Rings  
EDTA is hexadentate ligand. No. of Rings formed by a ligand = 5  
- MABCD Optically Active – Possess 2 stereo isomers
- 48.(ABD)** Square planar complex of  $\text{Ma}_2\text{b}_2$ , octahedral complexes of  $\text{Ma}_4\text{b}_2$ ,  $\text{Ma}_3\text{b}_3$  and  $\text{M}(\text{AA})_2\text{a}_2$  exhibit geometrical isomerism.
- 49.(ABD)**  $[\text{CoCl}_2(\text{NH}_3)_2\text{en}]^+$  has 3 geometrical isomers out of which, one is optically active to give d and l forms.
- 50.(ABC)** Absence of unpaired electron.
- 51.(ABC)** To exhibit coordination isomerism, both cation and anions should be complex ions
- 52.(ABCD)**  $[\text{Ag}(\text{NH}_3)_2]^+$  is linear with sp hybridization.  
 $\text{NiCl}_4^{2-}$ ,  $\text{VO}_4^{3-}$ ,  $\text{MnO}_4^-$  - tetrahedral geometry  
 $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ,  $[\text{Pt}(\text{NH}_3)_4]^{2+}$ ,  $[\text{Ni}(\text{CN})_4]^{2-}$  - Square planar  
 $\text{Fe}(\text{CO})_5$  - trigonal bipyramidal
- 53.(BC)** Based on oxidation state and crystal field theory.
- 54.(ABCD)** It is expected that there is charge transfer between Fe and NO.
- 55.(CD)** From Werner's theory the formula of the given complex in  $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
- 56.(ABD)** (A)  $\Delta_o \propto$  ligand field strength  
(B)  $\text{Ti}^{3+} : 3d^1$   
 $= t_2g^1$   
 $\text{CFSE} = -0.4\Delta_o$   
(C)  $\text{NiCl}_4^{2-}$  is paramagnetic while  $\text{Ni}(\text{CO})_4$  is diamagnetic  
(D) correct
- 57.(D)** (A), (B) and (C) are colourless due to  $d^0$  configuration
- 58.(ACD)** (A)  $\text{Co}^{3+} : 3d^6$   
- It form low spin complex with  $\text{NH}_3$ ,  $\text{CN}^-$  and  $\text{NO}_2^-$   
- diamagnetic  
-  $d^2sp^3$   
(B)  $[\text{Zn}(\text{NH}_3)_4]^{2+} : 3d^{10}$  configuration  
-  $sp^3$  hybridized  
(C)  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} :$   
-  $\text{Fe}^{+3} : 3d^5$   
- Paramagnetic with 5 unpaired electron  
-  $sp^3d^2$  hybridized  
-  $\mu = 5.92$



-Form low spin complex with  $\text{CN}^-$

-No. of unpaired electron = 1

- $\mu = 1.73$

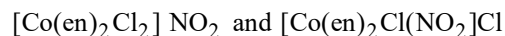
(D)  $\text{K}_4[\text{MnF}_6]$  and  $\text{K}_3[\text{FeF}_6]$  have 5 unpaired electron  $\mu = 5.92 \text{ BM}$

**59.(ABCD)** Crystal Field Theory explain Ionic Interactions between metal and ligands.

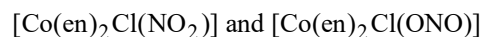
According to this theory, ligands are point charges and Metal is test charge. CFT considered Only Metal Ion d-orbitals in bonding.

**60.(ABCD)** Empirical Formula :  $[\text{Co}(\text{en})_2\text{Cl}_2](\text{NO}_2)$

- It can show ionization isomerism :



- It can show linkage Isomerism due to Presence of  $\text{NO}_2$



- it can show Geometrical Isomerism also i.e. Cis and Trans. Cis is Optically Active and trans Isomer is Optically Inactive.

**61.(BC)**  $S_1$  : Non Correct. Square planar complex are generally Optically inactive due to presence of plane of Symmetry.

$S_2$  : Correct.  $\Delta_t$  is less than  $\Delta_o$  as ligands are not interacting directly with metal.

$S_3$  : Correct

**62.(AB)**  $\Delta_o \propto Z_{\text{eff}}$  of Metal

$\propto$  ligand Field strength

(A) correct as  $\text{CN}^-$  is stronger ligand than  $\text{H}_2\text{O}$

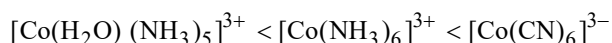
(B) correct as  $Z_{\text{eff}}$  of  $\text{Co}^{+3} > \text{Co}^{+2}$

(C) Not correct as  $Z_{\text{eff}}$  of Rh  $>$  Co

(D) not correct as  $\text{NH}_3$  is stronger ligand than  $\text{H}_2\text{O}$

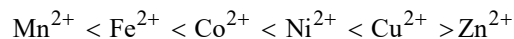
**63.(B)**  $[\text{Co}(\text{gly})_3]$  and Cis  $[\text{Co}(\text{en})_2\text{Cl}_2]$  show optical isomerism because of absence of symmetry element.

**64.(BD)** (A) Incorrect.  $\Delta_o \propto$  ligand Field strength correct order is:

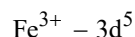


(B) Correct

(C) Incorrect :According to Irving William Series Order of Stability is:

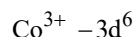
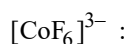


(D)  $[\text{Fe}(\text{H}_2\text{O})_6]^{3+} :$



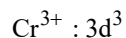
- No. of unpaired electrons = 4

- Magnetic moment =  $\sqrt{35}$



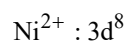
- No. of unpaired electrons = 4

- Magnetic moment  $\sqrt{24}$

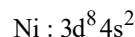


- No. of unpaired electrons = 3

- Magnetic moment =  $\sqrt{15}$



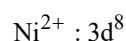
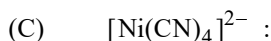
- tetrahedral and Paramagnetic



With CO as ligand, it gets converted to  $3d^{10}$

Hybridisation  $sp^3$

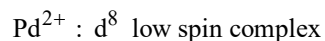
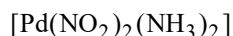
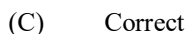
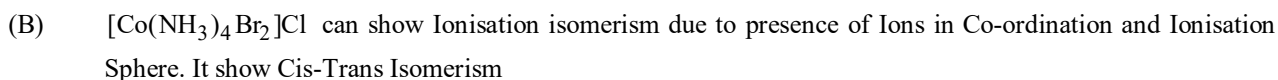
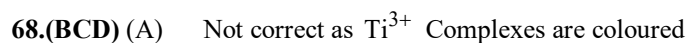
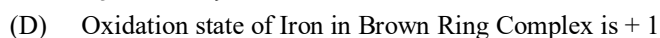
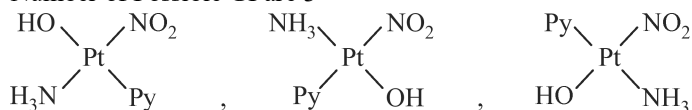
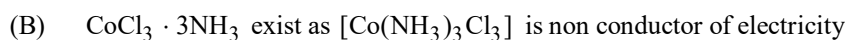
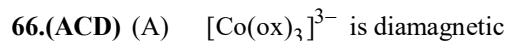
Tetrahedral and diamagnetic



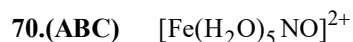
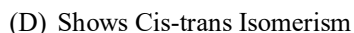
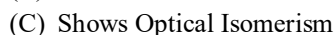
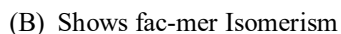
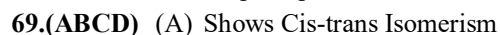
With  $\text{CN}^-$  as ligand, it forms low spin complex

Hybridisation =  $dsp^2$

Square planar and diamagnetic



: Square planar



- Here Fe is in +1 Oxidation state

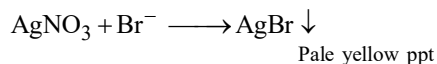
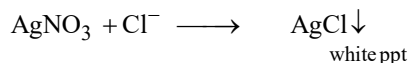
- Number of unpaired  $e^- = 3$

- Hybridisation =  $sp^3d^2$

71.(ABCD) (A) In Potassium Ferrocyanide  $K_4[Fe(CN)_6]$ , Iron is in +2 oxidation state and in potassium Ferricyanide  $K_3[Fe(CN)_6]$ , Iron is in +3 oxidation state.

(B) by adding aqueous solution of  $BaCl_2$ ,  $[Co(NH_3)_5Br]SO_4$  gives white ppt.

(C) Correct



(D) Correct

Conductance  $\propto$  No. of Ions

72.(ABD)  $[Fe(en)_2(H_2O)_2]^{2+} + en \longrightarrow [Fe(en)_3]^{2+}$   
low spin complex

$Fe^{2+} : 3d^6$  (diamagnetic)

It will not show Geometrical Isomerism but is Optically Active.

73.(AC) Acetylacetonato :

$$CH_3-C(=O)-CH=C(O^-)-CH_3$$

Oxalato ion : dianion

$$\begin{array}{c} ^-O-C=O \\ | \\ ^-O-C=O \end{array}$$

dimethyl glyoximatio :

$$\begin{array}{c} OH \\ | \\ C=N \\ / \quad \backslash \\ C=N \\ | \\ O^- \end{array}$$

74.(AB) Organometallic compounds have metal – carbon linkage. A and B are correct

75.(ABCD)  $[Co(NH_3)_5SO_4]Br$  and  $[Co(NH_3)_5Br]SO_4$  can be distinguished by

(A)  $BaCl_2$  :  $BaCl_2$  gives white ppt with  $SO_4^{2-}$  ion but will not give any ppt with  $Br^-$ .

(B)  $AgNO_3$  :  $AgNO_3$  gives pale yellow ppt with  $Br^-$  ion but will not give any ppt with  $SO_4^{2-}$

(C) It can be distinguished by conductance due to different size

(D) It can be distinguished by dipole moment measurement due to difference in EN

76.(AD) (A)  $MnCl_4^{2-}$  :

$Mn^{2+} : 3d^5$  configuration

:  $sp^3$  hybridized and Paramagnetic

(B)  $[Mn(CN)_6]^{2-}$  :

$Mn^{+4} (3d^3)$  : Octahedral and Paramagnetic

(C)  $[Cu(CN)_4]^{3-}$  has Square planar geometry and is diamagnetic due to  $d^{10}$  configuration.

77.(ABD) Given Isomers cannot be distinguished by Magnetic moment due to same number of unpaired electrons.

78.(BC) A and D are Optically Inactive due to presence of plane of Symmetry.

79.(C) To exhibit ionization isomerism, complex should have anionic ligand and anion in ionization sphere. Presence of ambidentate ligands produce linkage isomerism.

- 80.(B) 1)  $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$   
 $\text{Co}^{2+} : 3d^7$   
 : Paramagnetic  
 : exhibit geometrical Isomers i.e. Cis and Trans
- 2)  $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2] :$   
 $\text{Pt}^{2+} : 5d^8$   
 : exhibit geometrical Isomer i.e. Cis and trans
- 3)  $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl} :$   
 $\text{Co}^{2+} : 3d^7$   
 : Paramagnetic
- 4)  $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2 :$   
 $\text{Ni}^{2+} : 3d^8$   
 : Paramagnetic

1, 3, 4 gives ppt of AgCl by action of  $\text{AgNO}_3$  due to presence of  $\text{Cl}^-$  in Ionisation sphere.

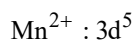
1, 3, 4 show conductance due to presence of ions.

- 81.(D) 1)  $[\text{MnCl}_6]^{2-} :$   
 $\text{Mn}^{4+} : 3d^3$   
 Hybridisation :  $sp^3d^2$
- 2)  $[\text{Fe}(\text{CN})_6]^{3-}$   
 $\text{Fe}^{3+} : 3d^5$   
 : low spin complex  
 :  $d^2sp^3$
- 3)  $[\text{Co}(\text{NH}_3)_6]^{3+}$   
 $\text{Co}^{3+} : 3d^6$   
 : low spin complex
- 4)  $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$   
 $\text{Fe}^{2+} : 3d^6$   
 : High spin complex

- 82.(B) 1)  $\text{PPh}_3$  – Monodentate  
 2) en – bidentate, chelating  
 3)  $\text{SCN}^-$  – Monodentate, Ambidentate  
 4) dmg – bidentate, chelating  
 - All polydentate ligands are chelating ligands

- 83.(D) 1)  $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$  - Brown Ring Complex  
 $\text{Fe}^{+1} : 3d^7$   
 : Octahedral ( $sp^3d^2$ )  
 : 3 unpaired electron
- 2)  $[\text{Mn}(\text{CN})_6]^{4-}$





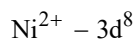
: forms low spin complex

: Contains 1 unpaired electron



- Trigonal bipyramidal

-  $dsp^3$  hybridized



- Diamagnetic

-  $dsp^2$  hybridized

84.(6) The given complex is an octahedral complex of formula  $[\text{Ma}_2\text{bc}_3]$  with b is ambidentate ligand. The possible number of structural and stereo isomers are six.

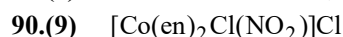
85.(4) Given complex is  $\text{Ma}_2\text{b}_2(\text{AA})$  which contain three geometrical isomers out of which one is optically active.

86.(5) The formed complex is  $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$

87.(3) While making complex we need to have at least 1 counter ion outside.

88.(2) Product may be fac or mer isomer only

89.(6)  $\text{EAN} = 42 + 2x = 54$ , so  $x = 6$



Isomers of  $[\text{Co}(\text{en})_2\text{Cl}(\text{NO}_2)]\text{Cl}$

Structural Isomers :  $[\text{Co}(\text{en})_2\text{Cl}(\text{NO}_2)]\text{Cl}$

$[\text{Co}(\text{en})_2\text{Cl}(\text{ONO})]\text{Cl}$

$[\text{Co}(\text{en})_2\text{Cl}_2]\text{NO}_2$

Each structural Isomer exist in 2 forms Cis and trans. Cis is Optically Active and trans is Inactive. Each structural Isomer show 3 stereoisomers.

Total Isomers = 9

91.(4) Optically Active Compounds exist in *d* and *l* form

$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+ \longrightarrow$  Optically Inactive

$[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \longrightarrow$  Optically Inactive

trans  $[\text{Co}(\text{en})_2\text{Cl}_2]^+ \longrightarrow$  Optically Inactive

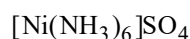
cis  $[\text{Co}(\text{en})_2\text{Cl}_2]^+ \longrightarrow$  Optically Active

$[\text{Cr}(\text{ox})_3]^{3-} \longrightarrow$  Optically Active

$[\text{Cr}(\text{en})_3]^{3+} \longrightarrow$  Optically Active

$[\text{Pt}(\text{en})_2\text{Cl}_2] \longrightarrow$  Cis form is Optically Active

92.(3) Paramagnetic octahedral compound



## Solutions - Advanced Problem Package | Chemistry

### Metallurgy

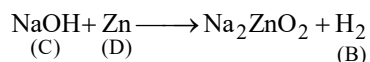
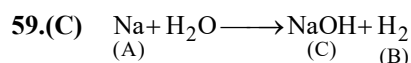
- 1.(B) Sulphide ores are concentrated by froth floatation method.
- 2.(C) Basic flux is used to remove acidic impurities. Lime stone produce CaO which is basic.
- 3.(C) Roasting is done to remove volatile matter as well as to convert ore to metal oxide.
- 4.(A) Silver in native form or in sulphide form is concentrated by dissolving in NaCN solution in presence of air.
- 5.(B) High electropositive elements such as alkali metals, alkaline earth metals and Al etc. are extracted by electrolytic reduction.
- 6.(C) Aluminium cannot be deposited from aqueous solution by passing electric current.
- 7.(C) Zone refining is a method of purification and it is used to obtain ultrapure metal.
- 8.(A) In thermite process reaction is started by the use of an ignition mixture containing magnesium and barium peroxide.
- 9.(B) Leaching is a chemical method of concentration of the ore.
- 10.(B) In smelting an oxide ore is heated with carbon in which metal oxide is reduced to metal.
- 11.(C) Roasting process is used to convert sulphide ore to metal oxide.
- 12.(B) Cupellation is a method of refining and it is used for Ag.
- 13.(B) Copper pyrite is  $\text{CuFeS}_2$  and after roasting  $\text{CuO}$  and  $\text{FeO}$  are formed.  $\text{FeO}$  is removed by adding  $\text{SiO}_2$  that form slag  $\text{FeSiO}_3$ .
- 14.(B)  $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$  is concentrated by chemical method known as leaching, in this method ore is dissolved in suitable solvent.
- 15.(B) Calamine ( $\text{ZnCO}_3$ ); Siderite( $\text{FeCO}_3$ )      Argentite ( $\text{Ag}_2\text{S}$ ); Cuprite( $\text{Cu}_2\text{O}$ )  
Zinc blende ( $\text{ZnS}$ ); Iron pyrite ( $\text{FeS}_2$ )      Malachite ( $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ )  
Azurite ( $2\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ )
- 16.(A) Silver and gold are extracted by cyanide process.
- 17.(C) Malachite is an ore of copper. It is  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ .
- 18.(D) Heating an ore in the absence of air is known as calcination. Roasting is heating of the ore in the presence of air. Smelting is heating in the presence of carbon.
- 19.(C) A molten mixture of  $\text{Al}_2\text{O}_3$  and  $\text{Na}_3\text{AlF}_6$ .
- 20.(D) Most electropositive metals are extracted by electrolytic reduction method.
- 21.(D) Blister copper contains the impurities of copper oxide which is removed by poling method.
- 22.(C)  $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$  slope  $= -\Delta S^\circ$
- 23.(B) Pitch blende ( $\text{U}_3\text{O}_8$ ) is an ore of uranium.
- 24.(D) Copper is extracted by self reduction process while silver is extracted by cyanide process.
- 25.(A)  $2\text{CuFeS}_2 + 4\text{O}_2 \rightarrow \text{Cu}_2\text{S} + \text{FeO} + 3\text{SO}_2$
- 26.(D)  $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
- 27.(C)  $\text{Cu}_2\text{S} + \text{O}_2 \rightarrow \text{Cu}_2\text{O} + \text{SO}_2$        $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$
- 28.(A) Poling process is a method of refining and it is used to remove  $\text{Cu}_2\text{O}$  from Cu.
- 29.(A) In electrowinning process impure copper acts as anode while a thin sheet of pure copper is used as cathode.
- 30.(ABCD)      Trona;  $\text{Na}_2\text{CO}_3 \cdot \text{NaHCO}_3 \cdot 2\text{H}_2\text{O}$       Malachite;  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$   
Dolomite;  $\text{CaCO}_3 \cdot \text{MgCO}_3$       Siderite;  $\text{FeCO}_3$

- 31.(ABC) Sulphide ores are concentrated by froth floatation process except argentite. Argentite is leached by NaCN in presence of air.
- 32.(AC) Calcination is used to convert hydrated oxide or hydroxide and carbonates into respective oxides.
- 33.(ABC) Self reduction process is used for less electro positive metals like Cu, Hg and Pb.
- 34.(ABCD) Copper pyrite;  $\text{CuFeS}_2$  Carnalite;  $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$   
Dolomite ;  $\text{CaCO}_3 \cdot \text{MgCO}_3$  Cryolite ;  $\text{Na}_3\text{AlF}_6$
- 35.(ACD) Roasting is carried out to convert sulphide ore to oxide ore. In this process volatile impurities and organic matters are also removed.
- 36.(ABC)  $\text{MgO} + \text{SiO}_2 \rightarrow \text{MgSiO}_3$   
 $3\text{MgO} + \text{P}_2\text{O}_5 \rightarrow \text{Mg}_3(\text{PO}_4)_2$   
 $\text{MgO} + \text{SO}_2 \rightarrow \text{MgSO}_3$
- 37.(AB) Sulphide ores are roasted.
- 38.(AD) The ores which are already in anhydrous oxide form are directly used for reduction i.e. Rutile ( $\text{TiO}_2$ ). Cassiterite ( $\text{SnO}_2$ ).
- 39.(C) The mineral bauxite is used for extraction of Al.
- 40.(ABCD) Galena; Froth floatation process  
Bauxite; Leaching  
Cassiterite; Gravity separation  
Magnetite; Electromagnetic separation
- 41.(BC) Silver and gold are concentrated by leaching process in which ore is dissolved in sodium cyanide solution in the presence of air to form soluble cyano complex.
- 42.(ABCD) Mond's process; Refining of Nickel  
Van Arkel's process; Ultrapure metal  
Cyanide process; Extraction of silver  
Froth-floatation process; Concentration of sulphide ores.
- 43.(ABC)  $\text{Ag}_2\text{S} + 4\text{NaCN} \rightarrow 2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Na}_2\text{S}$   
 $4\text{Na}_2\text{S} + 5\text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Na}_2\text{SO}_4 + 4\text{NaOH} + \text{S}$   
 $2\text{Na}[\text{Ag}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Ag}$
- 44.(ABCD)  $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$ ;  
 $\text{U}_3\text{O}_8 + 3\text{H}_2\text{SO}_4 + \frac{1}{2}\text{O}_2 \rightarrow 3\text{UO}_2\text{SO}_4 + 3\text{H}_2\text{O}$   
 $4\text{Au} + 8\text{KCN} + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Na}[\text{Au}(\text{CN})_2] + 4\text{NaOH}$   
 $4\text{Ag} + 8\text{NaCN} + 2\text{H}_2\text{O} + \text{O}_2 \rightarrow 4\text{Na}[\text{Ag}(\text{CN})_2] + 4\text{NaOH}$
- 45.(ABCD) Lead; self-reduction  
Boron; Decomposition of iodide  
Nickel; Decomposition of carbonyl complex  
Silver; Complex formation and displacement by metal
- 46.(ABC)  $\text{Ni}(\text{CO})_4 \xrightarrow{\Delta} \text{Ni} + 4\text{CO}$ ;  $2\text{BI}_3 \xrightarrow{\Delta} 2\text{B} + 3\text{I}_2$ ;  $\text{TiI}_4 \xrightarrow{\Delta} \text{Ti} + 2\text{I}_2$

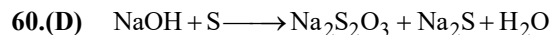
47. [A-r] → [B-p] → [C-s] → [D-q]  
Smelting is a process in which metal oxide is heated with carbon. Self reduction process is used for sulphide ores. Aluminium is extracted from bauxite by electrolytic reduction. In hydrometallurgy ore is concentrated by leaching process and bauxite is concentrated by leaching process. Copper glance ( $\text{Cu}_2\text{S}$ ), Silver glance ( $\text{Ag}_2\text{S}$ ), Haematite ( $\text{Fe}_2\text{O}_3$ ), Bauxite  $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
48. [A-r] → [B-r, s] → [C-p, q] → [D-q]  
Al & Au are purified by electrolysis. The process of extraction of Al is known as Baeyer's process. Self reduction is used for extraction of Pb. Impure metals like Bi, Sn, Pb etc. are purified by liquation method.
49. [A-r] → [B-s, t] → [C-p] → [D-q]  
Mond's process is used for purification of nickel. Van Arkel method is used for obtaining ultra pure metals. Titanium is purified by Van Arkel method. Distillation is used for those metals which are volatile. Zn, Cd and Hg are purified by distillation method. Cupellation is used to remove impurity of lead in silver.
50. [A-p] → [B-q] → [Cr] → [D-s]  
Fe and Sn are extracted by carbon reduction method. Cu and Pb are extracted by self reduction method from their sulphide ores. Mg and Al are extracted by electrolytic reduction of their fused salts. Ag and Au are extracted by cyanide process.
51. [A-p] → [B-q] → [C-r, s] → [D-t]  
Electrolytic reduction is used for extraction of more electropositive metals like Na, Mg and Al. Non metals are extracted by electrolytic oxidation from their salts. Iron is extracted by carbon reduction of its oxide while silver is extracted from its soluble cyano complex by reduction with Zn (chemical reduction). Sulphur is extracted by chemical oxidation method.
- $$\text{CaS} + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{S}$$
- $$2\text{H}_2\text{S} + \text{O}_2 \xrightarrow{\text{(air)}} 2\text{H}_2\text{O} + 2\text{S} \text{ or}$$
- $$2\text{Fe}_2\text{S}_3 + 3\text{O}_2 \xrightarrow{\text{(air)}} 2\text{Fe}_2\text{O}_3 + 6\text{S}$$
- 52.(6) [Oxygen 46.6%]
- 53.(6) Copperpyrite  $\text{CuFeS}_2$ ; Zinc blende  $\text{ZnS}$ ; Argentite  $\text{Ag}_2\text{S}$ ; Cinnabar  $\text{HgS}$ ; Galena  $\text{PbS}$ ; Iron pyrite  $\text{FeS}_2$ ; Haematite  $\text{Fe}_2\text{O}_3$ ; Malachite  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$ ; Dolomite  $\text{MgCO}_3 \cdot \text{CaCO}_3$
- 54.(8) Bauxite  $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$ ; Magnetite  $\text{Fe}_3\text{O}_4$ ; Cuprite  $\text{Cu}_2\text{O}$ ; Chromite  $\text{FeO} \cdot \text{Cr}_2\text{O}_3$ ; Cassiterite  $\text{SnO}_2$ ; Cryolite  $\text{Na}_3\text{AlF}_6$ ; Haematite  $\text{Fe}_2\text{O}_3$ ; Galena  $\text{PbS}$ ; Limestone  $\text{CaCO}_3$ ; Rutile  $\text{TiO}_2$ ; Pyrolusite  $\text{MnO}_2$
- 55.(3) Among metals, aluminium is the most abundant.
- 56.(1) Iron pyrite ( $\text{FeS}_2$ ) is known as fool's gold. Oxidation state of sulphur in  $\text{FeS}_2$  is -1.
- 57.(5) Na, K, Mg, Ca and Al are extracted by electrolysis of their fused salts because their oxides cannot be reduced easily with carbon at moderate temperatures.
- 58.(4) Cassiterite is  $\text{SnO}_2$ .
- 59.(9) All are correctly matched.
- 60.(9) All the steps are involved in the extraction of iron from magnetite.
- 61.(4) Siderite :  $\text{FeCO}_3$   
Calamine :  $\text{ZnCO}_3$   
Dolomite :  $\text{CaMg(CO}_3)_2$   
Malachite :  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$



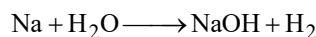
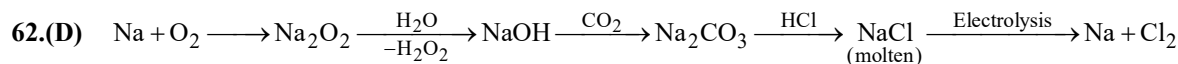
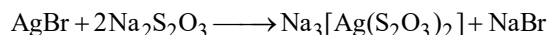
- 31.(B) Alkali and alkaline earth metals are extracted by electrolysis of their fused salt.
- 32.(D) LiCl is least stable due to more covalent character because of high polarization power of  $\text{Li}^+$ .
- 33.(B) Alkali metals are strong reducing agents.
- 34.(C)  $\text{Ca(OH)}_2$  can remove temporary hardness and forms  $\text{CaCO}_3$  on reaction with  $\text{CO}_2$ .
- 35.(B) Basic strength of alkaline earth metal oxide increases down the group.
- 36.(D)  $\text{Zn(OH)}_2$  is amphoteric
- 37.(C) K is more active than Mg.
- 38.(C) Malachite  $\text{CuCO}_3 \cdot \text{Cu(OH)}_2$  Cassiterite  $\text{SnO}_2$   
Carnallite  $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$  Gelena  $\text{PbS}$
- 39.(A) Be is less reactive than Mg.
- 40.(B)  $\text{BeCl}_2$  is least ionic because of high polarizing power of beryllium ion.
- 41.(D) Magnesium forms oxide and nitride.
- 42.(D)  $\text{Ca(H}_2\text{PO}_4)_2 \cdot 2\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  is super phosphate of lime.
- 43.(D) Because of high degree of hydration of alkaline earth metal ion.
- 44.(D) Alkaline earth metals are obtained by electrolysis of fused halides.
- 45.(B) It is due to oscillation of mobile valence electrons.
- 46.(D) Alkali metals are soluble in liquid ammonia  $\text{Na} + \text{NH}_3 \rightarrow \text{Na}_{(\text{am})}^+ + \text{e}_{(\text{am})}^-$ .
- 47.(C) Because of high ionic character.
- 48.(C)  $\text{NaOH} + \text{CO} \rightarrow \text{HCOONa}$
- 49.(A) NaOH is basic and  $\text{NaHCO}_3$  is acidic hence cannot exist together.  
 $\text{NaHCO}_3 + \text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ .
- 50.(A)  $\text{Na}_2\text{CO}_3 + 2\text{SO}_2 + \text{H}_2\text{O} \rightarrow 2\text{NaHSO}_3 + \text{CO}_2$
- 51.(B)  $\text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow 2\text{NaHCO}_3$
- 52.(A)  $\text{NaOH} + \text{AgNO}_3 \longrightarrow \text{NaNO}_3 + \text{AgOH}$   
 $2\text{AgOH} \xrightarrow{\Delta} \text{Ag}_2\text{O} + \text{H}_2\text{O}$
- 53.(C) Bleaching powder liberates  $\text{Cl}_2$ .  
 $\text{CaOCl}_2 \rightarrow \text{CaO} + \text{Cl}_2$
- 54.(B)  $2\text{Na} + \text{O}_2 \xrightarrow{300^\circ\text{C}} \text{Na}_2\text{O}_2 \xrightarrow[\text{(Y)}]{\text{CO}_2} \text{Na}_2\text{CO}_3 + \text{O}_2$
- 55.(A)  $\text{CaNCN} + 3\text{H}_2\text{O} \longrightarrow \text{CaCO}_{3(\text{s})} + 2\text{NH}_3$
- 56.(C) NaOH is hygroscopic absorb moisture and gets diluted.
- 57.(A)  $\text{Li}_{(\text{aq})}^+$  is biggest in size due to higher degree of hydration of  $\text{Li}^+$ .
- 58.(C) It is based on lattice energy and ionic character.



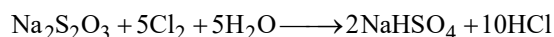
Sodium imparts golden yellow colour to flame.



61.(B)  $\text{Na}_2\text{S}_2\text{O}_3$  forms complex with  $\text{AgBr}$  and used for fixing of photographic film.

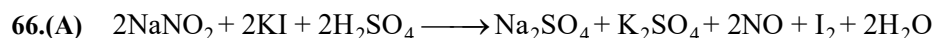
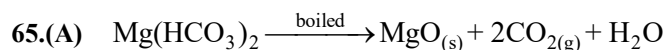


63.(B)  $\text{Na}_2\text{S}_2\text{O}_3$  is used as antichlor to remove  $\text{Cl}_2$ .



64.(D) Carnallite is  $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

K imparts violet colouration while Mg do not show flame test.



67.(D) Smaller alkali metal ion is bigger hydrated metal ion hence has least ionic conductance.

68.(B) Paramagnetic character decreases with concentration of solution.

69.(D) These solutions are strong reducing, dilute solution is paramagnetic while concentrated solution is diamagnetic.

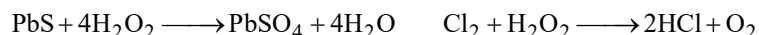
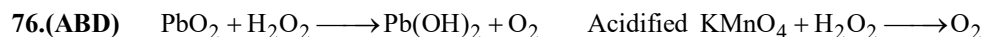
70.(C) Ammoniated solutions of alkali metals are used for reduction of aromatic compounds.

71.(D) Group I and II.

72.(D) No decomposition would occur because of high thermal stability.

73.(A) Magnesium oxide would be more stable because of comparable size of ions.

74.(D) Because nitrite anion is probably about the same size as the group I cations.



77.(ABC) Saline hydride are ionic in nature.

78.(BD) For  $\text{Na}_2\text{SO}_4$  hydration energy is more than lattice energy while for  $\text{BaSO}_4$  lattice energy is more than hydration energy.

79.(BD)  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  ions are exchanged with  $\text{Na}^+$ .    80.(ABD) Mg can also forms complex.

81.(ABD) Mg salt does not show flame test

82.(BCD)  $\text{NaOH}$  reacts with  $\text{Cl}_2$  and because of its hygroscopic nature it can't be used as primary standard in volumetric analysis.

83.(ACD) Li is much harder than the other group 1 metals.

- 84.(ACD)** Magnalium (95% Al + 5% Mg) is used in construction of airships and pistons of motor engines.
- 85.(ABC)** Basic nature is due to anionic hydrolysis. Conductivity of aqueous solution of alkali metals chloride increases down the group due to smaller size of hydrated alkali metal ion.
- 86.(BCD)**  $P + H_2O + NaOH \longrightarrow PH_{3(g)} + NaH_2PO_2$   
 $Al + NaOH \xrightarrow{\Delta} NaAlO_2 + H_2$   
 $Zn + NaOH \xrightarrow{\Delta} Na_2ZnO_2 + H_2$
- 87.(ABCD)** Only  $LiNO_3$  give  $NO_2$  and  $O_2$ . Only  $Li_2CO_3$  decompose to give  $CO_2$  and  $Li_2O$ . Only Li form  $Li_2O$ . Only lithium reacts with nitrogen.
- 88.(ACD)**  $Mg + CO_2 \longrightarrow MgO + CO$   
 $Mg + 2HNO_3 \longrightarrow Mg(NO_3)_2 + H_2$   
 $Mg + B_2O_3 \longrightarrow MgO + 2B$   
 $MgCl_2 \cdot 6H_2O \xrightarrow{\Delta} MgO + 5H_2O + 2HCl$
- 89.(ABD)** Beryllium shows diagonal relationship with Al hence its most of the chemistry resembles with Al.  $Be(OH)_2$  is amphoteric.
- 90.(ABC)** The reaction in which  $H_2O_2$  changes to  $O_2$ .
- 91.(ABD)** 50%  $H_2SO_4$  on electrolysis liberate  $H_2$  at cathode and  $H_2S_2O_8$  at anode.  
 $H_2SO_4 \rightarrow H^+ + HSO_4^-$   
 At cathode :  $2H^+ + 2e^- \rightarrow H_2$   
 At anode :  $2HSO_4^- \rightarrow H_2S_2O_8 + 2e^-$   
 $H_2S_2O_8 + 2H_2O \rightarrow 2H_2SO_4 + H_2O_2$
- 92.(ABC)** Different conformeric structures are possible for  $H_2O_2$ ,  $C_2H_6$  and  $N_2H_4$  due to free rotation about O–O, C–C and N–N bond.
- 93.(ABCD)**  $H_2O_2$  is non planar molecule having open book like structure.
- 94.(ABC)**  $O_3$  acts as oxidizing agent. It oxidises  $H_2O_2$ .  $H_2O_2 + O_3 \rightarrow H_2O + 2O_2$
- 95.(AC)** Hardness of water is due to  $HCO_3^-$  and  $SO_4^{2-}$  salt of calcium and magnesium.
- 96.(ABC)**  $H_2$  can reduce oxides of less active metals.
- 97.(BC)**  $CaCN_2 + 3H_2O \longrightarrow CaCO_3 + 2NH_3$        $Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$   
 $Ca_3P_2 + 6H_2O \longrightarrow 3Ca(OH)_2 + 2PH_3$        $Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$
- 98.(BD)**  $2Na + 2H_2O \longrightarrow 2NaOH + H_2$        $Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$   
 $Na_2O_2 + 2H_2O \longrightarrow 2NaOH + H_2O_2$        $NaNH_2 + H_2O \rightarrow NaOH + NH_3$   
 $Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2$        $CaO + H_2O \longrightarrow Ca(OH)_2$   
 $CaH_2 + 2H_2O \longrightarrow Ca(OH)_2 + H_2$
- 99.(BC)** Because of electron deficient nature  $BeH_2$  and  $BeCl_2$  exist in polymeric form.



- 100.(ABCD)** Hydrogen is nonmetallic element reacts with nonmetals and acts as reducing agent. Its electronic configuration is  $1s^1$
- 101.(ABCD)** Alkali metal forms salt like hydrides.
- 102.(ABD)** Calcium oxide and calcium hydroxide are basic hence soluble in acetic acid while  $\text{CaCO}_3$  is a salt of weak acid. Calcium oxalate is a salt of stronger acid.
- 103.(CD)**  $\text{Zn} + 2\text{NaOH} \longrightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$   
 $\text{Al} + \text{NaOH} \longrightarrow \text{NaAlO}_2 + \text{H}_2$   
 $\text{H}_2$  is combustible gas.  
 $\text{S} + \text{NaOH} \longrightarrow \text{Na}_2\text{S}_2\text{O}_3 + \text{Na}_2\text{S} + \text{H}_2\text{O}$   
 $\text{NH}_4\text{Cl} + \text{NaOH} \longrightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$
- 104.(BCD)** Nitride salt gives  $\text{NH}_3$  when come in contact with moisture.  
 $\text{Mg} + \text{N}_2 \longrightarrow \text{Mg}_3\text{N}_2$   
 $\text{Mg}_3\text{N}_2 + 6\text{H}_2\text{O} \longrightarrow 3\text{Mg}(\text{OH})_2 + 2\text{NH}_3$
- 105.(BC)** Calcium chloride and potassium fluoride are used to increase conductivity and to lower melting point.
- 106.** **[A-r] → [B-p, r, s] → [C-p, q, s] → [D-p, q, s]**  
 Li, Na and Cs shows flame test  
 Li is the strongest reducing agent  
 $\text{LiNO}_3 \longrightarrow \text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$   
 $\text{NaNO}_3 \xrightarrow{\Delta} \text{NaNO}_2 + \text{O}_2$   
 $\text{Ca} + 2\text{C} \xrightarrow{\Delta} \text{CaC}_2$   
 $\text{Ca}(\text{NO}_3)_2 \xrightarrow{\Delta} \text{CaO} + 2\text{NO}_2 + \text{O}_2$   
 $\text{Mg}(\text{NO}_3)_2 \xrightarrow{\Delta} \text{MgO} + 2\text{NO}_2 + \text{O}_2$
- 107.** **[A-p, q, s] → [B-p, s] → [C-p, q, r, s] → [D-p, q, s]**  
 Alkali metal sulphates are thermally stable. Alkali metals and alkaline earth metals when dissolved in liquid ammonia form blue coloured solution.
- 108.** **[A-q, r] → [B-p, s] → [C-p, q, r] → [D-p, s]**  
 Sulphur disproportionate in alkali.  $\text{Na}_2\text{S}$  and  $\text{Na}_2\text{SO}_3$  reduces ferric salts.  $2\text{FeCl}_3 + \text{Na}_2\text{S} \longrightarrow 2\text{FeCl}_2 + 2\text{NaCl} + \text{S}$
- 109.** **[A-p, q, r, s] → [B-p] → [C-p, q] → [D-r, s]**  
 $\text{Be}^{2+}$  has very strong tendency to form complex due to small size,  $\text{Mg}^{2+}$ ,  $\text{Ca}^{2+}$  and  $\text{Sr}^{2+}$  form complexes with EDTA.
- 110.** **[A-r, s] → [B-r] → [C-q, r] → [D-p, t]**  
 $\text{BeH}_2$  is polymeric hydride  $\text{AsH}_3$  is covalent hydride  
 $\text{B}_2\text{H}_6$  is Lewis acid and covalent hydride  $\text{LiAlH}_4$  is a complex species.
- 111.(8)** Eq. of base = Eq. of acid.

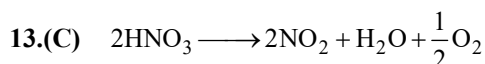
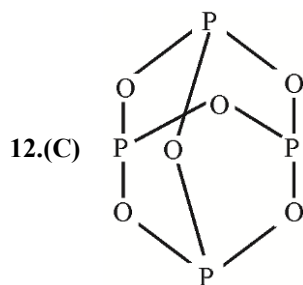


## Solutions - Advanced Problem Package | Chemistry

### p-Block Elements - I

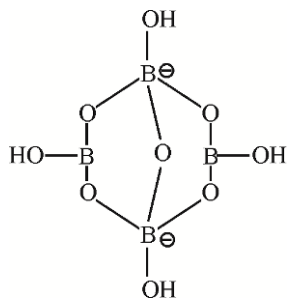
#### p-Block Elements - 1

- 1.(C)  $\text{AlCl}_3 + 3\text{H}_2\text{O} \longrightarrow \text{Al}(\text{OH})_3 + 3\text{HCl}(\text{g})$
- 2.(D) Anhydrous  $\text{AlCl}_3$  is covalent in nature and exist as dimer ( $\text{Al}_2\text{Cl}_6$ ) to complete octet.
- 3.(B) Due to presence of polar B—N bonds in borazole.
- 4.(A) Octet of Al in  $\text{AlCl}_3$  is complete due to dimeric structure.
- 5.(C) Non-metal hydroxides are acidic while metal hydroxides are basic or amphoteric.
- 6.(B) Diborane is a dimer of  $\text{BH}_3$  having 3c-2e bonds, as well as 2c-2e bonds. There are two bridged bonds and four terminal bonds.
- 7.(C) Due to dimeric structure having bridged 3c-2e bonds, B is  $\text{sp}^3$  hybridized.
- 8.(D) In graphite every carbon atom has one delocalizable electron.
- 9.(A) CO is every strong ligand and can form complexes with transition metal ions.
- 10.(B) The ability of a substance to assume two or more crystalline structures is called polymorphism
- 11.(D) (A)  $\text{H}_3\text{PO}_4$  (orthophosphoric acid) (B)  $\text{H}_3\text{PO}_2$  (hypophosphorous acid)  
(C)  $\text{HPO}_3$  (metaphosphoric acid) (D)  $\text{H}_4\text{P}_2\text{O}_7$  (pyrophosphoric acid)



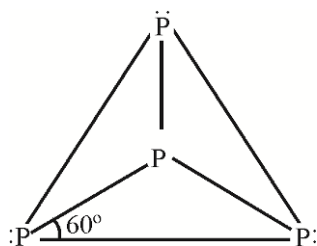
- 14.(C) Non-metal oxides are acidic while metal oxides are basic. Heavier members are metallic in nature.
- 15.(C) Due to larger size of central atom.
- 16.(A) Due to more difference in electronegativity of N and H.
- 17.(D) Both  $\text{N}_2\text{O}$  and  $\text{CO}_2$  have 22 electrons and are linear in shape.
- 18.(B)  $\text{NH}_3$  shows H-bonding therefore has high boiling point.  $\text{PH}_3$  has lowest boiling point because of small size and lower molecular mass.
- 19.(C) Because of more difference in electronegativity of N and F.
- 20.(A) Compound in higher oxidation state are less ionic due to more polarization of anion by cation.
- 21.(C)  $\text{N}_2\text{O} + \text{Cu} \longrightarrow \text{CuO} + \text{N}_2$
- 22.(A) Strongly electro positive metals evolve  $\text{H}_2$  on reaction with dil  $\text{HNO}_3$ .
- 23.(A) The biological reduction of nitrogen to ammonia and further conversion to  $\text{NO}_3^-$  is carried out by bacteria.
- 24.(A) Species having more electronegative central atom is strongly oxidizing agent.

- 25.(A) Producer gas :  $(\text{CO} + \text{N}_2)$  water gas :  $(\text{CO} + \text{H}_2)$
- 26.(D)  $\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_4$
- 27.(A)  $\text{PH}_4\text{I} + \text{NaOH} \longrightarrow \text{PH}_3 + \text{NaI} + \text{H}_2\text{O}$
- 28.(D) It has cross linked sheet structure hence less strain in the rings
- 29.(C)  $2\text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{B}_2\text{O}_3 + 3\text{H}_2\text{O}$
- 30.(B)  $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} \xrightarrow[-10\text{H}_2\text{O}]{\Delta} \text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\Delta} 2\text{NaBO}_2 + \text{B}_2\text{O}_3$   
 $\text{B}_2\text{O}_3 + \text{CuSO}_4 \xrightarrow{\Delta} \text{Cu}(\text{BO}_2)_2 + \text{SO}_3 \uparrow$
- 31.(B) Ultrapure form is obtained by zone refining and vapour phase refining.
- 32.(C) It is disproportionation reaction in which oxidation state of phosphorous changes from zero to -3 and +1.
- 33.(B)  $(\text{Na}_3\text{PO}_4)_3 + 2\text{H}_2\text{O} \longrightarrow \text{Na}_5\text{P}_3\text{O}_{10} + 4\text{NaOH}$  ;  $3\text{NaH}_2\text{PO}_4 \xrightarrow{\Delta} \text{Na}_3\text{P}_3\text{O}_9 + 3\text{H}_2\text{O}$   
 Triphosphate Sodium cyclotriphosphate
- 34.(A) Hypo phosphorus is monobasic oxoacid of phosphorus having two P—H bonds.
- 35.(B)  $\text{P}_4\text{O}_6 + 6\text{H}_2\text{O} \longrightarrow 4\text{H}_3\text{PO}_3$
- 36.(C)  $\text{Fe}_2\text{O}_3 + 2\text{Al} \longrightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$  ; Al is more electropositive than Fe and this reaction is highly exothermic.
- 37.(A) Anion in Egyptian blue is  $\text{Si}_4\text{O}_{10}^{4-}$  i.e.  $\text{Si}_{2n}\text{O}_{5n}^{2n-}$  hence it is a sheet silicate.
- 38.(D)  $6\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O} \xrightarrow{\Delta} \text{Na}_2[\text{Na}_4(\text{PO}_3)_6] + 6\text{NH}_3 + 30\text{H}_2\text{O}$
- 39.(C) Hydride of elements of group 13 are electron deficient.  $\text{H}_3\text{N} \cdot \text{BH}_3$  is 1 : 1 complex of  $\text{NH}_3$  and  $\text{BH}_3$ .
- 40.(D) It is complex anion of Borax  $\text{Na}_2[\text{B}_4\text{O}_5(\text{OH})_4] \cdot 8\text{H}_2\text{O}$ .

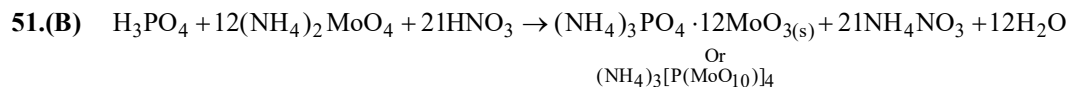
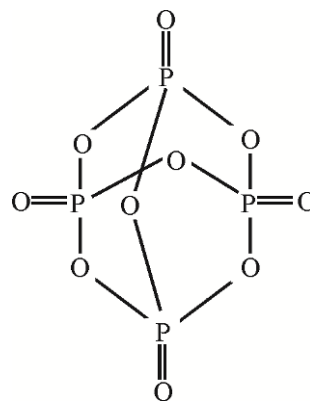


- 41.(A)  $(\text{CH}_3)_2\text{SiCl}_2$  form  $(\text{CH}_3)_2\text{Si}(\text{OH})_2$  and it on polymerization gives linear chain silicones.  $\text{CH}_3\text{SiCl}_3$  on hydrolysis gives  $\text{CH}_3\text{Si}(\text{OH})_3$  which on polymerization forms cross linked silicones.
- 42.(D)  $\text{BF}_3$  is planar because its dipole moment is zero.
- 43.(A) Borazole or Borazine is inorganic benzene and its structure resembles with benzene.
- 44.(C) In  $\text{B}_2\text{H}_6$  only four nonbridging H atoms can be substituted.
- 45.(B)  $\text{Al}_2\text{Me}_6$  because of  $3c-2e^-$  bond
- 
- 46.(C) Boric acid form complex with cis-diol for complete ionization.
- 47.(C) White phosphorus is tetratomic reactive form due to strain in the rings.
- 48.(D)  $\text{H}_2\text{P}_2\text{O}_7$  is non reducing due to absence of P—H or P—P bonds

49.(B)



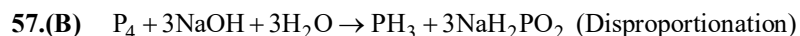
50.(D)

52.(C)  $\text{Fe}^{2+}$  and  $\text{S}^{2-}$  both will get oxidized.

53.(C) Concentrated nitric acid is a strong oxidizing agent and attacks most metals except noble metals such as gold and platinum. It also oxidizes non-metals and their compounds.

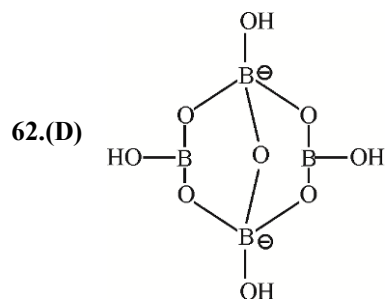
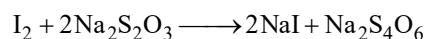
54.(A) Fe, Cr and Al becomes passive in conc. nitric acid because of the formation of a passive film of oxide on the surface.

55.(C) Because of more solubility of nitrate salts.

56.(C)  $\text{NH}_3$  is  $\text{sp}^3$  hybridized and better  $\sigma$  donor because of small size of nitrogen and more directional nature of hybrid orbital.58.(B) Bond angle in  $\text{NH}_3$  is  $107^\circ$  due to presence of lone pair electron.

59.(A) On increasing pressure equilibrium shift towards higher density solid.

60.(D) Layers can slide over each other due to weak Van der Waal's forces.

2 triangular units, 2 tetrahedral units, four  $-\text{OH}$  units.63.(B) One oxygen atom is shared,  $(\text{HO})_2\text{B}-\text{O}-\text{B}(\text{OH})_2$ .64.(A) In chain silicates two oxygen atoms of every  $\text{SiO}_4^{4-}$  unit are shared with neighbouring units. Hence formula is  $(\text{SiO}_3^{2-})_n$ .66.(C) Carbon suboxide is  $\text{C}_3\text{O}_2$  and its structural formula is  $:\ddot{\text{O}}=\text{C}=\text{C}=\ddot{\text{O}}:$  67.(B) Cyclic anhydride is formed.

68.(ABCD) Due to partial double bond character C—C bond order is higher in graphite.

69.(BCD) Hydrated aluminium chloride do not form anhydrous  $AlCl_3$  on heating.



70.(ABD)  $SO_3$  is soluble in water hence can't be collected over water.

71.(BCD) Hydrides of group 15<sup>th</sup> elements are poisonous and poisonous nature increases down the group, stability of Hydrides and basic strength decreases down the group.

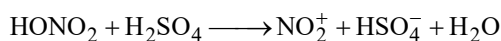
72.(ABCD) Trona is  $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$  or  $Na_3(CO_3)(HCO_3) \cdot 2H_2O$

Borax is  $Na_2B_4O_7 \cdot 10H_2O$  or  $Na_2[B_4O_5(OH)_4] \cdot 8H_2O$

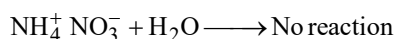
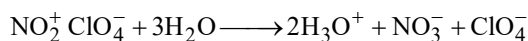
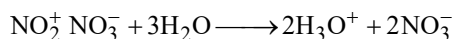
Graham's salt is sodium hexametaphosphate  $(NaPO_3)_6$  or  $Na_2[Na_4(PO_3)_6]$

Alum is  $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$  or  $[K(H_2O)_6]_2SO_4 \cdot [Al(H_2O)_6]_2(SO_4)_3$

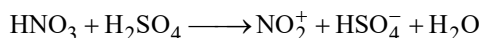
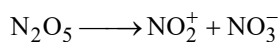
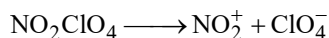
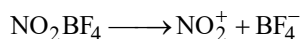
73.(C) Mixture of conc.  $HNO_3$  and conc.  $H_2SO_4$  is called as nitrating mixture and  $HNO_3$  acts as base (Bronsted base) in its reaction with  $H_2SO_4$ .



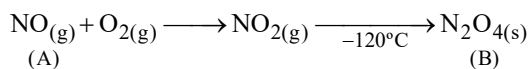
74.(ABC)  $NO_2^+ BF_4^- + 3H_2O \longrightarrow 2H_3O^+ + NO_3^- + BF_4^-$



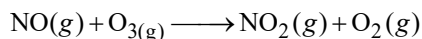
75.(ABCD) All are nitrating agents and are used for nitration of aromatic compounds.



76.(B) NO is colourless, gaseous, paramagnetic oxide



77.(B) Nitric oxide combine very rapidly with ozone

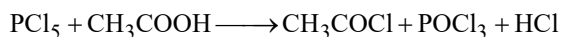


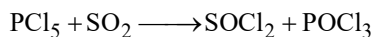
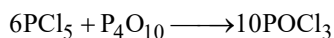
78.(B)  $N_2O_{4(s)} \xrightarrow{F_2} 2NO_2F(g) \xrightarrow{BF_3(g)} NO_2^+ BF_4^-(s) \xrightarrow{3H_2O} 2H_3O^+ + NO_3^- + BF_4^-$   
Colourless  
diamagnetic

79.(ABC) H-bond is formed with more electronegative atoms F, O and N.

80.(C)  $N_3H$  is acidic in nature,  $H - N \equiv \overset{+}{N} = \overset{-}{N}$

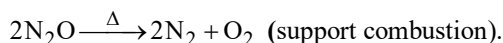
81.(ABCD) Compounds having —OH group and/or double bonded oxygen can reacts with  $PCl_5$ .



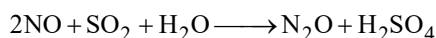
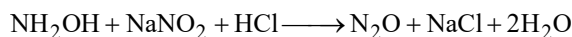
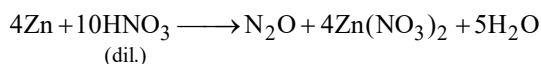


**82.(BCD)** Except orthosilicate all other silicates have one or more number of oxygen atoms of  $\text{SiO}_4^{4-}$  unit shared with neighbouring units.

**83.(ABCD)**  $\text{N}_2\text{O}$  is a good supporter of combustion as it provide more oxygen.  $\text{N}_2\text{O}$  and  $\text{CO}_2$  are isoelectronic as well as isostructural.



**84.(ABCD)**  $\text{NH}_4\text{NO}_3 \xrightarrow{\Delta} \text{N}_2\text{O} + 2\text{H}_2\text{O}$



**85.(ACD)** According to molecular orbital theory

**86.(ABCD)** White tin changes to grey tin at lower temperature

**87.(ABD)** Ammonium salt having oxidizing anion will not release  $\text{NH}_3$  on heating.

**88.(ABCD)** All carbon atoms in graphite are  $\text{sp}^2$  hybridized and in diamond all carbon atoms are  $\text{sp}^3$  hybridized.

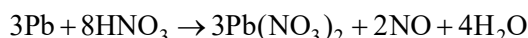
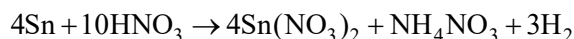
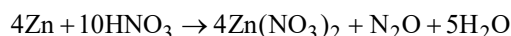
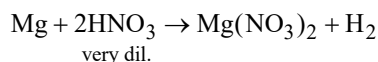
**89.** **[A-r] → [B-s] → [C-p, q] → [D-p]**

$\text{O}_2$  is absorbed by alkaline pyrogallol

$\text{NH}_3$  is absorbed by both water and  $\text{H}_2\text{SO}_4$

$\text{CO}$  is absorbed by ammonical  $\text{Cu}_2\text{Cl}_2$  solutions.

**90.** **[A-q] → [B-r] → [C-s] → [D-p]**



**91.** **[A-p, s, t] → [B-p, s] → [C-q, r] → [D-r, s, t]**

Hypophosphoric acid ( $\text{H}_4\text{P}_2\text{O}_6$ ) (Tetrabasic and reducing)

Pyrophosphoric acid ( $\text{H}_4\text{P}_2\text{O}_7$ ) (Tetrabasic)

Boric acid  $\text{H}_3\text{BO}_3$  (Lewis acid)

Hypophosphorus acid ( $\text{H}_3\text{PO}_2$ ) (Monobasic and reducing)

In all oxoacids of phosphorus, P is  $\text{sp}^3$  accept meta phosphoric acid.

**92.** **[A-p, q] → [B-q, t] → [C-p, r, s] → [D-q, t]**

Non metal oxides are mostly acidic except ( $\text{CO}$ ,  $\text{NO}$  and  $\text{N}_2\text{O}$ , these are neutral). Odd electron species are coloured and paramagnetic.

**93.** **[A-p, s, t] → [B-p, t] → [C-q, r, s] → [D-q, r]**

Borax and alum swells up on heating. Ammonium salt and aluminium salt reacts with  $\text{NaOH}$ .

**94.** **[A-p] → [B-q, r] → [C-q, r] → [D-r, s]**

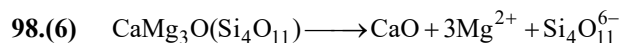
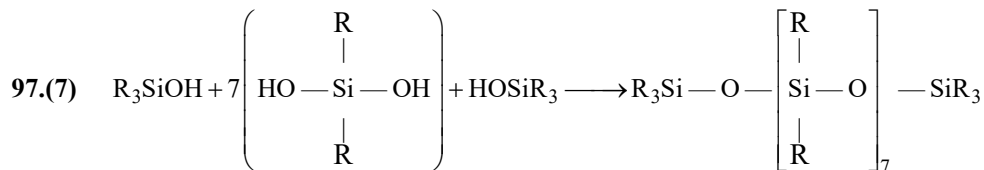
Non metal hydroxide are acidic while metal hydroxide are generally basic and some are amphoteric.

95. [A-p, r] → [B-p, q, s] → [C-p, q, t] → [D-p, q, t]

In all silicates co-ordination number of Si is four. Except ortho silicate in all other silicates one or more number of oxygen atoms are shared. Pyro silicate (one), single chain silicate and rings silicate two oxygen atoms are shared.

96. [A-p, q, r] → [B-p, q, r] → [C-p, q, r] → [D-s, t]

O<sub>3</sub>, HNO<sub>3</sub> & H<sub>2</sub>O<sub>2</sub> are oxidizing agent while H<sub>3</sub>PO<sub>3</sub> acts as reducing agent.



99.(5) Bigger size Lewis base are used for symmetrical cleavage of B<sub>2</sub>H<sub>6</sub>, (C<sub>2</sub>H<sub>5</sub>)<sub>3</sub>N, (CH<sub>3</sub>)<sub>3</sub>N, (C<sub>2</sub>H<sub>5</sub>)<sub>2</sub>NH, (i-Pr)<sub>3</sub>N, quinuclidine

100.(3) It is cyclic silicate in which two oxygen atoms of every SiO<sub>4</sub><sup>4-</sup> units are shared hence its empirical formula is SiO<sub>3</sub><sup>2-</sup>.

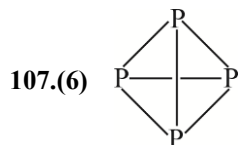
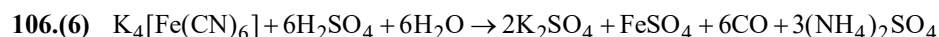
101.(5) Salt having non-oxidizing anion on heating evolve NH<sub>3</sub>. NO<sub>3</sub><sup>-</sup>, NO<sub>2</sub><sup>-</sup>, Cr<sub>2</sub>O<sub>7</sub><sup>2-</sup> and ClO<sub>4</sub><sup>-</sup> are oxidizing anions.

102.(2) B—H—B bond is 3c-2e bond hence it is formed by sharing of two electrons.

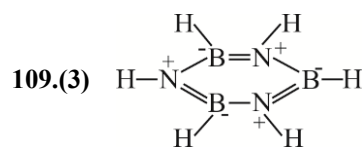
103.(1) In graphite one mobile or delocalizable electron is present on each carbon atom.

104.(2) Both phosphorus are sp<sup>3</sup> hybridized, hence there are two dπ - pπ bonds because of two P = O bonds.

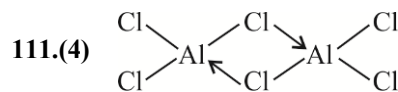
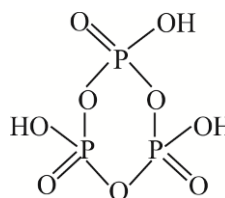
105.(1) Both Fe and NO have +1 oxidation state in brown ring complex.



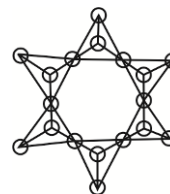
108.(1) MeSiCl<sub>3</sub> is used for synthesis of cross linked polymer.



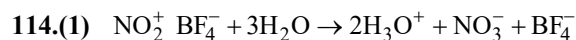
110.(3) (HPO<sub>3</sub>)<sub>3</sub> · (Trimer)



112.(2) It is a cyclic silicate hence two oxygen atoms of every unit are shared.



113.(1) Hydrazoic acid is acidic nitrogen hydride, N<sub>3</sub>H.



Mole of H<sub>3</sub>O<sup>+</sup> = mol of OH<sup>-</sup>

$$0.01 \times 2 = x \times \frac{20}{1000}$$

$$x = 1$$

115.(4) Silica is a network silicate in which every silicon atom is bonded with four oxygen atoms.



## Solutions - Advanced Problem Package | Chemistry

### p-Block Elements - II

- 1.(A) Hypochlorous acid is oxoacid and its anhydride is  $\text{Cl}_2\text{O}$ .
- 2.(C)  $\text{COCl}_2$  is phosgene and it does not contain metallic ion.
- 3.(A)  $\text{H}_2\text{O}$  because of H-bonding.
- 4.(A) HI is strongest reducing agent because  $\text{I}^-$  can be easily oxidized.
- 5.(C) Electron gain enthalpy decreases.
- 6.(D) Oxoacids having lower oxidation state of halogen are stronger oxidizing agent.
- 7.(C) Chlorate salts decomposes on heating,  $4\text{AgClO}_3 + 3\text{Cl}_2 \longrightarrow 6\text{ClO}_2 + 4\text{AgCl}$
- 8.(B) HCl is most volatile due to weakest inter molecular attractive forces. While HF is least volatile because of H-bonding.
- 9.(C) F do not show positive oxidation state.
- 10.(B)  $\text{Cl}_2$  is greenish yellow gas and  $\text{KClO}_3$  is used in safety matches.
- 11.(C)  $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \longrightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$
- 12.(C) I—I bond is the weakest bond.
- 13.(A) Salt of weakest acid. Acidic strength of HClO is minimum amongst oxoacid of chlorine.
- 14.(A) Acidic strength of non-metal oxide increases with increase in oxidation state.
- 15.(D) Because of Hydrogen bond between  $\text{F}^-$  and  $\text{H}-\text{F}$  hence mixture contains  $\text{K}^+\text{HF}_2^-$ .
- 16.(C) Xe form compounds with fluorine and oxygen and heavier noble gases can be easily liquefied.
- 17.(D) Ar is the most abundant rare gas in atmosphere and Radon is obtained as a decay product of radium.  

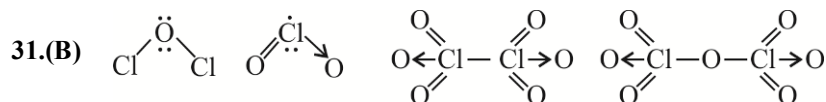
$${}_{88}^{226}\text{Ra} \longrightarrow {}_{86}^{222}\text{Rn} + {}_2^4\text{He}$$
 Helium is most abundant in universe.
- 18.(B)  $\text{XeF}_4$  is square planar and  $\text{sp}^3\text{d}^2$  hybridized.
- 19.(C) Ar is most abundant in the atmosphere while He is most abundant in universe.
- 20.(A)  $\text{Zn} + 2\text{NaOH} \longrightarrow \text{Na}_2\text{ZnO}_2 + \text{H}_2$  ;  $\text{Zn} + \text{H}_2\text{SO}_4 \longrightarrow \text{ZnSO}_4 + \text{H}_2$
- 21.(A)  $\text{KClO}_3(\text{s}) \xrightarrow{\Delta} \text{KClO}_4(\text{s}) + \text{KCl}(\text{s})$ ;  $\text{KClO}_4(\text{s}) \longrightarrow \text{KCl}(\text{s}) + 2\text{O}_2(\text{g})$
- 22.(A) Mercury loses its meniscus when come in contact with Hg.
- 23.(B)  $2\text{KI} + \text{H}_2\text{SO}_4 \longrightarrow \text{K}_2\text{SO}_4 + 2\text{HI}$  ;  $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
- 24.(D) More reactive halogen can displace less reactive halogen from their salts.
- 25.(B)  $\text{Cl}_2 + \text{H}_2\text{O} \longrightarrow \text{HOCl} + \text{HCl}$
- 26.(C)  $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \longrightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + \text{NaBr}$
- 27.(B) Total electron pairs around central atom are 5 hence its structure is trigonal bipyramidal.

28.(A)  $\text{Br}_2$  vapours will get dissolved in carbon tetrachloride and carbon disulphide while animal charcoal will adsorb  $\text{Br}_2$  vapours.

29.(D)  $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \longrightarrow 2\text{NaCl} + \text{SO}_2 + \text{S} + \text{H}_2\text{O}$  ;

$\text{Br}_2 + \text{SO}_2 + \text{H}_2\text{O} \longrightarrow 2\text{HBr} + \text{SO}_3$

30.(C)  $\text{H}_2\text{S}_2\text{O}_7$  (S–O–S bond) ;  $\text{H}_2\text{S}_2\text{O}_8$  (S–O–O–S bond) ;  $\text{H}_2\text{S}_2\text{O}_6$  (S–S bond) ;  $\text{H}_2\text{S}_2\text{O}_3$  (S=S bond)



32.(D) These are polar covalent compounds like  $\text{ICl}$ ,  $\text{ClF}_3$  etc.

33.(C)  $\text{H}_2\text{S}_2\text{O}_8 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{SO}_5$  (peroxomonosulphuric acid)

34.(D) Lone pair of electrons and the least electronegative Cl atom are in equatorial plane at  $120^\circ$  with more electronegative F atoms in axial position.

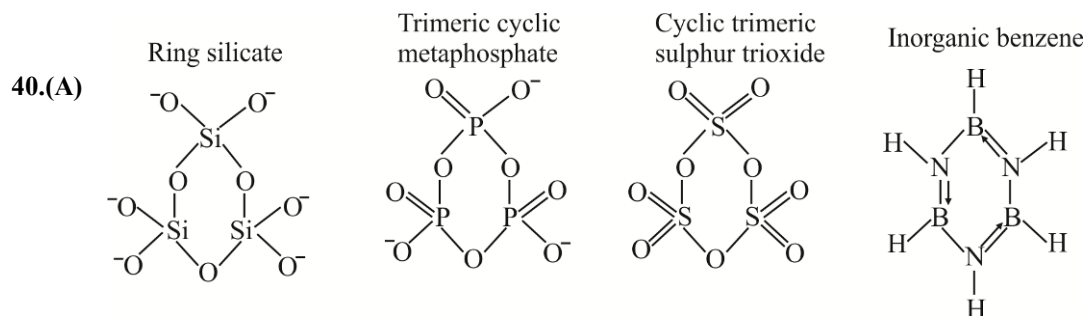
35.(D) Oxygen cannot form more than three bonds.

36.(D)  $\text{HSO}_3^-$  and  $\text{S}_2\text{O}_7^{2-}$  are formed in one of the step of reaction and then reused hence these are intermediates.

37.(C) Molar mass of  $\text{H}_2\text{O}$  is lower than that of  $\text{N}_2$  and  $\text{O}_2$  .

38.(D) Sulphur is used for vulcanization of rubber.

39.(B) Selenium is a poor conductor of electricity, but is a good photoconductor.



41.(D)  $4\text{XeO}_3 + 6\text{Ba}(\text{OH})_3 \rightarrow 3\text{Ba}_2\text{XeO}_6 + \text{Xe} + 6\text{H}_2\text{O}$

42.(D)  $\text{XeF}_6 + \text{BF}_3 \rightarrow \text{XeF}_5^+ \text{BF}_4^-$  (Flouride ion donor)

$\text{NaF} + \text{XeF}_6 \rightarrow \text{NaXeF}_7^-$  (Flouride ion acceptor)

43.(C) It is thermally unstable and decomposes to produce Xe and  $\text{O}_2$  .

44.(A) In  $\text{XeOF}_4$  there are total 6 electron pairs around central atom.

45.(B)  $2\text{H}_2\text{SO}_4 + \text{Cu} \rightarrow \text{CuSO}_4 + \text{SO}_2 + \text{H}_2\text{O}$

46.(B)  $\text{HNO}_3$  vapours condense below  $100^\circ\text{C}$  to form liquid  $\text{HNO}_3$  .

47.(D) Sulphur and sulphur containing compound on combustion form  $\text{SO}_2$  .

48.(C) Because formation of  $\text{SO}_3$  is reversible reaction and by remaining  $\text{SO}_3$  shift equilibrium toward forward direction.

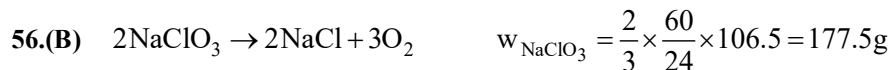
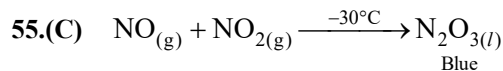
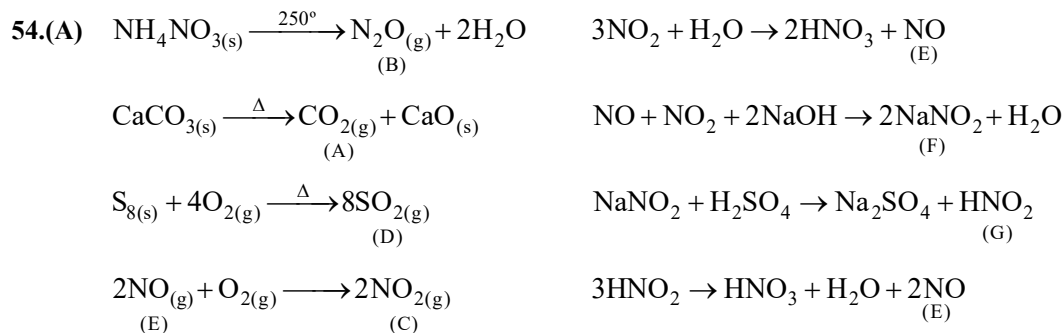
49.(B) Mole of water =  $\frac{\text{mass of solution} \times \% \text{ of water}}{18} = \frac{1.84 \times 1 \times 2}{18 \times 100}$

50.(A) Argon is chemically inert hence it is used to create inert atmosphere.

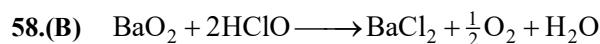
51.(C)  $\text{XeO}_3$  is pyramidal because of one lone pair and three bond pairs.

52.(C)  $\text{XeF}_4$  and  $\text{XeF}_6$  can act as oxidizing as well as reducing due to intermediate oxidation state.

53.(B)  $\text{N}_2\text{O}$  is used as propellant for whipped cream.

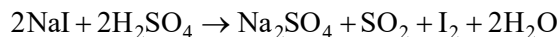
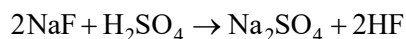
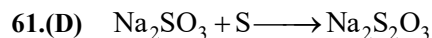


57.(A) Volume of oxygen produced is equal to area under the curve.

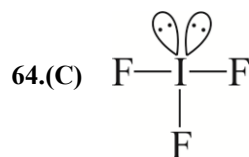
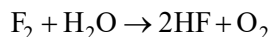


59.(A)  $\text{SO}_2$  can decolourise characteristic colouration of these solution while  $\text{CO}_2$  can't decolourise.

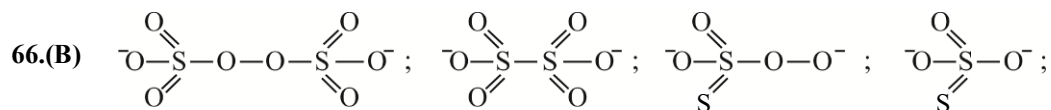
60.(B) Bleaching action of  $\text{SO}_2$  is due to its reducing nature. It reduces colouring matter and reduced colouring matter will restore its colour by oxidation from air.



63.(C)  $\text{F}_2$  do not disproportionate in water or alkali.

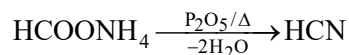
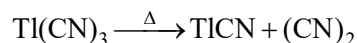
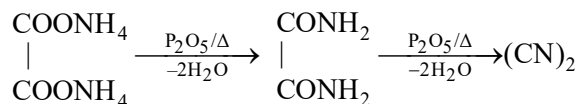
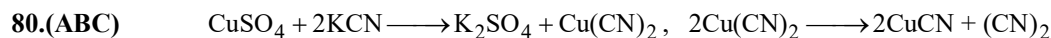


65.(AB)  $\text{P}_4\text{O}_{10}$  on complete hydrolysis gives  $\text{H}_3\text{PO}_4$ .



- 67.(ABC) Gallium have very low melting point (303K) it exist in liquid form at or near normal room temperature. Bromine and mercury are two well-known liquid substances.
- 68.(ABCD) Species having unpaired electrons are paramagnetic.  
 $O_2^-$ ,  $O_2$  and  $S_2$  are paramagnetic on the basis of molecular orbital theory. In  $ClO_2$  there is unpaired electron in chlorine atom.
- 69.(BD) Inter halogen compounds are all covalent molecules and are diamagnetic in nature. Their physical properties are intermediate between those of constituents halogens except that their m.p. and BP are little higher than expected. In general, inter halogen compounds are more reactive than halogens except fluorine.
- 70.(ABC) Comproportionation is reverse of disproportionation
- 71.(ABCD) Deacon's process:-  $4HCl + O_2 \xrightarrow[723K]{CuCl_2} 2Cl_2 + 2H_2O$ : it is also prepared by electrolysis of chloride salt solutions.
- 72.(ABCD)  $AgBr + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + NaBr$ ;  $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$   
 $Cl_2 + Na_2S_2O_3 + 2H_2O \longrightarrow 2NaHSO_4 + 2HCl$
- 73.(ABCD) **Element Allotropic forms**  
 Carbon Diamond, Graphite, Fullerene  
 Phosphorous White phosphorous, Red phosphorous and black phosphorous  
 Sulphur Monoclinic, Rhombic  
 Tin White tin, Grey tin
- 74.(ABC) Iodine is a molecular solid rest all are covalent solids.
- 75.(AB) Sulphur and Iodine are molecular solids
- 76.(ABCD)  $FeCl_3 \cdot 6H_2O + 6SOCl_2 \longrightarrow FeCl_3 + 6SO_2 + 12HCl$   
 $CH_3CH_2OH + SOCl_2 \longrightarrow CH_3CH_2Cl + SO_2 + HCl$   
 $CH_3COOH + SOCl_2 \longrightarrow CH_3COCl + SO_2 + HCl$   
 $P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$
- 77.(BC)  $C_2$  is diamagnetic while  $B_2$  and  $S_2$  are paramagnetic forms according to molecular orbital theory.
- 78.(D) White tin is stable at room temperature but at low temperature ( $<13.2^\circ C$ ) it transforms into grey tin on prolonged exposure at temperature well below this. Other allotropic transformations are possible at higher temperature.
- 79.(ABCD) 

| Molecular species                                                                                                                                                                                | No. of $\pi$ - bond        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| $B_2$                                                                                                                                                                                            | 1 {According to MO Theory} |
| $C_2$                                                                                                                                                                                            | 2 {According to MO Theory} |
| $\begin{array}{c} \cdot\dot{N} = \ddot{O}: \\ \cdot\ddot{C} \equiv \ddot{O}: \end{array}$ $\begin{array}{c} O = \ddot{Cl} - O^- \\ O = \ddot{S} = O \end{array}$                                 | 1                          |
| $\begin{array}{c} O \\    \\ O = S = O \\    \\ O \end{array}$ $\begin{array}{c} O \\    \\ O = Cl - O^- \\    \\ O \end{array}$ $\begin{array}{c} O \\    \\ O = Xe = O \\    \\ O \end{array}$ | 2                          |
| $\begin{array}{c} O \\    \\ O = S = O \\    \\ O \end{array}$ $\begin{array}{c} O \\    \\ O = Cl - O^- \\    \\ O \end{array}$ $\begin{array}{c} O \\    \\ O = Xe = O \\    \\ O \end{array}$ | 3                          |
| $\begin{array}{c} O \\    \\ O = Xe = O \\    \\ O \end{array}$                                                                                                                                  | 4                          |

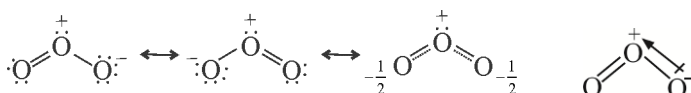


81.(ABCD)  $\text{O}_2$ ,  $\text{NO}$ ,  $\text{NO}_2$ , and  $\text{S}_2$  all are paramagnetic while  $\text{O}_3$ ,  $\text{N}_2\text{O}_3$ ,  $\text{N}_2\text{O}_4$ , and  $\text{S}_8$  all are diamagnetic.

82.(D)  $\text{OSCl}_2$  reacts vigorously with water and is particularly valuable for drying or dehydrating readily hydrolysable inorganic halides.  $\text{MX}_n \cdot m\text{H}_2\text{O} + m\text{OSCl}_2 \longrightarrow \text{MX}_n + m\text{SO}_2 + 2m\text{HCl}$ .

With conc.  $\text{H}_2\text{SO}_4$  sulphates and with  $\text{P}_4\text{O}_{10}$  phosphate and with  $\text{CaO}$  hydroxides may be precipitated.

83.(ABCD)

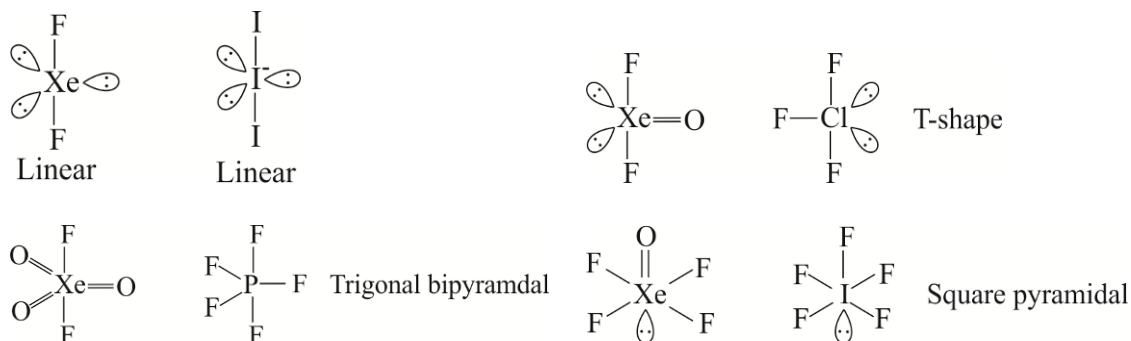


$\text{O}_2$  is paramagnetic while  $\text{O}_3$  is diamagnetic.  $\text{O}_3$  has nonzero dipole moment due to angular shape and presence of polar ends.

84.(ABCD)

85.(ABCD) All the substance in their elemental forms have standard free energy of formation equal is zero.

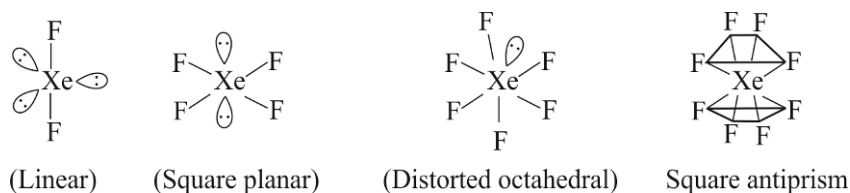
86.(ABCD)



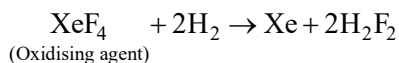
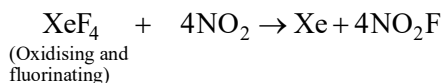
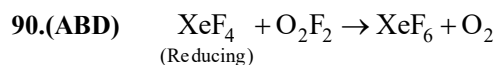
87.(ABCD) All are correctly matched refer NCERT text book.

88.(BC) Reactions mentioned in options (A) and (D) are non redox reactions. Any reaction in which a substance is present in its elemental form is a redox reaction and Xenon fluorides are strong oxidizing agents.

89.(ABCD)

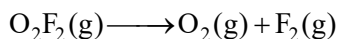
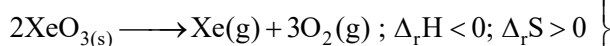
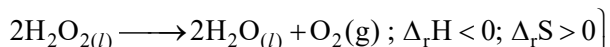
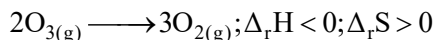


$\text{XeF}_8^{2-}$  is square antiprism, this shape does not provide a site for the lone pair on Xe.



91.(B) Since the formation of ozone from oxygen is an endothermic process, it is necessary to use a silent electrical discharge in its preparation to prevent its decomposition. Reaction of fluorine with water and  $\text{H}_2\text{O}_2$  gives  $\text{O}_2$ .

92.(ABCD) All these compounds are endothermic and their decomposition is favourable.



93.(ABC) Compounds of Xe with  $\text{O}_2$  and  $\text{F}_2$  are not formed by direct combination at 298K. Phosphorus readily burns in air to give  $\text{P}_4\text{O}_{10}$ .  $\text{N}_2$  also do not reacts with  $\text{O}_2$  at 298 K.

94.(C) Xenon oxides are endothermic ( $\Delta_f G^\circ > 0$ ) and cannot be prepared by direct interaction of the elements. The oxides are prepared by the hydrolysis of xenon fluorides.

95.(ABCD) HF cannot form three dimensional array of H-bonds as in water and so has less b.p. than water but it has stronger hydrogen bonds than water hence it is more viscous than water. It is stronger acid than water.

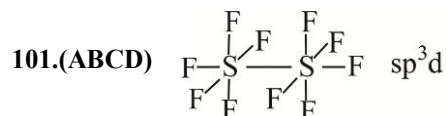
96.(ABCD)  $\text{OF}_2$  is not an oxide it is a fluoride.

97.(ABC)  $\text{I}^-$  can be oxidized by oxygen in acidic medium just the reverse of the reaction observed with fluorine.

98.(ABCD) All are common bleaching substances used for different bleaching purpose.  $\text{ClO}_2$  is used as a bleaching agent for paper pulp and textiles and in water treatment.

99.(ABCD) Chlorine oxides are highly reactive oxidizing agents and tend to explode.

100.(ABCD) All are correct facts refer NCERT text book.



102. [A-q]  $\rightarrow$  [B-p]  $\rightarrow$  [C-s]  $\rightarrow$  [D-r]

Order of bond energy  $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$

Order of b.p.  $\text{H}_2\text{O} > \text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$   
(H-bond)

Order of stability of monochloride.  $\text{TlCl} > \text{InCl} > \text{GaCl}$  [inert pair effect]

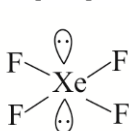
$\text{SiO}_2$  is a high m.p. network solid while  $\text{CO}_2$  is a gas.

Maximum bond dissociation enthalpy is for  $\text{Cl}_2$  not for  $\text{F}_2$  because of lone pair repulsions. Solubility of  $\text{F}_2$  is maximum due to its small size and ability to form hydrogen bond.  $\text{Br}_2$  is a liquid and maximum inter molecular distance is in  $\text{F}_2$  because of small Vander Waal's radii.

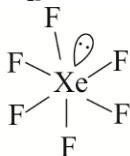
104.  $[A-p] \rightarrow [B-s] \rightarrow [C-q] \rightarrow [D-r]$

Potash solution absorb all acidic gases and  $\text{Cl}_2$ . Oxygen is absorb by alkaline pyrogallol solution.  $\text{FeSO}_4$  solution absorb nitric oxide (NO). Carbon monoxide is absorb by ammonical cuprous chloride solution.

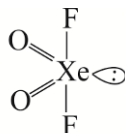
105.  $[A-s] \rightarrow [B-r] \rightarrow [C-p] \rightarrow [D-q]$



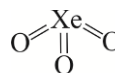
(Square planar)



(Distorted octahedral)



Trigonal bipyramdal



Pyramidal

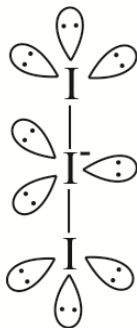
- 106.**      $[A-p, q, s] \rightarrow [B-q, r, t] \rightarrow [C-q, r] \rightarrow [D-p]$

All halogens are coloured substances. Although electron gain enthalpy of fluorine is less negative as compared to chlorine. Fluorine is a stronger oxidizing agent than chlorine to

- (1) Low enthalpy of dissociation of F – F bond and
- (2) High hydration enthalpy of  $F^-$ . Fluorine shows only  $-1$  and  $0$  oxidation states.

- 107.(3)** F(-1), Na(+1), Mg(+2)

- 108.(9)** There are total 9 lone pairs as shown in figure.



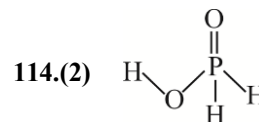
- 109.(2)** Acidified  $\text{KMnO}_4$  is an oxidizing agent and  $\text{CO}_3^{2-}$  and  $\text{SO}_4^{2-}$  cannot be oxidized.

- 110.(4)**  $\text{H}_4\text{P}_2\text{O}_7$  is formed when  $\text{H}_3\text{PO}_4$  is heated at  $250\text{--}260^\circ\text{C}$ . This acid contains 4—OH groups thus it can form four series of salts.

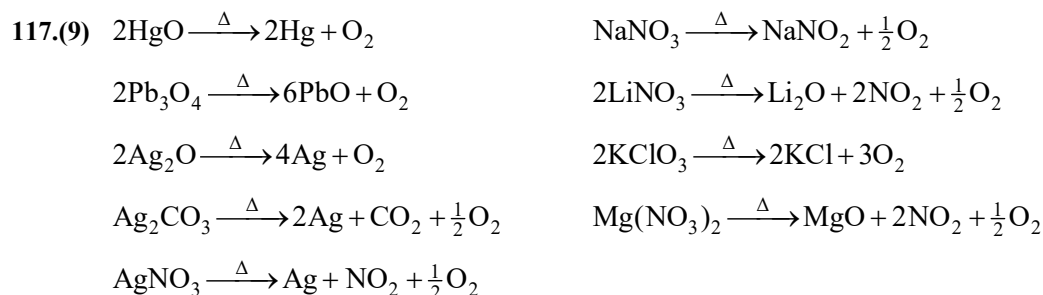
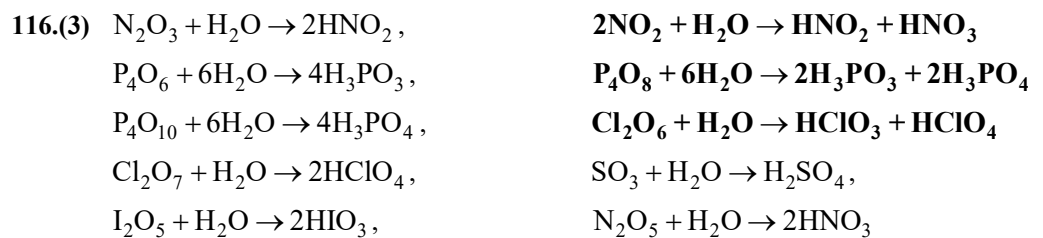
- 111.(2)** No. of  $lp = 16$ , No. of S-S bonds = 8, ratio =  $\frac{16}{8} = 2$

- 112.(5)**  $\text{CN}^-$ ,  $\text{SCN}^-$ ,  $\text{OCN}^-$ ,  $\text{CNO}^-$ ,  $\text{NNN}^-$  are pseudo halides.

- 113.(3)**  $\text{H}_3\text{BO}_3, \text{H}_3\text{PO}_2, \text{HPO}_3.$

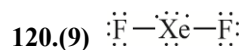


- 115.(8)**  $\text{BCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{BO}_3 + 3\text{HCl}$ ,  $\text{SiCl}_4 + 4\text{H}_2\text{O} \rightarrow \text{H}_4\text{SiO}_4 + 4\text{HCl}$   
 $\text{NCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{NH}_3 + 3\text{HOCl}$ ,  $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + 3\text{HCl}$   
 $\text{AsCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{AsO}_3 + 3\text{HCl}$ ,  $\text{SbCl}_3 + \text{H}_2\text{O} \rightarrow \text{SbOCl} + 2\text{HCl}$   
 $\text{SbCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SO}_2 + 4\text{HCl}$ ,  $\text{ICl}_3 + 2\text{H}_2\text{O} \rightarrow \text{HIO}_3 + 3\text{HCl}$



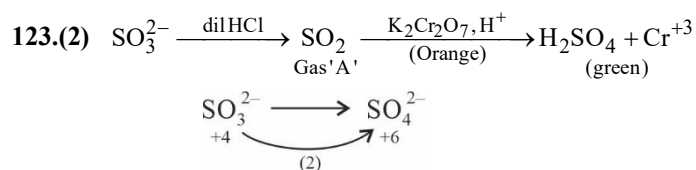
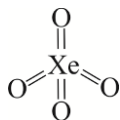
118.(7) As a general rule, only non-metal oxides are acidic but oxides of some metals in high oxidation state also have acidic character. The acidic oxides are  $\text{SO}_2$ ,  $\text{Cl}_2\text{O}_7$ ,  $\text{CO}_2$ ,  $\text{N}_2\text{O}_5$ ,  $\text{Mn}_2\text{O}_7$ ,  $\text{CrO}_3$  and  $\text{V}_2\text{O}_5$ .

119.(3)  $\text{CO}$ ,  $\text{NO}$  and  $\text{N}_2\text{O}$  are neutral oxides.



121.(5) The synthetic conditions indicate that formation of the higher halides is favoured by a higher proportion of fluorine and higher total pressure.  $\text{Xe} + \text{F}_2 \xrightarrow[1:5]{873\text{K}, 7\text{bar}} \text{XeF}_{4(s)}$

122.(4) In  $\text{XeO}_4$  central atom is  $\text{sp}^3$  hybridized hence all  $\pi$  bonds are  $\text{d}\pi - \text{p}\pi$  bonds.





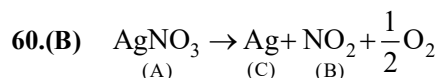
## Solutions - Advanced Problem Package | Chemistry

### d-Block Elements

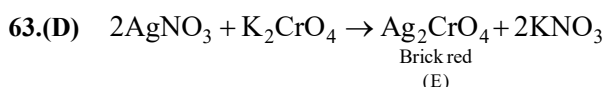
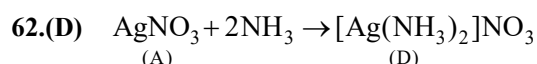
- 1.(B) Cu can displace Ag from  $\text{AgNO}_3$
- 2.(A) There is dynamic equilibrium between  $\text{CrO}_4^{2-}$  and  $\text{Cr}_2\text{O}_7^{2-}$  at  $\text{pH} = 4$ .  

$$2\text{CrO}_4^{2-} + 2\text{H}^+ \rightleftharpoons \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$$
- 3.(B) There is no unpaired electron in Mn but there is charge transfer from O to Mn.
- 4.(A) Transition elements are more metallic than representative elements because of availability of vacant d orbitals for bonding.
- 5.(A)  $\text{KMnO}_4$  is purple coloured solution and during reaction there is decolourization of purple colouration.
- 6.(A) The most common oxidation state of lanthanides is +3 involving loss of electrons from 5d and 6s orbitals.
- 7.(B) Blue colouration is due to formation of aqua complex and  $\text{CuI}_2$  is unstable and decompose to give  $\text{Cu}_2\text{I}_2$  and  $\text{I}_2$ .
- 8.(B) Electronic configuration of Zn is  $[\text{Ar}] 3d^{10} 4s^2$ . Hence because of completion of  $(n-1)d$  orbitals.
- 9.(D)  $\text{CuSO}_4 + 2\text{KCN} \rightarrow \text{Cu}(\text{CN})_2 + \text{K}_2\text{SO}_4$ ;  $2\text{Cu}(\text{CN})_2 \rightarrow 2\text{CuCN} + (\text{CN})_2$ ;  $\text{CuCN} + 3\text{KCN} \rightarrow \text{K}_3[\text{Cu}(\text{CN})_4]$
- 10.(A) F and O stabilize the higher oxidation states because of their strong oxidizing nature.
- 11.(B) Au and Ag are extracted by cyanide process.
- 12.(A)  $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$  has only one unpaired electron.
- 13.(D) It is an aqua complex,  $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$  having co-ordination number equal to four.
- 14.(B)  $\text{HgCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{HgO} + 2\text{NaCl} + \text{CO}_2$
- 15.(D)  $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$  is blue coloured and  $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$  is orange coloured.
- 16.(A) Manganese steel is used for making Railways because of its hardness.
- 17.(A) Measurement of magnetic moment in solid state can be used to determine oxidation state by knowing number of unpaired electrons.
- 18.(A) Due to the presence of vacant orbitals.
- 19.(B) Due to the presence of lone pair electron on Sn.
- 20.(D)  $\text{AgBr} + 2\text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + \text{NaBr}$
- 21.(A)  $\text{Cu} + \text{H}_2\text{O} \xrightarrow{\text{Red hot}} \text{CuO} + \text{H}_2$
- 22.(B)  $2\text{CuCl}_2 + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{Cu}_2\text{Cl}_2 + 2\text{HCl} + \text{H}_2\text{SO}_4$
- 23.(B)  $\text{AgNO}_3 \xrightarrow{\Delta} \text{Ag} + \text{NO}_2 + \frac{1}{2}\text{O}_2$
- 24.(D)  $\text{Hg}_2\text{Cl}_2 + 2\text{NH}_4\text{OH} \rightarrow \text{Hg} + \underbrace{\text{Hg}(\text{NH}_2)\text{Cl}}_{\text{Black}} + \text{NH}_4\text{Cl} + 2\text{H}_2\text{O}$
- 25.(B)  $\text{HgCl}_2 + \text{SnCl}_2 \longrightarrow \text{Hg}_2\text{Cl}_2 + \text{SnCl}_4$
- 26.(C) Zinc is a non-transition element. It has low enthalpy of atomization and is a volatile metal. It is not brittle.
- 27.(C) Mercury forms amalgam with metals except Fe and Pt hence can be transported in containers made up of iron.
- 28.(D) On heating molten mass of silver absorbs oxygen which is released on cooling?
- 29.(D) Because of covalent nature  $\text{CuCl}_2$  and  $\text{CuBr}_2$  exist in polymeric form.
- 30.(C)  $\text{Fe}^{2+}$  of Mohr's salt can be oxidized.
- 31.(A) Noble metals are not attacked by acid and alkalis.  $\text{AuCl}_3$  is used in toning of photographic film.
- 32.(A)  $\text{ZnO} + \text{BaO} \rightarrow \text{BaZnO}_2$   
Amphoteric      Basic
- 33.(A)  $\text{HgCl}_2$  is corrosive in nature and sublimes on heating and therefore, known as corrosive sublimate.

- 34.(D)  $\text{Zn(OH)}_2$  is soluble in excess of  $\text{NaOH}$ .
- 35.(C)  $2\text{CuSO}_4 + 4\text{KI} \rightarrow \text{Cu}_2\text{I}_2 + \text{I}_2 + \text{K}_2\text{SO}_4$
- 36.(D) Aqua regia releases nascent  $\text{Cl}$ .
- 37.(C)  $2\text{FeCl}_3 + 3\text{Na}_2\text{S}_2\text{O}_3 \rightarrow \text{Fe}_2(\text{S}_2\text{O}_3)_3 + 6\text{NaCl}$   
(violet colour)
- 38.(A) Some of the visible spectrum is being removed from white light as it passes through the sample of coloured substance, so the light that emerges is no longer white. The colour of the complex is complementary to that which is absorbed. The complementary colour is the colour generated from the wavelength left over. If orange-red light is absorbed by the complex, it appears blue.
- 39.(A)  $\text{Fe} + 2\text{HCl} \longrightarrow \text{FeCl}_2 + \text{H}_2$ . Liberation of  $\text{H}_2$  prevents the formation of  $\text{FeCl}_3$ .
- 40.(B)  $\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow[-7\text{H}_2\text{O}]{300^\circ\text{C}} 2\text{FeSO}_4 \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$   
(A) Green      White (B)      (C)      (E)      (D)
- 41.(C)  $6\text{FeSO}_4 + 6\text{HgCl}_2 \longrightarrow 3\text{Hg}_2\text{Cl}_2 + 2\text{Fe}_2(\text{SO}_4)_3 + 2\text{FeCl}_3$   
White
- 42.(D)  $\text{Fe}_2\text{O}_3 + 6\text{HCl} \longrightarrow 2\text{FeCl}_2 + 3\text{H}_2\text{O}$
- 43.(B) C is  $\text{Fe}_2\text{O}_3$
- $\text{Fe}_2\text{O}_3 + \text{HCl} \longrightarrow \text{FeCl}_3$   
Yellow
- $\text{FeCl}_3 + \text{H}_2\text{S} \longrightarrow \text{FeCl}_2 + \text{S}$   
Green      Turbidity
- 44.(C) Catalyst provide an alternate reaction mechanism that has lower activation energy.
- 45.(D) Overall there is no change in chemical composition of catalyst and it increases rate of reaction due to decrease in activation energy.
- 46.(A)
- 47.(B) Due to presence of unpaired electron.
- 48.(D)  $2\text{KMnO}_4 + 3\text{MnSO}_4 + 2\text{H}_2\text{O} \longrightarrow 5\text{MnO}_2 + \text{K}_2\text{SO}_4 + 2\text{H}_2\text{SO}_4$
- 49.(B)  $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \longrightarrow 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$       50.(B)  $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$   
green
- 51.(D) In  $\text{KMnO}_4$  there is no unpaired electron hence its purple colouration is due to charge transfer.
- 52.(B)  $2\text{Fe} + 3\text{Cl}_2 \xrightarrow{\Delta} 2\text{FeCl}_3$   
 $\text{Fe} + 2\text{HCl}_{(\text{g})} \xrightarrow{\Delta} 2\text{FeCl}_2 + \text{H}_2$  ( $\text{H}_2$  prevent formation of  $\text{FeCl}_3$ )
- 53.(D)  $\text{K}_3[\text{Fe}(\text{CN})_6] + \text{Fe}^{2+} \longrightarrow \text{Fe}_3[\text{Fe}(\text{CN})_6]_2$  (Turn bull's blue)
- 54.(A) Brown coloured ring is due to  $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]\text{SO}_4$ . On heating complex decomposes to release  $\text{NO}$ .
- 55.(D) Aqueous solution of  $\text{CO}_2$  is acidic and in acidic solution  $\text{Na}_2\text{CrO}_4$  changes to  $\text{Na}_2\text{Cr}_2\text{O}_7$ .  
 $\text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O} \rightleftharpoons 2\text{CrO}_4^{2-} + 2\text{H}^+$  ( $\text{pH} = 4$ )
- 56.(B)  $\text{Zn}$  is a strong reducing agent.
- 57.(D) Generally, the higher the oxidation state of the metal, the greater is the crystal field splitting.
- 58.(D)  $\text{Na}_2\text{CrO}_4 + \text{Ba}^{2+} \longrightarrow \text{BaCrO}_4$   
Yellow Ppt  
Barium salts give green colour to flame.
- 59.(A) Nitrate and bromide salt on heating evolve brown gas  $\text{NO}_2$  and  $\text{Br}_2$  respectively. Ammonical silver nitrate solution oxidises glucose.



61.(C)

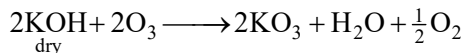
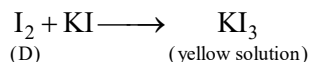
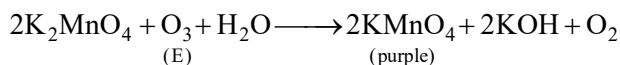
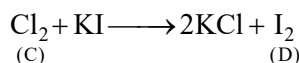
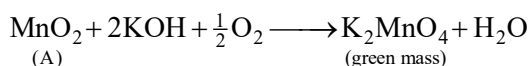
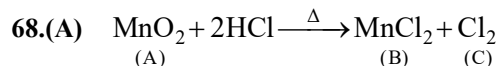


64.(B)

65.(B)

66.(C)

67.(B)



69.(ABC) Amalgam is an alloy of metal with mercury

70.(AB)

71.(AB)

72.(BC)

73.(ABC)

74.(ABC) Acidified  $\text{KMnO}_4$  can oxidise  $\text{SO}_2$ ,  $\text{H}_2\text{O}_2$  and  $\text{FeSO}_4$ .

75.(AB)

76.(ABCD) Due to completely filled orbitals. Uub is 112<sup>th</sup> element and it belong to 4<sup>th</sup> series of d-block elements.

77.(ABC) Due to partially filled orbitals in the atom or in their ions

78.(CD) Sc and Zn do not show variable oxidation states ( $\text{Sc}^{3+}$  and  $\text{Zn}^{2+}$ ) while Cu and Fe shows variable oxidation state.  
( $\text{Cu}^+$ ,  $\text{Cu}^{2+}$ ,  $\text{Fe}^{2+}$  and  $\text{Fe}^{3+}$ )

79.(ABCD) All are correct facts.

80.(BCD)

81.(BCD) Beyond Mn no metal has a trihalide except  $\text{FeX}_3$  and  $\text{CoF}_3$  in first series of d-block elements.

82.(ABCD) All are mixed oxides because oxidation state of metal is fractional which represent average of different oxidation state.

83.(AC) Fluorides are unstable in the lower oxidation state while iodides are unstable in higher oxidation states.

84.(ABC)  $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$  is 0.34V and  $\text{Cu}_{(\text{aq})}^{2+}$  is more stable than  $\text{Cu}_{(\text{aq})}^+$  because of the much more negative  $\Delta_{\text{hyd}}H^\circ$  of  $\text{Cu}_{(\text{aq})}^{2+}$  than  $\text{Cu}_{(\text{aq})}^+$ .

85.(BCD)  $\text{MnF}_7$  is not known because of steric hindrance.  $\text{VCl}_5$  undergo hydrolysis to give oxochloride.

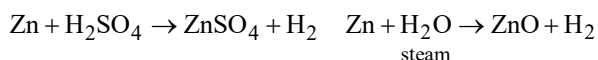
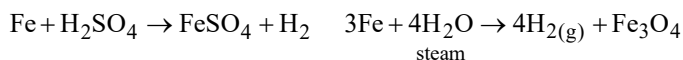
86.(ABD) Brass; Cu and Zn  
Bronze; Cu and Sn  
Zinc-amalgam; Zn and Hg  
German silver; Cu, Zn and Ni

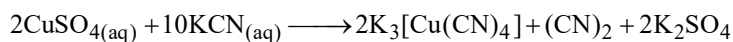
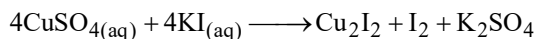
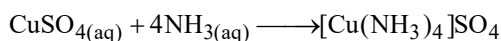
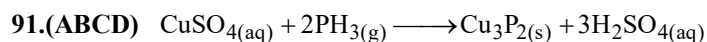
87.(ABD)  $\text{F}^-$  is weak field ligand.  $[\text{Fe}(\text{CNS})_6]^{3-}$  shows linkage isomerism while  $\text{FeF}_6^{3-}$  do not show isomerism.

88.(AB) Due to absence of unpaired electron.

89.(ABC) Li and Mg, Be and Al shows diagonal relationships hence shows similar chemical properties. While Hf and Zr possess similar atomic size because of lanthanide contraction hence shows similar chemical properties.

90.(AC) Steam when passed over hot metals like Zn, Fe, Ni, Mn, Co, Cr and Sn is decomposed to liberate hydrogen.



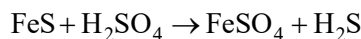
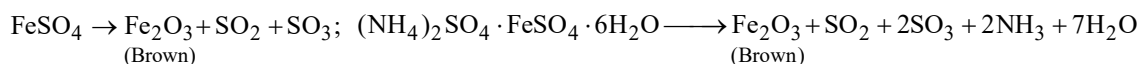


92.(ABCD)

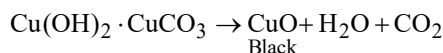
93.(AC)  $\text{Ce}^{4+}$  is a strong oxidizing agent because  $\text{Ce}^{3+}$  is its common oxidation state,  $\text{La}^{3+}$ ,  $\text{Ce}^{4+}$  and  $\text{Yb}^{2+}$  are diamagnetic due to absence of unpaired electrons.

94.  $[\text{A-r}] \rightarrow [\text{B-p, q, r, s}] \rightarrow [\text{C-p, r}] \rightarrow [\text{D-q}]$

Kipp's apparatus is used for preparation of  $\text{H}_2\text{S}$  in the laboratory. It contains  $\text{FeS}$  and  $\text{H}_2\text{SO}_4$ .



$\text{Cu}(\text{OH})_2 \cdot \text{CuCO}_3$  is malachite and it is green coloured.

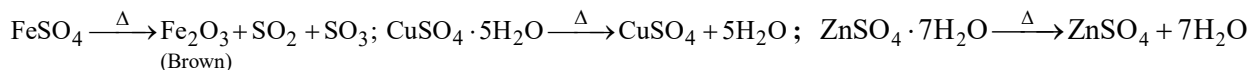


95.  $[\text{A-p, q, r, s, t}] \rightarrow [\text{B-p, q, r}] \rightarrow [\text{C-r}] \rightarrow [\text{D-p, q, s}]$

Transition metals shows variable oxidation state. Zn shows only one oxidation state +2. Fe becomes passive due to deposition of protective film of oxide on the surface. Aqua complexes of ions of Fe, Cu and Mn are coloured due to presence of unpaired electrons. Fe and Mn are mainly extracted from oxide ores while Zn and Cu are extracted from sulphide ores.

96.  $[\text{A-p, r, s}] \rightarrow [\text{B-q, r, s, t}] \rightarrow [\text{C-q, r, s}] \rightarrow [\text{D-q}]$

Hydrated ferric chloride can't be converted to anhydrous form just by heating due to hydrolysis.



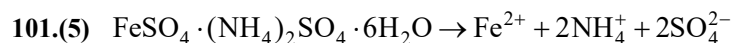
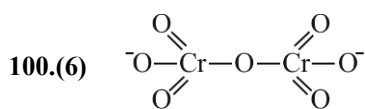
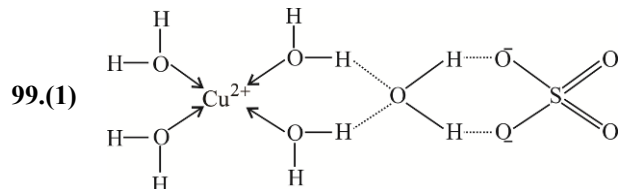
Except  $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$  all others are paramagnetic.

97.  $[\text{A-p, s}] \rightarrow [\text{B-q}] \rightarrow [\text{C-q, r, s}] \rightarrow [\text{D-q, t}]$

Fe and Pt do not form amalgam. Pd absorb large amount of  $\text{H}_2$  due to formation of interstitial hydrides.

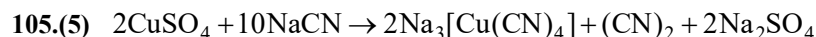
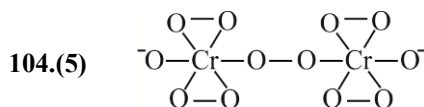
98.  $[\text{A-p, q, r, s}] \rightarrow [\text{B-p, q, r, s}] \rightarrow [\text{C-p, q, r, s}] \rightarrow [\text{D-r, s}]$

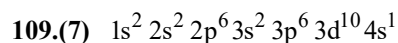
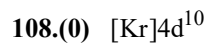
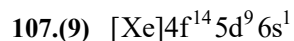
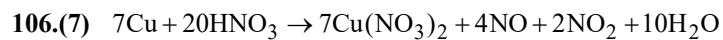
Acidified  $\text{KMnO}_4$ , Acidic  $\text{K}_2\text{Cr}_2\text{O}_7$ ,  $\text{FeCl}_3$  are oxidizing agent.  $\text{CuSO}_4$  can oxidize iodide to  $\text{I}_2$ .



102.(1)  $\text{BaCl}_2 \cdot 6\text{H}_2\text{O}$ , other undergoes hydrolysis to form oxide or oxohalide.

103.(4) Iron is 4<sup>th</sup> most abundant element in the earth's crust.



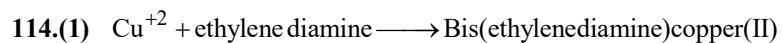


110.(6) Ion having unpaired electrons and these are  $\text{Ti}^{3+}$ ,  $\text{V}^{3+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Co}^{2+}$ ,  $\text{Cu}^{2+}$ .

111.(3) 59, 95, 102 are the atomic numbers of the inner transition elements, because last electron enters in f-orbital.

112.(2) Sc and Zn do not exhibit variable oxidation states. Sc(+3) and Zn(+2).

113.(5) Lanthanoid metal (~ 95%) and iron (~ 5%).



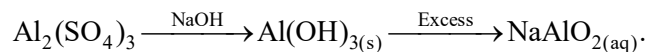
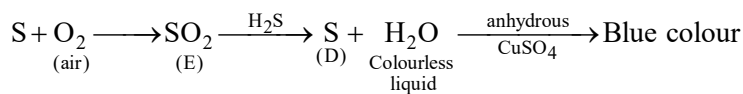
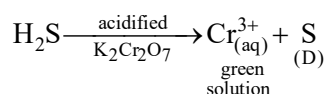
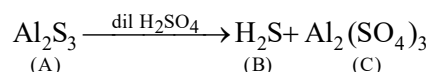
## Solutions - Advanced Problem Package | Chemistry

### Qualitative Analysis

- 1.(A) III group (Iron group) requires low concentration of hydroxide ion as precipitating reagent hence  $\text{NH}_4\text{Cl}$  is added to decrease concentration of  $\text{OH}^-$  ions.
- 2.(B) Violet vapours are due to  $\text{I}_2$ .
- 3.(B) Acidic solution of nitrite salt on reaction with iodide produce  $\text{I}_2$  that forms blue colouration with starch.
- 4.(B) Orange red vapours are due to chromyl chloride ( $\text{CrO}_2\text{Cl}_2$ ).
- 5.(D) Alkali metal ions and ammonium ion can't be precipitated.
- 6.(C) Due to hydrolysis of bismuth chloride,  $\text{BiCl}_3 + \text{H}_2\text{O} \rightarrow \text{BiOCl}_{(s)} + 2\text{HCl}$
- 7.(A) Bromide salt on heating with conc.  $\text{H}_2\text{SO}_4$  and  $\text{MnO}_2$  produce brown vapours of  $\text{Br}_2$ .
- 8.(B) Nitrate releases  $\text{NO}$  that is absorbed by  $\text{FeSO}_4$  to form brown coloured nitrosoferrous sulphate.
- 9.(C)  $\text{Al}(\text{OH})_3$  is soluble in  $\text{NaOH}$  solution.
- 10.(B)  $\text{Zn}^{2+} + 2\text{OH}^- \longrightarrow \text{Zn}(\text{OH})_{2(s)} \xrightarrow[\text{Excess}]{\text{NaOH}} \text{Na}_2\text{ZnO}_2$
- 11.(C) Due to oxidation of  $\text{H}_2\text{S}$ .  $\text{H}_2\text{S} \rightarrow \text{S}_{(s)} + 2\text{H}^+ + 2\text{e}^-$
- 12.(D) Fourth group radicals can't be precipitated by  $\text{H}_2\text{S}$  and  $\text{HCl}$ .  $\text{Ag}^+$  can be precipitate as  $\text{Ag}_2\text{S}$ .
- 13.(D)  $\text{Na}_2\text{CO}_3$  produce high concentration of  $\text{CO}_3^{2-}$  that may precipitate  $\text{Mg}^{2+}$  as  $\text{MgCO}_3$ .
- 14.(B) Alkali metals and alkaline earth metals except Be and Mg can be detected by the flame test.
- 15.(A) Second group radicals can't be separated by  $\text{H}_2\text{S}$  in dilute  $\text{HCl}$ .
- 16.(C)  $\text{AgI}$  is insoluble in ammonium hydroxide. Silver phosphate is soluble in dil  $\text{HNO}_3$  and  $\text{NH}_4\text{OH}$ . Silver chromate is red coloured.
- 17.(C) 18.(B)  $\text{AgCl}$  is soluble in  $\text{NH}_4\text{OH}$  due to formation of  $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
- 19.(B) Ferric salt forms Prussian blue with  $\text{K}_4[\text{Fe}(\text{CN})_6]$
- 20.(B) Insoluble chloride salts do not give positive chromyl chloride test?
- 21.(C)  $\text{PbS}$  is black while  $\text{CdS}$  is yellow. Sulphides of barium and strontium are soluble.
- 22.(D)  $\text{SnCl}_2 + \text{HgCl}_2 \rightarrow \text{SnCl}_4 + \underset{\text{White}}{\text{Hg}_2\text{Cl}_2}$   
 $\text{SnCl}_2 + \text{Hg}_2\text{Cl}_2 \rightarrow \text{SnCl}_4 + \underset{\text{(Grey)}}{2\text{Hg}}$
- 23.(D) Sulphides of basic radicals of II B are soluble in alkali.
- 24.(A) Violet colour is due to solution of  $\text{I}_2$  in chloroform.
- 25.(D) White precipitate is due to  $\text{ZnS}$ . Sulphides of barium, calcium and aluminium are soluble.
- 26.(C)  $\text{SO}_2$  produces turbidity with baryta water and also turns acidified dichromate solution to green.
- 27.(D) Sulphide, sulphite and thiosulphate salts decomposes on reaction with  $\text{H}_2\text{SO}_4$ .
- 28.(C)  $\text{BaCrO}_4$  starts precipitating first.

- 29.(B) CuS, CdS and Fe(OH)<sub>3</sub> will be precipitated.
- 30.(D) Hg<sub>2</sub>Cl<sub>2</sub> forms black precipitate with NH<sub>4</sub>OH due to formation of Hg and Hg(NH<sub>2</sub>)Cl.
- 31.(B) Green flame indicate presence of barium. 32.(A) HgS is insoluble in HNO<sub>3</sub>. It is soluble in aqua regia.
- 33.(A) Chromium (III) salt on reaction with NaOH and Na<sub>2</sub>O<sub>2</sub> form yellow coloured solution due to formation of CrO<sub>4</sub><sup>2-</sup>.
- 34.(D) PbCl<sub>2</sub> is soluble in hot water. 35.(A) PbCl<sub>2</sub> is insoluble while BaCl<sub>2</sub> is soluble.
- 36.(C) Sulphides of II B radicals are soluble in yellow ammonium sulphide.
- 37.(B) Ferric salt gives blood red colour with KCNS.
- 38.(C) Hydroxides of Al, Cr and Zn are amphoteric hence soluble in NaOH.
- 39.(B) Barium chromate is insoluble in acetic acid while calcium chromate is soluble in acetic acid.
- 40.(A)  $\text{MgSO}_4 + \text{NH}_4\text{OH} + \text{Na}_2\text{HPO}_4 \rightarrow \text{Mg}(\text{NH}_4)\text{PO}_{4(s)} + \text{H}_2\text{O} + \text{Na}_2\text{SO}_4$

41-43. 41.(C) 42.(B) 43.(D)



44-47. 44.(D) 45.(A) 46.(D) 47.(B)

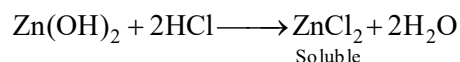
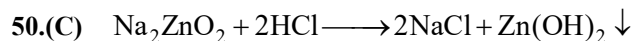
Bottle 4 - copper II sulphate.

Bottle 3 - sodium carbonate

Bottle 2 - Hydrochloric acid

Bottle 1 - lead nitrate

- 48.(D) Mixture is NaOH + ZnSO<sub>4</sub>. 49.(C)  $\text{NaOH} + \text{ZnSO}_4 \longrightarrow \underset{\text{Soluble}}{\text{Na}_2\text{ZnO}_2} + \text{Na}_2\text{SO}_4 + 2\text{H}_2\text{O}$



- 51.(B) A should contain Cl<sup>-</sup> ion since it gives a white precipitate with AgNO<sub>3</sub>, soluble in NH<sub>3</sub>.

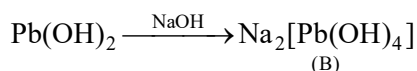
52.(B)

- 53.(A) 'A' gives black precipitate insoluble in NaOH with HCl and H<sub>2</sub>S.

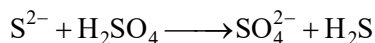
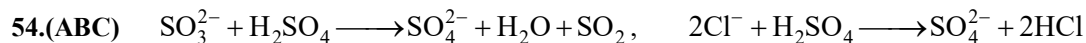
Hence, cation may be Hg<sup>2+</sup>, Pb<sup>2+</sup>, Bi<sup>3+</sup>

'A' gives yellow precipitate with K<sub>2</sub>CrO<sub>4</sub>, it confirms Pb<sup>2+</sup> ion.

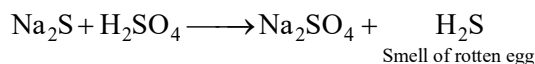
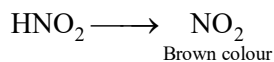
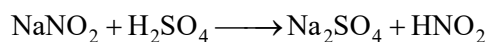
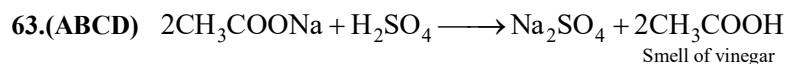
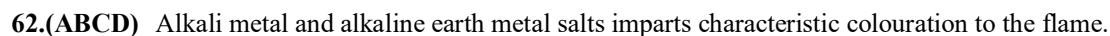
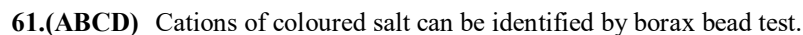
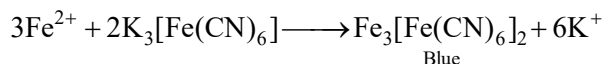
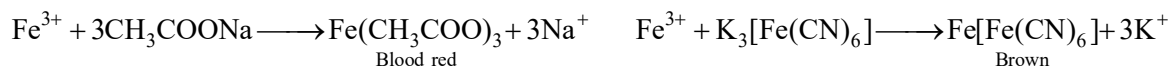
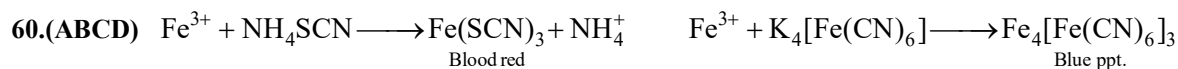
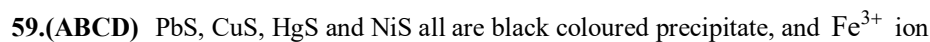
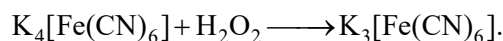
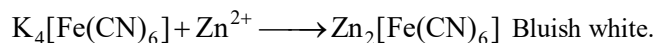
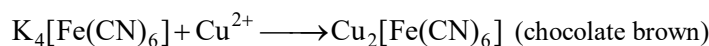
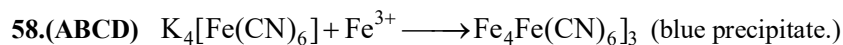
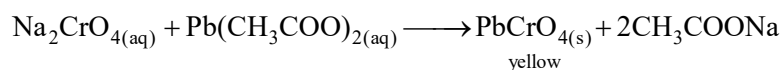
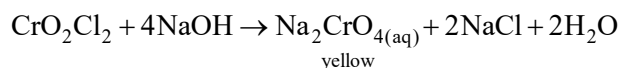
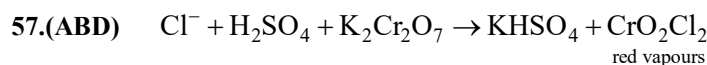
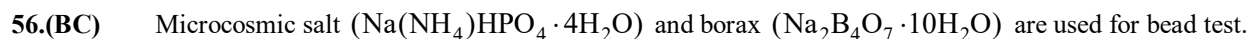
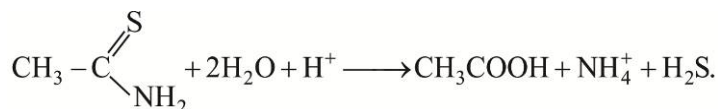
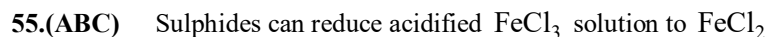
Thus, 'A' is PbCl<sub>2</sub>.  $\text{PbCl}_2 + 2\text{NaOH} \longrightarrow \underset{\text{White}}{\text{Pb}(\text{OH})_2} \downarrow + 2\text{NaCl}$



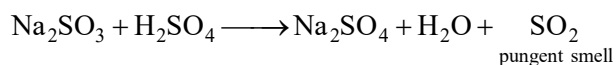
'A' on treatment with KI produces  $\text{PbI}_2$ , yellow in colour.



$\text{SO}_2$ ,  $\text{H}_2\text{S}$  and  $\text{HCl}$  all turns acidified  $\text{K}_2\text{Cr}_2\text{O}_7$  solution to green.







64.(ABCD)

| Group No. | Group Reagent                                                                          | Precipitate as |
|-----------|----------------------------------------------------------------------------------------|----------------|
| I         | Dil.HCl                                                                                | Chlorides      |
| III       | NH <sub>4</sub> Cl/NH <sub>4</sub> OH                                                  | Hydroxides     |
| IV        | NH <sub>4</sub> Cl/NH <sub>4</sub> OH + H <sub>2</sub> S                               | Sulphides      |
| V         | H <sub>4</sub> Cl/NH <sub>4</sub> OH + (NH <sub>4</sub> ) <sub>2</sub> CO <sub>3</sub> | Carbonates     |

65. [A-s] → [B-p, q, r, s] → [C-p, s] → [D-p]

S<sup>2-</sup>, NO<sub>2</sub><sup>-</sup> and SO<sub>3</sub><sup>2-</sup> decolourizes acidified KMnO<sub>4</sub> solution. NO<sub>2</sub><sup>-</sup> evolve NH<sub>3</sub> on reaction with Al and NaOH.

66. [A-p, q, r] → [B-r, s] → [C-r, s] → [D-r]

Pb<sup>2+</sup> can be precipitated by dil HCl as well as by H<sub>2</sub>S.

67. [A-p, q, s] → [B-p, r] → [C-r, s] → [D-q]

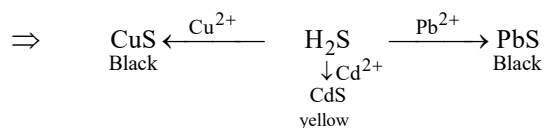
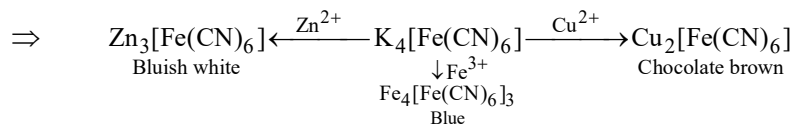
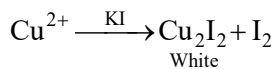
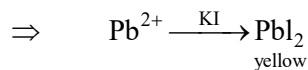
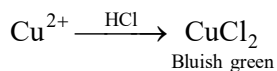
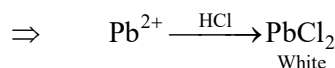
SO<sub>3</sub><sup>2-</sup> and S<sup>2-</sup> can decolourize acidified K<sub>2</sub>Cr<sub>2</sub>O<sub>7</sub> solution.

68. [A-q, s] → [B-p, q] → [C-r] → [D-s]

Pb<sup>2+</sup> and Ag<sup>+</sup> are precipitate as chlorides.

Cr<sup>3+</sup> is precipitate as hydroxide while Ni<sup>2+</sup> is precipitate as sulphide.

69. [A-p] → [B-p, q, s] → [C-p, r] → [D-r, s]



70. [A-s]  $\rightarrow$  [B-p]  $\rightarrow$  [C-p]  $\rightarrow$  [D-q]

$\text{Fe}(\text{OH})_3$  redish brown

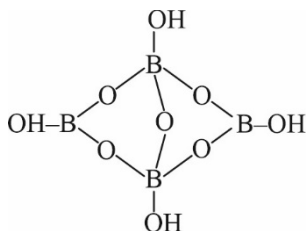
$\text{PbS}$  and  $\text{Bi}_2\text{S}_3$  Black

$\text{PbCl}_2$  and  $\text{AgCl}$  white

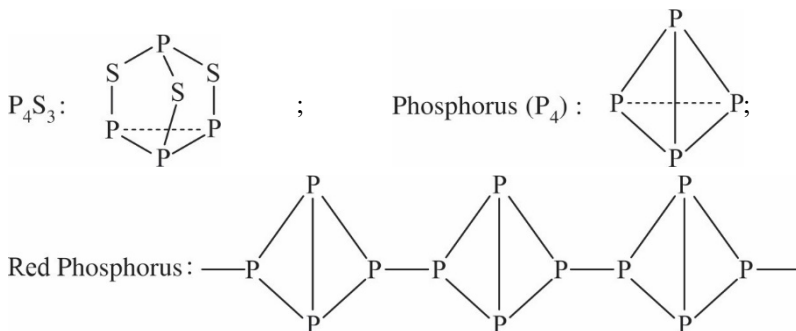
71.(6)  $\text{BCl}_3 + 2\text{LiAlH}_4 \longrightarrow \text{B}_2\text{H}_6$   
(X)

Compound (X) is  $\text{B}_2\text{H}_6$ . Maximum number of atoms of  $\text{B}_2\text{H}_6$  that lie in same plane are 6.

72.(2) In Borax, 2 Borons are  $\text{sp}^2$  hybridised and 2 are  $\text{sp}^3$  hybridised.



73.(3) Red phosphorous,  $\text{P}_4\text{S}_3$ ,  $\text{P}_4$



74.(5) i, iii, vii, viii, ix

$\text{Ag} + \text{Conc. HNO}_3 \longrightarrow \text{Ag} + \text{NO}_2 + \text{O}_2$

$\text{Sn} + \text{Cold dil HNO}_3 \longrightarrow \text{Sn}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3$

$\text{Cu} + \text{HNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}_2$

$\text{Mg} + \text{dil HNO}_3 \longrightarrow \text{Mg}(\text{NO}_3)_2 + \text{H}_2$

$\text{Mn} + \text{dil HNO}_3 \longrightarrow \text{Mn}(\text{NO}_3)_2 + \text{H}_2$   
(2%)

$\text{P}_4 + \text{Conc. HNO}_3 \longrightarrow \text{H}_3\text{PO}_4 + \text{NO}_2$

$\text{S} + \text{Conc. HNO}_3 \longrightarrow \text{H}_2\text{SO}_4 + \text{NO}_2$

$\text{Cu} + \text{dil HNO}_3 \longrightarrow \text{Cu}(\text{NO}_3)_2 + \text{NO}$

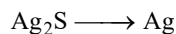
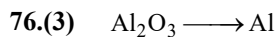
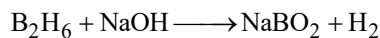
Cr is passive towards conc.  $\text{HNO}_3$

75.(2) Al,  $\text{B}_2\text{H}_6$

(1)  $\text{Al} + \text{HCl} \longrightarrow \text{AlCl}_3 + \text{H}_2$

$\text{Al} + \text{NaOH} \longrightarrow \text{NaAlO}_2 + \text{H}_2$

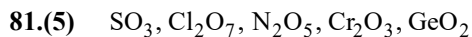
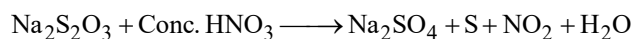
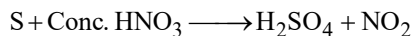
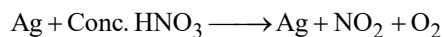
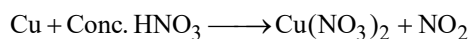
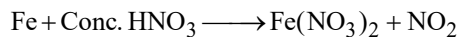
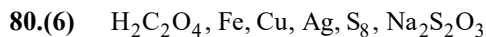
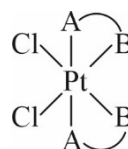
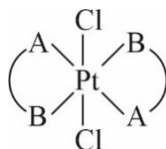
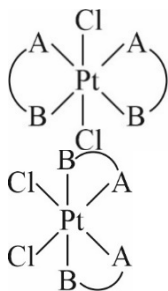
(2)  $\text{B}_2\text{H}_6 + \text{HCl} \longrightarrow \text{B}_2\text{H}_5\text{Cl} + \text{H}_2$



Au



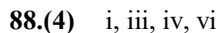
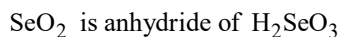
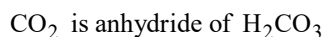
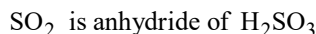
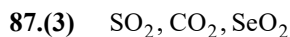
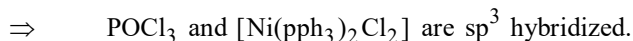
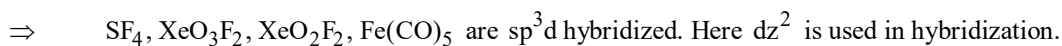
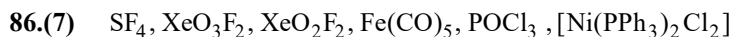
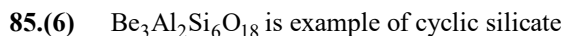
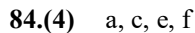
78.(5)



$\text{SO}_3, \text{Cl}_2\text{O}_7, \text{N}_2\text{O}_5$  being acidic reacts with NaOH.  $\text{Cr}_2\text{O}_3$  and  $\text{GeO}_2$  are amphoteric oxides and thus soluble in NaOH.



Sulphide ores are extracted by Self Reduction method.



Transition metal ions are coloured because of d-d transition.  $\text{d}^0$  and  $\text{d}^{10}$  species are colourless

- 89.(3)  $\text{Al}^{3+}$  and  $\text{Hg}^{2+}$   
 $\text{Cd}^{2+}$  and  $\text{Zn}^{2+}$   
 $\text{Fe}^{3+}$  and  $\text{Cu}^{2+}$
- 90.(1)  $\text{Cu} : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$   
 3s orbital has 2 radial nodes
- 91.(4)  $\text{NaCl}$ ,  $\text{MgCl}_2$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{CaCl}_2$
- 92.(7)  $\text{NO}_2$ ,  $\text{CO}_2$ ,  $\text{P}_4\text{O}_6$ ,  $\text{CrO}_3$ ,  $\text{Mn}_2\text{O}_7$ ,  $\text{I}_2\text{O}_5$ ,  $\text{OsO}_4$
- 93.(6)  $\text{Ba}^{2+} + \text{CrO}_4^{2-} \longrightarrow \text{BaCrO}_4 \downarrow$  (yellow ppt)  
 $\text{Cd}^{2+} + \text{Na}_2\text{S} \longrightarrow \text{CdS} \downarrow$   
 $\text{Ag}^+ + \text{Br}^- \longrightarrow \text{AgBr} \downarrow$  (Pale yellow)  
 $\text{Na}_3\text{PO}_4 + (\text{NH}_4)_2\text{MoO}_4 \longrightarrow (\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$  (yellow)  
 $\text{NH}_4^+ (\text{aq}) + [\text{PtCl}_6]^{2-} (\text{aq}) \longrightarrow (\text{NH}_4)_2[\text{PtCl}_6]$
- 94.(5)  $\text{P}_4$ ,  $\text{Al}$ ,  $\text{Be}$ ,  $\text{Fe}$ ,  $\text{Zn}$
- 95.(3)  $\text{NO}_3^-$ ,  $\text{IF}_7$ ,  $\text{SO}_4^{2-}$
- 96.(6)  $[\text{Mn}(\text{CN})_6]^{4-}$ ,  $[\text{Ni}(\text{CN})_4]^{2-}$ ,  $[\text{PdCl}_4]^{2-}$ ,  $[\text{Pd}(\text{CN})_4]^{2-}$ ,  $[\text{Co}(\text{SCN})_4]^{2-}$ ,  $[\text{AgF}_4]^-$
- |                                   | Hybridisation |
|-----------------------------------|---------------|
| $[\text{Mn}(\text{CN})_6]^{4-}$   | $d^2 sp^3$    |
| $[\text{Ni}(\text{NH}_3)_6]^{2+}$ | $sp^3 d^2$    |
| $[\text{Co}(\text{NO}_2)_6]^{4-}$ | $sp^3 d^2$    |
| $\text{AgF}_4^-$                  | $dsp^2$       |
| $[\text{Ni}(\text{CN})_4]^{2-}$   | $dsp^2$       |
| $[\text{PdCl}_4]^{2-}$            | $dsp^2$       |
| $[\text{Pd}(\text{CN})_4]^{2-}$   | $dsp^2$       |
| $[\text{Co}(\text{SCN})_4]^{2-}$  | $dsp^2$       |
- 97.(5) Here Fe is in +3 and +2 oxidation state. Total sum = 5
- 98.(0)  $\text{Mn}^{2+} + \text{S}_2\text{O}_8^{2-} \longrightarrow \text{MnO}_4^- + \text{SO}_4^{2-} + \text{H}^+$   
 (X)  
 Number of unpaired electrons = 0  
 Magnetic moment = 0
- 99.(4) i, ii, iii, v
- 100.(3)  $\text{V}_2\text{O}_5$ ,  $\text{Cr}_2\text{O}_3$ ,  $\text{ZnO}$
- 101.(6)  $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + 6\text{SOCl}_2 \longrightarrow \text{FeCl}_3 + 6\text{SO}_2 + 12\text{HCl}$
102. (a)  $0.02355 \text{ L HCl} \times \frac{0.195 \text{ mol}}{\text{L}} = 0.00459 \text{ mol H}^+ = 0.00459 \text{ mol Cl}^-$
- (b)  $0.00459 \text{ mol Cl}^- \times \frac{35.5 \text{ g}}{\text{mol}} = 0.163 \text{ g Cl}^-$

$$0.396 \text{ g MCl} - 0.163 \text{ g Cl}^- = 0.233 \text{ g} \times \frac{1}{0.00459 \text{ mol}} = \frac{50.8 \text{ g}}{\text{mol}}$$

$$0.396 \text{ g MCl} - 0.163 \text{ g Cl}^- = 0.233 \text{ g} \times \frac{1}{0.00230 \text{ mol}} = \frac{101.3 \text{ g}}{\text{mol}}$$

$$0.396 \text{ g MCl} - 0.163 \text{ g Cl}^- = 0.233 \text{ g} \times \frac{1}{0.00153 \text{ mol}} = \frac{152.3 \text{ g}}{\text{mol}}$$

(c)  $50.8 \text{ g mol}^{-1}$  most likely V VCl unlikely to be stable

$101.3 \text{ g mol}^{-1}$  most likely Ru  $\text{RuCl}_2$  stable

$152.3 \text{ g mol}^{-1}$  most likely Eu  $\text{EuCl}_3$  stable

103. (a) The element above element 114 is lead, Pb ; above element 116 is polonium, Po; above element 118 is radon, Rn
- (b) The ionization energy is expected to increase from element 114 to 116 to 118. This due to increasing nuclear charge density thus resulting in greater attraction for outer electrons. The ionization energy of elements 114, 116 and 118 are expected to be lower than those of the elements directly above them due to the presence of electrons in higher energy levels, those with higher value of n
- (c) The oxidation states for element 114 are predicted to be +2 (loss of electrons in p orbitals) and +4 (loss of electrons in both s and p orbitals). The oxidation state +2 is likely more stable ; lower oxidation states are usually more stable
- (d) Even values of Z are usually more stable due to pairing of protons. Odd values of Z are less stable
104. (a)  $n = PV/RT = (700.0 \text{ mm Hg}/760 \text{ mm Hg/atm}) (0.1575 \text{ L}) / [(0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1})(298 \text{ K})]$   
 $= 0.00593 \text{ mol CO}_2$   
 Alternate approach  $0.6500 \text{ g} - 0.3891 \text{ g} = 0.2609 \text{ g CO}_2$   
 $0.2609 \text{ g CO}_2 / 44.0 \text{ g/mole} = 0.00593 \text{ mole CO}_2$
- (b)  $(0.500 \text{ mol/L}) (0.03860 \text{ L}) - 0.01930 \text{ mol HCl}$   
 $(0.01930 \text{ mol HCl}) (1 \text{ mol MO}/2 \text{ mol HCl}) = 0.00965 \text{ mol MO}$
- (c)  $(0.3891 \text{ g MO})/0.00965 \text{ mol MO} = 40.32 \text{ g/mol MO}$   
 $(40.32 \text{ g/mol MO} - 16.0 \text{ g/mol O} = 24.32 \text{ g/mol M}) \quad \text{M} = \text{Mg}^{2+}$
- (d) Parts a-c give three constraints on the number of moles of MgO and  $\text{MgCO}_3$  present.  
 From a, the total mass of the sample was 0.6500 g, so  
 $(10.32 \text{ g/mol}) \times (n[\text{MgO}]) + (84.32 \text{ g/mol}) \times (n[\text{MgCO}_3]) = 0.6500 \text{ g}.$   
 From b, the total number of moles of magnesium =  $0.00965 \text{ mol } n(\text{MgO}) + n(\text{MgCO}_3).$   
 From c,  $0.00593 \text{ mol CO}_2$  implies  $n(\text{MgCO}_3) = 0.005928 \text{ mol}.$   
 Thus  $n(\text{MgO}) = 0.00965 \text{ mol} - 0.005928 \text{ mol} = 0.00372 \text{ mol}.$  Since one has three linear equations in two unknowns, one may use any two of the constraints to solve :  
 $n(\text{MgO}) = 0.00372 \text{ mol}, n(\text{MgCO}_3) = 0.00593 \text{ mol}$   
 $\text{Mol \% MgO} = 0.00372 \text{ mol MgO} / 0.00965 \text{ moles total} = 0.385 \times 100\% = 38.5 \text{ mol\% MgO}$   
 $\text{Mol \% MgCO}_3 = 0.00593 \text{ mol MgCO}_3 / 0.00965 \text{ moles total} = 0.615 \times 100\% = 61.5 \text{ mol\% MgCO}_3$
105. (a) (i) When you descend families on the periodic table, the ionization of the electron occurs from higher n values that are (usually) further from the nucleus.
- (ii) The 3d subshell fills just before Ga. There is no 2d subshell to fill prior to Al. The 10 added protons are not completely screened by the 10d electrons, resulting in a higher effective nuclear charge and smaller radius than expected for Ga, hence a higher IE than expected. For Tl, the newly added subshell is the 4f subshell, and a similar effect is observed. [Relativistic effects also contribute to the higher IE of Tl]

- (b) (i) The +1 ion corresponds to the loss of the  $np^1$  electrons, while +3 corresponds to the loss of the  $ns^2$  and  $np^1$  electrons. If 2 electrons were lost, a partially filled subshell would be present, which would interfere with bonding and would thus not be energetically favorable.
- (ii) As one descends a column, the s electrons are more penetrating, held more tightly, and the therefore more difficult to lose. This is often called the "inert pair effect".
- (c) (i) In a Lewis acid-base reaction, the Lewis base donates a pair of electrons to the Lewis acid, forming a new bond. Example (many others possible) :  $BCl_3 + NH_3 \longrightarrow BCl_3 - NH_3$
- (ii) The effectiveness of orbital overlap decreases down the family, so the bond between Lewis acid and Lewis base would weaken as the central atoms of the acid gets larger.
106. (a) In water, both HCl and HBr are equally strong, because both are 100% ionized. Water is a "good enough" base to pull the hydrogen ion from those two molecules to about 100% efficiency (this is called "leveling"). In 100% (glacial) acetic acid the acetic acid is a much weaker base than water. HCl only partially protonates acetic acid, while HBr more fully protonates it.
- (b)  $H_2SO_4 + SO_3 \longrightarrow H_2S_2O_7$  (fuming sulfuric acid)
- $H_2S_2O_7$  is a stronger acid than  $H_2SO_4$  because there are more oxygens around the S holding the  $-OH$  bonds so the effective polarity of the  $-OH$  bond increases, making the molecule a stronger acid.
- (c) In  $H_2SO_3$ , the two hydrogens to be ionized are initially bonded to different atoms. Thus the negative charge left behind when the first  $H^+$  is lost is (on average) farther away from the second  $H^+$  and attracts it more weakly. In  $H_2S$ , both hydrogens to be lost are bonded to the same atoms.
107. (a)  $(1310 \text{ s}) (1.24 \text{ A}) / (96485 \text{ C/mol}) = 0.0168 \text{ mol electrons}$
- $0.292 \text{ g Cr} / 52.00 \text{ g/mol} = 5.62 \times 10^{-3} \text{ mol Cr}$
- $0.0168 \text{ mol electrons} / (5.62 \times 10^{-3} \text{ mol Cr}) = 2.99 \text{ mol electrons/mol Cr}$
- Thus Cr must be in the +3 oxidation state,  $n = 3$ .
- (b)  $3.000 \text{ g} - 1.783 \text{ g} = 1.217 \text{ g } H_2O / (18.02 \text{ g/mol}) = 0.06754 \text{ mol } H_2O$
- $1.783 \text{ g } CrCl_3 / (158.35 \text{ g/mol}) = 0.01126 \text{ mol } CrCl_3$
- $m = 0.06754 \text{ mol } H_2O / 0.01126 \text{ mol } CrCl_3 = 6$
- (c) (i) During the titration,  $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
- At the endpoint,  $2Ag^+(aq) + CrO_4^{2-}(aq) \rightarrow Ag_2CrO_4(s)$
- (ii)  $\text{Mol Cr salt} = 0.3000 \text{ g} / (266.47 \text{ g/mol}) = 1.126 \times 10^{-3} \text{ mol}$
- $\text{Mol } Ag^+ \text{ added} = \text{mol } Cl^- \text{ present} = (0.400 \text{ mol/L}) \times (2.81 \times 10^{-3} \text{ L}) = 1.12 \times 10^{-3} \text{ mol}$
- There is thus 1.00 mol  $Cl^-$  titrated per mol Cr.
- (d) Presumably two of the chlorides are bonded directly to the Cr(III) ion and therefore do not react rapidly with  $Ag^+(aq)$ ; the salt is likely best formulated  $[CrCl_2(H_2O)_4]Cl \cdot 2H_2O$
108. (a)  $1s^2 2s^2 2p^6 3s^2 3p^5$  or  $[Ne] 3s^2 3p^5$
- (b)  $\begin{array}{c} \cdot\cdot \\ :\text{Cl} - \text{Cl}: \\ \cdot\cdot \end{array}$
- (c) Cl has a higher ionization energy than  $Cl_2$ . The highest-lying electrons in  $Cl_2$  are  $\pi^*$ , so they are higher in energy, and hence require less energy to ionize, than the 3p electrons in atomic chlorine. (The experimental values are 13.0 eV for Cl, 11.5 eV for  $Cl_2$ )
- (d)  $Cl^-$  would have a larger radius, since it has an additional electron.

- (e) Cl has 7 valence electrons. An oxoanion with the formula  $\text{ClO}_5^-$  would require an oxidation state of +9 for Cl, which would require removing core electrons. This is not energetically feasible.

109.(a)  $0.5637\text{g CO}_2 \times (12.01\text{ g C} / 44.01\text{ g CO}_2) = 0.1538\text{ g C}, 15.38\% \text{ C}$

$0.6924\text{ g H}_2\text{O} \times (2.016\text{ g H} / 18.016\text{ g H}_2\text{O}) = 0.07748\text{ g H}, 7.75\% \text{ H}$

(b)  $\% \text{O} = 100\% - (15.38\% + 7.75\% + 35.89\%) = 40.98\% \text{ O}$

So in 100 g fertilizer there would be :

$15.38\text{ g C} / (12.01\text{ g mol}^{-1}) = 1.28\text{ mol C}$

$7.748\text{ g C} / (1.008\text{ g mol}^{-1}) = 7.69\text{ mol H}$

$35.89\text{ g N} / (14.01\text{ g mol}^{-1}) = 2.56\text{ mol N}$

$40.98\text{ g O} / (16.00\text{ g mol}^{-1}) = 2.56\text{ mol O}$

Dividing by 1.28 mol gives an empirical formula  $\text{CH}_6\text{N}_2\text{O}_2$ .

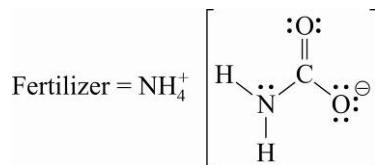
(c)  $2.38^\circ\text{C} / (1.86^\circ\text{C} / \text{m}) = 1.28\text{m solution}$

$(1.28\text{ mol} / 1000\text{ g H}_2\text{O}) \cdot (20\text{ g H}_2\text{O}) = 0.0256\text{ mol solute}$

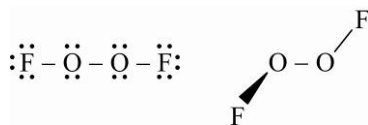
$1.000\text{ g solute} / 0.0256\text{ mol solute} = 39.1\text{ g mol}^{-1}$ .

From the empirical formula, the formula mass is  $78.1\text{ g mol}^{-1}$ ! This says that the molar mass appears to be half the formula mass ; the only way this is possible is if each mol of fertilizer gives rise to two moles of particles. Thus, the fertilizer is likely an ionic compound.

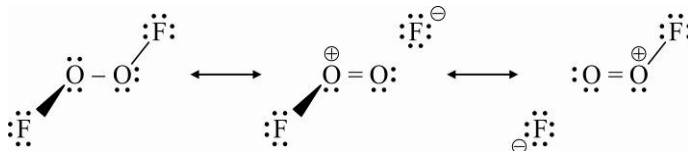
- (d) The most likely cation given the formula is ammonium ion, which leaves an ion with the formula  $\text{CH}_2\text{NO}_2$ . Several chemically plausible structure can be written, but the only one where the anion is not so basic that it would deprotonate the ammonium ion is carbamate,  $\text{NH}_2\text{CO}_2^-$ . (The only other chemical reasonable alternative is hydrazinium formate,  $(\text{NH}_2\text{NH}_3^+)(\text{HCOO}^-)$ ). Ammonium carbamate is a common fertilizer.



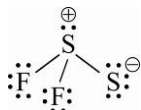
- 110.(a) In FOOF, each oxygen is bent, and the two FOO planes are roughly perpendicular to one another :



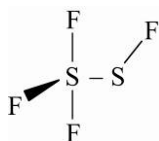
- (b) Electron donation from the oxygen lone pairs in FOOF into the  $\text{O}-\text{F}$   $\sigma^*$  orbitals ("negative hyperconjugation") lengthens the  $\text{O}-\text{F}$  bond (and shortens the  $\text{O}-\text{O}$  bond, which at  $121.7\text{ pm}$  is much shorter than the  $\text{O}-\text{O}$  single bond in  $\text{H}_2\text{O}_2$ ,  $147.4\text{ pm}$ , and is almost the same length as the double bond in  $\text{O}_2$ ,  $121\text{ pm}$ )!



- (c) The stable isomer of  $S_2F_2$  is pyramidal :



- (d) The four-coordinated sulphur has a "sea-saw" geometry, with the equatorial fluorine distinct from the two axial fluorines. (Obviously the fluorine on the divalent sulfur is distinct from the other three as well)! The two axial fluorines are not equivalent because the S–F bond on the divalent sulfur is pointed towards one of them and away from the other one :



- (e)  $SF_4$ , because of its sea-saw geometry, has a dipole moment, while  $SF_6$ , which is octahedral, does not. The favourable dipole-dipole interactions increase the boiling point of  $SF_4$  compared to  $SF_6$ . (The greater number of electrons in  $SF_6$  would be expected to give it greater London dispersion forces compared to  $SF_4$ , but the lower polarisability due to the higher oxidation state in  $SF_6$  may make this effect smaller than one would expect).
- 111.(a) At  $25^\circ C$ , some liquid water is present, so the pressure of water is equal to the vapor pressure, 23.80 mm Hg. Thus the partial pressure of  $O_2$  is  $48.65 \text{ mm Hg} - 23.80 \text{ mm Hg} = 24.85 \text{ mm Hg}$ . From the ideal gas law,

$$n = PV/RT$$

$$n = (24.85 \text{ mm Hg}) (1.000 \text{ L}) / (62.36 \text{ L [mm Hg]} \text{ mol}^{-1} \text{ K}^{-1}) (298.15 \text{ K})$$

$$n = 1.337 \times 10^{-3} \text{ mol}$$

- (b) At  $200^\circ C$ , the total moles of gas are given by

$$n = (355.0 \text{ mm Hg}) (1.000 \text{ L}) / (62.36 \text{ L [mm Hg]} \text{ mol}^{-1} \text{ K}^{-1}) (473.15 \text{ K})$$

$$n = 0.01203 \text{ mol}$$

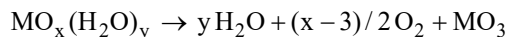
The number of moles of water vapor –  $(0.01203 \text{ mol total gases}) (1.337 \times 10^{-3} \text{ mol } O_2)$

- (c) The total mass of gaseous products

$$= (18.032 \text{ g mol}^{-1}) (0.01069 \text{ mol } H_2O) + (32.00 \text{ g mol}^{-1}) (1.337 \times 10^{-3} \text{ mol } O_2) = 0.2355 \text{ g}.$$

Thus 0.7645 g  $MO_3$  remains.

- (d) The balanced equation for this reaction is :



Since  $x$  is an integer, the smallest amount of  $O_2$  that can be produced per mol  $MO_3$  is 0.5, of any integer multiple of this could be produced.

If  $x = 4$ , then  $2(1.337 \times 10^{-3} \text{ mol}) \text{ mol } MO_3$  is present, and the molar mass of  $MO_3$  is

$$(0.7645 \text{ g}) / (2.674 \times 10^{-3} \text{ mol } MO_3) = 285.9 \text{ g mol}^{-1}.$$

This would imply that the atomic mass of  $M$  is  $285.9 - 3(16.00) = 237.9 \text{ g mol}^{-1}$ .

This is the atomic mass of U.

If  $x = 5$ , then only  $1.337 \times 10^{-3} \text{ mol}$  of  $MO_3$  would be present, and the molar mass of  $MO_3$  would be twice as high, and the atomic mass of  $M$  would be over twice that of uranium. This is impossible. (Higher values of  $x$  would give even higher values of the atomic mass of  $M$ ). Therefore  $M = U$ . Since  $x = 4$ , the 8 : 1 mol ratio of  $H_2O$  to  $O_2$  produced implies that  $y = 4$  as well.



- (e) Naively, one would expect that  $\text{UO}_4(\text{H}_2\text{O})_4$  would have U(VIII). But this is impossible : since U has only 6 valence electrons, it cannot possibly have an oxidation state greater than +6!. The only reasonable formulation is that some of the oxygens in studtite are in the form of peroxide,  $\text{O}_2^{2-}$ , with an oxidation state of  $-1$  for oxygen (not the  $-2$  of oxide). Since it is unlikely that a strongly oxidizing species such as peroxide would coexist with a reduced oxidation state of uranium, uranium must be in its highest oxidation state, +6. Studtite should then be formulated as  $\text{UO}_2(\text{O}_2)(\text{H}_2\text{O})_4$ . This mineral and its partially dehydrated form metastudtite,  $\text{UO}_2(\text{O}_2)(\text{H}_2\text{O})_2$ , are the only known peroxide-containing minerals.

- 112.(a) (i) The electron ionized in Be is a 2s electron, while Ba loses a 6s electron. The higher value of n corresponds to a higher energy, requiring less energy to remove.
- (ii) Since both Be and Ba have an  $ns^2$  configuration, an added electron must enter a new subshell, which is higher in energy. The 2p subshell that would be occupied for Be is much higher in energy, but the next available orbital for Ba is a 5d orbital, which is closer in energy to the 6s orbital.
- (iii) Ba is more electropositive than Be, so it costs much less energy to remove its valence electrons to form the +2 ion. This difference outweighs the somewhat higher lattice energy of  $\text{BeCl}_2$  than  $\text{BaCl}_2$ .
- (iv) Be can only form 4 bonds, so it can only adopt the left structure. The larger  $\text{Ba}^{2+}$  ion can easily achieve the coordination number of 8 required by the right-hand structure.



- (c) Let  $P_M$  = partial pressure of monomeric  $\text{BeCl}_2(\text{g})$   
 $P_D$  = partial pressure of dimeric  $\text{Be}_2\text{Cl}_4(\text{g})$

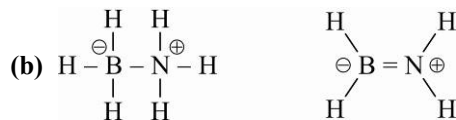
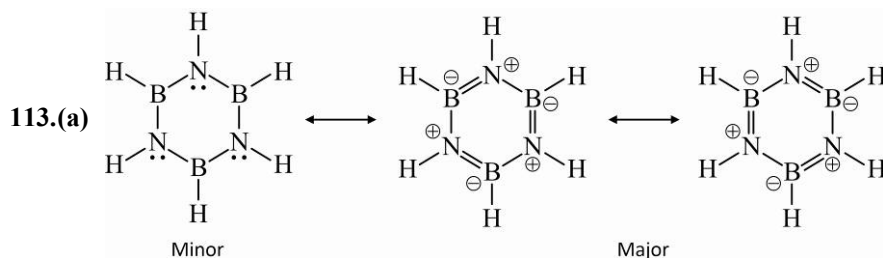
$$P_D / P_M^2 = 2.9$$

$$0.100 - P_M = 2.9 P_M^2$$

$$2.9 P_M^2 + P_M - 0.100 = 0$$

Solving gives  $P_M = 0.081$  bar, and therefore  $P_D = 0.100 - 0.081$  bar = 0.019 bar

The mole fraction of the dimer is  $P_D / 0.100$  bar = 0.19



- (c)  $\text{BH}_3\text{NH}_3$ , 156.4 pm  
 $\text{B}_3\text{N}_3\text{H}_6$ , 142.9 pm  
 $\text{BH}_2\text{NH}_2$ , 139.1 pm

From the Lewis structures, the bond order increases (from 1 to 1.5 to 2) in the order listed, with the bond lengths decreasing accordingly.

- (d) In piperazine, the N – H hydrogen can act as a hydrogen bond donor to another nitrogen lone pair. Changing the H to CH<sub>3</sub> thus decreases the amount of hydrogen bonding and weakens the intermolecular forces. This effect outweighs the increase in London dispersion forces in the –CH<sub>3</sub> derivative.
- (e) Borazine has no hydrogen bond acceptors! There are not really any lone pairs on nitrogen since they are involved in  $\pi$  donation to the boron atoms. So the N – H hydrogens do not participate in hydrogen bonding, and converting them to CH<sub>3</sub> groups leads only to an increase in London dispersion forces and a corresponding increase in boiling point.